Abstracts

of Papers Presented to the American Mathematical Society

Volume 41, Number 3, Issue 201 Summer 2020

Editorial Committee

Carla D. Savage, Chair
Georgia Benkart
Brian D. Boe
Michael L. Lapidus
Steven H. Weintraub

Abstracts for

Fall Central Virtual Sectional Meeting,
September 12–13, 2020 881

Fall Eastern Virtual Sectional Meeting,
October 3–4, 2020 924

ISSN 2689-4831 (online) Pages 881–988

AMS American Mathematical Society

SUBMISSION INFORMATION

ABSTRACTS PUBLISHED IN THIS JOURNAL are those submitted by authors who intend to present them at AMS meetings. The abstracts for each meeting are sorted by the two-digit 2010 Mathematics Subject Classification chosen by the author and appear in the order received. See the 2010 Mathematics Subject Classification at

mathscinet.ams.org/msc/msc2010.html.

THE SUITABILITY OF A PAPER for presentation to the Society is judged from the abstract as outlined in Article X, Section 6 of the bylaws:

> Papers intended for presentation at any meeting of the Society shall be passed upon in advance by a program committee appointed by or under the authority of the Council; and only such papers shall be presented as shall have been approved by such committee. Papers in a form unsuitable for publication, if accepted for presentation, shall be referred to on the program as preliminary communications or reports.

In addition, the editors of the *Abstracts* have adopted the following policy: In order to be accepted for publication, an abstract must have mathematical content. It should not contain libelous, defamatory, or tasteless remarks, commercial promotions, nor political or religious arguments. The AMS assumes no responsibility for the content or inappropriate remarks in any published abstract.

GENERAL INFORMATION ON ABSTRACTS is found at http://www.ams.org/journals/abs.

ABSTRACTS ARE PUBLISHED ELECTRONICALLY from copy submitted by the author. Web submission is the required electronic format. To access the interactive template, visit the *Abstracts* submission page on the AMS website at

http://www.ams.org/cgi-bin/abstracts/abstract.pl.

Step-by-step submission instructions are included in the template. No knowledge of LaTeX is necessary; however, any mathematical displays or accent marks in text or in names must be coded in LaTeX. Requests for general information concerning abstracts may be sent to abs-coord@ams.org.

THE ABSTRACT RECEIPT DEADLINES FOR ALL MEETINGS will be strictly enforced. Late papers cannot be accommodated. When all talks have been scheduled for a given meeting, the attendant abstracts will be available for viewing on the AMS website through the program display for that meeting.

NOTATIONS IN THIS JOURNAL are the following:

* Indicates who will present the paper at the meeting.

SUBSCRIPTION INFORMATION

CHANGES OF ADDRESS AND ALL GENERAL CORRESPONDENCE should be sent to Member and Customer Services, American Mathematical Society, 201 Charles Street, Providence, RI 02904-2213 USA (email: amsmem@ams.org or fax: 401-455-4026); dues payments and orders for AMS publications should be addressed to the Society at P.O. Box 845904, Boston, MA 02284-5904 USA; all correspondence relating to advance registration for meetings should be addressed to the Society, 201 Charles Street, Providence, RI 02904-2213 USA.

Abstracts of Papers Presented to the American Mathematical Society (ISSN 2689-4831 (online)) is published electronically four times a year by the American Mathematical Society at 201 Charles Street, Providence, RI 02904-2213. The subscription price for Volume 41 (2020) is US$212 list, US$169.60 institutional member, US$127.20 individual member. Subscription renewals are subject to late fees. See www.ams.org/customers/macs-faq.html#journal for more information.

© 2020 by the American Mathematical Society. All rights reserved.

PAPERS PRESENTED AT MEETINGS

THIS CALENDAR lists meetings of the Society which have been approved by the Council at which papers may be presented. Programs of Annual Meetings appear in the *Notices* and on the AMS website; programs for sectional meetings appear on the AMS Web pages in the Meetings & Conferences section, and are electronically archived in the *Notices* section on the AMS website.

MEETING #	DATE	PLACE	ABSTRACT DEADLINE	ABSTRACT ISSUE
1161	October 10–11, 2020	Virtual Meeting	EXPIRED	Vol. 41, No. 4
1162	October 24–25, 2020	Virtual Meeting	EXPIRED	Vol. 41, No. 4
1163	January 6–9, 2021	Virtual JMM	EXPIRED	Vol. 42, No. 1
1164	March 13–14, 2021	Virtual Meeting	January 19	Vol. 42, No. 2
1165	March 20–21, 2021	Virtual Meeting	January 19	Vol. 42, No. 2
1166	April 17–18, 2021	Virtual Meeting	February 16	Vol. 42, No. 2
1167	May 1–2, 2021	Virtual Meeting	February 16	Vol. 42, No. 2
1168	July 5–9, 2021	Grenoble, France	TBA	NONE
1169	July 19–23, 2021	Buenos Aires, Argentina	TBA	NONE
1170	September 18–19, 2021	Buffalo, NY	July 27	Vol. 42, No. 3
1171	October 9–10, 2021	Omaha, NE	August 10	Vol. 42, No. 3
1172	October 23–24, 2021	Albuquerque, NM	August 24	Vol. 42, No. 4
1173	November 20–21, 2021	Mobile, AL	September 21	Vol. 42, No. 4
1174	January 5–8, 2022	Seattle, WA	TBA	TBA
1175	March 11–13, 2022	Charlottesville, VA	January 18	TBA
1176	March 19–20, 2022	Medford, MA	January 18	TBA
1177	March 26–27, 2022	West Lafayette, IN	January 25	TBA
1178	May 14–15, 2022	Denver, CO	March 15	TBA

Fall Central Virtual Sectional Meeting, September 12–13, 2020

Abstracts of the 1159th Meeting.

00 ► General

1159-00-50 **Mohammad Mahabubur Rahman*** (mohammad-mahabu.rahman@ttu.edu), Department of Mathematics and Statistics, Texas Tech University, Lubbock, TX 79409. *Serrin-type regularity criteria for the two-dimensional Kuramoto-Sivashinsky equation.*

The Kuramoto-Sivashinsky equation (KSE) draws the attention of many mathematicians and scientists for its chaotic behavior and its similarity to Navier-Stokes equations. In this present study, Serrin-type regularity of two-dimensional Kuramoto-Sivashinsky equation is studied. (Received July 26, 2020)

1159-00-109 **Francis Biney*** (fbiney@utep.edu) and **Maria C. Mariani** (mcmariani@utep.edu). *Comparing predictive performance of statistical learning models on medical data.*

This work investigates the predictive performance of 10 Machine learning models on three medical data including Breast cancer, Heart disease and Prostate cancer. Furthermore, we use the models to identify risk factors that contribute significantly to these diseases.

The models considered ranged from Logistic regression to Deep-feedforward neural network. The models were grouped according to their similarities and learning style; Linear and non-Linear models.

The model selection and hyperparameter tuning were done using bias-variance tradeoff and cross-validation. The model's performance was improved for each method, by applying early stopping, dropout and removed non-significant variables to avoid overfitting. Different predictive performance measures were used including prediction accuracy, sensitivity and specificity depending of the nature of the response distribution whether balanced or imbalanced to compared the models.

The result show that the non-linear models; SVM-GK, RF and FFNN gave the best predictive performance for the breast cancer data. The linear models; LR-PLS, LR-PC ,LR-Ridge and LR-Enet were preferred for heart disease and mixture of linear and non-linear models; LR-Lasso, LR-Enet, RF and GBM best describes the Prostate cancer predictions. (Received August 02, 2020)

1159-00-172 **Wenbo Wu*** (wenbo.wu@utsa.edu). *On sufficient variable screening using log odds ratio filter.*

For high-dimensional data, variable screening is an important step to reduce the scale of the problem, hence, to improve the estimation accuracy and efficiency. In this paper, we propose a new dependence measure which is called the log odds ratio filter under the sufficient variable screening framework. In addition, we propose an ensemble approach to combine the proposed log odds ratio filter with the Kolmogorov filter to achieve supreme performance. We establish the sure screening property of the log odds ratio filter in both marginal screening and sufficient screening methods. Extensive simulations and a real data analysis are provided to demonstrate the usefulness of the proposed log odds ratio filter and the sufficient variable screening procedure. (Received August 04, 2020)

1159-00-193 **Wesley K Thompson*** (wktwktwkt@gmail.com), 130 West Le Roy Avenue, Arcadia, CA 91007. *Covariate Modulated False Discovery Rate.*

Large-scale hypothesis testing, such as genome-wide association studies (GWAS), is facing the challenge of properly controlling for type I error. Traditional multiple-comparison procedures tend to be underpowered. Procedures that control false discovery rate (fdr) are more powerful; yet most of proposed methods treat all hypothesis tests as exchangeable and independent, ignoring auxiliary covariates that may influence the distribution of the test statistics and ignore the prior knowledge of the correlations structures of Single nucleotide polymorphisms (SNPs). The current work incorporates Hidden Markov Random Field (HMRF) into a Bayesian two-group mixture model for a covariate-modulated local false discovery rate (cmfdr) on SNPs that are not independent. The simulations and real data application are carried out to compare the performance in terms of sensitivity, specificity and false discovery proportion (FDP) between the new method and the independent cmfdr, as well as the correlation-based fdr without covariates. (Received August 04, 2020)

05 ► Combinatorics

1159-05-32 **Matthias Beck** (mattbeck@sfsu.edu), San Francisco, CA, **Benjamin Braun** (benjamin.braun@uky.edu), Lexington, KY , and **Andrés R. Vindas Meléndez*** (andres.vindas@uky.edu), Lexington, KY. *Decompositions of Ehrhart h^*-Polynomials for Rational Polytopes.*

The Ehrhart quasipolynomial of a rational polytope P encodes the number of integer lattice points in dilates of P, and the h^*-polynomial of P is the numerator of the accompanying generating function. We provide two decomposition formulas for the h^*-polynomial of a rational polytope. The first decomposition generalizes a theorem of Betke and McMullen for lattice polytopes. We use our rational Betke–McMullen formula to provide a novel proof of Stanley's Monotonicity Theorem for the h^*-polynomial of a rational polytope. The second decomposition generalizes a result of Stapledon, which we use to provide rational extensions of the Stanley and Hibi inequalities satisfied by the coefficients of the h^*-polynomial for lattice polytopes. Lastly, we apply our results to rational polytopes containing the origin whose duals are lattice polytopes. (Received July 12, 2020)

1159-05-33 **Justin Chen**, **Christopher Eur*** (chrisweur@gmail.com), **Greg Yang** and **Mengyuan Zhang**. *Free resolutions in learning theory.*

Function classes are collections of Boolean functions on a finite set, which are fundamental objects of study in theoretical computer science. We study algebraic properties of ideals associated to function classes previously defined by the third author. For a wide family of function classes, we show that the multigraded betti numbers are the Mobius numbers of associated posets. We also discuss some open directions. (Received July 13, 2020)

1159-05-44 **Marcel Celaya**, **Georg Loho** and **Chi Ho Yuen*** (chi_ho_yuen@brown.edu). *Oriented Matroids from Triangulations of Products of Simplices.*

We introduce a construction of oriented matroids from a triangulation of a product of two simplices. For this, we use the structure of such a triangulation in terms of polyhedral matching fields. The oriented matroid is composed of compatible chirotopes on the cells in a matroid subdivision of the hypersimplex, which might be of independent interest. We also derive a topological representation of the oriented matroid using a variant of

Viro's patchworking. If time permits, we will mention the extension to matroids over hyperfields and potential applications in complexity theory. (Received July 21, 2020)

1159-05-46 **Quang Dao**, **Christina Meng**, **Julian Wellman**, **Zixuan Xu**, **Calvin Yost-Wolff** and **Teresa Yu*** (twyu@umich.edu). *Polytopal Realizations and Faces of Extended Nestohedra.*

Motivated by Laurent phenomenon algebras developed by Lam and Pylyavskyy, we introduce a family of simple convex polytopes called *extended nestohedra*. These polytopes extend the well-studied nestohedra and generalize the graph cubeahedra introduced by Devadoss, Heath and Vipismakul. We show that extended nestohedra share many important properties with nestohedra, such as being realizable as the boundaries of simple polytopes and having non-negative γ-vectors when the polytopes are flag. We also identify isomorphisms between duals of extended nestohedra and duals of nestohedra. (Received July 23, 2020)

1159-05-48 **Nicolau C Saldanha*** (saldanha@puc-rio.br), R Mq S Vicente 225, Depto de Matematica, PUC-Rio, Rio de Janeiro, 22451900, Brazil. *Domino tilings in dimension 3.*

Domino tilings in dimension 2 have been extensively studied and there are several deep and remarkable theorems. Almost without exception, similar problems in dimension 3 or higher are much harder.

In this talk we consider the simplest local move among domino tilings of a given compact region: a flip consists of removing two dominoes and placing them back in a different position. In dimension 2, Thurston proved that any two tilings of a simply connected region can be joined by a finite sequence of flips.

In dimension 3, the question is far subtler. There exists an integer valued invariant under flips, the twist. There are also tilings which admit no flip, and these give us examples of pairs of tilings with the same twist but in different connected components under flips. We prove that, under suitable hypotheses, it is almost always true that, if two tilings have the same twist then they are in the same connected component. Furthermore, sizes of connected components follow a normal distribution.

This includes joint work with C. Klivans, J. Freire and P. Milet.

References:

arXiv:1912.12102, arXiv:2007.09500, arXiv:2007.08474, arXiv:1702.00798, arXiv:1410.7693. (Received July 22, 2020)

1159-05-59 **Kevin M Marshall*** (kmarsh729@ku.edu). *A Hopf Monoid on Set Families.* Preliminary report.

We study a Hopf monoid **SetFam** whose underlying objects are arbitrary set families, with product given by join and coproduct defined by a general notion of restriction and contraction, analogous to that for matroids. Of particular interest is the Hopf submonoid **LOI** spanned by families $J(P)$ of order ideals of finite posets P. We present progress on a topological approach, inspired by the work of Aguiar and Ardila on generalized permutahedra, toward a more general cancellation-free formula for the antipode in **LOI**. (Received July 27, 2020)

1159-05-64 **Sheila Sundaram*** (shsund@comcast.net). *The reflection representation in the homology of subword order.* Preliminary report.

This paper examines the homology representation of the symmetric group S_n on rank-selected subposets of subword order. We prove that the action on the rank-selected chains is a nonnegative integer combination of tensor powers of $S_{(n-1,1)}$, and show that its Frobenius characteristic is h-positive and supported on the set $T_1(n) = \{h_\lambda : \lambda = (n-r, 1^r), r \geq 1\}$.

We give an explicit formula for the homology module for words of bounded length, as a sum of tensor powers of the S_n-irreducible $S_{(n-1,1)}$ indexed by the partition $(n-1, 1)$. This recovers, as a special case, a theorem of Björner and Stanley for words of length at most k. We exhibit a curious duality in homology in the case when one rank is deleted.

We also show that in many cases, the rank-selected homology modules, modulo one copy of the reflection representation, are h-positive and supported on the set $T_2(n) = \{h_\lambda : \lambda = (n-r, 1^r), r \geq 2\}$. (Received July 28, 2020)

1159-05-67 **Pamela E. Harris** (peh2@williams.edu), **Marissa Loving** (mloving6@gatech.edu), **Juan Ramirez** (jjramirez8@uh.edu), **Joseph Rennie** (rennie2@illinois.edu), **Gordon Rojas Kirby** (gkirby@math.ucsb.edu), **Eduardo Torres Davila** (etdavila10@gmail.com) and **Fabrice Odny Ulysse*** (fulysse@nd.edu). *Visualizing the Support of Kostant's Weight Multiplicity Formula for the Rank Two Lie Algebras.*

The multiplicity of a weight in a finite-dimensional irreducible representation of the rank 2 Lie algebras can be computed via Kostant's weight multiplicity formula. This formula consists of an alternating sum over a

finite group and involves Kostant's partition function. Our main result describes the terms that contribute nontrivially to Kostant's weight multiplicity formula. By taking a geometric and combinatorial approach, we provide concrete visualizations of these sets for all pairs of integral weights lambda and mu of all rank 2 Lie algebras and show that the diagrams associated to our main result present new and surprising symmetry. This research was conducted during the 2019 Mathematical Sciences Research Institute Undergraduate Program in Berkeley, CA under the direction of Dr. Pamela E. Harris. (Received July 28, 2020)

1159-05-73 **Jean A Guillaume*** (guillaumej@sacredheart.edu), 933 East 79th Street, Brooklyn, NY 11236. *The NGD-Circulant Graphs of the Form $C_n(1,k)$.*

Nordhaus and Gaddum showed, for any given graph G, that $\chi(G)+\chi(\overline{G}) \leq n+1$, where χ denotes the chromatic number and n the number of vertices. Collins and Trenk established an analogous result for the distinguishing chromatic number. They proved, for any graph G, that $\chi_D(G) + \chi_D(\overline{G}) \leq n + D(G)$, where $D(G)$ is the distinguishing number of the graph. They called the class of graphs that satisfy equality in this bound NGD-graphs after Nordhaus and Gaddum. In this talk, we present our investigation of the distinguishing chromatic number for the complements of circulant graphs $G = C_n(1,k)$. We also discuss our characterization of the NGD-graphs for this particular class of graphs as well as our improvement of the bound of Collins and Trenk for this family. Lastly, we show how to extend our investigation of the distinguishing chromatic number to a larger class of circulant graphs $G = C_n(1, k_1, \cdots, k_m)$.
(Received July 29, 2020)

1159-05-74 **Alex McDonough*** (amcd@math.brown.edu). *A Combinatorial Mapping for the Higher-Dimensional Matrix-Tree Theorem.*

For a natural class of matroids that are equipped with a multiplicity function, we provide a family of combinatorially meaningful maps from the sandpile group of a matroid to its bases such that the size of the preimage of every basis is its multiplicity squared. This generalizes a bijection given by Backman, Baker, and Yuen and extends work by Duval, Klivans, and Martin. Based on work from https://arxiv.org/abs/2007.09501 (Received July 29, 2020)

1159-05-76 **Carrie Frizzell*** (cfrizz@ksu.edu). *The Tree Growing Sequence.*

Spanning trees, G-parking functions, and the Tutte polynomial are familiar characters in combinatorics. In an effort to generalize bijective correspondences between them, we define an object called a tree growing sequence. The definition is inspired by "splitting" operations arising from the recursive formula for computing the Tutte polynomial. One can apply a tree growing sequence to a G-parking function in a way that simultaneously assigns to it both a spanning tree and a monomial. We construct maps from a few known algorithms to the collection of tree growing sequences, and discuss commutative diagrams involving these maps. A relation to splitting operations for zonotopal tilings will be briefly described. (Received July 29, 2020)

1159-05-88 **Lei Xue*** (lxue@uw.edu). *A Proof of Grünbaum's Lower Bound Conjecture for general polytopes.*

In 1967, Grünbaum conjectured that any d-dimensional polytope with $d+s \leq 2d$ vertices has at least

$$\phi_k(d+s,d) = \binom{d+1}{k+1} + \binom{d}{k+1} - \binom{d+1-s}{k+1}$$

k-faces. In the talk we will discuss the proof of this conjecture and also characterize the cases in which equality holds. (Received July 31, 2020)

1159-05-93 **Anastasia M Chavez***, amrchavez@ucdavis.edu, and **Carolina Benedetti** and **Daniel Tamayo Jiménez**. *Characterizing quotients of positroids.*

We characterize quotients of specific families of positroids. Positroids are a special class of representable matroids introduced by Postnikov in the study of the nonnegative part of the Grassmannian. Postnikov defined several combinatorial objects that index positroids. In this talk, we make use of two of these objects to combinatorially characterize when certain positroids are quotients. Furthermore, we conjecture a general rule for quotients among arbitrary positroids on the same ground set. This is joint work with Carolina Benedetti and Daniel Tamayo Jiménez. (Received July 31, 2020)

1159-05-104 **Margaret Bayer**, **Bennet Goeckner**, **Su Ji Hong**, **Tyrrell McAllister**, **McCabe Olsen**, **Casey Pickney**, **Julianne Vega*** (jvega30@kennesaw.edu) and **Martha Yip**.
Lattice polytopes from Schur and symmetric Grothendieck polynomials.

In this presentation we will consider Newton polytopes arising from two families of polynomials in algebraic combinatorics: Schur polynomials and inflated symmetric Grothendieck polynomials. In both cases, we show that these polytopes have integer decomposition property. (Received August 02, 2020)

1159-05-112 **Jesus A De Loera*** (deloera@math.ucdavis.edu), Dept. of Mathematics, University of California, Davis, Davis, CA 95616, and **Christos Athanasiadis** and **Zhenyang Zhang**.
On the CW-complex of Monotone Polyhedral Paths. Preliminary report.

Given a polytope P and linear functional, we study the directed paths inside the oriented graph of P (oriented by the linear function). We are interested in the properties of these monotone paths. For example, how many are there?

The monotone paths have a metric space structure which can be used to count how many are there or to generate them randomly. This comes from a very natural graph (the flip-path graph) and CW complex structures (cellular-strings complex) on the set of monotone paths. These objects are important in topological combinatorics (they were introduced and by Billera, Sturmfels, Gelfand, Kapranov, Zelevinsky and others in the early 1990's) and in discrete optimization (In particular the complexity of the simplex method with work going back to Klee).

I will discuss bounds on the number of monotone paths, and on the the diameter of the space of monotone paths (how far are two monotone paths from each other?). The picture is fairly complete in dimension three, but plenty of open problems remain for high dimensional polytopes..

The new theorems presented in my talk come from a recent paper (in Arxiv), joint work with Christos Athanasiadis (U. Athens) and Zhenyang Zhang (UC Davis). (Received August 02, 2020)

1159-05-125 **Galen Dorpalen-Barry*** (dorpa003@umn.edu). *Whitney Numbers for Poset Cones.*

Hyperplane arrangements dissect $\mathbb{R}^n$ into connected components called chambers, and a well-known theorem of Zaslavsky counts chambers as a sum of nonnegative integers called Whitney numbers of the first kind. His theorem generalizes to count chambers within any cone defined as the intersection of a collection of halfspaces from the arrangement, leading to a notion of Whitney numbers for each cone. Cones within the braid arrangement, consisting of the reflecting hyperplanes $x_i = x_j$ inside $\mathbb{R}^n$ for the symmetric group, thought of as the type A_{n-1} reflection group. Here, cones correspond to posets, chambers within the cone correspond to linear extensions of the poset, and the Whitney numbers of the cone interestingly refine the number of linear extensions of the poset. We interpret this refinement for all posets as counting linear extensions according to a statistic that generalizes the number of left-to-right maxima of a permutation. This is joint work with Vic Reiner and Jang Soo Kim, see 1906.00036. (Received August 03, 2020)

1159-05-128 **Anton Dochtermann*** (dochtermann@txstate.edu), **Eli Meyers**, **Raghav Samavedam** and **Alex Yi**. *Cycle and circuit chip-firing on graphs.*

Motivated by the notion of chip-firing on the dual graph of a planar graph, we consider 'cycle chip-firing' on an arbitrary graph G. The firing rule is governed by $\mathcal{L}^*(G)$, the dual Laplacian of G determined by choosing a basis for the lattice of integral flows. We show that any graph admits such a basis so that $\mathcal{L}^*(G)$ is an M-matrix, leading to a firing rule on these basis elements that is avalanche finite. This follows from a more general result on bases of integral lattices that may be of independent interest. Our results provide a notion of 'z-superstable' cycle configurations that are in bijection with the set of spanning trees of G. We show that for planar graphs, as well as for the graphs K5 and K3,3, one can find such an M-basis whose elements correspond to circuits of the underlying graph. We consider the question for arbitrary graphs and address some other open questions. Much of this work was done as part of a Mathworks summer program at Texas State University. (Received August 03, 2020)

1159-05-141 **Federico Castillo**, **Joseph Doolittle**, **Bennet Goeckner*** (goeckner@uw.edu), **Michael Ross** and **Li Ying**. *Type polytopes and products of simplices.*

For polytopes P and Q, we say that Q is a *weak Minkowski summand* of P if $Q + R = \lambda P$ for some polytope R and scalar λ. The type polytope Type(P) encodes all of the weak Minkowski summands of P. In general, combinatorially isomorphic polytopes can have different type polytopes. We show that if P is combinatorially isomorphic to a product of simplices, then Type(P) is a simplex. In particular, the type polytope of any combinatorial cube is a simplex. We also provide constructions of n-gons that maximize the f-vector and n-gons that minimize the f-vector of the type polytope for each n. (Received August 03, 2020)

1159-05-150 **Federico Castillo***, Inselstraße 22 04103, Leipzig, Germany, and **Jeremy Martin** and **Jose Samper**. *Hopf monoids of ordered simplicial complexes.*

We study a class of simplicial complexes over an ordered set defined by the property that the restriction to every initial segment is pure. This class contains ordered matroids, shifted complexes, broken circuit complexes, and many more. We show that this class gives rise to a Hopf Monoid and compute its antipode in some particular cases. This is joint work with Jeremy Martin and Jose Samper. (Received August 03, 2020)

1159-05-188 **Alexander Lazar** and **Michelle L Wachs*** (wachs@math.miami.edu). *New combinatorial models for the Genocchi and median Genocchi numbers.*

The Genocchi numbers and median Genocchi numbers are classical sequences of numbers, which have been defined in many ways. In 1974, Dumont gave an interpretation of the Genocchi numbers in terms of permutations now known as Dumont permutations, and in 1981 Barsky and Dumont did the same for the median Genocchi numbers. They also obtained generating function formulas for these numbers. Here we present some new models for these numbers in terms of permutations that are related to the Dumont permutations. These models arose from studying the chromatic polynomials of certain bipartite graphs known as Ferrers graphs. We also obtain generating function formulas for the chromatic polynomials, which reduce to those of Barsky and Dumont. (Received August 04, 2020)

1159-05-201 **Hailong Dao*** (hdao@ku.edu), 405 Snow Hall, 1460 Jayhawk Blvd, Lawrence, KS 66045, and **Justin Lyle** and **Joseph Doolittle**. *Minimal Cohen-Macaulay complexes.*

We define and study the notion of a minimal Cohen-Macaulay simplicial complex. We prove that any Cohen-Macaulay complex is shelled over a minimal one in our sense, and we give sufficient conditions for a complex to be minimal Cohen-Macaulay. We show that many interesting examples of Cohen-Macaulay complexes in combinatorics are minimal, including Rudin's ball, Ziegler's ball, the dunce hat, and recently discovered non-partitionable Cohen-Macaulay complexes. We further provide various ways to construct such complexes (Received August 04, 2020)

1159-05-204 **Mark R Pederson***, Mark R Pederson, Department of Physics, El Paso, TX 79968. *Determination of classical representations of quantum phenomena through data-enabled high throughput searchers.* Preliminary report.

Fermi-Lowdin orbitals provide a localized representation of quantum wave functions through the use of a quasi classical electronic geometry referred to as Fermi-Orbital descriptors (FODs).[1,2] These orbitals were specifically derived because an improvement of density functional theory, known as the self-interaction correction, required a mechanism for determining unitary matrices that is explicit dependent on and derived from an electronic density that itself is constructed from single-particle wave functions. Determination of the so-called FODs, which define the electronic geometry used for creating these local orbitals ,require a high-throughput search strategy. This strategy will be discussed along with a presentation of the applications of this method to molecules, clusters and qubits.

[1] Fermi orbital derivatives in self-interaction corrected density functional theory. Applications to closed shell atoms, M.R. Pederson, J. Chem. Physics. 142, 0641 (2015).

[1]A multiferroic molecular magnetic quart, A.I. Johnson, F.Islam, C.M. Canali and M.R. Pederson, JCP 151,1741-4 (2019. (Received August 05, 2020)

11 ► Number theory

1159-11-174 **Alison Beth Miller*** (alimil@umich.edu). *Number theory questions related to Seifert matrices and Alexander modules.*

This talk will use knot theory and Seifert matrices to motivate some interesting asymptotic counting problems in number theory. These questions give one way of answering quantitatively "How much information is contained in the Alexander module which is not already present in the Alexander polynomial?" (Received August 04, 2020)

13 ► Commutative rings and algebras

1159-13-4 **Adam Lee Boocher*** (aboocher@sandiego.edu). *Large Lower Bounds for Betti Numbers of Graded Modules.*

Let M be a finitely-generated graded module over a polynomial ring. I'll discuss the state of the art concerning lower bounds for the betti numbers of M including recent results that give large lower bounds for the first half of the betti numbers in many cases of interest. (Received February 25, 2020)

1159-13-20 **Delio Jaramillo** and **Rafael H Villarreal*** (vila@math.cinvestav.mx). *The v-number of edge ideals.*

The aim of this work is to study the v-number of edge ideals of clutters. We relate the v-number with the regularity of edge ideals and classify W_2 graphs. If the edge ideal of a graph has second symbolic power Cohen–Macaulay, we show that the graph is edge-critical. (Received June 29, 2020)

1159-13-27 **Thiago H Freitas**, **Victor Hugo Jorge Perez**, **Roger A Wiegand** and **Sylvia M Wiegand***, Math, UNL, Lincoln, NE 685880130. *Vanishing of Tor over fiber products.*

Let $(S, \mathfrak{m}, k)$ and $(T, \mathfrak{n}, k)$ be local rings, and let R denote their fiber product over their common residue field k. Inspired by work of Nasseh and Sather-Wagstaff, we explore consequences of vanishing of $\operatorname{Tor}_m^R(M, N)$ for various values of m, where M and N are finitely generated R-modules. (Received July 06, 2020)

1159-13-34 **Uwe Nagel*** (uwe.nagel@uky.edu), University of Kentucky, Department of Mathematics. *Stabilization in Sequences of Symmetric Ideals and Modules.*

Consider a sequence of compatible symmetric ideals or symmetric modules over polynomial rings in more and more variables. We discuss a framework for establishing stabilization results and studying asymptotic properties of objects in such a sequence. (Received July 14, 2020)

1159-13-39 **Sonja Mapes*** (smapes1@nd.edu), 255 Hurley Hall, Notre Dame, IN 46556, and **Kuei-Nuan Lin**. *Computing projective dimension of monomial ideals via associated hypergraphs and lcm-lattices.*

Given a square-free monomial ideal I in a polynomial ring R over a field k, we would like to know the projective dimension of R/I. We recall the definition of the lcm-lattice of a monomial ideal introduced by Gasharov, Peeva and Welker, and the definition of the dual hypergraph of a square-free monomial ideal introduced by Kimura, Terai and Yoshida. Our work focuses on the relationship between the lcm-lattice and the dual hypergraph of a given square-free monomial ideal. We use the properties of lcm-lattice to find whether two different dual hypergraphs have the same projective dimension, and thus are able to extend some of the results by Lin and Mantero which compute the projective dimensions for ideals with certain hypergraphs. (Received July 16, 2020)

1159-13-49 **Nursel Erey*** (nurselerey@gtu.edu.tr), **Jürgen Herzog**, **Takayuki Hibi** and **Sara Saeedi Madani**. *Squarefree powers of edge ideals.*

Let G be a graph with edge ideal $I(G)$. The kth squarefree power of $I(G)$ is generated by the monomials of the form $\prod_{e\in M} e$ where M is a matching in G of size k. In this talk, we will give some bounds for the regularity of squarefree powers of edge ideals. We will also discuss the question of when such powers are linearly related or have linear resolution. (Received July 23, 2020)

1159-13-58 **Kuei-Nuan Lin*** (kul20@psu.edu) and **Yi-Huang Shen**. *Blow-up algebras of rational normal scrolls and their secant varieties.*

We show that the special fiber rings of general rational normal scrolls are Cohen–Macaulay and compute their Castelnuovo–Mumford regularities. Then we study the secant varieties of balanced rational normal scrolls. We describe the defining equations of their associated Rees rings and compute the Castelnuovo–Mumford regularities of their special fiber rings. (Received July 27, 2020)

1159-13-60 **Caitlyn Booms*** (cbooms@wisc.edu), **Daniel Erman** and **Jay Yang**. *Heuristics for ℓ-torsion in Veronese Syzygies.* Preliminary report.

We explore the question, "To what extent should we expect the syzygies of Veronese embeddings of projective space to depend on the characteristic of the field?" Using an heuristic model involving random flag complexes, we produce conjectures towards answering this question. (Received July 27, 2020)

1159-13-65 **Claudiu Raicu*** (craicu@nd.edu), **Steven V Sam** and **Jerzy Weyman**. *Equations and syzygies for powers of binary forms.*

For $a, b \geq 1$, we consider the variety X_b^a of a-th powers of binary forms of degree b, with coefficients in a field of characteristic zero. We show that the defining ideal of X_b^a is generated in degree $b+1$ and has a linear minimal free resolution, by establishing the surjectivity of a generalized Howe map for SL_2-representations in the appropriate range, extending results of Abdesselam and Chipalkatti that hold when $b = 2$. Joint work with Steven V Sam and Jerzy Weyman. (Received July 28, 2020)

1159-13-75 **Steven Dale Cutkosky*** (cutkoskys@missouri.edu), Dept. Math., University of Missouri, Columbia, MO 65211. *The Minkowski equality of filtrations.*

Suppose that R is an analytically irreducible or excellent local domain with maximal ideal m_R. We consider multiplicities and mixed multiplicities of R by filtrations of m_R-primary ideals. We show that the theorem of Teissier, Rees and Sharp, and Katz, characterizing equality in the Minkowski inequality for multiplicities of ideals, is true for divisorial filtrations, and for the larger category of bounded filtrations. This theorem is not true for arbitrary filtrations of m_R-primary ideals. (Received July 29, 2020)

1159-13-81 **Todd Morra***, tmorra@clemson.edu. *Canonical modules of Stanley-Reisner rings of certain independence complexes and related free resolutions.* Preliminary report.

We use results of D'Alì, Fløystad, and Nematbakhsh [DFN] to explicitly describe the canonical modules of Stanley-Reisner rings of certain Cohen-Macaulay simplicial complexes. We will also report on progress towards finding a DG-algebra structure for a resolution given in [DFN] when restricting to such complexes. (Received July 30, 2020)

1159-13-84 **Alessandra Costantini*** (alessanc@ucr.edu). *The Rees algebra of almost linearly presented modules.* Preliminary report.

Let E be an n-generated module over a Noetherian ring R. The Rees algebra $\mathcal{R}(E)$ of E can be described as a quotient of a polynomial ring in n variables with coefficients in R modulo an ideal $\mathcal{J}$, which is called the *defining ideal* of $\mathcal{R}(E)$. Determining the defining ideal of Rees algebras is in general difficult, even in the case of Rees algebras of ideals. However, the problem becomes treatable when enough information is known on a presentation matrix of E as an R-module.

In this talk we will provide a description of $\mathcal{J}$ in the case when E is a module of projective dimension one admitting a presentation matrix where all but one column are linear. Our result generalizes a theorem of Boswell and Mukundan for perfect ideals of height two. The talk is based on an ongoing project, which is currently available on arXiv (arXiv:1811.08402v1). (Received July 30, 2020)

1159-13-85 **Alessandra Costantini*** (alessanc@ucr.edu). *Cohen-Macaulay property of the fiber cone of a module.* Preliminary report.

Let R be a Noetherian local ring and let E be a finite R-module. The fiber cone of E is the graded algebra $\mathcal{F}(E)$ defined by tensoring the Rees algebra $\mathcal{R}(E)$ with the residue field of R. In 2003 Simis, Ulrich and Vasconcelos showed that the study of the Cohen-Macaulay property of the Rees algebra $\mathcal{R}(E)$ can be reduced to the case of Rees algebras of ideals, by means of the so called *generic Bourbaki ideals*. The Cohen-Macaulay property of Rees algebras and fiber cones are usually unrelated. However, in this talk I will show that sometimes generic Bourbaki ideals can effectively be used in order to study the Cohen-Macaulay property of the fiber cone $\mathcal{F}(E)$ as well. I will also provide classes of modules whose fiber cone is Cohen-Macaulay, generalizing results of Corso, Ghezzi, Polini and Ulrich and of Montano. (Received July 30, 2020)

1159-13-87 **Laura Felicia Matusevich**, Department of Mathematics, Mailstop 3368, Texas A&M University, College Station, TX 77843-3368, and **Aleksandra Sobieska*** (ola@math.tamu.edu), UW-Madison Department of Mathematics, Van Vleck Hall, 480 Lincoln Hall, Madison, WI 53706. *Toward Free Resolutions Over Scrolls.*

Free resolutions over the polynomial ring have a storied and active record of study. However, resolutions over other rings are much more mysterious; even simple examples can be infinite! In these cases, we look to any combinatorics of the ring to glean information. This talk will present a minimal free resolution of the ground field over the semigroup ring arising from rational normal 2-scrolls, and (if time permits) a computation of the Betti numbers of the ground field for all rational normal k-scrolls. (Received July 30, 2020)

1159-13-89 **Susan Cooper**, **Sabine El Khoury**, **Sara Faridi*** (faridi@dal.ca), **Sarah Mayes-Tang**, **Susan Morey**, **Liana Sega** and **Sandra Spiroff**. *Morse resolutions of powers of square free monomial ideals of projective dimension 1.* Preliminary report.

A monomial ideal I is said to have a free resolution supported on a simplicial or cellular complex Δ if the topological chain complex of Δ can be homogenized, using the generators of I, to give a free resolution of I. For example, the Taylor resolution is a simplex on q vertices supporting a free resolution of any monomial ideal with q generators.

It is known that if I is a monomial ideal of projective dimension 1, then it has a minimal free resolution supported on a one-dimensional simplicial complex: a graph which is a tree. In this talk, using discrete Morse theory, we will show that all powers of I have resolutions supported on a CW complex, when I is a square-free monomial ideal.

We will explain the basic ideas behind discrete Morse theory and the construction of the CW complex.

This talk is part of a project that was started during a "Women in Commutative Algebra" meeting in Banff in 2019, and is directly related to the two talks below which will be given in the session "Commutative Algebra" by collaborators:

Cellular resolutions of powers of ideals of projective dimension 1, Part I (Sabine El Khoury);

Cellular resolutions of powers of ideals of projective dimension 1, Part II (Liana Sega). (Received July 31, 2020)

1159-13-91 **Monica Lewis*** (malewi@umich.edu). *The local cohomology of a parameter ideal with respect to an arbitrary ideal.*

Let R be a regular ring, let J be an ideal generated by a regular sequence of codimension at least 2, and let I be an ideal containing J. We give an example of a module $H^3_I(J)$ with infinitely many associated primes, answering a question of Hochster and Núñez-Betancourt in the negative. In fact, for $i \leq 4$, we show that under suitable hypotheses on R/J, $\operatorname{Ass} H^i_I(J)$ is finite if and only if $\operatorname{Ass} H^{i-1}_I(R/J)$ is finite. Our proof of this statement involves a novel generalization of an isomorphism of Hellus, which may be of some independent interest. The finiteness comparison between $\operatorname{Ass} H^i_I(J)$ and $\operatorname{Ass} H^{i-1}_I(R/J)$ tends to improve as our hypotheses on R/J become more restrictive. To illustrate the extreme end of this phenomenon, at least in the prime characteristic $p > 0$ setting, we show that if R/J is regular, then $\operatorname{Ass} H^i_I(J)$ is finite for all $i \geq 0$. (Received July 31, 2020)

1159-13-94 **Bruce Olberding*** (bruce@nmsu.edu), Department of Mathematical Sciences, Las Cruces, NM 88011. *Unmixedness for finitely generated algebras over zero-dimensional rings.* Preliminary report.

We discuss how several foundational properties for the theory of finitely generated algebras over a field hold if the field is replaced by a commutative ring R of Krull dimension 0. The talk can be summarized with the slogan, "A complete intersection over a zero-dimensional ring is Cohen-Macaulay." In the non-Noetherian context, this is a slogan rather than a theorem since both "complete intersection" and "Cohen-Macaulay" admit multiple interpretations. We unpack this slogan and show that a number of fundamental properties hold regardless of whether the base ring R is Noetherian. (Received July 31, 2020)

1159-13-99 **Susan M. Cooper*** (susan.cooper@umanitoba.ca), Department of Mathematics, 476 Machray Hall, University of Manitoba, Winnipeg, MB R3T 2N2, Canada, and **Sabine El Khoury**, **Sara Faridi**, **Sarah Mayes-Tang**, **Susan Morey**, **Liana Sega** and **Sandra Spiroff**. *Simplicial Resolutions for Powers of Square-free Monomial Ideals.* Preliminary report.

In 1966 Diana Taylor established a method to construct a free resolution of an ideal I generated by p monomials using the simplicial chain maps of a simplex on p vertices. Work of Bayer, Peeva and Sturmfels later extended Taylor's work to show that as long as such a simplicial complex satisfies certain homological conditions, it can support a free resolution of I.

In this talk we construct a family of simplicial complexes where the 2nd complex in the family supports a free resolution of the second power I^2 where I is a square-free monomial ideal. We will also discuss the generalization of our work to higher powers of I. This project stems from work initiated at a BIRS "Women in Commutative Algebra" meeting in Fall 2019. (Received August 01, 2020)

1159-13-101 **Huy Tai Ha*** (tha@tulane.edu) and **Takayuki Hibi**. *Max Min vertex cover and the size of Betti tables.*

Let G be a finite simple graph and let $I(G)$ be its edge ideal. We discuss the question of what pairs of integers could be the projective dimension and the regularity of $I(G)$. We give a characterization for when the projective

dimension of $I(G)$ attains its minimal value. We also find the spectrum of the projective dimension when the regularity is 2. (Received August 01, 2020)

1159-13-103 **Youngsu Kim*** (youngsu.kim@csusb.edu), **Lance E Miller** and **Wenbo Niu**. *The log canonical threshold of a generic link.*

Linkage is a classical topic in algebraic geometry and commutative algebra. Fix an affine space A. We say two subschemes X, Y of A are linked if their union is a complete intersection in A, and X and Y do not have a common component. Two linked subschemes share several properties in common. Linkage has been studied by various people, Artin-Nagata, Peskine-Szprio, Huneke-Ulrich, to name a few.

In 2014, Niu showed that if Y is a generic link of a variety X, then $\mathrm{lct}(A, X) \leq \mathrm{lct}(A', Y)$, where lct stands for log canonical threshold and A' is an extension of A. In this talk, we show that if Y is a generic link of a determinantal variety X, then X and Y have the same log canonical threshold. This is joint work with Lance E. Miller and Wenbo Niu. If time permits, we will talk about an extension of this result. (Received August 01, 2020)

1159-13-105 **Jennifer Biermann**, **Hernan De Alba Casillas**, **Federico Galetto**, **Satoshi Murai**, **Uwe Nagel**, **Augustine O'Keefe**, **Tim Romer** and **Alexandra Seceleanu*** (aseceleanu@unl.edu). *Betti numbers of symmetric shifted ideals.*

An ideal of a polynomial ring is symmetric if it is fixed by the natural action of the symmetric group. We introduce a new class of monomial ideals, which we call symmetric shifted ideals, and which can be considered as an analogue of stable monomial ideals within the class of symmetric ideals. The homological properties of symmetric shifted ideals and their applications will be discussed. (Received August 02, 2020)

1159-13-106 **Jason McCullough** and **Alexandra Seceleanu*** (aseceleanu@unl.edu). *Three short stories featuring idealization.*

Idealizations (also known as trivial extensions) of modules have been introduced by Nagata to extend results about ideals to modules. This construction also comes up in combinatorics, where it is featured in a famous example due to Stanley of a symmetric h-vector which is not unimodal.

This talk will focus on the homological properties of graded idealizations and their Hilbert functions. Several examples of graded Artinian Gorenstein rings obtained by idealization and exhibiting surprising behaviors will be analyzed in regards to the Koszul property, the subadditivity of Betti numbers, and the unimodality of Hilbert functions. (Received August 02, 2020)

1159-13-107 **Mihai Fulger**, **Alexandra Seceleanu*** (aseceleanu@unl.edu), **Tomasz Szemberg** and **Justyna Szpond**. *Growth of the symbolic defect.* Preliminary report.

The symbolic defect of an ideal, introduced by Galetto-Geramita-Shin-Van Tuyl, is a measure of the discrepancy between its ordinary and symbolic powers.

This talk focuses on the case of ideals with Noetherian symbolic Rees algebra, where the symbolic defect grows quasi-polynomially. We explore the connections between the polynomial order of growth of the symbolic defect function and the notion of symbolic analytic spread recently introduced by Dao-Montano. We also consider the leading coefficient of the symbolic defect quasi-polynomial, determining it for ideals defining certain sets of general points in the plane. (Received August 02, 2020)

1159-13-113 **John Eagon**, **Ezra Miller*** (ezra@math.duke.edu) and **Erika Ordog**. *Minimal resolutions of monomial ideals.*

It has been an open problem since the 1960s to construct closed-form, canonical, combinatorial minimal free resolutions of arbitrary monomial ideals in polynomial rings. This talk explains how to solve the problem, in characteristic 0 and almost all positive characteristics, using sums over lattice paths of combinatorial data from simplicial complexes, one simplicial complex for each lattice point. Any minimal free resolution of any monomial ideal must—either implicitly or explicitly—produce homomorphisms between various homology groups of these simplicial complexes. Therefore an important aspect of the solution is an explicit way to write down canonical homomorphisms between these homology groups without choosing bases. Joint work with Jack Eagon and Erika Ordog. (Received August 02, 2020)

1159-13-119 **Keller VandeBogert*** (kelerlv@math.sc.edu). *Linear Strand and Minimal Free Resolutions for some Equigenerated Monomial Ideals.*

Let $R=k[x_1,...,x_n]$ denote the standard graded polynomial ring over a field k. We study certain classes of equigenerated (squarefree) monomial ideals obtained by deleting subsets of generators from the set of all (squarefree) monomials of fixed degree. After computing the graded Betti numbers of these ideals, we are able to

construct an explicit linear strand, and, in the case where the ideals under consideration have linear resolution, explicit minimal free resolutions. (Received August 03, 2020)

1159-13-120 **Ayah Almousa*** (aka66@cornell.edu), NY, and **Gunnar Floystad** and **Henning Lohne**. *Polarizations of powers of graded maximal ideals.*

We give a complete combinatorial characterization of all possible polarizations of powers of the graded maximal ideal $(x_1, x_2, \ldots, x_m)^n$ of a polynomial ring in m variables. We also give a combinatorial description of the Alexander duals of such polarizations. In the three variable case $m = 3$ and also in the power two case $n = 2$ the descriptions are easily visualized and we show that every polarization defines a (shellable) simplicial ball. We conjecture that any polarization of an Artinian monomial ideal defines a simplicial ball. (Received August 03, 2020)

1159-13-121 **John Eagon**, **Ezra Miller** and **Erika Ordog*** (erika.ordog@duke.edu). *Lattice paths from Koszul double complexes.*

It has been an open problem since the 1960s to construct closed-form, canonical, combinatorial minimal free resolutions of arbitrary monomial ideals in polynomial rings. The sylvan resolution solves the problem, in characteristic 0 and almost all positive characteristics, using sums over lattice paths of combinatorial data from simplicial complexes, one simplicial complex for each lattice point. Abstractly, our solution constructs resolutions by splitting a deceptively simple double complex whose rows and columns are all Koszul complexes. The algebra of spectral sequences for this Koszul double complex translates into combinatorics of "chain-link fences", which reflect matroidal data – higher dimensional versions of spanning tree information – descending along lattice paths. (Received August 03, 2020)

1159-13-126 **John Eagon**, **Ezra Miller** and **Erika Ordog*** (erika.ordog@duke.edu). *Sylvan structures on near-cones.* Preliminary report.

Sylvan resolutions are minimal resolutions of monomial ideals whose differentials are sums over lattice paths of weights associated to higher dimensional analogues of spanning trees. The differentials arise from splitting the boundary maps of the simplicial complexes that record the local structure of the ideal near each lattice point. When resolving stable ideals, these simplicial complexes are near-cones. By selecting certain splittings of the boundary maps of near-cones, the resulting sylvan resolution is the Eliahou-Kervaire resolution. (Received August 03, 2020)

1159-13-127 **Susan Cooper**, **Sabine El Khoury*** (se24@aub.edu.lb), **Sara Faridi**, **Sarah Mayes-Tang**, **Susan Morey**, **Liana Sega** and **Sandra Spiroff**. *Cellular resolutions of powers of ideals of projective dimension 1, Part I.*

One of the most fruitful directions in the study of monomial ideals is finding topological objects whose chain maps can be adapted to a free resolution of the ideal. While such resolutions exist for any monomial ideal (with Taylor's resolution being the starting example), they are mostly non-minimal. A square-free monomial ideal I of projective dimension one is known to have a minimal resolution supported on a (graph) tree G. Starting with this tree, we construct for any $r > 0$ a polyhedral cell complex $\overline{G^r}$, with vertices labeled by a set of monomials that minimally generate I^r. In this talk, we give a concrete geometric description of $\overline{G^r}$ and we describe the homogenized chain complex associated with it. The second part of this talk will show that this chain complex is a minimal free resolution of I^r.

Both parts of this talk are part of a project that was started during a "Women in Commutative Algebra" meeting in Banff, which also includes the material to be presented in the following talk in the session "Free Resolutions, Combinatorics, and Geometry":

Morse resolutions of powers of square-free monomial ideals of projective dimension 1 (Sara Faridi). (Received August 03, 2020)

1159-13-137 **Paolo Mantero*** (pmantero@uark.edu). *Free resolutions of symbolic powers of star configurations.*

Star configurations of points are sets X of points in projective space with strong combinatorial properties. There has been interest in understanding how "fattening" points affects the algebraic properties of these configurations or, in other words, understanding the symbolic powers $I_X^{(m)}$ of their defining ideals I_X.

In this talk we unveil the structure of these ideals $I_X^{(m)}$: we will describe their minimal generating sets and Betti tables. Our results apply to the more general setting of star configuration of hypersurfaces. (Received August 03, 2020)

1159-13-138 **Oana Veliche*** (o.veliche@northeastern.edu), 360 Huntington Ave, Boston, MA 02115-5005, and **Van Nguyen**. *A construction of a truncated minimal free resolution of the residue field.* Preliminary report.

In a paper from 1968, Golod proved that the Betti sequence of the residue field of a local ring attains the upper bound given by Serre if and only if the homology algebra of the Koszul complex of the ring has trivial multiplications and trivial Massey operations. This is the origin of the notion of Golod ring. Using the Koszul complex components as building blocks he also constructed a minimal free resolution of the residue field. In a recent project with Van Nguyen, we extend this construction up to the degree five and explicitly show how the multiplicative structure of the homology of the Koszul algebra is involved, including the triple Massey products. The talk will discuss about various consequences of this construction. (Received August 03, 2020)

1159-13-139 **Paolo Mantero*** (pmantero@uark.edu) and **Matthew Mastroeni** (mmastro@okstate.edu). *Betti numbers of Koszul algebras defined by four quadrics.*

Let I be an ideal generated by quadrics in a standard graded polynomial ring S over a field. A question of Avramov, Conca, and Iyengar asks whether the Betti numbers of $R = S/I$ over S can be bounded above by binomial coefficients on the minimal number of generators of I if R is Koszul. This question has been answered affirmatively for Koszul algebras defined by three quadrics and Koszul almost complete intersections with any number of generators.

We give a strong affirmative answer to the above question in the case of four quadrics by completely determining the Betti tables of Koszul algebras defined by 4 quadrics. (Received August 03, 2020)

1159-13-140 **Laura Felicia Matusevich***, Mailstop 3368, College Station, TX 77845, and **Byeongsu Yu**. *Standard pairs for monomial ideals in semigroup rings.*

Standard pairs are combinatorial structures that contain information about primary and irreducible decompositions of monomial ideals. We generalize standard pairs to the context of monomial ideals in semigroup rings, and show how to compute them. (Received August 03, 2020)

1159-13-142 **Andrew R. Kustin**, **Rebecca R.G.*** (rrebhuhn@gmu.edu) and **Adela Vraciu**. *The resolution of (x^N, y^N, z^N, w^N) over the hypersurface $k[x,y,z,w]/(x^n+y^n+z^n+w^n)$.*

We find the minimal graded resolution of $R/(x^N, y^N, z^N, w^N)$ over the hypersurface ring $R = k[x,y,z,w]/(x^n + y^n + z^n + w^n)$ as well as the related resolution of $P/(x^N, y^N, z^N, w^N) : (x^n + y^n + z^n + w^n)$ over $P = k[x,y,z,w]$, where k is a field of characteristic zero. To compute these resolutions, we make some useful reductions to the case of fewer variables and apply results on compressed Gorenstein rings. This is joint work with Andrew R. Kustin and Adela Vraciu. (Received August 03, 2020)

1159-13-145 **Hema Srinivasan*** (srinivasanh@missouri.edu), Department of Mathematics, University of Missouri, Columbia, MO 65211, and **Papri Dey** (pdbdn@missouri.edu), Department of Mathematics, University of Missouri, Columbia, MO 65211. *Principal Matrices of Numerical Semigroups.* Preliminary report.

Let $\mathbf{a} = (\mathbf{a_1}, \ldots, \mathbf{a_n})$ be a set of relatively prime positive integers minimally generating the numerical semigroup $< \mathbf{a} >$ For every $1 \leq i \leq n$, there is a smallest positive integer c_i such that $c_i a_i = \sum_{j \neq i} a_{ij} a_i$. Thus, there is an $n \times n$ matrix $D(\mathbf{a}) = (\mathbf{a_{ij}})_{\mathbf{n}\times\mathbf{n}}$ where the diagonal entries are $-c_i$. This is called a principal matrix of $D(\mathbf{a})$. Although $D(\mathbf{a})$ is not unique, certainly the diagonal entries of $D(\mathbf{a})$ are uniquely determined. The rank of $D(\mathbf{a}) \leq \mathbf{n-1}$. When the rank of $D(\mathbf{a})$ is n-1, we can recover $\mathbf{a}$ from it. We discuss the problem of characterizing the numerical semigroup when the rank is less than $n-1$. In particular, we prove that the rank of $D(\mathbf{a}) \geq \mathbf{n/2}$. This problem is intricate even when $n = 4$; we prove that when $n = 4$, the principal matrix will have rank 3 if it is not a gluing of two smaller numerical semigroups. We give examples to show that the converse is not true and why. (Received August 03, 2020)

1159-13-147 **Susan Cooper**, **Sabine El Khoury**, **Sara Faridi**, **Sara Mayes-Tang**, **Susan Morey**, **Liana Şega*** (segal@umkc.edu) and **Sandra Spiroff**. *Cellular resolutions of powers of ideals of projective dimension 1, Part II.*

Let I be a square-free monomial ideal. Its minimal free resolution is known to be supported on a tree G. For any integer $r > 0$, we constructed in Part I of this talk a labelled cellular complex $\overline{G^r}$ and its associated homogenized chain complex, denoted $\mathbb{F}^r$. In this talk we show that $\mathbb{F}^r$ is a minimal free resolution of I^r. Our approach identifies $\mathbb{F}^r$ with a strand of a Koszul complex, and gives a full description of the Rees algebra of I.

Both parts of this talk are part of a project that was started during a "Women in Commutative Algebra" meeting in Banff, which also includes the material to be presented in the following talk in the session "Free Resolutions, Combinatorics, and Geometry":

Morse resolutions of powers of square free monomial ideals of projective dimension 1 (Sara Faridi). (Received August 03, 2020)

1159-13-148 **Jennifer Biermann**, **Selvi Kara**, **Kuei-Nuan Lin** and **Augustine O'Keefe*** (aokeefe@conncoll.edu). *Algebraic invariants of weighted oriented graphs.*

Let $\mathcal{D}$ be a weighted oriented graph with vertex set $V(\mathcal{D}) = \{x_1, \ldots, x_n\}$ such that the vertex x_i has positive integer weight ω_i. We define the edge ideal of $\mathcal{D}$, denoted $I(\mathcal{D})$, to be the monomial ideal in $R = k[x_1, \ldots, x_n]$ generated by all monomials $x_i x_j^{\omega_j}$ such that $\mathcal{D}$ has a directed edge from x_i to x_j. In this paper we calculate the Castelnuovo-Mumford regularity and projective dimension of $R/I(\mathcal{D})$ for certain classes of weighted oriented graphs. (Received August 03, 2020)

1159-13-152 **James E Gossell*** (jgossel@clemson.edu). *Gorenstein Injective Modules under Flat Base Change.* Preliminary report.

For $A \to B$ a flat homomorphism of commutative noetherian rings and M an injective A-module, we can calculate the injective dimension of $M \otimes_A B$ as a B-module by calculating the injective dimensions of the fibers $F(\mathfrak{p}) = B_{\mathfrak{p}}/\mathfrak{p}$ for each $\mathfrak{p} \in \operatorname{Ass}(M)$ by the formula $\operatorname{id}_B(M \otimes_A B) = \sup_{\mathfrak{p} \in \operatorname{Ass}(M)} \operatorname{id}_{F(\mathfrak{p})} F(\mathfrak{p})$ (Foxby, 1975). This formula can be used to recover the fact that injective modules remain injective under certain flat base changes, such as a localization. We work towards a generalization of Foxby's Theorem to calculate the Gorenstein injective dimension of the base change of a Gorenstein-injective module. (Received August 03, 2020)

1159-13-153 **Anthony A. Iarrobino*** (a.iarrobino@neu.edu) and **Pedro Macias Marques**. *Jordan type for some Artinian local Gorenstein algebras.* Preliminary report.

The Jordan type of an Artinian algebra is the Jordan block partition associated to multiplication by a generic element of the maximal ideal. We study the Jordan type for Artinian Gorenstein (AG) local (non-graded) algebras A and the interaction of Jordan type with the symmetric decomposition of the (non-symmetric) Hilbert function H(A). We give examples where the family Gor(H) of AG algebras having Hilbert function H has two or more irreducible components. These examples result from the intersection of two filtrations of the family Gor(H): the filtration by Jordan type satisfies the usual orbit-closure dominance property; the second filtration, by symmetric decomposition, satisfies a known semicontinuity property. Their intersection can force there to be several irreducible components to Gor(H), in codimension three or higher. (Received August 03, 2020)

1159-13-155 **Hugh Roberts Geller*** (hgeller@g.clemson.edu). *Minimal Free Resolutions of Fiber Products.* Preliminary report.

In 2006 Visscher published an explicit construction for the minimal free resolution of the fiber product $k[\underline{x}] \times_k k[\underline{y}]$ where k is a arbitrary field and $\underline{x}$ and $\underline{y}$ are distinct lists of variables. In this talk, we briefly rephrase Visscher's construction in terms of Koszul complexes. From there we show how to use this construction along with the minimal free resolutions of $k[\underline{x}]/\mathcal{I}$ and $k[\underline{y}]/\mathcal{J}$, where $\mathcal{I} \subseteq \langle \underline{x} \rangle^2$ and $\mathcal{J} \subseteq \langle \underline{y} \rangle^2$, to obtain a minimal free resolution of $k[\underline{x}]/\mathcal{I} \times_k k[\underline{y}]/\mathcal{J}$. (Received August 03, 2020)

1159-13-161 **Joseph P. Brennan*** (joseph.brennan@ucf.edu), Department of Mathematics, University of Central Florida, 4393 Andromeda Loop N, Orlando, FL 32816-1364. *Application of the Italian Identities to the computation of Hilbert Series.* Preliminary report.

This talk will present tools for the computation of the Hilbert Series of Edge rings of graphs. As an example will will consider the problem of computation of the Hilbert series for the class of trees with the minimal number of leaves. (Received August 03, 2020)

1159-13-162 **Joseph W Skelton*** (jskelton@tulane.edu), 6823 St Charles Ave, Tulane University - Math Department, NEW ORLEANS, LA 70118. *Symbolic Powers of Cover Ideals and Koszul Property.*

We show that adding a whisker at the vertices of a cycle cover of a graph results in a new graph with the following property: all symbolic powers of its cover ideal are Koszul or, equivalently, componentwise linear. This extends previous work where the whiskers were added to all the vertices or to the vertices of a vertex cover of the graph. The presented results are from the joint work with Yan Gu, and Huy Tài Hà. (Received August 03, 2020)

1159-13-163 **Jay Schweig*** (jay.schweig@okstate.edu). *Poset-Borel ideals and poset-shifted complexes.*

Borel monomial ideals are those closed under a certain replacement operation. These ideals can be generalized by restricting the replacements allowed, based on a given poset. Shifted simplicial complexes can be thought of as an analogue of Borel ideals, and a similar poset generalization can be defined. Thus, a general question

becomes: What can we tell about an ideal or complex, given a (or the) related poset? I'll discuss some past work along these lines, as well as some newer results involving matroid complexes. (Received August 03, 2020)

1159-13-165 **Jay Yang*** (jkyang@umn.edu). *Virtual Resolutions of Monomial Ideals on Toric Varieties.* Virtual resolutions as introduced by Berkesch-Erman-Smith are free complexes for ideals in the Cox ring of a toric variety which can be simpler and more geometrically meaningful than the minimal free resolution. In the case of B-saturated monomial ideals in a smooth projective space, I prove there exists virtual resolutions with length bounded by the dimension of the ambient toric variety. This can be viewed as an analog of Hilbert's syzygy theorem and provides evidence suggesting such a statement should be true for virtual resolutions of ideals in smooth toric varieties (Received August 04, 2020)

1159-13-167 **Christine Berkesch**, **C.Y. Jean Chan**, **Patricia Klein**, **Laura Matusevich**, **Janet Page** and **Janet Vassilev*** (jvassil@math.unm.edu). *The ring of differential operators on graded quotients of affine semigroup rings.* Preliminary report.

For any ring R we denote the ring of differential operators on R by $D(R)$. For any $J \subseteq R$, the idealizer of J in $D(R)$, $\mathbb{I}(J)$, is the subset of differential operators which when applied to elements of J remain inside of J. If R is a regular ring and $J \subseteq R$ is an ideal, the ring of differential operators of R/J is equal to $\mathbb{I}(J)/JD(R)$. If R is not regular, we are able to give an alternate discription of the differential operators of R/J. Then using the description of the ring of differential operators on an affine semigroup ring given by Saito and Traves, we are able to describe the ring of differential operators on quotients of affine semigroup rings by certain multigraded radical ideals. (Received August 04, 2020)

1159-13-180 **Jack Jeffries*** (jack.jeffries@unl.edu) and **Anurag K. Singh**. *Differential operators on classical invariant rings.*

Levasseur and Stafford described the rings of differential operators on vari- ous classical invariant rings of characteristic zero; in each of the cases they considered, the differential operators form a simple ring. Towards an attack on the simplicity of rings of differential operators on invariant rings of linearly reductive groups over the complex numbers, Smith and Van den Bergh asked if differential operators on the corresponding rings of positive prime characteristic lift to characteristic zero differential operators. We prove that, in general, this is not the case for determinantal hypersurfaces, as well as for Pfaffian and symmetric determinantal hypersurfaces. We also prove that, with few exceptions, these hypersurfaces do not admit a mod p^2 lift of the Frobenius endomorphism. (Received August 05, 2020)

1159-13-182 **Sankhaneel Bisui*** (sbisui@tulane.edu), 7532 Hampson Street, New Orleans, LA 70118. *Harbourne–Huneke Containment and Bounds on Waldschmidt Constant.*

In order to provide a counterexample to Hilbert's 14th problem, Nagata raised a question: Given a finite set of points $X = \{P_1, \dots P_s\} \subset \mathbf{P}^N_{\mathbb{C}}$ what is the minimal degree, $\alpha_x(m)$ of a hyper-surface that passes through the points with multiplicity at least m? Chudnovsky provided a conjectural answer to the above question involving a lower bound of the Waldschmidt constant of the ideal corresponding to the points. Later on Demailly generalized the bound. The approach we use is via stable containment. Harbourne and Huneke gave containment conjectures which give positive answers to the above-mentioned conjectures. We study stable versions of that containment and as a consequence, we get positive answers to Chudnovsky and Demailly's conjecture for a sufficiently large number of general points. In this talk, I will introduce Chudnovsky's conjecture, Demailly's conjecture, and the containment conjectures. I will also present the results from the joint work with Eloísa Grifo, Huy Tài Hà, and Thái Thành Nguyễn. (Received August 04, 2020)

1159-13-200 **Ali Alilooee*** (aalilooee@fsmail.bradley.edu) and **Arindam Banerjee**. *Packing properties of cubic squarefree monomial ideals.*

The symbolic powers, in general, are not equal to the ordinary powers. Therefore, one interesting question here is for what classes of ideals ordinary and symbolic powers coincide? The answer to this question for squarefree monomial ideals may be packing property. In this paper, we classify all cubic path ideals for those the symbolic and ordinary powers that coincide. We find a criterion to see when the second symbolic and ordinary powers of the edge ideal of a 3-partite hypergraph are equal. Also, we give the formula to compute the Waldschmidt constant of the path ideals of a cycle. (Received August 04, 2020)

1159-13-202 **Hailong Dao***, 405 Snow Hall, 1460 Jayhawk Blvd, Lawrence, KS 66045, **Ryo Takahashi**, Nagoya, 464-8602, Japan, and **Toshinori Kobayashi**, Nagoya, 464-8602, Japan. *Burch ideals and Burch rings.*

We introduce the notion of Burch ideals and Burch rings. They are easy to define, and can be viewed as generalization of many well-known concepts, for example integrally closed ideals of finite colength and Cohen–Macaulay rings of minimal multiplicity. We give several characterizations of these objects. We show that they satisfy many interesting and desirable properties: ideal-theoretic, homological, categorical. We relate them to other classes of ideals and rings in the literature. (Received August 04, 2020)

14 ► Algebraic geometry

1159-14-17 **Vida Milani*** (vida_milani@yahoo.com), Utah State University, Mathematics department, 4205 old Main Hill, Logan, UT 84322, and **Seyed M-H Mansourbeigi** (phy_math_ee@yahoo.com), Department of Electrical Engineering and, Computer Science, College of Engineering, Utah State University, 4205 old Main Hill, Logan, UT 84322. *Mathematical Modeling for Viral Disease Distributed Applications.* Preliminary report.

The goal of the present article is to demonstrate a mathematical modeling for viral disease distributed applications. The present paper applies tools from topology and sheaf theory as an appropriate mathematical modeling to reflect interactions among elements of resources. The methodology is applied to study the efficiency of the resources to prevent spread of viral diseases in populations of interacting individuals. Heterogeneous data are gathered from variety of tools for Full blood count (a viral infection may raise or reduce the white cell count), C-reactive protein (CRP) test, biopsy, electron microscopy, resistance testing for viral mutation. (Received June 26, 2020)

15 ► Linear and multilinear algebra; matrix theory

1159-15-208 **Christina Pospisil***, pospisil.christina@gmx.de. *Generalization Theory of Linear Algebra I+ II.*

An algorithm for multiplying and adding matrices regardless of dimensions via an embedding and inverses for non-injective mappings in one dimension are presented (this first part of the project Generalization Theory of Linear Algebra I was presented at JMM 2019). Then the talk continues and presents inverses for non-injective mappings in multiple dimensions, inverses for non-surjective mappings in one and multiple dimensions and introduces a general determinant theory (second part was presented at JMM 2020). In future work there will be further operations and applications to physics and other natural sciences be explored. (Received August 06, 2020)

17 ► Nonassociative rings and algebras

1159-17-15 **Igor Ya. Subbotin*** (isubboti@nu.edu), National University, Department of Mathematics and Natural Sci, Los Angeles, CA 90045, and **Leonid A. Kurdachenko**. *From Groups to Leibniz Algebras: Common Approaches, Parallel Results.*

The specifics of Leibniz algebras, the features that distinguish them from Lie algebras, can be seen from the description of Leibniz algebras of small dimensions. However, this description concerns algebras over fields of characteristic zero. Some reminiscent of the theory of groups are immediately striking, precisely with its period when the theory of finite groups was already quite developed, while the theory of infinite groups was at the beginning level of creation. Therefore, the idea of using the experience gained in group theory naturally arises. It is clear that we cannot talk about some kind of similarity of results; we can talk about approaches and problems, about application of group theory philosophy. Moreover, any theory has a number of natural problems that arise in the process of its development, and these problems quite often have analogues in other disciplines. In this survey, we want to focus on such issues, our goal is to see which parts of the picture involving the general structure of Leibniz algebras have already been drawn, and this will allow us to see which parts of this picture, in our opinion, should be developed further. We also show significant distinguishing futures of Leibniz algebras from Lie algebras. (Received June 26, 2020)

20 ► Group theory and generalizations

1159-20-157 **Gordon Rojas Kirby*** (girkirby@gmail.com). *Pure Salvetti complexes and Euclidean Artin groups.*

Artin groups are a far-reaching generalization of the much better-known braid groups. For most Artin groups it is still unknown how to solve their word problem. This talk will introduce a new approach to the word problem for Artin groups that lies at the intersection of topology, combinatorics, and geometry. My work centers on a combinatorially defined cell complex that is homotopy equivalent to an intermediate covering space and decomposing it into understandable pieces. I will reduce the word problem to a simpler problem that can be stated about compact, combinatorially defined, zonotopal cell complexes, give brief illustrations of some work from my dissertation, and give concrete solutions to word problem for specific Euclidean Artin groups. Lastly, I will discuss the promise and difficulties of generalizing this approach to general Artin groups. (Received August 03, 2020)

30 ► Functions of a complex variable

1159-30-5 **Gerardo R. Chacon*** (gerardo.chacon@gallaudet.edu) and **Gerardo A Chacon** (gerardoachg@uan.edu.co). *Variable exponent function spaces.*

Variable exponent spaces are a generalization of Lebesgue spaces in which the exponent is a measurable function. Most of the research done in this topic has been situated under the context of real functions. In this work, we present two examples of variables exponent spaces of analytic functions: variable exponent Hardy spaces and variable exponent Bergman spaces. We will introduce the spaces together with some basic properties and the main techniques used in the context. We will show that in both cases, the boundedness of the evaluation functionals plays a key role in the theory. We also present a section of possible directions of research in this topic. (Received April 28, 2020)

34 ► Ordinary differential equations

1159-34-12 **Mingji Zhang*** (mingji.zhang@nmt.edu), **Peter Bates** and **Zhenshu Wen**. *Effects on I-V relations from small permanent charge and channel geometry via classical Poisson-Nernst-Planck equations with multiple cations.*

We analyze a quasi-one-dimensional Poisson-Nernst-Planck system for ionic flow through a membrane channel with nonzero but small permanent charge. The system includes *three* ion species, two cations with the same valences and one anion. The cross-section area of the channel is included in the system, which provides certain information of the geometry of the three-dimensional channel. This is critical for our analysis. We treat the model as a boundary value problem of a singularly perturbed differential system. Under the framework of geometric singular perturbation theory, together with specific structures of the model, the existence and local uniqueness of solutions to the boundary value problem for small permanent charges is established. Furthermore, treating the permanent charge as a small parameter, via regular perturbation analysis, we are able to derive an approximation of the I-V (current-voltage) relations explicitly, from which the permanent charge and channel geometry effect on ionic flows are analyzed in details. Critical potentials are identified and their roles in characterizing the ionic flow properties of interest are studies. (Received June 25, 2020)

1159-34-158 **Liangbing Luo*** (liangbing.luo@uconn.edu), 341 Mansfield Rd, Storrs, CT 06268. *Logarithmic Sobolev Inequalities on Non-isotropic Heisenberg Groups.*

A Heisenberg group is the simplest non-trivial example of a sub-Riemannian manifold. In this talk, we will discuss the dimension (in)dependence of the constants in logarithmic Sobolev inequalities on non-isotropic Heisenberg groups. In this setting, a natural Laplacian is not an elliptic but a hypoelliptic operator. The argument relies on viewing the logarithmic Sobolev inequality as a limiting case of Sobolev inequalities and tensorization of the logarithmic Sobolev inequalities. (Received August 03, 2020)

35 ► Partial differential equations

1159-35-9 **Michael P Presho*** (mpresho@semo.edu), Michael Presho, and **Michael Hill**. *An Accurate Multiscale Method for Modeling Two-Phase Flow with Capillary Pressure.*

In this presentation we investigate a numerical method used for solving a two-phase flow model within the context of a physics-based two-level operator splitting scheme. The governing equations involve an elliptic pressure equation coupled with a convection-dominated parabolic equation. The underlying permeability of the porous medium is assumed to exhibit high variation over many length scales which leads to poor approximation properties in standard multiscale finite element spaces. In order to effectively solve the system we construct an approach that combines a mass-conservative generalized multiscale finite element method with an operator splitting scheme. In particular, we split the coupled system into three subsystems: the elliptic, the hyperbolic, and the parabolic pieces. To solve the elliptic equation we propose the use of a mass-conservative generalized multiscale finite element method (GMsFEM-FV), for the parabolic equation we use a mass-conservative multiscale finite element method (MsFEM-FV), and for the hyperbolic equation we use a standard non-oscillatory finite-volume scheme. The performance of the methodology is validated through a variety of numerical experiments. (Received May 21, 2020)

1159-35-22 **Behzad Djafari Rouhani** (behzad@utep.edu), Department of Mathematical Sciences, University of Texas at El Paso, 500 W. University Avenue, El Paso, TX 79968, **Gisèle Ruiz Goldstein** (ggoldste@memphis.edu), Department of Mathematical Sciences, University of Memphis, Memphis, TN 38152, and **Jerome A Goldstein*** (jgoldste@memphis.edu), Department of Mathematical Sciences, University of Memphis, Memphis, TN 38152. *Ergodic limits for inhomogeneous evolution equations.*

We show that the ergodic limit

$$\frac{1}{t}\int_0^t u(s)ds$$

exists and can be calculated explicitly where u satisfies one of a class of linear inhomogeneous differential equations which includes various wave equations of first or second order in time, involving operator groups and/or operator cosine functions. (Received July 03, 2020)

1159-35-23 **Gisèle Ruiz Goldstein*** (ggoldste@memphis.edu), Department of Mathematical Sciences, University of Memphis, Memphis, TN 38152, **Jerome A Goldstein** (jgoldste@memphis.edu), Department of Mathematical Sciences, University of Memphis, Memphis, TN 38152, **Ismail Kömbe** (ikombe@ticaret.edu.tr), Department of Mathematics, Faculty of Humanities and Social Sciences, Istanbul Commerce University, Beyoglu, Istanbul, Turkey, and **Reyhan Tellioglu** (reyhan.tellioglu@istanbulticaret.edu.tr), Department of Mathematics, Faculty of Humanities and Social Sciences, Istanbul Commerce University, Beyoglu, Istanbul, Turkey. *Nonlinear Parabolic Equations with Robin Boundary Conditions and Hardy-Leray Type Inequalities.*

Of concern is the absence of positive solutions in the sense of distributions of the mixed problem

$$\begin{aligned}\frac{\partial u}{\partial t} &= \Delta(u^m) + V(x)u^m + \lambda u^q, \quad \text{on } \Omega\times(0,T),\\ u(x,0) &= u_0(x) \geq 0 \quad \text{in } \Omega,\\ \frac{\partial u}{\partial \nu} &= \beta(x)u \quad \text{on } \partial\Omega\times(0,T),\end{aligned}$$

for $x \in \Omega \subset \mathbb{R}^N$ and $t \geq 0$. Here $0 < m < 1$, so the equation includes *fast diffusion,* and $q > 0$ so there is a Fujita type term. In the case where $m = 1, \lambda = 0$, the Dirichlet boundary condition replaces the Robin one, $0 \in \Omega$ and $V(x)$ is the inverse square potential, then this result reduces to the 1984 Baras-Goldstein nonexistence result. (More general conditions will be discussed in the talk.) When $0 < m < 1$ and $\lambda = 0$, this extends the nonexistence result of Goldstein-Kombe (2003). The complicated analysis involves new refinements of Hardy's inequality with new hidden energy terms. (Received July 04, 2020)

1159-35-29 **Wesley Hamilton*** (wham@live.unc.edu), **Jeremy L Marzuola** and **Hau-tieng Wu**. *On the behavior of 1-Laplacian Ratio Cuts on nearly rectangular domains.*

Given a connected set $\Omega \subset \mathbb{R}^2$ define a sequence of sets $(\Omega_n)_{n=0}^{\infty}$ where Ω_{n+1} is the subset of Ω_{n+1} where the first eigenfunction of the (properly normalized) Neumann p-Laplacian $-\Delta^{(p)}\phi = \lambda_1|\phi|^{p-2}\phi$ is positive (or negative). For $p = 1$, this is also referred to as the Ratio Cut of the domain. We conjecture that, unless Ω_0 is an

isosceles right triangle, these sets converge to the set of rectangles with eccentricity bounded by 2 in the Gromov-Hausdorff distance as long as they have a certain distance to the boundary $\partial\Omega_0$. We establish some aspects of this conjecture for $p = 1$ where we prove that (1) the 1-Laplacian spectral cut of domains sufficiently close to rectangles of a given aspect ratio is a circular arc that is closer to flat than the original domain (leading eventually to quadrilaterals) and (2) quadrilaterals close to a rectangle of aspect ratio 2 stay close to quadrilaterals and move closer to rectangles in a suitable metric. We also discuss some numerical aspects and pose many open questions. (Received July 07, 2020)

1159-35-40 **Salvatore Stuvard*** (stuvard@math.utexas.edu) and **Yoshihiro Tonegawa**. *An existence theorem for Brakke flow with fixed boundary conditions.*

Brakke flow is a measure-theoretic generalization of the mean curvature flow which exploits the flexibility of geometric measure theory in order to describe the evolution by (generalized) mean curvature of surfaces exhibiting singularities, such as, for instance, a planar network with multiple junctions. The primary goal of this talk is to discuss the proof of the following result: given any n-dimensional rectifiable subset Γ_0 of a strictly convex bounded domain $U \subset \mathbb{R}^{n+1}$ such that $U \setminus \Gamma_0$ is not connected, there exists a Brakke flow of surfaces $\Gamma(t)$ such that $\Gamma(0) = \Gamma_0$ and with the additional property that the boundary $\partial\Gamma(t)$ coincides with $\partial\Gamma_0$ at all times $t \geq 0$. Furthermore, $\Gamma(t)$ subconverge, as $t \to \infty$ and in a suitable sense, to a (generalized) minimal surface in U with the prescribed boundary $\partial\Gamma_0$, thus providing a dynamical solution to Plateau's problem. If time permits, I will also discuss recent developments concerning the relationship between the singularities of the initial surface Γ_0 and the uniqueness of the flow. (Received July 16, 2020)

1159-35-47 **Andrea Cianchi*** (andrea.cianchi@unifi.it), Viale Morgagni 67/a, 50134 Firenze, Italy, and **Vit Musil** and **Lubos Pick**. *Moser inequalities in Gauss space.*

The sharp constants in a family of exponential Sobolev type inequalities in Gauss space are exhibited. They constitute the Gaussian analogues of the Moser inequality in the borderline case of the Sobolev embedding in the Euclidean space. Interestingly, the Gaussian results have common features with the Euclidean ones, but also reveal marked diversities. (Received July 22, 2020)

1159-35-52 **Jiahong Wu*** (jiahong.wu@okstate.edu), 401 Mathematical Sciences, Stillwater, OK 74078. *A partially dissipated incompressible Oldroyd-B model.*

The Oldroyd-B equations govern the motion of complex fluids such as a solvent with particles suspended in it. This talk focuses on a partially dissipated Oldroyd-B system with dissipation only in the equation of stress tensor. We present recent results on the small data global well-posedness and large-time behavior problem. Since there is no kinematic dissipation, the velocity satisfies a forced incompressible Euler equation and its Sobolev norms could potentially grow rather rapidly. This fact makes the global existence and stability problem very difficult. By discovering and exploiting the extra smoothing due to the coupling and interaction in this Oldroyd-B model, we are able to obtain the desired small data global well-posedness. In addition, sharp large-time behavior estimates are also established. (Received July 23, 2020)

1159-35-57 **Jungang Li*** (jungang_li@brown.edu), **Guozhen Lu** and **Qiaohua Yang**. *Higher order Brezis-Nirenberg problems on hyperbolic spaces.*

The classical Brezis-Nirenberg problem treats second order semilinear equations whose nonlinearity is associated with the critical Sobolev exponent. In this talk, we will discuss a higher order version of the Brezis-Nirenberg problem on hyperbolic spaces. Such problem is closely related to a recent progress on higher order Hardy-Sobolev-Mazya inequalities on upper half spaces (due to G. Lu and Q. Yang). We establish both existence and nonexistence results of solutions and the Fourier analysis on hyperbolic spaces plays an important role in our study. Meanwhile we investigate the hyperbolic symmetry of positive solutions. This is a joint work with G. Lu and Q. Yang. (Received July 26, 2020)

1159-35-62 **Zachary Bradshaw*** (zb002@uark.edu), **I. Kukavica** and **T.-P. Tsai**. *The Navier-Stokes equations in weighted spaces.*

We discuss recent results on global existence of solutions to the Navier-Stokes equations in weighted spaces which allow local L^2 norms to grow at spatial infinity. Similar results in this direction were developed independently by Fernandez-Dalgo and Lemarie-Rieusset as well as the speaker, Kukavica and Tsai. (Received July 28, 2020)

1159-35-68 **Florin Catrina*** (catrinaf@stjohns.edu), St. John's University, 8000 Utopia Pkwy, Queens, NY 11439. *An energy balance identity for nonlinear ODE's.*

We discuss implications of an energy balance identity on properties of solutions for some second order ODE's with Hamiltonian structure. Variational proofs of existence of solutions for semilinear elliptic PDE's are often

based on compactness of embeddings between the appropriate functional spaces. In the case of critical or supercritical nonlinearities, the loss of compactness is manifested in the concentration of minimizing sequences at critical points or at singularities of the potential. This identity is able to capture the loss of compactness for radial functions in some special cases of rotationally symmetric PDE's. (Received July 29, 2020)

1159-35-69 **Bruno Franchi*** (bruno.franchi@unibo.it), Department of Mathematics, University of Bologna, 4026 Bologna, BO, Italy. *Mathematical Models for Alzheimer's Disease.*

(in collaboration with M. Bertsch, V. Meschini, M.C. Tesi & A. Tosin) We propose a mathematical model for the onset and progression of Alzheimer's disease based on transport and diffusion equations. We treat brain neurons as a continuous medium and structure them by their degree of malfunctioning. Three different mechanisms are assumed to be relevant for the temporal evolution of the disease:

- diffusion and agglomeration of soluble peptides of Amyloid beta,
- neuron-to-neuron prion-like transmission of the disease and
- effects of phosphorylated tau protein.

We model these processes by a system of Smoluchowski equations for the Amyloid beta concentration, an evolution equation for the dynamics of tau protein and a kinetic-type transport equation for the distribution function of the degree of malfunctioning of neurons. The latter equation contains an integral term describing the random onset of the disease as a jump process localized in particularly sensitive areas of the brain. We are particularly interested in investigating the effects of the synergistic interplay of Amyloid beta and tau on the dynamics of Alzheimer's disease. (Received July 29, 2020)

1159-35-72 **Nguyen Hoang Lam*** (nlam@grenfell.mun.ca), Memorial University of Newfoundland, Grenfell Campus, 20 University Drive, Corner Brook, NL A2H6R7, Canada. *p-Bessel pairs and Hardy type inequalities.*

In this talk, we discuss the notion of p-Bessel pairs and its applications to Hardy identities. We will also present several examples of p-Bessel pairs and the Hardy type inequalities with remainder terms and virtual optimizers. This is joint work with Nguyen Tuan Duy, Joshua Flynn and Guozhen Lu. (Received July 30, 2020)

1159-35-78 **Alberto Cialdea*** (cialdea@email.it), Dipartimento di Matematica, Informatica ed, Economia, Università della Basilicata, Via dell'Ateneo Lucano, 10, 85100 Potenza, Italy. *Functional dissipativity of second order differential operators with complex coefficients.*

In this talk I will present some recent results obtained with Vladimir Maz'ya. They concern the Dirichlet problem for the second order differential operator $E = \nabla(A\nabla)$, where A is a matrix with complex valued L^∞ entries. We have introduced the concept of dissipativity of E with respect to a given function $\varphi : \mathbb{R}^+ \to \mathbb{R}^+$. This means that

$$Re \int_\Omega \langle A\nabla u, \nabla(\varphi(|u|)\, u)\rangle \, dx \geq 0$$

for any complex valued $u \in \mathring{H}^1(\Omega)$ such that $\varphi(|u|)\, u \in \mathring{H}^1(\Omega)$, Ω being a domain in $\mathbb{R}^N$. Under the assumption that the $Im\, A$ is symmetric, we have proved that the condition

$$|s\, \varphi'(s)|\, |\langle Im\, A(x)\, \xi, \xi\rangle| \leq 2\, \sqrt{\varphi(s)\, [s\, \varphi(s)]'}\, \langle Re\, A(x)\, \xi, \xi\rangle$$

(for almost every $x \in \Omega$ and for any $s > 0$, $\xi \in \mathbb{R}^N$) is necessary and sufficient for the functional dissipativity of E. (Received July 30, 2020)

1159-35-86 **J L Flynn*** (joshua.flynn@uconn.edu). *Sharp L^2-Caffarelli–Kohn–Nirenberg-Type Inequalities for the Grushin operator and Iwasawa Groups.*

Sharp L^2-Caffarelli-Kohn-Nirenberg inequalities are established for the Grushin vector fields and for Iwasawa groups (i.e., the boundary group of a real rank one noncompact symmetric space). For all but one parameter case, this is done by introducing a generalized Kelvin transform which is shown to be an isometry of certain weighted Sobolev spaces. For the exceptional parameter case, the best constant is found for the Grushin vector fields by introducing Grushin cylindrical coordinates and solving the transformed Euler-Lagrange equation. (Received July 30, 2020)

1159-35-95 **Vu Hoang***, duynguyenvu.hoang@utsa.edu. *Local Existence and Breakdown of Solutions for Relativistic Fluids with Viscosity.*

Relativistic hydrodynamics describes the motion of fluids in regimes where relativistic effects are important, including flow velocities close to the speed of light (e.g the relativistic quark-gluon plasma) and fluids interacting with strong gravitational fields (e.g. in neutron star mergers and black hole accretion disks). Relativistic fluid descriptions serve as an essential tool in high-energy nuclear physics, cosmology, and astrophysics. Several of these

phenomena are not adequately modeled by perfect fluids, but require viscous fluids. Recently, the longstanding problem of treating viscosity in the framework of the relativistic fluid equations has received renewed attention. In this talk, I will discuss recent results concerning the local existence of solutions for relativistic viscous fluids with shear and the breakdown of smooth solutions. This is joint work with M. Disconzi (Vanderbilt) and M. Radosz (UTSA). (Received August 01, 2020)

1159-35-100 **Andrea Giorgini*** (agiorgin@iu.edu), Rawles Hall, 831 East 3rd St., Bloomington, IN 47405. *Incompressible two-phase flows: well-posedness for the AGG model.*

The Diffuse Interface theory is nowadays widely employed in Fluid Mechanics to describe two-phase flows. This approach combines in a unified framework the concept of transition interface with the energy-based formalism from thermodynamics and statistical mechanics. In the last years, particular interest has been devoted to theoretical and numerical studies of different DI models for mixtures of two incompressible fluids with different densities. Among these systems, the Abels-Garcke-Grün (AGG) model consists in a Navier-Stokes-Cahn-Hilliard type system, which is characterized by concentration-dependent density and viscosity, and an additional flux term due to concentration diffusion at the interface. In literature the analysis of the AGG has been focused so far on the existence of weak solutions. In this talk I will present some recent results concerning the existence and uniqueness of strong solutions in two dimensions. (Received August 01, 2020)

1159-35-110 **Hwan Goh*** (hwan.goh@utexas.edu), **Sheroze Sheriffdeen** (sheroze@oden.utexas.edu) and **Tan Bui-Thanh** (tanbui@oden.utexas.edu). *Solving Forward and Inverse Problems with Model-Aware Autoencoders.* Preliminary report.

This work develops a model-aware autoencoder networks as a new method for solving scientific forward and inverse problems. Autoencoders are unsupervised neural networks that are able to learn new representations of data through appropriately selected architecture and regularization. The resulting mappings to and from the latent representation can be used to encode and decode the data. In our work, we set the data space to be the parameter space of a parameter of interest we wish to invert for. Further, as a way to encode the underlying physical model into the autoencoder, we enforce the latent space of an autoencoder to be the space of observations of physically-governed phenomena. In doing so, we leverage the well known capability of a deep neural network as a universal function approximator to simultaneously obtain both the parameter-to-observation and observation-to-parameter map. The results suggest that this simultaneous learning interacts synergistically to improve the inversion capability of the autoencoder. (Received August 02, 2020)

1159-35-111 **Fausto Ferrari*** (fausto.ferrari@unibo.it), Piazza di Porta S.Donato, 5, 40126 Bologna, Italy. *Few remarks about Alt-Caffarelli-Friedman monotonicity formula.*

We will discuss some results concerning the existence of a monotonicity formula in the simplest Heisenberg group. (Received August 02, 2020)

1159-35-116 **Phuc C Nguyen*** (pcnguyen@math.lsu.edu). *A viewpoint on the Navier-Stokes system and some results from it.*

I will discuss some results on the Navier-Stokes system of equations that involve sharp existence in the stationary case and conditional regularity in the non-stationary case. The common theme is that the quadratic nonlinearity and the pressure are both treated as weights generally belonging to a Sobolev space of negative order. This talk is based on joint work with Tuoc Phan and Cristi Guevara. (Received August 03, 2020)

1159-35-168 **Cecilia F Mondaini*** (cf823@drexel.edu) and **Nathan E Glatt-Holtz**. *Numerical approximation for invariant measures of the 2D Navier-Stokes equations.*

We consider the problem of approximating statistically steady states of the 2D stochastic Navier-Stokes equations (SNSE) via an approximating sequence of measures generated from a space-time discretization of the 2D SNSE. More specifically, we consider a spectral Galerkin spatial discretization and a semi-implicit Euler time scheme. We show that successive iterations of the Markov semigroup associated to the discretized system, starting from any initial probability distribution, converge to the invariant measure of the continuous system. The proof is obtained with two main steps: a spectral gap result for the discretized system which is independent of the discretization parameters, and finite time L^2-convergence of the discretized system towards the continuous one. Most importantly, this approach allows us to obtain explicit rates of convergence with respect to the number of iterations in the Markov chain, up to (also explicit) numerical discretization error. This is a joint work with Nathan Glatt-Holtz (Tulane U). (Received August 04, 2020)

1159-35-171 **Geordie Richards*** (geordie.richards@usu.edu), **Nathan Glatt-Holtz** and **Vincent Martinez**. *Asymptotic coupling and unique ergodicity for damped stochastic KdV.*

We will review a straightforward asymptotic coupling strategy for establishing uniqueness of the invariant measure for stochastic PDEs, of particular utility for models from fluid mechanics. As a novel application, we will provide a concise proof of unique ergodicity for damped stochastic KdV. This is a joint work with Nathan Glatt-Holtz (Tulane) and Vincent Martinez (Hunter). (Received August 04, 2020)

1159-35-183 **Alexey Cheskidov*** (acheskid@uic.edu) and **Xiaoyutao Luo**. *Nonuniqueness of weak solutions for the transport equation at critical space regularity.*

We consider the linear transport equations driven by an incompressible flow in dimensions $d \geq 3$. For divergence-free vector fields $u \in L^1 W^{1,q}$, the celebrated DiPerna-Lions theory of the renormalized solutions established the uniqueness of the weak solution in the class $L^\infty L^p$ when $\frac{1}{p} + \frac{1}{q} \leq 1$. For such vector fields, we show that in the regime $\frac{1}{p} + \frac{1}{q} > 1$, weak solutions are not unique in the class $L^1 L^p$. One crucial ingredient in the proof is the use of both temporal intermittency and oscillation in the convex integration scheme. (Received August 04, 2020)

1159-35-192 **Juraj Foldes*** (foldes@virginia.edu) and **Mouhamadou Sy**. *Invariant measures for equations of fluid dynamics.*

Using a fluctuation dissipation method, we construct an invariant measure for the surface quasi-geostrophic equation (SQG) and 3D Euler equation. Since the support of the measure contains entire solutions, we obtain a manifold containing solutions that do not blow-up. This complements results in which a blow-up solutions for SQG and grow up solutions for Euler are constructed. The method of the proof relies on an addition of a stochastic forcing and a small dissipation to the equation. For such stochastic equation, one can construct an invariant measure and by passing the strength of the forcing and the dissipation to zero, we obtain the desired invariant measure. We also discuss the size of the support of the measure, which relies on the number of conservation laws for the particular equation.

This is a joint project with Mouhamadou Sy. (Received August 04, 2020)

1159-35-196 **Joao Marcos Bezerra do O*** (jmbo.pq@gmail.com), Universidade Federal da Paraiba, Centro de Ciencias Exatas e da Natureza, Departamento de Matematica, Joao Pessoa, PB 58051900, Brazil. *Trudinger-Moser inequalities on the upper half-space.*

We establish some sharp embedding results, including trace embeddings from a specific Sobolev space, defined on the upper half-space, into the Lebesgue and Orlicz spaces. Moreover, we prove some Trudinger-Moser type inequalities for functions defined on the upper half-space. As an application, we show the existence of solutions for a class of elliptic problems involving nonlinear boundary conditions. (Received August 04, 2020)

1159-35-207 **Changfeng Gui** (changfeng.gui@utsa.edu) and **Amir Moradifam*** (amirm@ucr.edu). *The Sphere Covering Inequality and Its Applications.*

We show that the total area of two distinct Gaussian curvature 1 surfaces with the same conformal factor on the boundary, which are also conformal to the Euclidean unit disk, must be at least 4π. In other words, the areas of these surfaces must cover the whole unit sphere after a proper rearrangement. We refer to this lower bound of total areas as the Sphere Covering Inequality. This inequality and it's generalizations are applied to a number of open problems related to Moser-Trudinger type inequalities, mean field equations and Onsager vortices, etc, and yield optimal results. In particular we confirm the best constant of a Moser-Truidinger type inequality conjectured by A. Chang and P. Yang in 1987. This is a joint work Changfeng Gui. (Received August 05, 2020)

37 ► Dynamical systems and ergodic theory

1159-37-38 **Jorge Rebaza*** (jrebaza@missouristate.edu), Missouri State University, Springfield, MO 65897. *Stability Analysis of a Multipatch Model of COVID-19 Dynamics.* Preliminary report.

A model of COVID-19 in an interconnected network of communities is studied. The model considers the dynamics of susceptible, asymptomatic and symptomatic individuals, deceased but not yet buried people, as well as the dynamics of the virus or pathogen at connected nodes or communities. People can move between communities carrying the disease to any node in a region of n communities (or patches). The model considers both virus direct (person to person) and indirect (contaminated environment to person) transmission. Using either matrix and graph-theoretic methods and some combinatorial identities, appropriate Lyapunov functions are constructed

to study global stability properties of both the disease-free and the endemic equilibrium of the corresponding system of 5n differential equations. (Received July 15, 2020)

1159-37-117 **Vincent R Martinez*** (vrmartinez@hunter.cuny.edu), 695 Park Ave, New York, NY 10065. *On the controlliability of the damped-driven KdV equations and applications.* Preliminary report.

This talk will discuss, in the context of the damped-driven KdV equations, an approach recently developed by GlattHoltz, Herzog, and Mattingly for establishing controllability. In particular, this approach provides a versatile framework for constructing controls that achieve global approximate controllability and exact controllability in finite-dimensional subspaces. We highlight the difficulties one faces with this approach in the case of the damped-driven KdV equation, how they can be overcome, as well as discuss applications of the result in the stochastically forced case to unique ergodicity and non-degeneracy of the Malliavan matrix. This is joint work with Nathan Glatt-Holtz (Tulane) and Geordie Richards (Utah State). (Received August 05, 2020)

42 ► Fourier analysis

1159-42-11 **William Beckner*** (beckner@math.utexas.edu), Department of Mathematics, University of Texas at Austin, Austin, TX 78712. *Embedded Symmetry & Encoding Information.* Preliminary report.

Symmetry is intrinsic to the framework of Fourier Analysis. Our purpose is to formulate a new paradigm for analysis on Lie groups that moves beyond the Euclidean domain starting with SL(2,R) invariance and the Heisenberg group. These basic groups are a natural playing field to explore the laws of symmetry and the interplay of geometric structure. New results illustrate connections among Sobolev embedding on the Heisenberg group, Stein-Weiss integrals with mixed homogeneity, and Riesz potentials on hyperbolic space with insight gained from application – vortex dynamics, Keller-Segel model for chemotaxis, and the global Biot-Savart law in physics. Fundamental identities, formulas, and inequalities are essential for developing analysis on a geometric manifold – explicit, elegant, influential – plus they encode information about the manifold. (Received June 25, 2020)

1159-42-133 **Jianxiong Wang*** (jianxiong.wang@uconn.edu), U1009, 341 Mansfield Road, Storrs, CT 06269. *Hardy-Sobolev, log-Sobolev and Moser-Onofri-Beckner inequalities with monomial weights.*

Recently, there have been increasing interests in studying geometric inequalities with monomial weights.

In this talk, we give some improved functional and geometric inequalities in the presence of both monomial and radial weights, where monomial weights is defined by $|x|^A = |x_1|^{A_1} \cdots |x_n|^{A_n}$. In particular, I will start from Hardy-Sobolev inequality, and then establish log-Sobolev inequality and derive the general Moser-Onofri-Beckner inequality with monomial weights. (Received August 05, 2020)

1159-42-189 **Taryn C Flock***, tflock@macalester.edu, and **Andrea Nahmod**. *A singular bilinear operator with a determinant like kernel.* Preliminary report.

We consider a biliner operator arising in the study of PDE's. We show that in two dimmensions the L^p-L^q mapping properties of this operator are completely understood through its association to a trilinear form studied by Valdimarson and Gressman, He, Kovač, Street, Thiele, Yung. We will further discuss Sobolev estimates and, time permitting, extensions to higher dimensions. (Received August 04, 2020)

43 ► Abstract harmonic analysis

1159-43-134 **Guozhen Lu*** (guozhen.lu@uconn.edu), University of Connecticut, Storrs Mansfield, CT 26069. *Hardy-Sobolev-Maz'ya and Adams inequalities on complex hyperbolic spaces.*

Our aim is to establish the Hardy-Sobolev-Maz'ya inequality and Adams inequality on complex hyperbolic space. By considering differential operators on complex hyperbolic space which are closely related to some CR invariant differential operators on the Heisenberg group and CR sphere and by using, among other things, the Fourier analysis techniques on complex hyperbolic space, we establish the Hardy-Sobolev-Maz'ya inequality on Siegel domain $\mathcal{U}^n$ and unite ball $\mathbb{B}^n_{\mathbb{C}}$. Finally, we establish the sharp Hardy-Adams inequalities and sharp Adams type inequalities on Sobolev spaces of any positive fractional order on the complex hyperbolic spaces. This is joint work with Qiaohua Yang. (Received August 04, 2020)

46 ► Functional analysis

1159-46-136 **Geraldo Botelho**, **Qingying Bu**, **Donghai Ji** and **Khazhak Varazdat Navoyan***, Khazhakanush@gmail.com. *The positive Schur property on spaces of regular multilinear operators.* Preliminary report.

In this paper we give necessary and sufficient conditions for the space of regular multilinear operators from the product of Banach lattices to a Dedekind complete Banach lattice to have the positive Schur property. We also characterize the positive Schur property on the positive projective m-fold tensor product of Banach lattices, $m \in \mathbb{N}$, and on its dual. (Received August 03, 2020)

1159-46-154 **B. Bin Dehaish*** (bbindehaish@yahoo.com), jeddah. *Fixed point iteration for a class of asymptotically nonexpansive mappings in modular function spaces.*

Abstract: In our talk, we will extend the fundamental results of Schu to a class of asymptotically nonexpansive mappings in modular function spaces. In particular, we will investigate the convergence of the Fibonacci–Mann iteration associated with a monotone asymptotic nonexpansive mapping defined in a modular function space. (Received August 03, 2020)

1159-46-166 **Parin Chaipunya*** (parin.cha@kmutt.ac.th), 126 Pracha Uthit Rd., Bang Mod, Thung Khru, Bangkok, 10140, Thailand. *On the conditioning of equilibrium problems under nonpositive curvatures.*

One of the most common condition used in the theory of equilibrium problems in linear spaces is the convexity/concavity of the equilibrium bifunction in question. When the underlying space is equipped with the geometry of nonpositive curvature, it turns out that the traditional convexity/concavity condition prevents the equilibrium problems to naturally generalize to variational inequalities. In this talk, we replace such convexity/concavity condition allows the equilibrium problems to generalize to variational inequalities and allows the use of the famous proximal methods. (Received August 04, 2020)

1159-46-177 **Abdul Rahim Khan*** (arahim@kfupm.edu.sa), Department of Mathematics and Statistics, King Fahd University of Petroleum & Minerals, Dhahran 31261, Dhahran, Eastern 31261, Saudi Arabia, and **Dolapo Muhammed Oyetunbi** (g201706250@kfupm.edu.sa), Department of Mathematics and Statistics, King Fahd University of Petroleum & Minerals, Dhahran 31261, Dhahran, Eastern 31261, Saudi Arabia. *Unique common fixed point for Asymptotically regular mappings.*

F. E. Browder and W. V. Petryshyn [The solution by iteration of nonlinear functional equations in Banach spaces, Bull. Amer. Math. Soc. ,72(1966), 571-575] introduced the concept of asymptotic regularity for single-valued mappings. This notion is significant since several contractive mappings are asymptotically regular; see for example, M. A. Geraghty [On contractive mappings, Proc. Amer. Math. Soc. ,40 (1973), 604-608]. In this talk, we present a common fixed point result for two asymptotically regular mappings, which are neither orbital continuous nor k-continuous on a complete metric space. Our work significantly improves and complements comparable results in the literature; in particular, the main result of J. Gornicki [On some mappings with a unique fixed point, J. Fixed Point Theory Appl., 22(2020):8] follows as a special case. (Received August 04, 2020)

1159-46-184 **Khairul Saleh***, Dept. of Math and Stat, KFUPM, Dhahran, Eastern , Saudi Arabia. *On asymptotically nonexpansive mappings with nonconvex domain.*

In this talk, we discuss the existence of fixed point of an asymptotically nonexpansive mapping on nonconvex domain in uniformly convex Banach space. We also introduce the notion of λ-firmly asymptotically nonexpansive mapping and prove the existence of fixed point of the mapping in nonconvex domain. (Received August 04, 2020)

1159-46-191 **Rawan Alharbi*** (1870730@uj.edu.sa), Makkah, Saudi Arabia. *Approximating fixed points for some generalized nonexpansive mappings in uniformly convex Banach spaces.*

Among this talk, we will investigate convergence fixed points theorems for some generalized nonexpansive mappings in uniformly convex Banach spaces. We will obtain weak convergence theorem under relevant conditions and strong convergence theorem by adding some necessary condition on the mappings. Our results extend and improve some recent results in the literature. (Received August 04, 2020)

47 ► Operator theory

1159-47-61 **Tomas Dominguez Benavides*** (tomasd@us.es), Facultad de Matematicas, Tarfia s/n, Sevilla, Spain, and **Pepa Lorenzo** (ploren@us.es). *Measures of noncompactness and fixed point for set-valued nonexpansive mappings in modular spaces.*

In this talk we will show the existence of a fixed point for a class of set-valued nonexpansive mappings defined in a modular space. The notion of uniform ρ-noncompact convexity with respect to a measure of noncompactness, as defined in [1], and the reflexivity of the Banach space associated to the corresponding Luxemburg norm of the modular space, are basic tools for our results.

[1] M. A. Khamsi, W. M. Kozlowski, Fixed Point Theory in Modular Function Spaces, Birkäuser: Basel, Switzerland, 2015 (Received July 28, 2020)

1159-47-160 **Monther Rashed Alfuraidan** and **Eduardo Daniel Jorquera Alvarez*** (eduardo.jorqueraa@userena.cl), La Serena, Chile, and **Mohamed Amine Khamsi**. *Fixed Point Theorems for Monotone Caristi Inward Mappings.*

In this paper, We introduce the monotone Caristi inward mappings. As an example, We show that monotone inward contraction mappings are monotone Caristi inward mappings. A general fixed point theorem for such mappings is given. A multivalued version of these results is also introduced. (Received August 03, 2020)

1159-47-175 **Adrian Petrusel***, Kogalniceanu Street, no. 1, 400084 Cluj-Napoca, Cluj, Romania, and **Gabriela Petrusel**, Kogalniceanu Street, no. 1, 400084 Cluj-Napoca, Cluj, Romania. *Existence and approximation for the fixed points of some nonself operators.*

In this talk we will present some fixed point results for nonself operators in complete metric spaces. We will discuss some existence and approximation theorems for the fixed points of nonself operators in a metric and a topological setting. The case of single-valued and multi-valued operators will be considered. Some applications are also suggested. The talk is based on several papers such as:

[1] A. Petrusel: Local fixed point results for graphic contractions, J. Nonlinear Variational Anal. 3(2019), no. 2, 141-148

[2] A. Petrusel, R. Trusca: Iterative approximations for non-self operators, Proc. 21st Intern. Symposium on Symbolic and Numer. Algorithms for Sci. Computing (SYNASC) 2019, IEEE Computer Soc. Conference Publ. Serv. 2019, 307-310

[3] A. Petrusel, G. Petrusel, J.-C. Yao: Multi-valued graph contraction principle with applications, Optimization, 69 (2020), no. 7-8, 1451-1556

[4] A. Petrusel, Radu Precup, Marcel-Adrian Serban: On the approximation of fixed points for non-self mappings on metric space, Discrete and Continuous Dynamical Systems-Series B, 25(2020), No. 2, 733-747 (Received August 04, 2020)

49 ► Calculus of variations and optimal control; optimization

1159-49-26 **Andrew M Pownuk*** (ampownuk@utep.edu), Department of Mathematical Sciences, The University of Texas at El Paso, 500 West University Avenue, El Paso, TX 79968. *Solution of Algebraic Equations by Using Autonomous Computational Methods.*

In order to find the answers to the qualitative research questions which are described by algebraic equations it is necessary to find exact or approximate solutions of appropriate algebraic equations. Solution method (if exists) is determined only by available algebraic operations and the form of given equation. Finding a solution by using only algebraic properties without knowing any prior knowledge about the solution procedure is a very complex mathematical problem. By using autonomous computational method, it is possible to find a step-by-step solution of selected algebraic equation by using information only about algebraic operations and the form of the equations. Selected results will be presented. Solutions created by the computational algorithms can be used in the future calculations in order to solve future mathematical problems. In some cases, it is possible to speed up the calculations by using machine learning techniques. Autonomous computational methods can be applied for autonomous development of scientific theories which are based on finite number of mathematical operations. Autonomous computational methods reduce the number of possible errors and allow processing of large amount of scientific data in large scale, parallel, and distributed way. (Received July 05, 2020)

1159-49-79 **Rafal Zalas*** (zalasrafal@gmail.com), Haifa, Israel, and **Simeon Reich**. *Error Bounds for the Method of Simultaneous Projections with Infinitely Many Subspaces.*

We investigate the properties of the simultaneous projection method as applied to countably infinitely many closed and linear subspaces of a real Hilbert space. We establish the optimal error bound for linear convergence of this method, which we express in terms of the cosine of the Friedrichs angle computed in an infinite product space. In addition, we provide estimates and alternative expressions for the above-mentioned number. Furthermore, we relate this number to the dichotomy theorem and to super-polynomially fast convergence. We also discuss polynomial convergence of the simultaneous projection method which takes place for particularly chosen starting points. For more details see <arxiv.org/abs/2007.10664>. (Received July 30, 2020)

1159-49-97 **Giorgio Bornia*** (giorgio.bornia@ttu.edu). *On a class of boundary optimal control problems in mechanics.*

We discuss Dirichlet boundary optimal control problems for either the Navier-Stokes equations or the elasticity equations together with incompressibility constraints. Such constraints impose integral compatibility conditions on full Dirichlet controls. We compare two different approaches for the treatment of these compatibility conditions: one that is based on the use of a scalar Lagrange multiplier, the other one that uses lifting functions to treat boundary controls as distributed controls. The differences between the two formulations are described. Numerical results are presented for the finite element approximation of the optimality systems arising from the first-order necessary conditions. (Received August 01, 2020)

1159-49-118 **Ouayl Chadli*** (o.chadli@uiz.ac.ma), Department of Economics, Ibn Zohr University, B.P. 8658, Poste Cité Dakhla, Agadir, Morocco. *Optimal Control of Quasi-Equilibrium Problems under monotonicity-type conditions with Applications.*

The main goal of this talk is to present our recent results on the study of the existence of solutions for optimal control problems governed by mixed quasi-equilibrium problems under monotonicity type conditions. More precisely, the state control system takes the general form of a mixed quasi-equilibrium problem described by the sum of a maximal monotone bifunction and a pseudomonotone bifunction in the sense of Brézis. As applications, we study the existence of solutions for optimal control problems governed by quasi-variational inequalities. Particularly, we consider the optimal control of an evolutionary quasivariational inequality described by a p-Laplacian type operator. The results obtained are new and can be applied to the study of optimal control of a variety of systems whose formulations can be presented as a mixed equilibrium problem. (Received August 03, 2020)

1159-49-181 **M K RIAHI*** (mohamed.riahi@ku.ac.ae), Abu Dhabi. *Time split control: A divide and conquer approach foroptimal control with PDE constraints.*

We proposed in [3] a hybrid Newton-Gradient approach and consider parallelization of the control through an on-off time-windowing decomposition. We show that our algorithm significantly improves the computational time compared with recognized methods. Convergence analysis of the new optimal control algorithm is provided for an arbitrary choice of partition. Numerical experiments are presented to illustrate the efficiency and the rapid convergence of the method. This work is currently subject of extension to further investigate the adaptive time windowing and coupling with parareal in time algorithm such as in [2]. (Received August 04, 2020)

51 ► Geometry

1159-51-173 **Alvaro Pampano***, alvaro.pampano@ehu.eus. *Construction of Rotational Constant Skew Curvature Surfaces in Space Forms.*

We introduce and study the exponential type curvature energy

$$\mathbf{\Theta}_\mu(\gamma) = \int_\gamma e^{\mu\kappa}$$

acting on curves immersed in 2-space forms $M^2(\rho)$. Here, κ denotes the curvature of the curve γ and the energy parameter μ is a constant.

Using a technique involving Killing vector fields along curves and the binormal evolution procedure, we construct rotational surfaces in 3-space forms $M^3(\rho)$ whose profile curves are critical for $\mathbf{\Theta}_\mu$.

It turns out that these binormal evolution surfaces have constant skew curvature, i.e. their principal curvatures satisfy a relation

$$\kappa_1 = \kappa_2 + c\,,$$

where c is a constant related with the energy parameter μ.

Conversely, any rotational surface with constant skew curvature in $M^3(\rho)$ can be locally described as a binormal evolution surface with prescribed velocity and, as a consequence, their profile curves are critical for $\mathbf{\Theta}_\mu$. (Received August 04, 2020)

52 ► Convex and discrete geometry

1159-52-63 **R. Amzi Jeffs** (rajeffs@uw.edu), Department of Mathematics, University of Washington, Seattle, WA 98195-4350, and **Isabella Novik*** (novik@uw.edu), Department of Mathematics, University of Washington, Seattle, WA 98195-4350. *Convex union representable complexes.*

Motivated by problems related to convex neural codes as well as by the notion of representable complexes, we introduce the class of convex union representable complexes. We discuss some of the fascinating properties that this class has (e.g., collapsibility of such complexes as well as of their Alexander duals, constructible-like behavior, etc.). We also present many open questions. (Received July 28, 2020)

53 ► Differential geometry

1159-53-18 **Anthony D Gruber*** (anthony.gruber@ttu.edu). *Codazzi tensors with parallel mean curvature.*

We address the local-to-global structure of submanifolds in space forms admitting a tensor which satisfies Codazzi equations. In particular, it is shown that simply connected submanifolds of nonnegative sectional curvature which admit a Codazzi tensor with parallel trace are product manifolds with factors determined by the spectrum of this tensor. The relationship of this result to submanifolds with parallel mean curvature is discussed, as well as its consequences for Einstein submanifolds and submanifolds with parallel Ricci tensor. (Received June 29, 2020)

1159-53-30 **Thomas A Ivey*** (iveyt@cofc.edu), College of Charleston, Charleston, SC 29424. *A brief survey of austere submanifolds.* Preliminary report.

Suppose X is a Riemannian manifold such that T^*X carries a canonical Ricci-flat Kahler metric (e.g., $X = \mathbb{R}^n$, S^n, or a compact rank-one symmetric space). A submanifold $M \subset X$ is *austere* if its conormal bundle N^*M is a special Lagrangian submanifold of T^*X. In this talk I will give a short survey of austere submanifolds, mostly for $X = \mathbb{R}^n$ Euclidean space, where the austerity conditions (first computed by Harvey and Lawson) imply that M is minimal but also include higher-order polynomial conditions on the eigenvalues of the second fundamental form of M. I will summarize classification results for low-dimensional submanifolds in $\mathbb{R}^n$, due to Bryant and Ionel-Ivey, and constructions of austere submanifolds due to Bryant, Dajczer and collaborators. I will also discuss the austere conditions for submanifolds of $\mathbb{C}P^n$, and recent work with Karigiannis on a variation of the austere construction where the fiber of N^*M is 'twisted' away from the zero section by adding a 1-form. I will conclude with a list of open questions. (Received July 10, 2020)

1159-53-37 **Stephen E McKeown*** (stephen.mckeown@utdallas.edu), Dept. of Mathematics, FO 35, UT Dallas, 800 W. Campbell Road, Richardson, TX 75080. *Extrinsic Curvature and Conformal Gauss-Bonnet for Four-Manifolds with Corner.*

Building on work of Gilkey, Branson, Chang, and Qing, I present a rewriting of the Gauss-Bonnet formula for Riemannian four-manifolds with codimension-two corner for which each integrand has interesting conformal transformation properties; in particular, I define a new linear second-order conformally invariant differential operator that parametrizes the conformal change of the corner integrand. (Received July 15, 2020)

1159-53-56 **Rafael Lopez*** (rcamino@ugr.es), Dpt. Geometria y Topologia, University of Granada, 18071 Granada, Spain. *Surfaces with constant curvature in Euclidean space: old problems and new results.*

Surfaces with constant Gauss curvature and constant mean curvature play a central role in classical differential geometry of Euclidean space $\mathbb{R}^3$. In order to find explicit examples of such surfaces, it is natural to impose some geometric property on the surface, such for instance, that the surface is rotational or that is ruled. This idea appears already at the beginning of the classical theory: it is enough to mention works of Euler, Meusnier, Scherk, Schwarz and Riemann. In this talk, we address two techniques that were developed in the XIXth century and that have been, in part, forgotten over in the course of time.

A first class of surfaces are the translation surfaces, which are surfaces can be locally written as the sum of two space curves. We classify all translations surfaces with constant Gauss curvature. In case of minimal translation surfaces, we give a procedure to find many examples.

The second family of surfaces are those ones that can be expressed by separation of variables $f(x) + g(y) + h(z) = 0$. We give a full classification of separable surfaces with constant Gaussian curvature and if the mean curvature is a non-zero constant, we prove that the surface is rotational.

This is a joint work with T. Hasanis and O. Perdomo. (Received July 26, 2020)

1159-53-90 **Annalisa Baldi***, Dipartimento di Matematica, Piazza di Porta S. Donato 5, 40126 Bologna, Italy. *Poincaré and Sobolev inequalities for differential forms in Heisenberg groups.*

In this talk we present some recent results obtained in collaboration with B. Franchi and P. Pansu about Poincaré and Sobolev inequalities for differential forms in Heisenberg groups (some results are new also for Euclidean spaces). For L^p, $p > 1$, the estimates are consequence of singular integral estimates. I would like to concentrate the seminar, in particular, to the limiting case L^1, where the exterior Rumin-differential of a differential form is measured in L^1 norm. Unlike for L^p, $p > 1$, the estimates are doomed to fail in top degree. In the limiting case, the singular integral estimates are replaced with inequalities which go back to Bourgain-Brezis and Lanzani-Stein in Euclidean spaces, and to Chanillo-Van Schaftingen and Baldi-Franchi-Pansu in Heisenberg groups. Also the case $p = \infty$ is considered. (Received July 31, 2020)

1159-53-123 **Sigurd B Angenent** and **Dan Knopf*** (danknopf@math.utexas.edu). *Ricci solitons, conical singularities, and nonuniqueness.*

In dimension n=3, there is a complete theory of weak solutions of Ricci flow — the singular Ricci flows constructed from Ricci flow spacetimes by Kleiner and Lott. These are unique across singularities, as was proved by Bamler and Kleiner. In joint work with Angenent, we show that uniqueness should not be expected to hold for Ricci flow spacetimes (weak solutions) in dimensions n>4. Specifically, we exhibit a discrete family of asymptotically conical gradient shrinking soliton singularity models, each of which admits non-unique forward continuations by gradient expanding solitons. (Received August 03, 2020)

1159-53-132 **Alex Mramor*** (amramor1@jhu.edu), 311 Krieger Hall, Baltimore, MD. *Compactness and finiteness results for rotationally symmetric self shrinkers.*

In this talk I'll discuss some recent results on rotationally symmetric self shrinking tori, with an eye towards the question of uniqueness of Angenent's shrinking donut. (Received August 03, 2020)

55 ► Algebraic topology

1159-55-108 **Michal J. Kukiela** (mckuk@mat.umk.pl) and **Bernd S W Schroeder*** (bernd.schroeder@usm.edu). *Fixed Simplexes of Group Actions on Link Collapsible Simplicial Complexes.*

Once the fixed point property is established for a class of ordered sets, it is natural to ask which properties of the class are inherited by the fixed point sets of order-preserving maps.

A simplicial complex is called link collapsible iff it has one vertex, or, there is a simplicial retraction onto a one-vertex-deleted subcomplex such that the subcomplex as well as the link of the retracted vertex are link collapsible. We prove that, for every ordered set with a link collapsible chain complex, the (topological realization of the) fixed point set of every order-preserving function is contractible. Examples show that this result is near optimal in terms of the property of the fixed point set.

The proof focuses on the fixed simplex sets of group actions on link collapsible simplicial complexes. On one hand, this approach shows that, following the canonical translation from ordered sets to comparability graphs to chain/clique complexes to their topological realizations, there appears to be a significant change of behavior when we generalize from graphs to simplicial complexes that are not clique complexes. On the other hand, the generality of the result allows us to confirm Kahn, Saks and Sturtevant's evasiveness conjecture for link collapsible simplicial complexes. (Received August 02, 2020)

1159-55-185 **Adam Clay*** (adam.clay@umanitoba.ca), **Tyrone Ghaswala** and **Jason Bell**. *The L-space conjecture, left-orderability and circular orderability of fundamental groups.*

The L-space conjecture posits a relationship between the Heegaard-Floer homology of a 3-manifold, whether or not the manifold supports a coorientable taut foliation, and left-orderability of its fundamental group. Motivated by this conjecture, my talk will focus on the third of these three, that is, on left-orderability of the fundamental

groups of 3-manifolds and on a generalization of left-orderability known as circular-orderability. It happens that the generalized notion of circular-orderability often arises more naturally in this setting, and recent advances in left-ordering fundamental groups mostly tend to rely on a cohomological analysis in order to pass from circular to left-orderings. This talk will outline some of the classical techniques for doing this, as well as a few new ones. This is joint work with Jason Bell and Tyrone Ghaswala. (Received August 04, 2020)

57 ► Manifolds and cell complexes

1159-57-41 **Ross Staffeldt*** (ross@nmsu.edu), MSC 3MB, PO Box 30001, Las Cruces, NM 88003-0131, and **Rama Mishra** and **Hitesh Raundal**. *Probability distributions associated with weaving knots.* Preliminary report.

In 2016 Champanerkar, Kofman, and Purcell proved asymptotically efficient bounds on the hyperbolic volume of the complements of general weaving knots. Motivated by this result, we started to investigate other invariants of weaving knots for asymptotic behavior. Running computer experiments on knots $W(3,n)$, the weaving knots on 3 strands with $2n$ crossings, having up to 300 crossings we discovered that the ranks of Khovanov homology groups appeared to be normally distributed asymptotically. The phenomenon resembles the well-understood relationship between the discrete binomial distribution and the continuous normal distribution. As these knots are alternating, once one knows the signature, the ranks of their Khovanov homology and Heegaard-Floer homology groups are determined by the coefficients of the Jones polynomial and the coefficients of the Alexander polynomial, respectively. The focus of the report will be on recursive formulas for the coefficients of the Alexander polynomials of $W(3,n)$ and progress toward identifying continuous distributions that approximate the discrete data. We will also describe experiments performed by two of the authors indicating the asymptotic phenomena persist as the number of strands increases. (Received July 17, 2020)

1159-57-42 **Francis Bonahon*** (fbonahon@usc.edu), **Helen Wong** and **Tian Yang**. *Asymptotics of quantum invariants of surface diffeomorphisms.* Preliminary report.

The Kashaev-Murakami-Murakami Volume Conjecture asserts that, for a knot K in the 3–sphere S^3, the evaluation $J_n^K\left(\mathrm{e}^{\frac{2\pi\mathrm{i}}{n}}\right)$ of the n–th colored Jones polynomial of K at the root of unity $\mathrm{e}^{\frac{2\pi\mathrm{i}}{n}}$ grows exponentially as $n\to\infty$, with exponential growth rate related to the hyperbolic volume $\mathrm{vol}(S^3-K)$ of the knot complement. We will discuss a closely related conjecture for diffeomorphisms of surfaces, based on the representation theory of the quantum Teichmüller space and/or the Kauffman bracket skein algebra of the surface. We will present experimental evidence for this conjecture, and describe partial results in work in progress. (Received July 18, 2020)

1159-57-98 **Allison N. Miller*** (allison.miller@rice.edu). *Satellite operators on knot concordance.*

I'll start with a brief discussion of why 4-manifold topology is interesting and unusual, and the ways that knot theory offers powerful tools to help us better understand it. Next, I'll sketch the basics of knot concordance, focusing on geometrically nice operators coming from the classical satellite construction. I'll go on to state some results and open questions in this area, and then close by discussing recent work (joint with P. Feller and J. Pinzón-Caicedo) on how various 4-dimensional measures of knot complexity change under the satellite operation. (Received August 01, 2020)

1159-57-122 **Susan Hermiller*** (hermiller@unl.edu), Department of Mathematics, University of Nebraska, Lincoln, NE 68588-0130. *Computation in fundamental groups of 3-manifolds.*

For all fundamental groups of compact 3-manifolds, the word problem is solvable; that is, for each 3-manifold group there is an algorithm which, upon input of a word in the generators, determines whether the word represents the identity. In the case that the 3-manifold has no Nil or Sol prime factors, the word problem is solvable in quadratic time. In this talk I will discuss solutions the word problem for 3-manifold groups using finite state automata (computers with a finite amount of memory), and methods to build these algorithms. Based on joint projects with M. Brittenham and T. Susse, and with D. Holt, S. Rees, and T. Susse. (Received August 03, 2020)

1159-57-130 **Luis G Valdez-Sanchez*** (lvsanchez@utep.edu), University of Texas at El Paso, Department of Mathematical Sciences, 500 West University Ave, El Paso, TX 79968. *The Kakimizu complex for a genus one hyperbolic knot in the 3-sphere.* Preliminary report.

For a knot $K \subset \mathbb{S}^3$ with exterior $X_K = \mathbb{S}^3 \setminus \operatorname{int} N(K)$, the Kakimizu complex $MS(K)$ is the simplicial complex with simplices the collections of isotopy classes of minimal genus Seifert surfaces in X_K that can be realized by mutually disjoint and non-parallel such surfaces.

The complex $MS(K)$ is flag, finite dimensional and contractible, and in the case of a hyperbolic knot K it is a finite complex with diameter bounded by a quadratic function of its genus.

In this talk we consider the complex $MS(K)$ for genus one hyperbolic knots $K \subset \mathbb{S}^3$, in which case the optimal bounds for its dimension and diameter are known to be 5 and 2, respectively. We refine these results by showing that, in the top dimension, the complex $MS(K)$ consists of at most 2 simplices, and exactly one 5-simplex in the 5-dimensional case. We also provide infinitely many examples of such knots for which $MS(K)$ consists of exactly one or exactly two top-dimensional simplices. (Received August 03, 2020)

1159-57-135 **J Scott Carter*** (carter@southalabama.edu), 16003 Windroot St., Austin, TX 78728. *Amusing Permutation Representations of Group Extensions.*

This work is joint with Byeorhi Kim and Yungju Bae. Herein we demonstrate a number of diagrammatic ways of depicting the 2-fold central extensions of the 3-dimensional motion groups. Some other groups will be considered. For example, the hyper-octahedral groups have similar depictions. The methods are based upon a consequence of the Krasner-Kaloujnine Theorem. The talk will be replete with pictures. (Received August 03, 2020)

1159-57-143 **Mark Brittenham*** (mbrittenham2@unl.edu), Department of Mathematics, Avery 203, University of Nebraska, Lincoln, NE 68588, and **Susan Hermiller** (hermiller@unl.edu), Department of Mathematics, Avery 203, University of Nebraska, Lincoln, NE 68588. *Minimal crossing knot diagrams and unknotting number.*

In the 1990s Bernhard and Jablan independently conjectured that the unknotting number of a knot can be computed using minimal crossing number diagrams and crossing changes. This computational method has proved to have a remarkable track record of correctly predicting the unknotting numbers of knots, but in recent work we showed that it cannot always succeed; it is sometimes too high. We will discuss this result, as well as what the data amassed in searching for counterexamples can say about other open problems on unknotting number. (Received August 03, 2020)

1159-57-170 **Kenneth L Baker*** (kenken@math.miami.edu), 1365 Memorial Drive, 515 Ungar, Mathematics, Coral Gables, FL 33146. *The Morse-Novikov number of knots: connected sums and cabling.*

We show the Morse-Novikov number of knots in S^3 is additive under connected sum and unchanged by cabling, answering a question of M. Boileau and C. Weber as communicated by A. Pajitnov. (Received August 04, 2020)

58 ► Global analysis, analysis on manifolds

1159-58-131 **Yunfeng Zhang*** (yunfeng.zhang@uconn.edu). *Size of Laplacian eigenfunctions on compact symmetric spaces.*

We discuss estimates on the L^p-norm of Laplacian eigenfunctions on compact globally symmetric spaces. We present a result on general compact globally symmetric spaces, and a better result for the special case of compact Lie groups. These estimates resemble those on tori as established by Bourgain, and serve as improvement of the general estimates on compact manifolds as established by Sogge. (Received August 03, 2020)

60 ► Probability theory and stochastic processes

1159-60-6 **Osei K Tweneboah*** (oktweneboah@miners.utep.edu) and **Maria C Mariani**. *Modeling earthquake series dependencies by using coupled system of stochastic differential equations.* Preliminary report.

In this work, we model the influence of any natural or physical phenomena and the effect on each other, by using a coupled system of two stochastic differential equations. We will describe the correlation of earthquake series from four regions in the country Chile by using coupled system of Ornstein-Uhlenbeck stochastic differential equations. The objective is to model the dependency and effects of earthquake series occurring at the four

different regions within the same geographical area. We will test our approach and present a numerical example by using real world earthquake series. (Received May 07, 2020)

1159-60-8 **Md Al Masum Bhuiyan*** (bhuiyanm@apsu.edu), 103 King Cole Dr., Clarksville, TN 37042, **Maria C. Mariani**, 500 W University Ave., El Paso, TX 79968, **Maria P. Beccar Varela**, 500 W University Ave, El Paso, TX 79968, and **Ionut Florescu**. *ANALYSIS OF HIGH FREQUENCY FINANCIAL DATA USING DYNAMIC FOURIER MODELS.*

This work deals with the modeling of high frequency financial stock market data. A sequence of return values of stock market in each minute and on each day is studied in this work. It has been observed that there is a correlation among the numbers of data points at successive time intervals. The mean-reverting behavior, stationary behavior, and long memory behavior of financial data are analyzed. These behaviors help to characterize the autocorrelation of data at different levels. We study a Discrete Fourier Transform (DFT) to determine the power spectrum of returns in a stationary condition. A tapering process with DFT technique has been used to avoid the spectral leakage or discontinuity in the sequence of financial data. To estimate the power spectrum, the smoothed periodograms of data are computed by using the Daniell kernel window. Our results suggest that the power spectrums are effective in making difference between the minute-based data and the day-based data corresponding to their frequencies. The frequency-based power spectrum is important as well in detecting the financial crash of stock market, which facilitates prediction. (Received May 09, 2020)

1159-60-10 **Michael J Roberts***, 1126 College St N, 203, Fargo, ND 58102. *Application of sequential jump hypothesis testing to oil prices.*

In this presentation, a sequential hypothesis test for the detection of general jump size distribution is investigated. Infinitesimal generators for the corresponding log-likelihood ratios are presented and analyzed. Bounds for infinitesimal generators in terms of super-solutions and sub-solutions are computed. This is shown to be implementable in relation to various classification problems for a crude oil price data set. Machine and deep learning algorithms are implemented to extract a specific deterministic component from the crude oil data set, and the deterministic component is implemented to improve the Barndorff-Nielsen and Shephard model, a commonly used stochastic model for derivative and commodity market analysis. (Received May 24, 2020)

1159-60-19 **Chuntian Wang***, Department of Mathematics, The University of Alabama, Tuscaloosa, AL 35487-0350, and **Yuan Zhang**, **Andrea L Bertozzi** and **Martin Short**. *Stochastic-Statistical Modeling of Criminal Behavior.*

Residential crime is one of the toughest issues in modern society. A quantitative, informative, and applicable model of criminal behavior is needed to assist law enforcement. We have made progress to the pioneering statistical agent-based model of residential burglary (Short et al., Math. Models Methods Appl., 2008) in two ways. (1) In one space dimension, we assume that the movement patterns of the criminals involve truncated Levy distributions for the jump length. Furthermore (2), in two space dimensions, we used the Poisson clocks to govern the time steps of the evolution of the model, rather than a discrete-time Markov chain with deterministic time increments used in the previous works. Poisson clocks are particularly suitable to model the times at which arrivals enter a system. Introduction of the Poisson clock not only produces similar simulation output, but also brings in theoretically the mathematical framework of the Markov pure jump processes, e.g., a martingale approach. The martingale formula leads to a continuum equation that coincides with a well-known mean-field continuum limit. Moreover, the martingale formulation together with statistics quantifying the relevant pattern formation leads to a theoretical explanation of the finite size effects. (Received June 29, 2020)

1159-60-21 **Parisa Fatheddin*** (fatheddin.1@osu.edu), 1465 Mt Vernon Ave, Marion, OH 43302, and **Zhoyang Qiu** (zhqmath@163.com) and **Yanbin Tang** (tangyb@hust.edu.cn). *Large deviations for two-dimensional stochastic Navier-Stokes and stochastic Boussinesq equations.*

We consider the large deviation principle by the classical Azencott method for two-dimensional stochastic Navier-Stokes equation and as applications derive the law of the iterated logarithm and exit problem. Large deviations by weak convergence approach is also presented for two-dimensional stochastic Boussinesq equation. (Received July 01, 2020)

1159-60-24 **William Kubin*** (wkubin@miners.utep.edu), **Maria C. Mariani** (mcmariani@utep.edu), **Peter K. Asante** (pkasante@miners.utep.edu) and **Osei K. Tweneboah** (oktweneboah@miners.utep.edu). *Relationship between Self-Similar Models and The Lévy Model.*

Many real-world time series are known to exhibit memory effects of some sort. These memory effects may be short or long, temporal or permanent. Herein, we make use of time series that exhibit long-memory behavior. Barany et al. (2012) described an inverse relationship between the Hurst exponent and the parameter of the Lévy model. The purpose of this study is to describe a mathematical relation between self-similar models (Detreneded Fluctuation Analysis and Diffusion Entropy Analysis) and the Lévy model. Using financial and geophysical time series that exhibit long-memory behavior, this relation is confirmed numerically. To investigate the presence of long-memory behavior in the time series, we estimate scaling parameters of DFA (H), DEA (δ) and the Lévy Model (α). We conclude that an inverse relation is generated between the self-similar parameters and the Lévy model parameter characterizing the data. i.e. $(H,\ \delta) \approx \frac{1}{\alpha}$ (Received July 05, 2020)

1159-60-25 **PETER K ASANTE*** (pkasante@miners.utep.edu), TX, and **Maria C Mariani**, **William Kubin** and **Osei K Tweneboah**. *Superposed Inverse-Gaussian (IG(a,b)) Ornstein-Uhlenbeck model applied to geophysics and financial data.*

Understanding stock behaviors benefits not only investors but also economic growths of countries. This speaks to the many existing and ongoing research on financial indices. On the other hand, earthquake occurrences though not directly linked to stock markets, may pose unprecedented risks including loss of human lives and even the fall of stock markets as experienced during and after the 2011 tohoku earthquake and tsunami in Japan. Research works in these areas have shown that in most cases data arising from financial indices and geophysics usually deviate from normal behavior. Thus, they are best modeled with non-Gaussian processes (Mariani et. al (2020)). In Mariani et. al (2015) and Habtemicael et. al (2014), the background driving process (BDLP) of the SDE was a gamma process. In this work we propose a superposed stochastic differential equation of Ornstein Uhlenbeck (OU) type driven by an Inverse Gaussian process (BDLP). We apply this model on high frequency financial indices and geophysical data. We seek to draw a comparison between the performance of the superposed $\Gamma(a, b)$ OU and the superposed IG(a, b) OU applied to both the financial indices and the geophysical data. (Received July 05, 2020)

1159-60-31 **Hakima Bessaih*** (bessaih@uwyo.edu), 1000 E University Ave, Depts. 3036 & 3332, Ross Hall 210, Laramie, WY 82071, and **Benedetta Ferrario**. *Invariant measures for stochastic damped 2D Euler equation.*

We study a two-dimensional Euler equations, damped by a linear term and driven by an additive noise. The existence of weak solutions has already been studied; pathwise uniqueness is known for solutions that have bounded vorticity. We prove the Markov property and the existence of an invariant measure in the space of bounded vorticity. Since this is not a Polish space, we cannot apply the classical tools found in the literature but we develop a Krylov-Bogoliubov's type method working with the weak star and the bounded weak star topologies. (Received July 10, 2020)

1159-60-35 **Xia Chen** (xchen@math.utk.edu), Department of Mathematics, University of Tennessee Knoxville, 813 Swanston Street, Knoxville, TN 37996, **Aurelien Deya** (aurelien.deya@univ-lorraine.fr), Institut Elie Cartan, University of Lorraine, Nancy, France, **Cheng Ouyang*** (couyang@math.uic.edu), Depart of Math, Statistics and Compute Sci, University of Illinois at Chicago, 851 S Morgan St, Chicago, IL 60607, and **Samy Tindel** (stindel@purdue.edu), Department of Mathematics, Purdue University, 150 N. University Street, West Lafayette, IN 47907. *Moment estimates for some renormalized parabolic Anderson models.*

We consider some parabolic Anderson models with Gaussian noise whose space time covariance function is singular. We shall give some information about the moments of the solution when the stochastic heat equation is interpreted in the Skorohod sense. Of special interest is the critical case, in which one observes a phase transition (in time) for the existence of moments. The talk is based on a joint work with Xia Chen, Aurelien Deya and Samy Tindel. (Received July 14, 2020)

1159-60-45 **Paul Glasserman** and **Dan Pirjol*** (dpirjol@gmail.com), School of Business, Hoboken, NJ 07030, and **Qi Wu**. *Tail risk monotonicity under temporal aggregation in GARCH(1,1) models.*

The stationary distribution of a GARCH(1,1) process has power law tails, which makes it appropriate for modeling financial time series for heavy tailed variables. A GARCH process observed at one frequency can be

approximated with another observed at a lower frequency and different parameters. We study the change in the tail exponent under temporal aggregation of the time series, and derive conditions under which the tail exponent increases under temporal aggregation. These conditions cover most relevant combinations of parameters and innovation distributions, but we also note the existence of counterexamples near the edge of the parameter space. (Received July 22, 2020)

1159-60-129 **Ming-Ying Leung*** (mleung@utep.edu). *Scan statistics for finding inversion clusters in genomic sequences.*

Scan statistics have been applied in various biomedical studies to identify genomic regions containing nonrandom clusters of chromosomal translocation breakpoints, viral DNA integration sites, or other specific word patterns in molecular sequences. In these applications, the positions of the sites of interests are modeled as independent and identically distributed random points on the unit interval. The scan statistics are then defined either as the maximum count of points contained in a scanning window of fixed length or the minimum aggregated spacing between a fixed number of consecutive points. We have applied the scan statistics to characterize palindrome clusters in DNA and search for inversion clusters in RNA sequences. An inversion is a string of nucleotide bases followed closely by its inverse complementary sequence with a gap of g intervening nucleotides in between, and a palindrome is a special case of inversions with g = 0. Several palindrome clusters found on herpesvirus genomes are associated with viral replication origins. Inversion clusters with g ranging from 0 to 8 have also been identified in genomic RNAs of coronaviruses. Such inversions are essential for RNA secondary structure prediction, which is a computationally intensive problem for long RNA sequences. (Received August 03, 2020)

1159-60-164 **Si Tang*** (sit218@lehigh.edu), **Rick Durrett**, **Matthew Junge** and **Steven Lalley**. *Phase-transition and critical behavior in the chase-escape Model.*

Chase-Escape is a simple stochastic model that describes a predator-prey interaction. In this model, there are two types of particles, red and blue. Red particles colonize adjacent empty sites at an exponential rate λ_R, whereas blue particles take over adjacent red sites at exponential rate λ_B, but can never colonize empty sites directly. Numerical simulations suggest that there is a critical value p_c for the relative growth rate $p := \lambda_R/\lambda_B$. When $p < p_c$, red particles will always go extinct, and when $p > p_c$ mutual survival of both types of particles becomes possible. In particular, $p_c \approx 0.50$ for the square lattice case ($\mathbb{Z}^2$). In this talk, we will discuss the critical behavior near p_c as well as a few special cases where we can prove the existence of such a phase transition and find the precise value of p_c. (Received August 04, 2020)

1159-60-209 **Tonghui Wang*** (twang@nmsu.edu), Department of Mathematical Sciences, Las Cruces, NM 88003, and **Ziwei Ma**. *Decomposition of Quadratic Forms under Skew-normal Setting.*

The normality is a fundamental assumption in many natural and social science theories such as economics, finance, population genetics, just to name a few. However, the real data, which exhibit asymmetry, are very common. The more flexible model is desired and necessary to archive robust and reliable analysis. The family of skew-normal distributions, which is an extension of the normal distributions and is enable to model skewed observations. In this talk, (i) the properties of skew normal family in both univariate and multivariate forms are discussed; (ii) the noncentral skew chi-square distribution is introduced; (iii) the de-compositions of noncentral skew chi-square distribution and of noncentral skew Wishart distributions are obtained. Examples are given for the illustration of the main results. (Received August 06, 2020)

62 ► Statistics

1159-62-43 **Treena Basu***, 1600 Campus Road, Los Angeles, CA 90041, and **Kanadpriya Basu**, **Ron Buckmire** and **Nishu Lal**. *"WILL THEY OR WON'T THEY?": PREDICTIVE MODELS OF STUDENT COLLEGE COMMITMENT DECISIONS USING MACHINE LEARNING.*

Every year, academic institutions invest considerable effort and substantial resources to influence, predict and understand the decision-making choices of applicants who have been offered admission. In this study, we applied several supervised machine learning techniques to four years of data on 11,001 students, each with 35 associated features, admitted to a small liberal arts college in California to predict student college commitment decisions. By treating the question of whether a student offered admission will accept it as a binary classification problem, we implemented a number of different classifiers and then evaluated the performance of these algorithms using the metrics of F-measure and area under the receiver operator curve. The results from this study indicate that the logistic regression classifier performed best in modeling the student college commitment decision problem,

i.e., predicting whether a student will accept an admission offer, with an AUC score of 79.6%. The significance of this research is that it demonstrates that many institutions could use machine learning algorithms to improve the accuracy of their estimates of entering class sizes, thus allowing more optimal allocation of resources and better control over net tuition revenue. (Received July 20, 2020)

1159-62-54 **Bowen Zhao**, **Xi Xiao**, **Wanpeng Zhang**, **Bin Zhang** and **Guojun Gan*** (guojun.gan@uconn.edu), 341 Mansfield Rd, U-1009, Storrs, CT 06269, and **Shutao Xia**. *Self-Paced Probabilistic Principal Component Analysis for Data with Outliers.*

Principal Component Analysis (PCA) is a popular tool for dimension reduction and feature extraction in data analysis. Probabilistic PCA (PPCA) extends the standard PCA by using a probabilistic model. However, both standard PCA and PPCA are not robust, that is, they are sensitive to outliers. To alleviate this problem, we propose a novel method called Self-Paced Probabilistic Principal Component Analysis (SP-PPCA) by introducing the Self-Paced Learning mechanism into PPCA. Furthermore, we design the corresponding optimization algorithm based on an alternative search strategy and an expectation-maximization algorithm, so that SP-PPCA uses an iterative procedure to find the optimal projection vectors and filter out outliers. In this talk, I will present the proposed method and some experimental results based on both synthetic and real data. (Received July 25, 2020)

1159-62-55 **Lei Liu*** (lei.liu@wustl.edu), 660 S. Euclid Ave., St. Louis, MO 63141, and **Lili Liu**, **Mae Gordon**, **Michael Kass**, **Philip Miller**, **Lu Lin** and **Shujie Ma**. *Capturing Heterogeneity in Repeated Measures Data by Fusion Penalty.*

In this paper we are interested in capturing heterogeneity in clustered or longitudinal data. Traditionally such heterogeneity is modeled by either fixed effects or random effects. In fixed effects models, the number of degree of freedom for the heterogeneity equals the number of clusters/subjects minus 1, which could result in less efficiency. In random effects models, the heterogeneity across different clusters/subjects is described by e.g., a random intercept with 1 parameter (for the variance of random intercept), which could lead to oversimplification and biases (for the estimates of subject-specific effects). Our "fused effects" model stands in between these two approaches: we assume that there are unknown number of distinct levels of heterogeneity, and use the fusion penalty approach for estimation and inference. We evaluate and compare the performance of our method to the fixed and random effects models by simulation studies. We apply our method to the Ocular Hypertension Treatment Study (OHTS) to capture the heterogeneity in the progression rate toward primary open-angle glaucoma of left and right eyes of different subjects. (Received July 25, 2020)

1159-62-66 **Chamila Perera*** (chamila.perera4@gmail.com), **Xiaogang Su**, **Mae Gordon**, **Michel Kass**, **Philip Miller** and **Lei Liu**. *Precision Medicine: Subgroup Identification in Ocular Hypertension Treatment Study.*

Subgroup identification is a major topic in precision medicine specially when the treatment effect is significant only for some subjects. Here, we are interested in identifying patient subgroups responding differently to the topical treatment on the risk of primary open angle glaucoma (POAG) in Phase I of the Ocular Hypertension Treatment Study (OHTS I). Due to a large portion of missing values for some variables, we first conducted multiple imputation resulting in five imputed data sets. Then we applied an interaction tree (IT) procedure which uses recursively partitioning methods to identify subgroups with homogeneous treatment effect. We identified some subgroups with differential treatment effect. (Received July 28, 2020)

1159-62-114 **Michael Orosz*** (mcorosz@miners.utep.edu), 500 W University Ave, El Paso, TX 79968, **Michael Pokojovy** (mpokojovy@utep.edu), 500 W University Ave, El Paso, 79968, and **Maria C. Mariani** (mcmariani@utep.edu), 500 W University Ave, El Paso, TX 79968. *Quantitative stock market modeling using multivariate geometric random walk.* Preliminary report.

We propose a new stock market model based on multivariate geometric random walk without imposing any parametric assumptions (such as Gaussianity) or structural assumptions (such as ellipticity) on the increments and establish connection with continuous-time Geometric Brownian Motion. Our approach can be applied to modeling a variety of stocks traded at multiple stock exchanges and can adequately account for heavy tails and other departures from Gaussianity. Both calibration and prediction steps are genuinely nonparametric. Model calibration involves multivariate imputation to account for partially overlapping trading hours, while prediction is performed using resampling from imputed observations. Per-comparison and simultaneous prediction regions for multiple stocks and prediction intervals for stock portfolios are constructed. Finally, our model is applied to a wide selection of stocks traded at NYSE and LSE. Analyzing three months' worth of closing stock price

data collected from over 300 stocks sampled every minute, respective prediction regions are constructed and successfully "backtested" on a one-month horizon of historical data. (Received August 02, 2020)

1159-62-124 **Andrews Tawiah Anum***, atanum@miners.utep.edu. **Robust Statistical Inference For The Gaussian Distribution**.

The aim of robust statistics is to develop statistical procedures which are not unduly influenced by outliers or observations that are not representative of the underlying "true" data generating process. This talk focuses on an estimator with this characteristic and the function f for the divergence function is the univariate normal distribution and the parameter α is chosen between [0,1]. The estimator fails when we rely on the classic Newton's method to converge to the minimum of the density power divergence (MDPD) function. There is a tendency of such estimator never to approach this minimum and thus we implement the MDPD estimator using the Gradient Descent with Armijo's Rule. The MPDP estimator is compared to one of the most prominent competitors in the area of robust estimation, the univariate Minimum Covariance Determinant (MCD) estimator, to examine the performance of the estimator under discussion. For a real and fair comparison of the results, breakdown points for the MPDP estimator and the MCD are matched. The implementation of the MDPD estimator was efficient in the sense that taking a data set with a given number of outlying values, the numerical implementation was in correspondence with the theoretical breakdown point. (Received August 03, 2020)

1159-62-176 **Jun Zhang** (juz30@pitt.edu), **Greg J Siegle** (gsiegle@pitt.edu), **Wendy D'Andrea** (dandreaw@newschool.edu) and **Robert T Krafty*** (rkrafty@pitt.edu). *Interpretable Principal Component Analysis for Multilevel Multivariate Functional Data.*

Many studies collect functional data from multiple subjects that have both multilevel and multivariate structures. An example of such data comes from popular neuroscience experiments where participants' brain activity is recorded using modalities such as EEG and summarized as power within multiple time-varying frequency bands within multiple electrodes, or brain regions. In this talk, we discuss a novel approach to conducting interpretable principal components analysis on multilevel multivariate functional data that decomposes total variation into subject-level and replicate-within-subject-level (i.e. electrode-level) variation, and provides interpretable components that can be both sparse among variates (e.g. frequency bands) and have localized support over time within each frequency band. Smoothness is achieved through an roughness penalty, while sparsity and localization of components are achieved by solving an innovative rank-one based convex optimization problem with block Frobenius and matrix L1-norm based penalties. The method is used to analyze data from a study to better understand reactions to emotional information in individuals with histories of trauma and the symptom of dissociation. (Received August 04, 2020)

1159-62-179 **Suneel Babu Chatla*** (sbchatla@utep.edu) and **Galit Shmueli**. *Functional Regression with the COM-Poisson Distribution: a Study of Modeling COVID-19.* Preliminary report.

We develop a functional regression model for the Conway-Maxwell-Poisson(CMP) distribution (Shmueli et al., 2005) which is a flexible two-parameter generalizationof the Poisson distribution and can model both underdispersion and overdispersion in thecount data. For model estimation, we consider the component-wise boosting algorithm(Hothorn et al., 2014). The advantage of this approach is that it can provide model selection and handle large number of covariate effects which could be more than the number ofobservations. One of the noted drawbacks of boosting algorithm is its tendency to select a relatively high number of non-informative covariates. To overcome this limitation, our approach also incorporates stability selection to investigate the importance of covariates by repeatedly subsampling the data. We illustrate the usefulness of our method by modelingthe countrywide daily confirmed deaths due to COVID-19 (Received August 04, 2020)

1159-62-187 **Xiaogang Su***, 500 W. University Ave., El Paso, TX 79968, and **Youngjoo Cho**, **Liqiang Ni**, **Lei Liu** and **Elise Dusseldorp**. *Refined Moderation Analysis with Categorical Outcomes.* Preliminary report.

Concerning moderation analysis with categorical outcomes, we start with an interesting observation, which shows that heterogeneous treatment effects could be equivalently estimated via a role exchange between the outcome and the treatment variable. The result holds for both experimental data and observational data, yet with an important difference in interpretation. On this basis, we explore two methods of obtaining more efficient estimators of the moderating effects. One approach is to combine two estimators while the other involves a joint estimator. Refined statistical inference can be made accordingly. The improved efficiency is helpful in addressing the lack-of-power problem that is common in search for moderators. We investigate the proposed method by simulation and provide an illustration with data from a randomized trial concerning wart treatment. (Received August 04, 2020)

1159-62-198 **Noelle I. Samia*** (n-samia@northwestern.edu), 2006 Sheridan Road, Evanston, IL 60208. *Climate-driven context-dependent structure of vole population: Analysis using Bayesian Mixture Model.*

We develop a Bayesian mixture model for analyzing data on grey-sided voles derived from a monitoring program in Hokkaido, Japan. The number of voles caught at a given location and time is assumed to follow a binomial distribution that depends on the number of traps and probability of being trapped. This probability depends on a latent time series modeled as a mixture that accommodates covariates. Estimation and inference are performed using Markov chain Monte Carlo techniques.

This is based on a joint work with Michael Bertolacci, Ori Rosen, Osnat Stramer, Takashi Saitoh, and Nils Chr. Stenseth. (Received August 05, 2020)

65 ► Numerical analysis

1159-65-28 **Minah Oh*** (ohmx@jmu.edu). *Multigrid in H(div) on Axisymmetric Domains.*

In this talk, we will present a multigrid algorithm that can be applied to weighted H(div) problems on a two-dimensional domain. These problems arise after performing a dimension reduction to a three-dimensional axisymmetric H(div) problem with general data. We will use recently developed Fourier finite element spaces that can be applied to H(div) problems on axisymmetric domains. We present mathematical results that show that if the axisymmetric domain is convex, then the multigrid V-cycle will converge uniformly with respect to the mesh size. We will also show numerical results consistent with the mathematical theory. (Received July 06, 2020)

1159-65-83 **Vladimir Yushutin** and **Annalisa Quaini*** (aquaini@central.uh.edu), Department of Mathematics, University of Houston, 3551 Cullen Blvd, Houston, TX 77204, and **Maxim Olshanskii**. *Numerical modelingof phase separation on dynamic surfaces.*

We present a model of lateral phase separation in a two component material surface. The resulting fourth order nonlinear PDE can be seen as a Cahn-Hilliard equation posed on a time-dependent surface. Only elementary tangential calculus and the embedding of the surface in $\mathbb{R}^3$ are used to formulate the model, thereby facilitating the development of a fully Eulerian discretization method to solve the problem numerically. A hybrid method, finite difference in time and trace finite element in space, is introduced. The method avoids any triangulation of the surface and uses a surface-independent background mesh to discretize the equation. Thus, the method is capable of solving the Cahn-Hilliard equation numerically on implicitly defined surfaces and surfaces undergoing strong deformations and topological transitions. We assess the approach on a set of test problems and apply it to model spinodal decomposition and pattern formation on colliding surfaces. (Received July 30, 2020)

1159-65-96 **Sai-Mang Pun*** (smpun@math.tamu.edu), Department of Mathematics, Mailstop 3368, College Station, TX 77843. *Computational methods and model reductions for multiscale problems in heterogeneous porous media.*

In this talk, we introduce the Constraint Energy Minimizing Generalized Multiscale Finite Element Method and illustrate how it can be employed to solve problems with multiscale features arising in engineering applications. The proposed method makes use of the idea of energy minimization with suitable constraints to generate efficient basis functions for the variables related to governed partial differential equations. These basis functions are constructed by solving a class of local auxiliary optimization problems based on eigenfunctions containing local information on the heterogeneity. Techniques of oversampling are adapted to enhance the computational performance. A number of numerical examples will be provided to illustrate the performance of the proposed methodology. (Received August 01, 2020)

1159-65-156 **Matthias Maier*** (maier@math.tamu.edu). *Finite-element computation of the conductivity feedback of nanoscale optical devices.*

In the terahertz frequency range, the effective (complex-valued) surface conductivity of atomically thick 2D materials such as graphene has a positive imaginary part that is considerably larger than the real part. This feature allows for the propagation of slowly decaying electromagnetic waves, called surface plasmon-polaritons (SPPs), that are confined near the material interface with wavelengths much shorter than the wavelength of the free-space radiation. SPPs are promising ingredients in the design of novel optical applications promising "subwavelength optics" beyond the diffraction limit. There is a compelling need for controllable numerical schemes which, placed on firm mathematical grounds, can reliably describe SPPs in a variety of geometries.

In this talk we present an a finite element approach for the simulation of the nanoscale conductivity response of complex optical devices with nanoscale 2D material inclusions. The approach is based on a homogenization theory of layered optical heterostructures. We conclude by shortly describing a spectral analysis of the underlying cell problem that quantitatively describes resonances in the effective material parameters of plasmonic crystals. (Received August 03, 2020)

1159-65-169 **Cuiyu He***, 452 BOYD GSRC, ATHENS, GA 30602, and **Erik Burman** and **Mats Larson**. *Shape optimization methods for ill-posed Bernoulli Problem using CutFEM method.*

In this talk, we discuss a level set approach for the identification of an unknown boundary in a computational domain. The problem takes the form of a Bernoulli problem where only the Dirichlet datum is known on the boundary that is to be identified, but additional information on the Neumann condition is available on the known part of the boundary. The approach uses a classical constrained optimization problem, where a cost functional is minimized with respect to the unknown boundary, the position of which is defined implicitly by a level set function. To solve the optimization problem a steepest descent algorithm using shape derivatives is applied. In each iteration the cut finite element method is used to obtain high accuracy approximations of the pde-model constraint for a given level set configuration without re-meshing. We considered three different shape derivatives.The theoretical discussion is illustrated with a series of numerical examples showing that all three approaches produce similar result on the proposed Bernoulli problem. (Received August 04, 2020)

1159-65-186 **Jeonghun J. Lee***, Sid Richardson Science Building, One Bear Place #97328, Waco, TX 76798. *Numerical methods for time-transient wave propagation in acoustic metamaterials.*

Waves in metamaterials show unconventional propagation patterns and there are many potential applications of metamaterials. In this talk we consider mathematical models and numerical methods for wave propagation in acoustic metamaterials. We discuss finite element discretization, the error analysis, and applications. (Received August 04, 2020)

1159-65-195 **Sibusiso Mabuza***, School of Math and Stat. Sciences, O-110 Martin Hall, Clemson, SC 29634. *Iterative limiters for continuous Galerkin discretization of hyperbolic conservation laws.*

In this brief talk, various nodal limiters for continuous Galerkin discretization of hyperbolic systems are reviewed. The design and application of these limiters follows the algebraic flux correction paradigm. First, the introduction of low-order artificial dissipation and mass lumping is performed on the semi-discrete scheme of the Euler equations of gas dynamics to get a first order scheme. Then, antidiffusion which is the difference between the standard Galerkin scheme and the first order scheme is constrained using an element based limiter in a conservative fashion to get a high resolution scheme. The element based limiters are computed from a linearity preserving nodal limiter. Numerical simulations are performed to illustrate the performance of the scheme using implicit or explicit time steppers. (Received August 04, 2020)

68 ► Computer science

1159-68-199 **James E Arthur*** (jearthur@miners.utep.edu), 910 N Oregon, El Paso, TX 79902, and **Maria Mariani** and **Osei Tweneboah**. *Using Machine Learning for the prognosis and diagnosis of an Imbalanced Cancer dataset.*

With an estimated 1.4 million cancer diagnosis worldwide and the increasing death of cancer patients. It is very important to investigate methods and techniques used for predicting and diagnosing of cancer so that a holistic approach can be used to curb or reduce false predictions, increase exact predictions and also provide meticulous prognosis information.

We will show here that this problem of cancer prognosis and diagnosis can be efficiently tackled with the aid of machine learning techniques and the best, feasible and efficient technique can be used to reduce this cancer menace. Cancer has been characterized as a heterogeneous disease consisting of many different subtypes. The early diagnosis and prognosis of a cancer type have become a necessity in cancer research, as it can facilitate the subsequent clinical management of patients.

Even though it is evident that the use of Machine Learning methods can improve our understanding of cancer progression, an appropriate level of validation is needed in order for these methods to be considered in the everyday clinical practice. In this work, what we present are view of recent ML approaches employed in the modeling of cancer prognosis and diagnosis. (Received August 04, 2020)

70 ► Mechanics of particles and systems

1159-70-146 **Indon Yeom***, RISE Group, Cresskill, NJ , and **Richard Kyung**, RISE Group, Cresskill, NJ. *Dynamics of piston in an internal combustion engine using a mathematical model and computational analysis.*

To develop a new engine for a vehicle, the work performed by the engine as well as the power requirements of the vehicle in various driving environments must be estimated. In this paper, parameters such as compression ratio, expansion coefficient, and maximum pressure are set to calculate the powers of the presented vehicles. To maintain a proper engine power with the ability to overcome aerodynamic resistance and rolling resistance, accelerations and maximum powers were estimated. In addition, to determine the static and dynamic characteristics of the engine as well as the common configuration of the slider crank mechanism, the angle and the length of the connecting rod, the angle of the crankshaft, and the radius of the crankshaft were considered. To analyze the applied forces acting on the engine for the study of the kinematics of the engine system, constant angular frequency, specific angular frequency, and a steady-state operation of the engine were assumed. Computational experiments applied to different vehicle engines for the crank angle of rotations, varying from 0 degree to 720 degree, prove that an angle about 350 degree 400 degree is ideal for the crank angle to produce high power. (Received August 03, 2020)

1159-70-206 **V M Krushnarao Kotteda** (kvmkrao@gmail.com), 1000 E University Ave, Department of Mechanical Engineering, Laramie, WY 82070, **Arturo Rodriguez*** (arodriguez123@miners.utep.edu), 500 W university Ave, Department of Mechanical Engineering, El Paso, TX 79968, **Vinod Kumar** (vkumar@utep.edu), 500 W University Ave, Department of Mechanical Engineering, El Paso, TX 79968, and **Natasha Sharma** (nssharma@utep.edu), 500 W University Ave, Department of Mathematical Science, El Paso, TX 79968. *Automating Mesh Adaptivity with Neural Network in Volume of Fluids methods for Multiphase Flows.*

Multiphase flow is prevalent in fossil, renewable, and nuclear energy as well as many other processes such as coal gasifier reactors, fluidized bed reactors, particle receivers, or even aerosol/virus transmission. Understanding the behavior of such flows can assist in devising better post- and pre-combustion CO2 capture technologies, reducing CO2 emission to the atmosphere, and improving the understanding of aerosol transmission. The flow regime in such multiphase systems often involves physics and physical phenomena occurring at multiple spatial and temporal scales. Analytic solutions to such complex multiphysics problems are rarely available. Numerical approaches are often used to investigate the underlying physics. In order to obtain a "good" approximation of the true solution, a high-fidelity approximation (such as high-resolution meshes) can be used. However, this leads to extremely large scale problems on the discrete level which pose many challenges to numerical computations due to prohibitively high computational expense or poorly-conditioned linear systems. We will present analysis work for Partially Observable Markov Decision Processes, a Reinforced Machine Learning framework based on the Markov Decision Process for automating the mesh adaptivity of multiphase flows. (Received August 05, 2020)

74 ► Mechanics of deformable solids

1159-74-70 **Tagir Farkhutdinov**, , Canada, **Francois Gay-Balmaz**, Paris, France, and **Vakhtang Putkaradze*** (vakhtang.putkaradze@atco.com), 5302 Forand St SW, Calgary, AB T3E8B4, Canada. *Geometric variational methods in deformable porous media, with applications to active materials.*

Many biological organisms are comprised of deformable porous media, with additional complexity of an embedded muscle, see e.g. Ludeman et al., BMC Evol. Biol. 2014 for 'sneezing' dynamics of a freshwater sponge. Using geometric variational methods, we derive the equations of motion of a for the dynamics of such an active porous media (*i.e.*, a deformable porous media that is capable of applying a force to itself with internal muscle work. We then proceed to extending this theory by computing the case when both the active porous media and the fluid are incompressible, with the porous media still being deformable, which is often the case for biological applications. For a particular case of uniform initial state, we rewrite the equations of motion in terms of two coupled telegraph-like equations for the material (Lagrangian) particles expressed in the Eulerian frame of reference, particularly suitable for numerical simulations, formulated for both the compressible media/incompressible fluid case and the doubly incompressible case. We derive interesting conservation laws for the motion, perform numerical

simulations in both cases and show the possibility of self-propulsion of a biological organism due to particular running wave-like application of the muscle stress. (Received July 29, 2020)

76 ► Fluid mechanics

1159-76-14 **Luan T Hoang***, Department of Mathematics and Statistics, Texas Tech University, 1108 Memorial Circle, Lubbock, TX 79409. *Asymptotic expansions for the Lagrangian trajectories from solutions of the Navier-Stokes equations,.*

Consider any Leray-Hopf weak solution of the three-dimensional Navier-Stokes equations for incompressible, viscous fluid flows. We prove that any Lagrangian trajectory associated with such a velocity field has an asymptotic expansion, as time tends to infinity, which describes its long-time behavior very precisely. (Received June 26, 2020)

1159-76-71 **Loic Cappanera*** (lmcappan@central.uh.edu). *Momentum based approximation of incompressible flows with variable density and viscosity. Applications to magnetohydrodynamics.*

We present a new approximation method for the incompressible Navier-Stokes equations with variable density and viscosity. This method uses the momentum $\mathbf{m} := \rho\mathbf{u}$, with ρ the density and $\mathbf{u}$ the velocity, as dependent variable for the Navier-Stokes equations which results in a time-independent mass matrix that is suitable for spectral methods. The incompressibility condition of the flow is enforced via an artificial compression method. To improve the efficiency of the method for large scale computations, the stiffness matrix is made time independent by rewriting appropriately the diffusion operator and the grad-div operator that enforces the incompressibility of the flow. A level set method is applied to reconstruct the density and viscosity of the fluid. After establishing the stability of the algorithm, we study its convergence properties numerically with manufactured solutions involving a large range of ratios of density and viscosity. Comparisons with a pressure-correction projection method are also provided. Eventually, we extend the method to conducting fluids with variable electrical conductivity to study magnetohydrodynamics problems such as the metal pad roll instability. (Received July 29, 2020)

86 ► Geophysics

1159-86-151 **Sebastian Jaroszewicz*** (sebasjaro@gmail.com), Av. General Paz 1499, 1650 San Martin, Bs. As., Argentina, and **Maria C. Mariani**, **Osei K. Tweneboah** and **Maria P. Beccar-Varela**. *High frequency study of EL Niño Southern Oscillation index.*

The El Niño-Southern Oscillation (ENSO) phenomenon is recognized as the main source of seasonal and interannual short-term climatic variability in the tropics and in much of the mid-latitudes, causing changes in the interaction between the ocean and the atmosphere. The interaction of the ocean and the atmosphere in the Tropical Pacific Ocean fluctuates somewhat irregularly between the warm phase (SST warmer than normal) called El Niño, and the opposite phase known as La Niña. The Southern Oscillation Index (SOI) is often used as an input for seasonal climate forecasting in ENSO sensitive areas around the world. In this work, we analyse daily data of the SOI index from 1991 to 2020. We study distribution that characterizes SOI increments and by means of the Detendred Fluctuation Analysis (DFA) the presence of long-range correlations for periods less than 80 days was revealed. Additionally, in order to study the multifractal behavior of the signal we apply a generalization of the DFA method called Multifractal Detendred Fluctuation Analysis (MF-DFA). The application of this technique to the SOI time series allowed us to find the generalized Hurst exponents, and the multifractal spectrum of the signal. (Received August 03, 2020)

91 ► Game theory, economics, social and behavioral sciences

1159-91-7 **Indranil SenGupta*** (indranil.sengupta@ndsu.edu), Department of Mathematics, North Dakota State University, Fargo, ND 58108-6050. *Machine learning-based refinement of stochastic models and oil data analysis.*

A commonly used stochastic model for derivative and commodity market analysis is the Barndorff-Nielsen and Shephard (BN–S) model. Though this model is very efficient and analytically tractable, it suffers from the

absence of long-range dependence and many other issues. In this presentation, we propose a machine-learning-based refinement of the BN–S model. Empirical applications validate the efficacy of the proposed model for long-range dependence. The model will be implemented for various oil price data to show the significance of the analysis. (Received May 08, 2020)

1159-91-51 **Ionut Florescu***, 1 Castle Point Terrace, Babbio Center 544, Hoboken, NJ 07030, **Thiago Winkler Alves**, 1 Castle Point Terrace, Hoboken, NJ 07030, **Dragos Bozdog**, 1 Castle Point Terrace, Hoboken, NJ 07030, and **George Calhoun**. *Using SHIFT a laboratory environment to test changes to financial markets.* Preliminary report.

In this work we will document a novel system designed to replicate a complete financial market exchange. We demonstrate the use of the system by showcasing an implementation of the Frequent Batch Auctions (FBA) which is only at the literature stage at the moment. We are not only able to consider the differences from a regular equity market as implemented today but also able to highlight requirements if FBA would be implemented in practice. (Received July 23, 2020)

1159-91-53 **Jun Deng**, **Huifeng Pan**, **Shuyu Zhang** and **Bin Zou***, 341 Mansfield Road U1009, Storrs, CT 06269. *Minimum-Variance Hedging of Bitcoin Inverse Futures.*

We formulate an optimal hedging problem of Bitcoin inverse futures under the minimum-variance framework. We obtain the optimal hedging strategy in closed forms for both short and long hedges, and compute hedging effectiveness under the optimal strategy. Our empirical analyses show that the optimal hedging strategy achieves superior effectiveness in reducing risk and outperforms the naïve hedge in all scenarios. (Received July 24, 2020)

1159-91-144 **Olympia Hadjiliadis*** (olympia.hadjiliadis@gmail.com), 695 Park Ave, New York, NY 10065-5024. *Trends and trades.*

We discuss the intricacies of real-time trading and distinguish the types of orders. We review different strategies and the concept of statistical arbitrage. We present a trend following algorithm based on the sequential statistical rule known as the cumulative sum (CUSUM). We draw connections between these statistics and the problem of online statistical surveillance and quality control. We build a trading strategy based on the CUSUM stopping rule and apply it to high-frequency tick data from 5-year and 30-year US Treasury notes sold at auction. We analyze the performance of the proposed trend following strategy in detail. In particular, it is seen that the proposed trading rule is most profitable during times of market instability and long trends. We further calculate in closed form the expected value of the gain of the proposed strategy for a class of random walk models. Not surprisingly, it is seen that the suggested strategy is most profitable in biased random walks but is indifferent to the direction of the bias. We also examine the performance of the proposed strategy in simulated data from a variety of random walk models and analyze this behavior in relation to the analytical results and the results of the performance of the strategy on the actual data. (Received August 03, 2020)

92 ► Biology and other natural sciences

1159-92-13 **Tuan A Phan*** (tuanpa86@nmsu.edu), Department of Mathematical Sciences, 1290 Frenger Mall, MSC 3MB / Science Hall 236, Las Cruces, NM 88003-8001, and **Dang H Nguyen** and **Jianjun P Tian**. *Deterministic and stochastic modeling for PDGF-driven gliomas reveals a classification of gliomas.*

Motivated by our study of in filtrating dynamics of immune cells into tumors, we propose a stochastic model in terms of Ito stochastic differential equations to study how two parameters, the chemoattractant production rate and the chemotactic coefficient, influence immune cell migration and how these parameters distinguish two types of gliomas. We conduct a detailed analysis of the stochastic model and its deterministic counterpart. The deterministic model can differentiate two types of gliomas according to the value range of the chemoattractant production rate as two equilibrium solutions, while the stochastic model can also differentiate two types of gliomas according to the value ranges of the chemoattractant production rate and chemotactic coefficient with thresholds as one non-zero ergodic invariant measure and one weak persistent state when the noise intensities are small. When the noise intensities are large comparing with the chemotactic coefficient, there is only one type of glioma that corresponds to a non-zero ergodic invariant measure. Using our experimental data, numerical simulations are carried out to demonstrate the properties of our models, and we give medical interpretations and implications for our analytical results and numerical simulations. (Received June 26, 2020)

1159-92-16 **Casey J. Mills*** (casey.j.mills@ttu.edu), **Raegan Higgins** and **Angela Peace**. *A Time Scales Approach for Modeling Intermittent Hormone Therapy for Prostate Cancer.*

Prostate cancer is a common cancer among males in the United States and is most frequently treated by intermittent androgen deprivation therapy. This therapy requires a patient to alternate between periods of androgen suppression treatment and no treatment. Prostate-specific antigen levels are used to track relative changes in tumor volume of prostate cancer patients undergoing intermittent androgen deprivation therapy. During this therapy, there is a pause between treatment cycles. Traditionally, continuous ordinary differential equations are used to estimate prostate-specific antigen levels. In this paper, we use dynamic equations to estimate prostate-specific antigen levels and construct a novel time scale model to account for both continuous and discrete time simultaneously. This allows us to account for breaks between treatment cycles. Using empirical data sets of prostate-specific antigen levels, a known bio-marker of prostate cancer, across multiple patients, we fit our model and use least squares to estimate two parameter values. We then compare our model to the data and find a resemblance on treatment intervals similar to our time scale. (Received June 26, 2020)

1159-92-92 **Jonathon E Mohl*** (jemohl@utep.edu), 500 W University, El Paso, TX 79968, and **Thomas Gerken** and **Ming-Ying Leung**. *Using derived protein feature enhancement values to calculate the propensity for mucin-type O-glycosylation with the ISOGlyP program.*

Mucin-type O-glycosylation is a post-translational modification (PTMs) of proteins. This glycosylation is initiated by the addition of the sugar N-acetylgalactosamine (GalNAc) onto protein Ser and Thr residues by a family of polypeptide GalNAc transferases. In humans there are 20 isoforms that are differentially expressed across tissues that serve multiple important biological roles. Aberrant O-glycosylation or other PTMs have links to many developmental diseases or cancers. Using random peptide substrates, isoform and position specific amino acid preferences were obtained in the form of enhancement values (EV). The EV product (EVP) is the product of the EVs and highlights the propensity for O-glycosylation to occur. These initial calculations have been incorporated within the ISOGlyP O-glycosylation prediction program. The inclusion of EV protein substrate features (such as secondary structure and surface accessibility) was found to increase sensitivity with minimal loss of specificity, when tested with three different published in vivo O-glycoproteomics data sets, thus increasing the overall accuracy of the ISOGlyP predictions. The generation of new EV tables and the flexible use of an extended EVP calculation incorporating the additional features is the focus of this talk. (Received July 31, 2020)

1159-92-194 **Hamid Mofidi*** (hamidreza.mofidi@gmail.com), 951 Arkansas Street, Apartment K6, Lawrence, KS 66044, and **Weishi Liu**. *Effects of Diffusion Coefficients and Permanent Charge on Reversal Potentials in Ionic Channels.*

In this work, the dependence of reversal potentials and zero-current fluxes on diffusion coefficients are examined for ionic flows through membrane channels. The study is conducted for the setup of a simple structure defined by the profile of permanent charges with two mobile ion species, one positively charged (cation) and one negatively charged (anion). Numerical observations are obtained from analytical results established using geometric singular perturbation analysis of classical Poisson–Nernst–Planck models. For 1:1 ionic mixtures with arbitrary diffusion constants, Mofidi and Liu (arXiv:1909.01192) conducted a rigorous mathematical analysis and derived an equation for reversal potentials. We summarize and extend these results with numerical observations for biological relevant situations. The numerical investigations on profiles of the electrochemical potentials, ion concentrations, and electrical potential across ion channels are also presented for the zero-current case. Moreover, the dependence of current and fluxes on voltages and permanent charges is investigated. (Received August 04, 2020)

1159-92-203 **Nayoung Ko**, CRG NJ, Cresskill, NJ , and **Richard Kyung*** (nycrick@gmail.com), CRG NJ, Cresskill, NJ. *Study on thermodynamic stability and geometrical optimization of phytoncide molecules.*

Based on variational method for density functional theory, the correlation between flavanon molecules and the effectiveness in antioxidant activity was studied in this research. For the flavanon or phytoncide molecules to have better effect, molecules with higher thermodynamic stability, better geometrical optimization and higher activity are needed for their anti-inflammatory effects and effective activity in radical scavenging. In this project, the geometrical and molecular properties of different molecules in phytoncides were studied. Electrostatic potential maps were also used to interpret the activity of the molecules. A molecular software was used to measure the optimization energy and the dipole moment of each molecule tested in this experiment. Molecules with lower optimization energies were predicted to be more thermodynamically stable than those with higher optimization

energies. Molecules with higher dipole moment were predicted to be more active than those with lower dipole moment. The purpose of this research was to determine the safety of the components used in the phytoncides for their antioxidant activity by checking whether there was any noticeable difference in molecular energies and activities of the different molecules or structural isomers. (Received August 04, 2020)

1159-92-205 **A Jajoo***, ajajoo@math.uh.edu, and **O Hirschi**, **K Schulze**, **Y Guan** and **N Hanchard**. *Patterns of Correlated DNA Methylation.*

Genome-scale assessments of DNA methylation have been used to identify disease-contributing loci in variety of human diseases, cancers, and traits. Mainly focusing upon identifying differences in methylation level between disease and non-disease states. Methylation sites in close proximity of each other are known to have sustainable correlation structures (CSs). Very little is known about CSs consistency across individuals and tissues, genomic context, and relevance to disease. Here, we develop a novel Image Clustering Method (ICM) that leverages image conversion and correlation matrices, to assess patterns of correlated methylation sites. We apply our method to 31 publicly-available datasets, spanning 27 tissues and >11,000 individuals. We identify 207 clusters of contiguous correlation between adjacent sites. We also observe significant positive and negative correlation between contiguous clusters extending as far as 250kb. These non-contiguous clusters are bound within topology-associated domains (TADs) and suggest coordinated methylation between CpG sites at gene-clusters, as well as across gene bodies. Finally, we identify significant differences in cluster correlation between diseased and non-disease states, including cancer, with disease-specific patterns of disruption. (Received August 05, 2020)

97 ► Mathematics education

1159-97-36 **Cody L Patterson*** (codypatterson@txstate.edu), Department of Mathematics, 601 University Drive, San Marcos, TX 78666, **Lino Guajardo** (lrg74@txstate.edu), Department of Mathematics, 601 University Drive, San Marcos, TX 78666, and **Emma Yu** (emmayyu15@gmail.com). *Building on recursive thinking: Combinatorial game problems as a window into students' understanding of proof by induction.*

In this talk we will describe the generalization and justification practices of students in an informal summer mathematics program as they explore a sequence of problems from combinatorial game theory. We find that while study participants readily generate examples and reason recursively when analyzing Nim-like two-player combinatorial games and are able to reach valid conclusions about winning strategies in these games, some obstacles occur in the process of formally justifying these winning strategies. We use study participants' work on these combinatorial game problems as a window into their proof practices in an unfamiliar problem domain and into their understanding of mathematical induction as a strategy for generalizing and formalizing recursive reasoning. (Received July 14, 2020)

1159-97-77 **Kyeong Hah Roh*** (khroh@asu.edu), School of Mathematical & Statistical Sciences, Arizona State University, Tempe, AZ 85287-1804, and **Yong Hah Lee** (yonghah@ewha.ac.kr), Department of Mathematics Education, Ewha Womans University, Seoul, South Korea. *Undergraduate Students' Logical Consistency in Mathematical Contexts and its Relation to Their Experience of Proof-Oriented Mathematics.*

By logical consistency we mean individuals' capacity to self-organize their thinking to have no logical contradiction. We developed an online instrument to assess undergraduate students' logical consistency while they are evaluating statements and validating arguments in mathematics contexts. The online instrument was administered to 167 undergraduate students from 3 universities in the United States with 8 different mathematics courses at the end of the semester either in Fall 2019 or in Spring 2020. We first detail the process of designing the online instrument and scoring rubrics. We then report the survey results, focusing on how differently students displayed logical consistency in terms of (1) the number of proof-oriented courses they had taken, (2) their grades of the most advanced mathematics course that they had taken, and (3) their major areas of study. We also discuss implications of this study for the teaching and learning of proof-oriented mathematics courses. (Received July 29, 2020)

1159-97-80 **Allison Dorko*** (allison.j.dorko@okstate.edu). *How Students Engage with Online Homework: Multiple Tries and the 'See Similar Example' Feature.*

Online homework provides particular promise for supporting student learning because technology allows for multiple attempts, immediate feedback, hints, and other 'help' features. I will report on the results from two studies that focused on how students leverage some of these benefits. In the first study, I investigated the nature of student activity while doing online homework. I will report on how students leveraged multiple tries as a

form of formative assessment that guided their learning and sense-making. In the second study, I explored how students used the 'see similar example' feature in an online homework platform and how that use reflected their learning goals. Findings indicate students used solutions to similar problems to troubleshoot, to check if they are on the right track, and to see the steps to solve a problem or the form of the answer. Students also sought to unpack the reasoning in solution steps, used solutions as templates for solving their own problems, and sometimes simply copied answers. Students' goals included simply completing the homework, maximizing their homework score, and understanding the content. I will discuss the implications of these findings for instructors seeking to improve students' learning from online homework. (Received July 30, 2020)

1159-97-82 **Erika David Parr*** (erika.parr@mtsu.edu) and **Robert Ely** (ely@uidaho.edu). *A hypothetical learning trajectory for conceptualizing distances on Calculus graphs.* Preliminary report.

A Cartesian coordinate pair (x, y) is not just a name for a point in the plane; it is a description of the horizontal and vertical (signed) distances between the origin and that point. It is crucial to view coordinates as representing distances between points in order to understand many Calculus ideas, including areas under and between curves (e.g., $f(x) - g(x)$ is a vertical length), volumes of solids of rotation, and the definition of the derivative. Yet research has shown that undergraduate students, even those who have taken multiple Calculus courses, may not conceive of distances when reasoning about related expressions on graphs (Parr, 2020). We present a hypothetical learning trajectory for developing student reasoning about distances in the Cartesian coordinate system with variable and function notation. We will discuss our plans to try out and refine this hypothetical learning trajectory, as well as directions for future research. (Received July 30, 2020)

1159-97-102 **Elsa Q Villa*** (evilla@utep.edu), Center for Educ Research & Policy Studies, College of Education, 500 W. University, El Paso, TX 79968, **M. Christina Mariani**, Department of Mathematical Sciences, College of Science, 500 W. University, El Paso, TX 79968, **Erika Mein**, College of Education, 500 W. University, El Paso, TX 79968, **Christina Convertino**, College of Education, 500 W. University, El Paso, TX 79968, and **Nada Al-Hanna**, Department of Mathematical Sciences, College of Science, 500 W. University, El Paso, TX 79968. *Findings from a Pre-Calculus Intervention to Shift Pre-Calculus to a Gateway Course.*

To address the systemic gap between Whites and students of color in STEM degree attainment, The University of Texas at El Paso (UTEP) implemented an intervention in its pre-calculus course to shift the course from a gatekeeper to gateway course over a 3-year period. With funding from the U.S. Department of Education, a team of researchers and practitioners focused on the 2-hour workshop of the 5-hour pre-calculus course in order to promote the creation of learning communities, which was especially critical for UTEP, a minority-serving institution with over 85% of its undergraduate population being of Hispanic origin. This presentation presents the qualitative and quantitative findings of the project, which indicate the intervention had a positive effect on student pass rates, their sense of belonging, and their success in the subsequent Calculus 1 course. (Received August 01, 2020)

1159-97-149 **Allison Dorko*** (allison.j.dorko@okstate.edu), **John Paul Cook**, **Josiah Ireland** (josiah.ireland10@okstate.edu), **William Jaco** (william.jaco@okstate.edu), **Michael Oehrtman** (michael.oehrtman@okstate.edu) and **Michael Tallman** (michael.tallman@okstate.edu). *Fostering Academic Success Skills in Entry-Level Mathematics Courses.*

Recent state and national efforts to provide entry-level mathematics options relevant to diverse degree programs and replace remedial courses with corequisite support for students placed directly in credit-bearing courses have significantly improved student success rates. In addition to structural changes in course offerings, faculty have called for resource to help foster students' productive beliefs, identity, and persistence. We have engaged mathematics instructors from all 26 public institutions of higher education in Oklahoma via The Mathematical Inquiry Project, a five-year NSF-funded project that supports faculty collaboration to address the incorporation of such academic success skills in entry-level courses. Building on the outcomes of a week-long workshop of Oklahoma college mathematics faculty, this session will engage meeting participants in a conversation about goals and strategies for supporting academic success skills in entry-level mathematics courses. We will focus specifically on areas Oklahoma faculty identified as critical, including growth mindsets; problem-solving and critical thinking; productive struggle, persistence, and perseverance; lowering mathematics anxiety; motivation and interest; beliefs about mathematics; and developing classroom community. (Received August 03, 2020)

1159-97-178 **Paul Christian Dawkins*** (pcd27@txstate.edu) and **Kyeong Hah Roh** (khroh@asu.edu). *How logical relationships may depend on context for learners.*

A key aspect of logic in advanced mathematics is that it describes content-independent relationships. However, research reveals little as of yet about how students segue content-independent logic with their content-specific reasoning. To learn more about this, we have been conducting experiments guiding students to reinvent key logical relationships – specifically contrapositive equivalence – in various mathematical contexts to see how the two interact. We will share some insights from one student's reasoning about proofs by contraposition (which we provided to her to read). We found that she initially questioned whether such proofs proved the given theorem until she constructed a relationship of necessity between the categories in the proof (e.g., why a multiple of 6 must also be a multiple of 3). This pattern repeated three times and though she thrice constructed her logical justification for why a contrapositive proof justified a conditional claim, she remained unsure of its generality for all conditional claims. We explore some of the implications of this phenomenon for the learning of logic in proof-oriented mathematics courses. (Received August 04, 2020)

1159-97-190 **Sepideh Stewart***, 601 Elm Ave, Norman, OK 73019, and **Jonathan Epstein**, **Jonathan Troup** and **Aidan Powers**. *Linear Algebra Thinking in the Embodied, Symbolic and Formal Worlds.*

To understand linear algebra concepts, one needs to be familiar with many different modes of thinking. Mathematicians often move between these modes of thinking fluently and expect that students will pick up the main ideas along the way. Yet, a majority of students don't see the need or do not have the desire to perform the move that is so natural to mathematicians. Employing Tall's (2013) three-world model as both a theoretical perspective and a pedagogical tool, we analyzed an instructor journal, two student surveys, and a student interview. Our working hypothesis is that by creating opportunities to move between the worlds, students may be exposed to multiple modes of thinking which results in richer conceptual understanding. The results of the study revealed that a majority of students preferred the symbolic world. Their reasoning for their choices will be discussed. (Received August 04, 2020)

1159-97-197 **James A. Mendoza Álvarez*** (james.alvarez@uta.edu), Box 19408, Department of Mathematics, The University of Texas at Arlington, Arlington, TX 76019-0408, and **Andrew Kercher** and **Kyle Turner**. *Learning from Lessons that Focus on Applications to Teaching in Undergraduate Abstract Algebra.*

The Mathematical Education of Teachers as an Application of Undergraduate Mathematics project provides lessons integrated into various mathematics major courses that incorporate school mathematics teaching connections as a legitimate application area of undergraduate mathematics. In Abstract Algebra, undergraduates engaged in three different lessons aiming to enhance understanding of connections between undergraduate mathematics and school mathematics. In this paper, we use thematic analysis of hour-long interviews with four faculty and 20 undergraduates across four different institutions to describe what undergraduates who use the lessons understand about the connections and about the mathematics in the lesson as well as what faculty who use the lessons understand about these connections. We explore evidence that undergraduates understood the abstract algebra concepts and made connections between abstract algebra and secondary school mathematics. We also see examples that convey that faculty understood the connections. Using these lessons strengthened their understanding of how to incorporate applications to teaching into an undergraduate course in abstract algebra and the potential for these applications to support undergraduate student learning in a robust manner. (Received August 04, 2020)

00 ► General

1160-00-185 **Felicia Magpantay*** (felicia.magpantay@queensu.ca), Jeffery Hall, 48 University Avenue, Kingston, ON K7L 3N6, Canada. *Challenges in modeling the transition period of childhood diseases from the pre-vaccine to vaccine era.* Preliminary report.

Mathematical models of childhood diseases often employ homogeneous time-dependent transmission rates. These models can provide good agreement with data in the absence of significant changes in population demography or levels of transmission, such as in the case of pre-vaccine era measles in industrialized countries. However, accurate modeling and forecasting of transient dynamics after the start of mass vaccination has proved more challenging. This is true even in the case of measles which has a well understood natural history and a very effective vaccine. Here, we demonstrate how the dynamics of homogeneous and age-structured models can be similar in the absence of vaccination, but diverge after vaccine roll-out. We also propose methods to fit such models to long term epidemiological data with imperfect covariate information. (Received August 10, 2020)

03 ► Mathematical logic and foundations

1160-03-12 **Mehmet M. Dalkilic*** (dalkilic@iu.edu), 700 North Woodlawn Dr, Office 3056, Bloomington, IN 47408, and **Kurban Hasan** (hakurban@gmail.com), 700 North Woodlawn Dr, Bloomington, IN 47408. *Teaching an Old Dog New Tricks: Making EM work with Big Data using Heaps.*

Contemporary data mining algorithms are easily overwhelmed with truly big data. While parallelism, improved initialization, and ad hoc data reduction are commonly used and necessary strategies, we note that (1) continually revisiting data and (2) visiting all data are two of the most prominent problems especially for iterative learning techniques like expectation-maximization algorithm for clustering (EM-T). To the best of our knowledge, there is no freely available software that specically focuses on improving the original EM-T algorithm in the context of big data. We demonstrate the utility of CRAN package DCEM that implements an improved version of EM-T which we call EM* (EM star). DCEM provides an integrated and minimalistic interface to EM-T and EM* algorithms, and can be used as either (1) a stand-alone program or (2) a pluggable component in existing software. We show that EM* can both eectively and efficiently cluster data as we vary size, dimensions, and separability. (Received February 28, 2020)

1160-03-88 **Michele Dolce*** (michele.dolce@gssi.it), Via Criano, 15, 02100 Rieti, Rieti, Italy. *Linear stability analysis of the planar Couette flow for a compressible or a stably stratified fluid.* Preliminary report.

In the last decade, the classical problem of studying the stability of shear flows has received renewed attention, especially regarding the incompressible case. In the talk, we will mainly focus on the linear stability of the Couette flow, which is a shear flow with a linear velocity profile. We first consider perturbations around the Couette flow with constant density in the compressible case. For an inviscid fluid, we show a generic Lyapunov type instability for the density and the irrotational part of the velocity, namely their L^2 norm grows as $\sqrt{t}$. Instead, the solenoidal component of the velocity experience inviscid damping, meaning that it decays to zero even in the absence of viscosity. For a viscous compressible fluid, we show that the perturbations decay exponentially fast on a faster time-scale with respect to the standard diffusive one, a phenomenon also called enhanced dissipation. This is a joint work with P. Antonelli and P. Marcati.

Then, we deal with the linear stability of shear flows close to Couette in a stably stratified fluid in the inviscid case. Under the Miles-Howard criterion, we show an inviscid damping result for the velocity and density. This is a joint work with R. Bianchini and M. Coti Zelati. (Received August 06, 2020)

05 ► Combinatorics

1160-05-14 **Brandt Kronholm*** (brandt.kronholm@utrgv.edu), University of Texas Rio Grande Valley, School of Mathematics and Statistical Science, Edinburg, TX 78539. *Congruences and Cranks for $p(n, m)$.*

In this presentation we establish infinite families of cranks witnessing infinite families of congruences for the function $p(n, m)$ which enumerates partitions of n into at most m parts.

This is joint work with Acadia Larsen (UC Davis) and Dennis Eichhorn (UC Irvine). (Received June 27, 2020)

1160-05-25 **James A Sellers*** (jsellers@d.umn.edu). *Relating the Crank of a Partition and Smallest Missing Parts.*

The primary goal of this talk is to demonstrate a natural connection between the smallest missing part of an integer partition and the concept of the crank of a partition. After providing a brief history of the crank of a partition a la Dyson as well as Andrews and Garvan, we will utilize straightforward generating function manipulations to make this connection. We close by strengthening a result of Yuefei Shen comparing the number of partitions with smallest missing part odd and those with smallest missing part even. This is joint work with Brian Hopkins. (Received July 19, 2020)

1160-05-27 **Ben Cameron*** (ben.cameron@uoguelph.ca), **Chính T. Hoàng** and **Joe Sawada**. *Dichotomizing k-vertex-critical H-free graphs for H of order four.*

A graph is H-free if it does not contain H as an induced subgraph and it is k-vertex-critical if it has chromatic number k but each of its induced subgraphs can be colored with at most $k-1$ colors. In this talk we will show that for all $k \geq 3$ and $\ell \geq 0$, there is a finite number of k-vertex-critical $(P_2 + \ell P_1)$-free graphs. Together with previous research, this completes a dichotomy for k-vertex-critical H-free graphs when H is a graph of order four. We then improve upon previous research to show tight upper bounds on the order and independence number of k-vertex-critical $(P_3 + P_1)$-free graphs that allow for exhaustive generation of these graphs for all $k \leq 7$. We will also discuss remaining open problems and how our results imply the existence of new polynomial-time certifying algorithms for deciding the k-colorability of $(P_2 + \ell P_1)$-free graphs for fixed k. (Received July 20, 2020)

1160-05-30 **Shane Chern*** (chenxiaohang92@gmail.com), Department of Mathematics, Penn State University, University Park, PA 16802. *The EGZ Theorem and a formula of Vladeta Jovovic.*

The famous EGZ (Erdos-Ginzburg-Ziv) Theorem states that each set of $2n-1$ integers contains some subset of n elements the sum of which is a multiple of n. When the $2n-1$-member set becomes $\{1, 2, \ldots, 2n-1\}$, a closed form of the number of such n-member subsets was conjectured by Vladeta Jovovic and proved by Max Alekseyev. In this talk, we will consider several extensions of Jovovic's formula. (Received July 24, 2020)

1160-05-52 **John Machacek***, jmachacek.math@gmail.com. *Sign variation and poset topology.*

We will consider two families of posets of projective sign vectors and present various results on the topology of their order complexes. The totally nonnegative Grassmannian is usually defined in terms of Plücker coordinates, but it can alternatively be defined using sign variation. We give a generalization of the totally nonnegative Grassmannian using sign variation which is conjecturally related to boundary measurement on surfaces with boundary. One of our families of posets describes the topology of this generalization in the case of real projective space, and in joint work with Bergeron and Dermenjian we find "new Eulerian number of type D" as their h-vectors. The other family of posets gives a decomposition of the $m = 1$ amplituhedron from work of Karp and Williams. We will show the $m = 1$ amplituhedron is shellable in joint work with Karp. (Received August 01, 2020)

1160-05-57 **Chris Jennings-Shaffer**, **Shashank Kanade*** (shashank.kanade@du.edu), **Matthew C. Russell** and **Ali K Uncu**. *Search for multi-colour partition identities.* Preliminary report.

I will present an overview of a search performed jointly with Matthew C. Russell to find new multi-colour partition identities. The outcomes are currently being investigated further jointly with Russell, Chris Jennings-Shaffer and Ali K. Uncu. All of this is a work in progress. (Received August 02, 2020)

1160-05-124 **Haixu Tang***, 700 N. Woodlawn Avenue, Bloomington, IN 47408, and **Wazim Mohammed Ismail** and **Etienne Nzabarushimana**. *Clonal reconstruction from time course genomic sequencing data.*

Long-term evolution experiments (LTEEs) reveals the change of clonal composition in an evolving bacterial population over time. Accurately inferring the haplotypes - the set of mutations that identify each clone, as well as the clonal frequencies and the clonal evolutionary history in a bacterial population is useful for the characterization of the evolutionary pressure on multiple correlated mutations instead of that on individual mutations. Here, we study the computational problem of reconstructing the haplotypes of bacterial clones from the variant allele frequencies observed in a LTEE. We formalize the problem using a maximum likelihood function, which is defined under the assumption that mutations occur spontaneously, and thus the likelihood of a mutation occurring in a specific clone is proportional to the frequency of the clone in the population when the mutation occurs. We will present a series of greedy heuristic algorithms to address the maximum likelihood inference and will discuss the combinatorial nature of the problem. We show through simulation experiments and real-world LTEE data that the algorithms are fast and achieve near optimal accuracy that is practically plausible under the maximum likelihood framework. (Received August 07, 2020)

1160-05-125 **Gary Gordon*** (gordong@lafayette.edu). *Permutations of finite subsets of $\mathbb{R}^2$ generated by Euclidean distances.* Preliminary report.

Given a finite set of points $S = \{P_1, P_2, \ldots, P_n\}$ and two vantage points V_1 and V_2, all in the Euclidean plane, define d_k to be $(d(V_1, P_k) + d(V_2, P_k))/2$, the average of the distances from P_k to the two vantage points. This distance function induces an ordering of the points in S, with P_i preceding P_j if $d_i < d_j$. Fixing S and allowing the two vantage points to move, we get a number of different permutations of S. Our goal is to count the number of permutations for some special point configurations. In particular, when the points of S are collinear, we find upper bounds for the number of possible permutations. (Finding closed form solutions, even for the linear case, seem to be difficult.)

This problem is motivated by voting preference problems in social science. This is joint work with Brittany Shelton, Alvaro Carbonero, Beth Anne Castellano, Charles Kulick, and Karie Schmitz as part of an REU project in 2019. (Received August 07, 2020)

1160-05-129 **Bhadrachalam Chitturi*** (chalam@utdallas.edu). *Counting cardinalities of subsets of S_n with k adjacencies and their applications.*

A permutation is a sequence over the corresponding alphabet Σ where every symbol of Σ occurs precisely once. Permutations over Σ are denoted by S_n. In genetics, a permutation models a genome where a gene is distinct. A mutation of a particular type is modeled by the respective operation (a set of generators). If a given operation is closed under inverses then it is *symmetric*. We call three symmetric operations of prefix transposition, suffix transposition and transposition as block-moves. The symmetric *distance* between α and β in S_n is defined as the minimum number of generators of a given operation (i.e. block-move) that are required to transform α into β or vice versa.

If $\pi = (\pi_1, \pi_2, \pi_3, ..., \pi_n)$ is a permutation over $\Sigma = 0, 1, ..., n-1$ then π_i , and π_{i+1} form an *adjacency* if $\pi_i + 1 = \pi_{i+1},\ 1 \leq i \leq n-1$. Variations of adjacency have been defined. The cardinalities of subsets of S_n with exactly k adjacencies (of a given type) are efficiently computed with recurrence relations and dynamic programming.These cardinalities help one accurately estimate the expected genomic dissimilarity between a pair of genomes in S_n. (Received August 10, 2020)

1160-05-135 **Alexander Kunin**, **Caitlin Lienkaemper*** (cul434@psu.edu) and **Zvi Rosen**. *Neural codes, oriented matroids, and their ideals.*

A combinatorial neural code describes a pattern of neural activity in terms of which neurons fire together. Many types of neurons respond to a set of stimuli, called the neuron's receptive field. The neural ideal is a pseudomonial ideal which encodes the relationships between receptive fields entailed by a neural code. Oriented matroids are combinatorial objects arising from hyperplane arrangements in much the same way that combinatorial codes arise from receptive fields. In this talk, we relate combinatorial neural codes to oriented matroids and neural ideals to oriented matroid ideals. We define a contravariant functor from the category of oriented matroids to the category of neural codes, and show that it commutes with the functors taking oriented matroids and neural codes to their respective monomial ideals. This lets us view the neural ideal as a generalization of the oriented matroid ideal, and lets us see the receptive field relationships of a neural code as a generalization of the circuits of an oriented matroid. (Received August 08, 2020)

1160-05-143 **Nathan R. T. Lesnevich** and **Peter R. W. McNamara*** (peter.mcnamara@bucknell.edu), Bucknell University, Department of Mathematics, 1 Dent Drive, Lewisburg, PA 17837. *Positivity among P-partition generating functions.*

We seek simple conditions on a pair of labeled posets that determine when the difference of their (P, ω)-partition enumerators is F-positive, i.e., positive in Gessel's fundamental basis. This is a quasisymmetric analogue of the extensively studied problem of finding conditions on a pair of skew shapes that determine when the difference of their skew Schur functions is Schur-positive. We determine necessary conditions and separate sufficient conditions for F-positivity, and show that a broad operation for combining posets preserves positivity properties. We conclude with classes of posets for which we have conditions that are both necessary and sufficient. (Received August 08, 2020)

1160-05-148 **Alexander Garver*** (alexander.garver@gmail.com) and **Thomas McConville**. *Chapoton triangles and nonkissing complexes.*

For any finite real reflection group W, Chapoton defined three polynomials enumerating combinatorial objects associated with W: the F-triangle $F(x, y)$, the H-triangle $H(x, y)$, and the M-triangle $M(x, y)$. In particular, $F(x, y)$ enumerates faces of the cluster complex associated with W. Chapoton conjectured certain identities satisfied by $F(x, y)$ and $H(x, y)$ and by $F(x, y)$ and $M(x, y)$, which were later proved by Thiel and Athanasiadis, respectively. We present analogues of these three polynomials given the initial data of a nonkissing complex in the sense of Petersen, Pylyavskyy, and Speyer. The cluster complex associated with the symmetric group is a special case of the nonkissing complex. We prove the analogue of Chapoton's $F(x, y)$ and $H(x, y)$ identity and conjecture the analogue of his $F(x, y)$ and $H(x, y)$ identity. This is joint work with Thomas McConville. (Received August 08, 2020)

1160-05-152 **William J Keith*** (wjkeith@mtu.edu). *On Bergeron's $bc - ad$ unimodality conjecture.*

In 2016 Bergeron offered a conjecture which included the positivity and unimodality of the differences $\begin{bmatrix} b+c \\ c \end{bmatrix}_q - \begin{bmatrix} a+d \\ d \end{bmatrix}_q$ under the condition that $bc = ad$, $c > d$. Using the KOH theorem, Stanley and Zanello were able to prove this claim for small d. In this talk we prove positivity by an explicit injection when the difference between c and d is small. (Received August 09, 2020)

1160-05-157 **Véronique Bazier-Matte**, **Guillaume Douville**, **Alexander Garver**, **Rebecca Patrias*** (patr0028@stthomas.edu), **Hugh Thomas** and **Emine Yildirim**. *Leading terms of SL_3 web invariants.*

We use Khovanov and Kuperberg's web growth rules to identify the minimal term in the invariant associated to an SL_3 web diagram, with respect to a particular term order. (Received August 09, 2020)

1160-05-158 **Pavel Galashin**, **Steven N. Karp*** (karp.steven@courrier.uqam.ca) and **Thomas Lam**. *Regularity theorem for totally nonnegative flag varieties.*

The classical example of a totally positive space is the set of $n \times n$ totally positive matrices, formed by matrices whose every submatrix has positive determinant. Total positivity has been studied extensively over the past century, and has seen a renewed interest in the past 30 years, initiated by work of Lusztig and of Fomin and Zelevinsky. My talk will focus on the topology of such spaces. Historically, the motivation for studying the topology of totally positive spaces is that they have cell decompositions which realize interesting posets in combinatorics, related to Bruhat orders. A new motivation comes from recent work in physics, which calls for understanding convex polytopes generalized from affine space into the Grassmannian, where the notion of convexity is replaced by total positivity. I will present new techniques, developed in joint work with Pavel Galashin and Thomas Lam, for establishing the homeomorphism type of totally positive spaces and their compactifications. In particular, we prove that the totally nonnegative part of a partial flag variety forms a regular CW complex, confirming conjectures of Postnikov and of Williams. (Received August 09, 2020)

1160-05-162 **Khrystyna Serhiyenko*** (khrystyna.serhiyenko@uky.edu), **Melissa Sherman-Bennett** and **Lauren Williams**. *Cluster structures in Schubert varieties.*

In 2006 Scott showed that the coordinate ring of the Grassmannian has a cluster algebra structure given by the combinatorics of plabic graphs. Since then it was conjectured that this result holds more generally for positroid varieties, and we prove this in the case of Schubert varieties. In this talk we will discuss the main ingredients of the proof that involves Leclerc's categorification in terms of representation theory of preprojective algebras and Karpman's construction of plabic graphs. (Received August 09, 2020)

1160-05-163 **Chris Fraser** and **Melissa Sherman-Bennett*** (m_shermanbennett@berkeley.edu).
Cluster structures on Schubert varieties from relabeled plabic graphs.

We give a family of cluster structures on Schubert (and more generally, positroid) varieties, generalizing the "target" and "source" cluster structures. Each of the cluster structures in this family has seeds given by face labels of *relabeled plabic graphs*, which are plabic graphs whose boundary is labeled by a permutation of 1, ..., n. For Schubert varieties, all cluster structures in this family *quasi-coincide*, which means they differ only by rescaling by frozen variables and their cluster monomials coincide. As part of our results, we show the target and source cluster structures quasi-coincide, confirming a conjecture of Muller and Speyer. One proof tool of independent interest is a permuted version of the Muller-Speyer twist map, which we use to prove many (open) positroid varieties are isomorphic. (Received August 09, 2020)

1160-05-169 **Elizabeth Kelley***, School of Mathematics, 206 Church St SE, Minneapolis, MN 55455, and **Man-Wai Cheung** and **Gregg Musiker**. *Scattering diagrams for generalized cluster algebras with reciprocal coefficients.* Preliminary report.

Cluster algebras, as defined by Fomin and Zelevinsky, are characterized by binomial exchange relations. A natural generalization of these algebras, introduced by Chekhov and Shapiro, relaxes this restriction and allows the exchange polynomials to have arbitrarily many terms. Scattering diagrams have been used as a powerful tool for proving structural properties of cluster algebras. Following the work of Gross, Hacking, Keel, and Kontsevich, we discuss the construction of scattering diagrams for generalized cluster algebras, focusing on the subclass of generalized cluster algebras with reciprocal exchange coefficients. We then discuss some applications for these generalized scattering diagrams. (Received August 09, 2020)

1160-05-172 **Nathan Reading*** (reading@math.ncsu.edu) and **Salvatore Stella**. *Cluster scattering diagrams of affine type.*

We construct cluster scattering diagrams (in the sense of Gross, Hacking, Keel, and Kontsevich) for initial data of acyclic affine type. Scattering diagrams are collections of codimension-1 rational cones, along with some algebraic data. Cluster scattering diagrams are special scattering diagrams that encode the corresponding cluster algebra. Each scattering diagram defines a fan, and the cluster scattering diagrams of finite type define Cambrian fans (normal fans of generalized associahedra). Thus cluster scattering diagrams constitute a vast generalization of the classical associahedron.

Our affine construction extends the construction of affine-type **g**-vector fans as doubled Cambrian fans, and also extends the construction of finite-type cluster scattering diagrams in terms of shards. The walls of the affine cluster scattering diagram correspond to the join-irreducible c-sortable elements and c^{-1}-sortable elements, with each wall given by inequalities arising from the combinatorics/geometry of Coxeter groups/root systems.

Work is in progress to determine all theta functions of acyclic affine type. The determination of theta functions rests on work in progress linking theta functions to the mutation fan in general. (Received August 09, 2020)

1160-05-175 **Esther Banaian*** (banai003@umn.edu) and **Lena Chen**. *Frieze Patterns from Dissected Annuli.* Preliminary report.

Frieze patterns from triangulated polygon, triangulated annuli, and dissected polygon have been investigated by Conway-Coxeter, Baur-Parsons-Tshabold, and Holm-Jørgensen respectively. We continue this trend by studying frieze patterns from dissected annuli. We also give a combinatorial interpretation for entries in frieze patterns from both dissected polygon and dissected annuli, which generalizes matching formula known for triangulation cases. (Received August 11, 2020)

1160-05-178 **Anna Felikson**, **John Lawson** and **Michael Shapiro*** (mshapiro@msu.edu), Deparment of Mathematics, Michigan State University, 619 Red Cedar Road C212 Wells Hall, East Lansing, MI 48824, and **Pavel Tumarkin**. *CLUSTER ALGEBRAS FROM SURFACES AND EXTENDED AFFINE WEYL GROUPS.*

We characterize mutation-finite cluster algebras of rank at least 3 using positive semi-definite quadratic forms. In particular, we associate with every unpunctured bordered surface a positive semi-definite quadratic space V , and with every triangulation a basis in V , such that any mutation of a cluster (i.e., a flip of a triangulation) transforms the corresponding bases into each other by partial reflections. Furthermore, every triangulation gives rise to an extended affine Weyl group of type A, which is invariant under flips. The construction is also extended to exceptional skew-symmetric mutation-finite cluster algebras of types E. This is a joint work with A.Felikson, J.Lawson, and P.Tumarkin. (Received August 10, 2020)

1160-05-199 **Kayla Wright***, kaylaw@umn.edu, and **Gregg Musiker**, musiker@umn.edu. *Mixed Dimer Model for the F-Polynomial in Type D Cluster Algebras.* Preliminary report.

In joint work with Gregg Musiker, we give a combinatorial model for F-polynomials for type D_n cluster algebras where the associated quiver is acyclic using dimers and double dimers. In particular, we give a graph theoretic recipe that describes which monomials appear in such F-polynomials, as well as a graph theoretic way to determine the coefficients of each of these monomials. To prove this formula, we utilize a combinatorial formula due to Thao Tran and provide explicit bijections between her combinatorial model and our own. (Received August 10, 2020)

1160-05-234 **Sunita Chepuri*** (chepuri@umich.edu) and **Feiyang Lin**. *Symmetric Group Action of the Geometric R-matrix.* Preliminary report.

The geometric R-matrix is a birational transformation that appears in the theory of geometric crystals, the study of total positivity in loop groups, and discrete dynamical systems. This R-matrix gives rise to an action of the symmetric group S_m on an m-tuple of vectors. While the geometric R-matrix is precisely the formula corresponding to the action of the simple transposition s_i, explicit formulas for the action of other permutations are generally not known. One particular case was studied by Lam–Pylyavsky as it relates to intrinsic energy. In this talk, we will discuss formulas for several additional cases as well as provide combinatorial interpretations for the functions that arise in our work. (Received August 10, 2020)

1160-05-243 **Emma Yu Jin** (yu.jin@univie.ac.at), Faculty of Mathematics, Oskar-Morgenstern-Platz 1, 1090 Vienna, Austria, and **Michael J. Schlosser*** (michael.schlosser@univie.ac.at), Faculty of Mathematics, Oskar-Morgenstern-Platz 1, 1090 Vienna, Austria. *Basic hypergeometric proofs of two quadruple equidistributions of Euler–Stirling statistics on ascent sequences.*

Employing an analytic technique used by George Andrews and Vít Jelínek, we establish a non-terminating ${}_4\phi_3$ transformation formula of base q, valid as an identity in a neighborhood around $q = 1$. We use special cases of this formula to deduce two different quadruple equidistribution results involving Euler–Stirling statistics on ascent sequences. One of them concerns a symmetric equidistribution, the other confirms a bi-symmetric equidistribution that was recently conjectured in a paper (published in JCTA) by Shishuo Fu, Emma Yu Jin, Zhicong Lin, Sherry H.F. Yan, and Robin D.B. Zhou. (Received August 10, 2020)

1160-05-267 **Nicholas J Early*** (earlnick@gmail.com), Perimeter Institute, 31 Caroline St N, Waterloo, Ontario N2L 2Y5, Canada. *Generalized Feynman diagrams, weakly separated collections and positroidal subdivisions of hypersimplices.*

In 1948, Richard Feynman introduced a formalism in Quantum Field Theory to organize the expansion of a scattering amplitude as a sum of elementary rational functions, labeled by graphs. These graphs, now called Feynman diagrams, specify which singularities of the amplitude are compatible and can appear in the same such function.

In the past year the question of compatibility has been given the following interpretation: given two matroid subdivisions of the second hypersimplex $\Delta_{2,n}$, when is their common refinement also a matroid subdivision?

Motivated in part by recent joint work with Cachazo, Guevara and Mizera, as well as Cachazo-Guevara-Umberto-Yong, I will discuss how basic constructions in tropical geometry allow this question to be reformulated – and generalized to arbitrary $\Delta_{k,n}$ – as a statement about weakly separated collections, hence plabic graphs, in the case when the maximal cells in the subdivisions are positroid polytopes. (Received August 10, 2020)

1160-05-271 **Mark Skandera*** (mas906@lehigh.edu). *Generating functions for induced characters of the hyperoctahedral group.*

Merris and Watkins interpreted results of Littlewood to give generating functions for symmetric group characters induced from one-dimensional characters of Young subgroups. Beginning with an n by n matrix X of formal variables, one obtains induced sign and trivial characters by expanding sums of products of certain determinants and permanents, respectively. We will look at a new analogous result which holds for hyperoctahedral group characters induced from the four one-dimensional characters of its Young subgroups. This requires a $2n$ by $2n$ matrix of formal variables and four combinations of determinants and permanents. (Received August 10, 2020)

1160-05-287 **Alicia R Lima*** (alima@uchicago.edu), 449 Lowell ave Apt 8, Newton, MA 02460. *Cluster algebra and Grassmannian Geometry.* Preliminary report.

Ian Le and I worked on Cluster algebra and Grassmannian Geometry. It is well-known that the homogeneous coordinate ring of any Grassmannian manifold carries a cluster algebra structure. Given the Grassmannian projective structure, Ian and I attempted to use basic projective geometry properties to "out-smart" Cluster

algebra. Cross-ratio, being a defining property of projective geometry, was our best tool to unfold the basic structure hidden in the combinatorics of the cluster algebra associated to the homogeneous coordinate ring of any Grassmannian manifold. We were able to show that by parameterizing the Plucker Cluster by Fock-Goncharov coordinates, we can always express those coordinates as a cross ratio in some configuration space we constructed. We did calculations outside of Plucker Cluster, but we could not come up with a general algorithm to construct the configuration space to calculate the cross ratios, which would represent the Fock-Goncharov coordinates. (Received August 11, 2020)

1160-05-296 **Michael Albert**, **Christian Bean**, **Anders Claesson**, **Jay Pantone*** (jay.pantone@marquette.edu) and **Henning Ulfarsson**. *Combinatorial Exploration: A New Approach to Enumeration.* Preliminary report.

Combinatorial structures are ubiquitous throughout mathematics. Graphs, permutations, words, and other such families of combinatorial objects often play a central role in work from many different fields. The study of enumerative combinatorics is concerned with the elucidation of structural properties of these families, including counting, classification, and limiting behavior.

Combinatorial Exploration is a framework that unifies the often ad-hoc methods used in enumerative combinatorics. In this talk we'll explain how Combinatorial Exploration works, how it can be automated, and how it's being applied to the study of pattern-avoiding permutations to prove new results and reprove dozens of old ones. (Received August 11, 2020)

1160-05-305 **Helen Jenne** (helen.jenne@gmail.com), **Tri Lai** (tlai3@unl.edu) and **Gregg Musiker*** (musiker@math.umn.edu). *Double-dimer configurations and quivers of dP3 (del Pezzo) type.* Preliminary report.

In this work in progress, the authors extend combinatorial interpretations for so called toric cluster variables as was previously studied by the second and third authors. In [LM 2017] and [LM 2020], most toric cluster variables were shown to have Laurent expansions agreeing with partition functions of dimers on subgraphs cut out by six-sided contours. However, the case of cluster variables parameterized by six-sided contours with a self-intersection eluded our techniques. In this talk we discuss our research rectifying this issue by using the first author's condensation results for the double-dimer model [J 2019]. While we focus on quivers of dP3 type of Model 1 and Model 4, we anticipate our techniques will extend to certain additional cluster algebras related to brane tilings. (Received August 11, 2020)

1160-05-312 **Megan Owen*** (megan.owen@lehman.cuny.edu). *Covering Tree-Based Phylogenetic Networks.*

A phylogenetic tree or network represents the evolutionary history of a set of organisms, like bacteria or marsupials. Tree-based phylogenetic networks, which may be roughly defined as leaf-labeled networks built by adding arcs only between the original tree edges, have elegant properties for modeling evolutionary histories. We answer an open question of Francis, Semple, and Steel about the complexity of determining how far a phylogenetic network is from being tree-based, including non-binary phylogenetic networks. We show that finding a phylogenetic tree covering the maximum number of nodes in a phylogenetic network can be be computed in polynomial time via an encoding into a minimum-cost flow problem.

This is joint work with Katherine St. John (Hunter College) and our Research Experience for Undergraduates (REU) students from Fall 2017. (Received August 11, 2020)

1160-05-332 **Michael T. Engen*** (engenmt@ufl.edu). *Efficient graph representations, adjacency labeling schemes, and universal permutations.* Preliminary report.

Graphs are ubiquitous in both mathematics and computer science. The study of *graph representations*, the different ways to describe a graph or set of graphs, has enjoyed attention from researchers for decades, with one more recent area of interest being *efficient* graph representations, which are those that require a relatively small amount of memory to store. Most prevalent among these efficient representations are *adjacency labeling schemes*, which represent a collection of graphs by viewing them all as particular subgraphs of one larger graph, with the aim to make this larger graph as small as possible. In other contexts, this containing graph is called a *universal graph*. We discuss popular adjacency labeling schemes, universal graphs, and the related recently growing topic of universal *permutations*. (Received August 11, 2020)

1160-05-344 **Angela M Berardinelli*** (aberardinelli@uncc.edu) and **C Austin Lamp**. *Algorithms for generating reduced words in complex reflection groups.* Preliminary report.

Reduced words have been an important tool in studying the representation theory and combinatorics of Coxeter groups for at least the last 30 years. This tool does not fully translate to complex reflection groups in all contexts,

however. For example, Coxeter groups famously have a unique element of maximal length, but complex reflection groups do not. In this talk, we will discuss some useful properties of reduced words that do extend to complex reflection groups. We will also explore newly developed algorithms for generating the list of reduced words for a given element in a reflection group and for generating the corresponding graph whose vertex set is the set of reduced words for the element and whose edges represent the commuting and braid relations of the group. (Received August 11, 2020)

08 ► General algebraic systems

1160-08-358 **Tianran Chen***, ti@nranchen.org, and **Robert Davis**. *Power-flow equations and tropical intersections.*

The power-flow equations describe the intricate balancing conditions on buses in a power network for generating and distributing electric power. They form nonlinear system of equations, and their solutions describe theoretical operating configurations for the underlying network. Algebraic methods, including methods from numerical algebraic have long been used to solve and study power-flow systems. In this talk, we explore new development in this direction using the tools from tropical and convex geometry. (Received August 12, 2020)

11 ► Number theory

1160-11-11 **Sungkon Chang*** (schang@georgiasouthern.edu), 11935 Abercorn St, SAVANNAH, GA 31419. *The space of convolution identities on divisor functions.*

In Ramanujan's well-known work, a set of convolution identities on divisor functions σ are introduced, and his method of using Eisenstein series inspired numerous works on convolution identities on divisor functions. However, little is known about convolution identities on divisor functions with modular congruence restrictions. In this talk we introduce the notion of the space of convolution identities on divisor functions with modular congruence restriction, and a set of linearly independent convolution identities. We conjecture that our set is a basis of the space, and its dimension is the number of partitions of the modulus into three parts. We also introduce further generalizations of this phenomenon using newly discovered elliptic functions. (Received February 21, 2020)

1160-11-20 **Theresa C Anderson*** (tcanderson@purdue.edu), **Angel V Kumchev** and **Eyvindur A Palsson**. *Discrete maximal functions over surfaces of higher codimension and sharp triangular counts.*

We unite two themes at the interface of analysis and number theory: intermediate codimensional integration and discrete operators. Along the way, we provide a circle method treatment of optimally counting triangles, in all dimensions greater than or equal to 7, thus complimenting work of Brandes, Raghavan and Ellenberg-Venkatesh using different techniques. We briefly describe how the unification of these themes leads to the triangular count as well as other connections. This is joint work with Eyvi Palsson and Angel Kumchev. (Received July 15, 2020)

1160-11-22 **Ayla Gafni*** (argafni@olemiss.edu). *Partitions into powers of primes.* Preliminary report.

Given a set $\mathcal{A} \subset \mathbb{N}$, the restricted partition function $p_{\mathcal{A}}(n)$ counts the number of integer partitions of n with all parts in $\mathcal{A}$. In this talk, we will explore the features of the restricted partition function $p_{\mathcal{P}_k}(n)$ where $\mathcal{P}_k$ is the set of k-th powers of primes. Powers of primes are both sparse and irregular, which makes $p_{\mathcal{P}_k}(n)$ quite an elusive function to understand. We will discuss some of the challenges involved in studying restricted partition functions and what is known in the case of primes, k-th powers, and k-th powers of primes. (Received July 15, 2020)

1160-11-23 **Chantal David**, **Alexandra Florea** and **Matilde Lalin*** (mlalin@dms.umontreal.ca). *Non-vanishing for cubic L-functions.*

Chowla's conjecture predicts that $L(1/2, \chi)$ does not vanish for Dirichlet L-functions associated with primitive characters χ. It was first conjectured for the case of χ quadratic. For that case, Soundararajan proved that at least 87.5% of the values $L(1/2, \chi)$ do not vanish , by calculating the first mollified moments,

For cubic characters, the first moment has been calculated by Baier and Young (on $\mathbb{Q}$), by Luo (for a restricted family on $\mathbb{Q}(\sqrt{-3})$), and on function fields by David, Florea, and Lalín.

In this talk we prove that there is a positive proportion of cubic Dirichlet characters for which the corresponding L-function at the central value does not vanish. We arrive at this result by computing the first mollified moment using techniques that we previously developed in our work on the first moment of cubic L-functions, and by obtaining a sharp upper bound for the second mollified moment, building on work of Lester and Radziwiłł, Harper, and Radziwiłł- Soundararajan. Our results are on function fields, but with additional work they could be extended to number fields, assuming GRH.

This is a joint work with Chantal David and Alexandra Florea. (Received July 15, 2020)

1160-11-26 **Brian Hopkins*** (bhopkins@saintpeters.edu). *Further Connections Between Partition Crank and Mex.*

The crank is a well-known statistic of integer partitions: requested by Dyson and eventually defined by Andrews and Garvan, it helps explain the Ramanujan partition congruences. The mex is a newer statistic, the minimal excluded positive part of a partition, with the name borrowed from combinatorial game theory. Building on recent independent work of Andrews & Newman and Hopkins & Sellers, we use a generalized notion of mex to connect to partitions having any upper or lower bound on their crank, which leads to a partition recurrence formula for the number of partitions with a given crank. We will also show related results concerning Frobenius symbols, generalizing a 2011 result of Andrews. This is joint work with James Sellers and Dennis Stanton. (Received July 20, 2020)

1160-11-28 **Tuan Ngo Dac*** (ngodac@math.univ-lyon1.fr), CNRS - Université Claude Bernard Lyon 1, Institut Camille Jordan (ICJ), 43 boulevard du 11 novembre 1918, 69622 Villeurbanne Cedex, France, and **Nathan Green**. *Log-algebraicity and characteristic p multiple zeta values.*

Based on the notion of Stark units we present a new approach to obtain refinements of log-algebraic identities for Anderson t-modules. As a consequence we recover many earlier results in this direction. Further, we devise a direct and conceptual way to get logarithmic interpretations for both MZV's and ν-adic MZV's. This generalizes the work of Anderson and Thakur for Carlitz zeta values. (Received July 21, 2020)

1160-11-33 **Ralf Schmidt*** (ralf.schmidt@unt.edu). *Packet Structure and Paramodular Forms.*

According to Arthur's theory, the discretely appearing automorphic representations of the group GSp(4) are classified into 6 different types of global packets. The structure of such packets, combined with knowledge about fixed vectors in local representations, leads to new results about Siegel modular forms of degree 2. Among such results is a strong multiplicity one theorem for paramodular forms. (Received July 26, 2020)

1160-11-39 **Sharon A Garthwaite*** (sharon.garthwaite@bucknell.edu). *Congruences and incongruences for generalized Frobenius partitions.*

In 1984, G. Andrews defined k-colored generalized Frobenius partitions, which are now known to exhibit many patterns analogous to the beautiful Ramanjuan congruences satisfied by the classical partition counting function $p(n)$. In this talk we narrow the search for when such congruences occur. For classical partitions, this is akin to studying why $p(5n+4)$ divisible by 5 for all integers n but $p(5n+3)$ is not. This is joint work with Marie Jameson at University of Tennessee. We appeal to the theory of modular forms, inspired by the work of C.-S. Radu, S. Ahlgren, B. Kim, N. Andersen, and S. Löbrich. (Received July 29, 2020)

1160-11-41 **Jon Aycock*** (jaycock@uoregon.edu), Fenton Hall, University of Oregon, Eugene, OR 97403. *Families of Differential Operators for Overconvergent Hilbert Modular Forms.*

We construct differential operators for families of overconvergent Hilbert modular forms by interpolating the Gauss–Manin connection on strict neighborhoods of the ordinary locus. This is related to work done by Harron and Xiao and by Andreatta and Iovita in the case of modular forms and has applications in particular to p-adic L-functions of CM fields. (Received July 30, 2020)

1160-11-43 **Akshat Mudgal*** (amudgal@purdue.edu). *Diameter free estimates for Vinogradov systems.*

A classical object of study in additive number theory has been the Vinogradov system, that is, the system defined by the equations

$$x_1^j + ... + x_s^j = y_1^j + ... + y_s^j \quad (1 <= j <= k).$$

In particular, given a finite set A of integers, finding sharp upper bounds for the number of solutions $J_{s,k}(A)$ to this system, when all the variables lie in the set A, has been an important topic of work. Recently, two major approaches have been developed to tackle this problem - the efficient congruencing method of Wooley, and the decoupling techniques of Bourgain-Demeter-Guth. Both these methods give upper bounds for $J_{s,k}(A)$ in terms

of s, k, and the cardinality $|A|$ of A, and the diameter X of A. In particular, when X is large in terms of $|A|$, say when $X \gg \exp(\exp(|A|))$, these bounds perform worse than the trivial estimates.

In this talk, we present new upper bounds for $J_{s,2}(A)$ which depend only on $|A|$ and s. These improve upon, and generalise, a previous result of Bourgain and Demeter. (Received July 30, 2020)

1160-11-45 **Brandon Hanson*** (brandon.w.hanson@gmail.com) and **Giorgis Petridis**. *A better-than-Plunnecke bound for $A + 2A$.*

If A is a finite set in an abelian group, we can measure the additive structure of A by the size of its sumset, $A + A$. One way of quantifying this structure is the doubling constant, $K = |A + A|/|A|$. Plunnecke's inequality lets us measure the size of iterated sumsets in terms of K, and in particular it tells us that $|A + A + A| \leq K^3|A|$. This inequality can be sharp. What about the set $A+2A = \{a+b+b : a, b \in A\}$? It is a subset of $A+A+A$ and so the upper bound $K^3|A|$ applies. Moreover, it may not be a proper subset, as is the case when $A = \{1, \dots, N\}$. However, in this case Plunnecke's inequality is not sharp. In this talk, I will describe recent work with G. Petridis where we prove that $|A + 2A| \leq K^{2.95}|A|$, answering a question of B. Bukh. (Received July 31, 2020)

1160-11-47 **Andrew Granville*** (and.granville@gmail.com), Dept Math, CP 6128 Succursale Centre-Ville, Montréal, Quebec H3C 3J7, Canada, and **Allysa Lumley** (allysalumley@gmail.com). *The distribution of primes in short intervals.* Preliminary report.

What is the maximum number of primes in an interval of length y? Here we are thinking of $y \ll (\log x)^2$, that is, y is not too big. We will present several conjectures (for different ranges of y) based on (a couple of) heuristic ideas, and investigate these conjectures with data from calculations of primes. There are one or two surprising issues that arise. (Received July 31, 2020)

1160-11-51 **Ertan Elma** (eelma@uwaterloo.ca) and **Yu-Ru Liu*** (yrliu@uwaterloo.ca). *Number of prime factors with a given multiplicity.*

In this talk, we study a variation of the ω function. More precisely, given the positive integer k, let $\omega_k(n)$ denote the number of distinct prime factors of n which occur with multiplicity k. We will prove that $\omega_1(n)$ has the normal order $\log \log n$, while $\omega_k(n)$ does not have normal order with $k \geq 2$. This is joint work with Ertan Elma. (Received August 01, 2020)

1160-11-54 **Jing Yu*** (yu@math.ntu.edu.tw), 10-th fr. No. 355, Zhi-Chiang Rd., Tamsui, New Taipei, Taiwan. *Analytic Homomorphisms into t-Modules Revisited.*

We shall report on some recent applications of a theorem of mine about analytic homomorphisms into t-Modules (1997, Annals of Math.). This is now used as one of the key tools by Chang-Mishiba (to appear in Invent. math. 2021) in studying multiple zeta values (MZV introduced previously by D. Thakur) in positive characteristic. Chang-Mishiba, together with Chen, manages to go further proving an analogue of a deep conjecture from Furusho (for classical MZV) which says that all algebraic relations among the real MZV's are also algebraic relations among the p-adic MZV's, for any prime p. (Received August 01, 2020)

1160-11-55 **Michael Filaseta** (filaseta@math.sc.edu), **Joseph C Foster** (josephcf@math.sc.edu) and **Jacob Juillerat*** (juillerj@email.sc.edu), 263 Stonewood Dr, West Columbia, SC 29170, and **Jeremiah Southwick** (southwij@math.sc.edu). *Irreducibility criteria of polynomials with non-negative coefficients and prime values.* Preliminary report.

Let $f(x)$ be a reducible polynomial with non-negative integer coefficients, and let $b > 1$ be an integer such that $f(b)$ is a prime. As shown by Filaseta and Gross for $b = 10$ and by Cole, Dunn, and Filaseta for $2 \leq b \leq 20$, the maximum coefficient and the degree of such an $f(x)$ each have nontrivial lower bounds depending on b. Filaseta, Foster, Southwick and I were able to extend these results to all integers $b > 2$. In this talk we will discuss how we were able to generalize the previous results by analyzing the region in the complex plane that represents the locations of possible roots of such an $f(x)$. (Received August 01, 2020)

1160-11-62 **Krishnaswami Alladi*** (alladik@ufl.edu), University of Florida, Department of Mathematics, Gainesville, FL 32611. *EULER'S PARTITION THEOREM AND REFINEMENTS WITHOUT APPEAL TO INFINITE PRODUCTS.*

We present a new proof of Euler's celebrated theorem that the number of partitions of an integer into odd parts equals the number of partitions of that integer into distinct parts without using infinite product representations but only certain "amalgamation" properties and dissections of their series generating functions. Our approach also yields a simpler and more direct proof of a well known refinement of Euler's theorem due to Fine. We then use the same ideas along with conjugation of partitions into distinct parts to improve Fine's theorem and also

obtain a refinement of Euler's theorem due to Bessenrodt. Finally we discuss links between our approach and the Rogers-Fine identity. (Received August 03, 2020)

1160-11-68 **Pengyong Ding*** (pud35@psu.edu). *On a variance associated with the distribution of $r_3(n)$ in arithmetic progressions.* Preliminary report.

There is a long history in studying the asymptotic formula for the variance

$$V(x,Q)=\sum_{q\le Q}\sum_{a=1}^{q}|A(x;q,a)-f(q,a)M(x)|^2,$$

where

$$A(x;q,a)=\sum_{\substack{n\le x\\ n\equiv a(\mathrm{mod}\,q)}}a_n,$$

and $f(q,a)$ and $M(x)$ approximately reflect the local and global properties of the real sequence $\{a_n\}$ respectively. In this talk, we will discuss the corresponding expression when $a_n=r_3(n)$, the number of ordered representations of n as the sum of three positive cubes. The Hardy-Littlewood Circle Method will be used to obtain the result, and some special examples will be discussed at the end of the talk. (Received August 03, 2020)

1160-11-69 **Anna, Ying Pun*** (annapunying@gmail.com) and **William Craig** (wlc3vf@virginia.edu). *Distribution properties for t-hooks in partitions (Part 1).*

Partitions, the partition function $p(n)$, and the hook lengths of their Ferrers-Young diagrams are important objects in combinatorics, number theory and representation theory. For positive integers n and t, we study $p_t^e(n)$ (resp. $p_t^o(n)$), the number of partitions of n with an even (resp. odd) number of t-hooks. Using the Rademacher circle method, we find an exact formula for $p_t^e(n)$ and $p_t^o(n)$.

In this talk, we will discuss how we use this exact formula to show the distribution properties of $p_t^e(n)$ and $p_t^o(n)$ which is far from uniform, and the signs of $p_t^e(n)-p_t^o(n)$ for large n. (Received August 03, 2020)

1160-11-71 **Trevor D Wooley*** (twooley@purdue.edu), Department of Mathematics, Purdue University, 150 N. University Street, West Lafayette, IN 47907-2067. *Subconvexity in the circle method via (the absence of) translation-dilation invariance.* Preliminary report.

The recent resolution of the main conjecture for Vinogradov's mean value theorem depends crucially on the system of Diophantine equations associated with the mean value being translation-dilation invariant. The absence of translation-dilation invariance is usually seen as an impediment to establishing the sharpest bounds that one might envision. In this talk we illustrate ideas by reporting on an example where the absence of translation-dilation invariance plays a critical role. Specifically, we establish a Hasse principle for the existence of lines having integral coordinates passing through a fixed integral point lying on an affine diagonal hypersurface of degree d whenever it possesses $d(d+1)$ or more variables. This conclusion, derived via the circle method, surmounts the conventional convexity ("square-root") barrier for this problem. (Received August 04, 2020)

1160-11-72 **Madeline L Dawsey*** (mdawsey@uttyler.edu) and **Dermot McCarthy**. *Modular forms, hypergeometric functions, and Ramsey numbers.*

For a fixed integer $k\ge 2$, let q be a prime power such that $q\equiv 1 \pmod{k}$ if q is even, or $q\equiv 1 \pmod{2k}$ if q is odd. The generalized Paley graph $G_k(q)$ is the graph with vertex set $\mathbb{F}_q$ and edge set $\{ab : a-b \text{ is a } k\text{-th power residue modulo } q\}$. We give a formula in terms of finite field hypergeometric functions for the number of complete subgraphs of order four contained in $G_k(q)$, and a formula in terms of Jacobi sums for the number of complete subgraphs of order three contained in $G_k(q)$, for all k. This generalizes work of Evans, Pullham, and Sheehan on the Paley graph with $k=2$. We apply these formulas to obtain lower bounds for the multicolor diagonal Ramsey numbers $R_k(4)$ and $R_k(3)$. We also examine connections to Fourier coefficients of modular forms, when q is prime. (Received August 05, 2020)

1160-11-77 **Cris Poor*** (poor@fordham.edu), **Ralf Schmidt** and **David S, Yuen**. *Examples of Paramodular newforms and supercuspidal representations.*

This work bridges the abstract representation theory of GSp(4) with recent computational techniques. We construct four examples of paramodular newforms whose associated automorphic representations have local representations at p=2 that are supercuspidal. We classify all relevant irreducible, admissible, supercuspidal representations of GSp(4, Q_2) and show that our examples occur at the lowest possible paramodular level, 16. The required theoretical and computational techniques include paramodular newform theory, Jacobi restriction, bootstrapping and Borcherds products. (Received August 05, 2020)

1160-11-81 **Timothy Huber*** (timothy.huber@utrgv.edu) and **Dongxi Ye**. *Ramanujan type congruences for modular forms of level p.*

For $pin5, 7, 11$, all holomorphic quotients of a certain class of Klein forms up to weight $(p-1)/2$ are determined satisfying an infinite class of Ramanujan type congruences modulo powers of p. The congruences and corresponding generating functions are formulated explicitly through the construction of matrices for relevant linear operators and corresponding eigenspaces. Congruences satisfied by generating functions for p-cores appear as special cases. The permutative action of $\Gamma_0(p)$ on each quotient implies an algebraic symmetry in which the quotients satisfying Ramanujan type congruences are divided into equivalence classes of size $(p-1)/2$. A number of novel byproducts are obtained from dissection identities. The techniques constitute a foundation for the study of congruences for modular forms of any weight and level. (Received August 05, 2020)

1160-11-89 **Peter Graef*** (peter.graef@iwr.uni-heidelberg.de). *Hyperderivatives on Drinfeld cusp forms via representation theory.*

Drinfeld cusp forms are certain function field analogs of classical cusp forms. Contrary to what is expected in the classical situation, spaces of Drinfeld cusp forms (of fixed level and varying weight) admit various interesting maps between them which behave well with respect to the Hecke actions. Prominent examples of this phenonemon are certain hyperderivatives studied by Bosser-Pellarin.

In joint work with G. Böckle and R. Perkins, we study this phenomenon from a representation-theoretic point of view. I will start by explaining the relevant representation-theoretic framework and construct maps therein which become the hyperderivatives of Bosser-Pellarin via an isomorphism due to Teitelbaum. This also provides a natural extension of the maps of Bosser-Pellarin to more general bases and levels. Moreover, I will explain how the Frobenius map on Drinfeld cusp forms can be interpreted naturally on the representation-theoretic side. Finally, I will present computational results on how the Frobenius and the hyperderivatives interact. (Received August 06, 2020)

1160-11-93 **Alan M Talmage*** (abt5217@psu.edu). *Forms of Diagonal Equations Over the Primes.*

The system of diagonal equations

$$\sum_{i=1}^{s} u_{ij} p_i^{d_j} = 0, \qquad (1 \le j \le r)$$

has prime solutions $(p_1, \dots p_s)$, assuming that the system permits local solutions and that $s \ge k(M+r)+M$, where $k = \max\{d_i\}$ and M is the largest number of equations with the same exponent d_i. This is proved using the Hardy-Littlewood circle method, using techniques of Brandes and Parsell for handling the repeated degrees and Bourgain, Demeter, and Guth's proof of the Vinogradov Mean Value Theorem for the necessary mean values. (Received August 06, 2020)

1160-11-94 **Fan Ge*** (fge@wm.edu). *On the index conjecture in zero-sum theory.*

In zero-sum theory, an interesting and important question is to determine all pairs (ℓ, m) for which every minimal zero-sum sequence of length ℓ modulo m has index 1. This question has been settled for all $\ell \neq 4$, and the methods in literature are of combinatorial nature. The case $\ell = 4$ is a bit more subtle and remains unsolved. In this talk, I will discuss some recent progress toward this problem. (Received August 06, 2020)

1160-11-100 **Steven M. Gonek*** (gonek@math.rochester.edu), Department of Mathematics, 915 Hylan Bldg, University of Rochester, Rochester, NY 14627, and **Fan Ge** and **Alex Iosevich**. *The mean square discrepancy in the circle and divisor problems.*

We discuss the mean square of the error term in the Gauss circle and Dirichlet divisor problems. A heuristic argument based on the consideration of off-diagonal terms in the mean square of the relevant Voronoi-type summation formulae leads to precise conjectures for the structure of the mean squares of these discrepancies. (Received August 06, 2020)

1160-11-101 **Jonathan W. Bober**, **Zhenchao Ge*** (gezhenchao@gmail.com) and **Micah B. Milinovich**. *Irregularities of Dirichlet L-functions and a gap parity bias of zeros.*

The integral of Hardy's Z-function from 0 to T measures the occurrence of its sign changes. Hardy proved that this integral is $o(T)$ from which he deduced that the Riemann zeta-function has infinitely many zeros on the critical line. More recently, Ivic conjectured this integral is $O(T^{1/4})$ and $\Omega_{\pm}(T^{1/4})$ as $T \to \infty$. These estimates were proved, independently, by Korolev and Jutila. In this talk, we will show that the analogous conjecture is false for the Z-functions of certain "special" Dirichlet L-functions. In particular, we show that the integral of the Z-function of a Dirichlet L-function from 0 to T is asymptotic to $c_\chi T^{3/4}$ and we classify precisely when the constant c_χ is nonzero. This constant equals zero for most primitive characters, however there is a thin (but

infinite) set of exceptions. Somewhat surprisingly, numerical evidence seems to suggest that the unexpectedly large mean value is caused by a currently unexplained parity bias in the gaps between the zeros of these "special" Dirichlet L-functions. This is joint work with Jonathan Bober and Micah Milinovich. (Received August 07, 2020)

1160-11-102 **Chien-Hua Chen*** (cxc813@psu.edu), 1400 Martin St. Apt.1051, State College, PA 16803. *Surjectivity of the adelic Galois representation associated to a Drinfeld Module of prime rank.*

In this talk, we show under certain condition that the adelic Galois representation

$$\rho_\phi : \mathrm{Gal}(\mathbb{F}_q(T)^{\mathrm{sep}}/\mathbb{F}_q(T)) \longrightarrow \varprojlim_{\mathfrak{a}} \mathrm{Aut}(\phi[\mathfrak{a}]) \cong \mathrm{GL_r}(\widehat{A})$$

associated to the Drinfeld module ϕ over $\mathbb{F}_q(T)$ of prime rank r, where ϕ is defined by $\phi_T = T + \tau^{r-1} + T^{q-1}\tau^r$, is surjective. (Received August 07, 2020)

1160-11-104 **Andreas Maurischat*** (maurischat@fh-aachen.de), Aachen, Germany. *Taylor coefficients of Anderson generating functions.*

Algebraic extensions of the rational function field $K = \mathbb{F}_q(\theta)$ can be obtained by adjoining torsion points of Drinfeld modules over K. The Anderson generating functions for such Drinfeld modules are defined as certain power series $\sum c_i t^i \in \bar{K}[[t]]$ which converge for all $|t| \leq 1$ with respect to (an extension of) the norm associated to the infinite place of K. In particular, they converge at all roots of unity. The main goal of this talk is to explain how the coefficients of the Taylor expansions of the Anderson generating functions at roots of unity give rise to prime power torsion points of the Drinfeld module. This is joint work with Rudy Perkins. (Received August 07, 2020)

1160-11-105 **Stanley Yao Xiao*** (syxiao@math.toronto.edu), Bahen Centre, 40 St. George St., Room 6290, Toronto, Ontario M5S 2E4, Canada. *On monic abelian cubics.*

We show that the number of monic, irreducible cubic polynomials $f(x) = x^3 + ax^2 + bx + c$ with integer coefficients satisfying $\max\{|a|, |b|, |c|\} \leq X$ and Galois over $\mathbb{Q}$ is $O(X(\log X)^2)$. An easy construction gives a lower bound of $2X + O(1)$, which shows that the exponent of X is sharp. (Received August 07, 2020)

1160-11-107 **Ernst-Ulrich Gekeler*** (gekeler@math.uni-sb.de), Germany. *Torsion fields of Drinfeld modules.*

Let A be the polynomial ring F[T] over a finite field F = F_q, K_{inf} the field of formal Laurent series F((1/T)) in 1/T , and L a finite field extension of K_{inf}. We study Drinfeld A-modules (of arbitrary rank r) Phi over L and the L-extension M generated by the period lattice of Phi, which agrees with the field L(tor(Phi)) generated by the torsion of Phi. Then M is finite over L, but its degree [M:L] may become arbitrarily large even for fixed r >=2. In contrast, the residual degree f(M—L) is bounded by a constant that depends only on r. As a consequence, the degree of the maximal constant extension in L(tor(Phi)) is bounded by a constant depending only on r. (Received August 07, 2020)

1160-11-109 **Cecile Armana*** (cecile.armana@univ-fcomte.fr) and **Fu-Tsun Wei**. *Sturm bounds for Drinfeld-type automorphic forms over function fields.*

Sturm bounds indicate how many successive Fourier coefficients suffice to determine a modular form. For classical modular forms, they also provide bounds for the number of Hecke operators generating the Hecke algebra. In this talk I will present Sturm bounds for Drinfeld-type automorphic forms over function fields, also known as harmonic cochains on the Bruhat-Tits tree, and for generators of the corresponding Hecke algebras. This is a joint work with Fu-Tsun Wei. (Received August 07, 2020)

1160-11-110 **Heidi Goodson*** (heidi.goodson@brooklyn.cuny.edu) and **Melissa Emory**. *Sato-Tate Distributions of $y^2 = x^p - 1$ and $y^2 = x^{2p} - 1$.*

Let $C_m : y^2 = x^m - 1$ be a smooth projective curve defined over Q with $m = p$ or $2p$. We would like to study the limiting distributions of the coefficients of the normalized L-polynomial for C_m. To determine the distributions, we study the Sato-Tate groups of the Jacobians of the curves. In this talk we give both general results and explicit examples of Sato-Tate groups for the curves C_m. We will use these groups to determine the limiting distributions of the coefficients of the normalized L-polynomial. (Received August 07, 2020)

1160-11-112 **Alexander Cowan** and **Kimball Martin***, Department of Mathematics, University of Oklahoma, Norman, OK 73019. *Small Galois orbits of modular forms.* Preliminary report.

Maeda's conjecture and generalizations predict that newforms have large Galois orbits or, equivalently, large rationality fields. On the other hand, small Galois orbits are of interest because they correspond to nice geometric

objects such as elliptic curves or abelian surfaces with real multiplication. We will discuss various predictions about sizes of Galois orbits and report on computational progress. (Received August 07, 2020)

1160-11-113 **Hugh L. Montgomery*** (hlm@umich.edu), Prof. Hugh L. Montgomery, 6140 Arrowhead Drive, Sylvania, OH 43560. *Lower bounds for moments.* Preliminary report.

Lower bounds for moments of the zeta function for abscissae greater than one-half will be derived by an elegant approach whose motivation is simple and clear. (Received August 07, 2020)

1160-11-115 **Nathan Green*** (n2green@ucsd.edu) and **Tuan Ngo Dac**. *Algebraic Independence of Zeta Values of Function Fields of Elliptic Curves and General Genus Curves.*

Classically, the transcendence of odd zeta values is widely conjectured, yet unproven. However, for zeta values defined over the rational function field, Jing Yu succeeded in proving their transcendence in 1991, and many other transcendence results (including algebraic independence) followed in the intervening years. In this work, joint with T. Ngo Dac, we prove the algebraic independence of zeta values defined over the function field of an elliptic curve. The main technique we use is to construct a Tannakian category of t-motives whose associated periods contain these zeta values - thus we may exploit the existence of a motivic Galois group to study the transcendence degree. We also discuss the progress in our ongoing project to prove algebraic independence for zeta values of function fields of arbitrary curves. (Received August 07, 2020)

1160-11-117 **Lindsay Dever*** (lmdever@brynmawr.edu) and **Djordje Milićević** (dmilicevic@brynmawr.edu). *Ambient prime geodesic theorems on compact hyperbolic 3-manifolds.* Preliminary report.

The study of hyperbolic 3-manifolds draws deep connections between number theory, geometry, topology, and quantum mechanics. Specifically, the closed geodesics on a manifold are intrinsically related to the eigenvalues of Maass forms via the Selberg trace formula and are parametrized by their length and holonomy. The trace formula for spherical Maass forms can be used to prove the Prime Geodesic Theorem, which provides an asymptotic count of geodesics up to a certain length. In 1999, Sarnak and Wakayama established a count of geodesics by both length and holonomy that shows that holonomies of geodesics of increasing lengths become equidistributed throughout the circle. In this talk, I will present new results including a count of geodesics in shrinking intervals of length and holonomy, which implies effective equidistribution of holonomy. (Received August 07, 2020)

1160-11-120 **Amita Malik*** (amita.malik@aimath.org) and **Neha Prabhu**. *Extremal primes for non-CM elliptic curves.* Preliminary report.

For an elliptic curve E over the field of rationals, the distribution of the number of points on E modulo a prime has been well-studied over decades. More recently, the objects of interests are the extremal primes for a given curve E. These are the primes p of good reduction for which the number of points on E mod p is either maximal or minimal. In this talk. we discuss bounds on the counting function for these primes in the non-CM case. This is joint work with Neha Prabhu. (Received August 07, 2020)

1160-11-121 **Chantal David***, Department of Mathematics, Concordia University, 1455 de Maisonneuve West, Montreal, Quebec H3G 1M8, Canada, and **Ahmet Muhtar Guloglu**. *Non-vanishing of cubic Dirichlet L-functions.*

A famous conjecture of Chowla predicts that $L(1/2,\chi) \neq 0$ for Dirichlet L-functions attached to primitive characters χ. It was conjectured first in the case where χ is a quadratic character, which is the most studied case. For quadratic Dirichlet L-functions, Ozluk and Snyder showed that at least 15/16 of the L-functions $L(1/2,\chi)$ attached to quadratic characters do not vanish, by computing the one-level density for the low-lying zeroes at $1/2 + it$ under GRH. Without assuming the GRH, Soundararajan then proved that at least 87.5% of the quadratic Dirichlet L-functions do not vanish at $s = 1/2$, using mollified moments.

We first explain in this talk the different approaches in the case of quadratic characters, and then consider the case of Dirichlet cubic L-functions. There are few papers in literature about Dirichlet cubic L-functions, compared to the abundance of papers on Dirichlet quadratic L-functions, as this family is more difficult, in part because of the cubic Gauss sums.

Controlling the behaviour of the cubic Gauss sums, we show how to get a positive density of non-vanishing for a natural sub-family of the cubic Dirichlet L-functions over the Eisenstein field, by computing the one-level density, assuming GRH. (Received August 07, 2020)

1160-11-122 **Amita Malik*** (amita.malik@aimath.org). *On the number of restricted partitions.* Preliminary report.

Hardy and Ramanujan gave an asymptotic formula for the number of unrestricted partitions of a positive integer, and claimed a similar asymptotic formula for the number of partitions into fixed powers, which was later proved by Wright. More recently,Vaughan established a similar result for partitions into primes. In this talk, we discuss the case of partitions into primes in an arithmetic progression. (Received August 07, 2020)

1160-11-126 **Jyothsnaa Sivaraman*** (jyothsnaa.sivaraman@utoronto.ca). *On the study of Euclidean ideal classes appearing in abelian extensions.*

In 1979, H. W. Lenstra showed that the ring of integers of a number field with unit rank at least one has a Euclidean ideal class if and only if the class group is cyclic provided the extended Riemann hypothesis is true. In 2013, Graves and Murty introduced techniques from sieve theory to prove unconditionally the above result for a family of fields of unit rank greater than or equal to 4. In this talk, we will present some details about the history of this problem, state our unconditional results in this context. (Received August 07, 2020)

1160-11-130 **Jacques Benatar**, **Alon Nishry** and **Brad Rodgers*** (brad.rodgers@queensu.ca). *On the distribution of polynomials with random multiplicative coefficients.* Preliminary report.

We will discuss ongoing work related to the distribution of random polynomials with multiplicative coefficients. We will discuss a proof that almost surely the distribution becomes normal, an analogue of a classical result of Salem and Zygmund. This result was previously obtained in unpublished work of Harper, making use of martingale methods; we will outline a proof which relies on the moment method and a counting argument of Vaughan and Wooley. If there is sufficient time we will discuss questions surrounding the sup-norm of such polynomials. (Received August 07, 2020)

1160-11-131 **Fu-Tsun Wei*** (ftwei@math.nthu.edu.tw). *On class number relations and intersections over function fields.*

In this talk, we shall discuss a function field analogue of the Hirzebruch-Zagier class number formula. More precisely, we establish a connection between class numbers of "imaginary" quadratic function fields and the corresponding intersections of "Hirzebruch-Zagier-type" divisors on the Drinfeld-Stuhler modular surfaces. The main bridge is the theta series associated to anisotropic quadratic spaces of dimension 4. The connection directly comes from two different expressions of the Fourier coefficients of the theta series, which can be viewed as a geometric Siegel-Weil formula in this particular case. This is a joint work with Jia-Wei Guo. (Received August 07, 2020)

1160-11-137 **Francesc Castella*** (castella@ucsb.edu), **Giada Grossi**, **Jaehoon Lee** and **Christopher Skinner**. *Iwasawa theory of elliptic curves at Eisenstein primes and applications.*

In the study of the Iwasawa theory of elliptic curves $E/\mathbb{Q}$, it is often assumed that p is a non-Eisenstein prime, meaning that $E[p]$ is irreducible as a $G_{\mathbb{Q}}$-module. Because of this, most of the recent results on the p-converse to the theorem of Gross–Zagier and Kolyvagin (following Skinner and Wei Zhang) and on the p-part of the Birch–Swinnerton-Dyer formula in analytic rank 1 (following Jetchev–Skinner–Wan) were only known for non-Eisenstein primes p.

In this talk, I'll explain some of the ingredients in a joint work with Giada Grossi, Jaehoon Lee, and Christopher Skinner in which we study the (anticyclotomic) Iwasawa theory of elliptic curves over $\mathbb{Q}$ at Eisenstein primes. As a consequence of our study, we obtain an extension of the aforementioned results to the Eisenstein case. In particular, for $p = 3$ this leads to an improvement on the best known results towards Goldfeld's conjecture in the case of elliptic curves over $\mathbb{Q}$ with a rational 3-isogeny. (Received August 08, 2020)

1160-11-141 **Preston Wake*** (wakepres@msu.edu). *Tame derivatives and the Eisenstein ideal.*

As was made famous by Mazur, the mod-5 Galois representation associated to the elliptic curve X_0(11) is reducible. Less famously, but also noted by Mazur, the mod-25 Galois representation is reducible. We'll talk about this kind of extra reducibility phenomenon more generally, for cuspforms of even weight k and prime level. We'll observe that the characters appearing in the reducible representation are related, on one hand, to an algebraic invariant (the 'tame deriviative' of an L-function), and, on the other hand, to an algebraic invariant (the 'tame L-invariant'). This type of 'algebraic=analytic' relation is predicted by a version of the Bloch-Kato conjecture for families of motives formulated by Kato. (Received August 08, 2020)

1160-11-142 **Cristian D Popescu*** (cpopescu@math.ucsd.edu), University of California, San Diego, Department of Mathematics, 9500 Gilman Drive, La Jolla, CA 92093-0112. *An Equivariant Tamagawa Number Formula for Drinfeld Modules and Applications.*

We will discuss the statement and idea of proof of an equivariant Tamagawa number formula for Drinfeld modules. This generalizes to a Galois equivariant setting Taelman's celebrated class-number formula for Drinfeld modules. As an application, we will discuss the statement and idea of proof of the Drinfeld module analogue of the classical refined Brumer–Stark conjecture. This is joint work with J. Ferrara, N. Green, and Z. Higgins. (Received August 08, 2020)

1160-11-145 **Özge Ülkem*** (oezge.uelkem@iwr.uni-heidelberg.de). *Uniformization of generalized $\mathcal{D}$-elliptic sheaves.*

Drinfeld introduced the notion of *elliptic modules*, which are now called Drinfeld modules, as an analogue of elliptic curves in the function field setting. To prove the local Langlands correspondence, Drinfeld studied moduli spaces of elliptic sheaves. Since then, many generalizations of elliptic sheaves have been studied, such as $\mathcal{D}$-elliptic sheaves defined by Laumon, Rapoport, Stuhler and Frobenius-Hecke sheaves by Stuhler. In this talk, I will introduce generalized $\mathcal{D}$-elliptic sheaves, present some basic results on their moduli spaces and on the analytic uniformization of the latter. This builds on work of Laumon-Rapoport-Stuhler, of Hartl and of Rapoport-Zink. (Received August 08, 2020)

1160-11-147 **Zhiyuan Ding***, 5-1-5 Kashiwanoha, Kashiwa, Chiba 2778583, Japan. *On analogues of modular units for the moduli stack of Drinfeld shtukas.*

The notion of shtukas, introduced by Drinfeld, is a generalization of the notion of Drinfeld modules. We construct certain rational functions on the moduli stack of shtukas. These rational functions are analogues of modular units for elliptic curves. Their divisors are supported on horocycles. We introduce the notion of toy shtukas, and use their moduli space to describe the space of principal horospherical divisors generated by these rational functions. This gives an explicit version of the Manin-Drinfeld theorem for function fields. (Received August 08, 2020)

1160-11-151 **Ahmad M El-Guindy*** (aelguindy@sci.cu.edu.eg). *Modular Identities of Hypergeometric Type.*

In the classical theory of modular forms, there are numerous connections with hypergeometric series. These appear, for instance, in questions related to solving modular differential equations, computing periods and supersingular polynomials, and inverting the j-invariant and other modular functions, to name a few. In this talk, we investigate analogous questions in the function field setting, exploring some of the parallels and limitations. (Received August 09, 2020)

1160-11-153 **Chi-Yun Hsu*** (cyhsu@math.ucla.edu), **Zhaorong Jin** and **Ryotaro Sakamoto**. *Construction of Euler systems for $\mathrm{GSp}_4 \times_{\mathrm{GL}_1} \mathrm{GL}_2$.*

An Euler system is a collection of first Galois cohomology classes with the Galois group varying over that of cyclotomic fields. These classes are required to be norm-compatible up to local L-factors. By constructing a non-trivial Euler system, one can bound certain Selmer groups. Following a similar strategy to the pioneering work of Lei–Loeffler–Zerbes, we construct Euler systems for the Galois representations coming from automorphic representations of $\mathrm{GSp}_4 \times_{\mathrm{GL}_1} \mathrm{GL}_2$. We will explain how the integral formula of L-functions, due to Novodvorsky in this case, plays an important role. (Received August 09, 2020)

1160-11-155 **Ari Shnidman*** (ari.shnidman@gmail.com), Hebrew University of Jerusalem. *Manin-Drinfeld cycles and L-functions.*

I'll present a formula relating the r-th central derivative of an automorphic L-function for PGL_2 over a function field to the intersection of a pair of algebraic cycles in the moduli space of shtukas with r legs. The intersection is between the Heegner-Drinfeld cycle considered by Yun-Zhang and a 'Manin-Drinfeld' cycle coming from the split torus. When the analytic rank is 0 this gives a new description of the Taylor series coefficients. When the analytic rank is greater than 0, the intersection pairing vanishes, suggesting that these Manin-Drinfeld cycles are either torsion in the Chow group or are linearly independent from the Heegner-Drinfeld classes. The key idea in the proof is a representation theoretic interpretation of Yun-Zhang's operator which plays the role of differentiation in the geometry of shtukas. (Received August 09, 2020)

1160-11-165 **Ram Murty*** (murty@mast.queensu.ca), Department of Mathematics and Statistics, Jeffery Hall, Queen's University, Kingston, Ontario K7L 3N6, Canada. *On the normal number of prime factors of sums of Fourier coefficients of Hecke eigenforms.* Preliminary report.

In 1984, Kumar Murty and I studied the normal number of prime factors of Fourier coefficients of modular forms. In recent joint work with Kumar Murty and Sudhir Pujahari, we study the normal number of prime factors of sums of Fourier coefficients of Hecke eigenforms using recent advances in the theory of ℓ-adic Galois representations. (Received August 09, 2020)

1160-11-167 **James G McLaughlin***, jmclaughlin2@wcupa.edu, and **Peter Zimmer**, pzimmer@wcupa.edu. *Further results on Vanishing Coefficients in infinite products of the form* $(q^b, q^{p-b}; q^p)_\infty^3 (q^{jb}, q^{2p-jb}; q^{2p})_\infty$.

This talk continues a topic initiated by Michael Hirschhorn, and deals with infinite q-products for which all coefficients in certain arithmetic progressions vanish. For example,

Let p be a prime of the form $p = 24t + 11$, so that $p = 2U^2 + 3V^2$ for positive integers U and V, and $2|U$. Let h and g be any integers such that

$$hU + 3gV = 1, \tag{1}$$

and set $x = 3g + U$ and $y = 3(h - V)$. Let

$$j = 2x(x - 3g) - y\left(h - \frac{y}{3}\right) = 2xU - yV.$$

(i) Let v $(0 \le v \le p - 1)$ and w be defined by

$$v \equiv -xV^{-1} (\mod p), \tag{2}$$

$$w = \frac{j + \chi p + 3}{2}, \text{ where } \chi = \begin{cases} 0, & j \text{ is odd,} \\ 1, & j \text{ is even.} \end{cases}$$

Let b be any integer and let the sequence $\{r_n\}$ be defined by

$$(q^b, q^{p-b}; q^p)_\infty^3 (q^{jb}, q^{2p-jb}; q^{2p})_\infty = \sum_{n=0}^{\infty} r_n q^n, \tag{3}$$

Then $r_{pn+vb^2+wb} = 0$ for all integers n.

(ii) A similar results (possibly only for even b) holds if

$$v \equiv -y(2U)^{-1} (\mod p).$$

Similar theorems are given for primes in the arithmetic progressions $24t + 1$, $24t + 5$ and $24t + 7$. (Received August 09, 2020)

1160-11-181 **Zheng Liu*** (zliu@math.ucsb.edu). *p-adic families of Yoshida lifts.* Preliminary report.

We construct a Hida family of Yoshida lifts for two given Hida families of modular forms, and compute the Petersson inner products of its specializations. The key step in the construction is to choose suitable Schwartz functions at p. The computation of the Petersson inner products can be viewed as a generalization of the computation in the works by Bocherer–Dummigan–Schulze-Pillot and Hsieh–Namikawa. Our computation makes use of an equivariant property of the chosen Schwartz functions at p for the action of U_p operators. This is an ongoing joint work with Ming-Lun Hsieh. (Received August 10, 2020)

1160-11-184 **Gebhard Böckle** and **Florian Breuer*** (florianbreuer@gmail.com), School of Mathematical and Physical Sciences, University of Newcastle, Callaghan, NSW 2308, Australia. *The generic monodromy of Drinfeld modular varieties in special characteristic.*

By combining theorems of Drinfeld and Strauch, we show that the monodromy representation on the special fiber of a Drinfeld modular variety, with level not divisible by the characteristic, is surjective. We illustrate this result in the special case of Drinfeld $\mathbb{F}_q[t]$-modules in level t, and apply this to show that the Kronecker factors of a Drinfeld modular polynomial in rank r are irreducible. (Received August 10, 2020)

1160-11-190 **Jennifer Balakrishnan**, **Will Craig** and **Ken Ono*** (ken.ono691@gmail.com), Dept. Mathematics, University of Virginia, Charlottesville, VA 22904, and **Wei-Lun Tsai**. *Variants of Lehmer's Conjecture for Ramanujan's tau-function.*

Lehmer's unresolved speculation on the nonvanishing of Ramanujan's tau-function, it is natural to ask whether a fixed integer is a value of $\tau(n)$ or is a Fourier coefficient $a_f(n)$ of any given newform $f(z)$. We offer a method, which applies to newforms with integer coefficients and trivial residual mod 2 Galois representation, that answers this question for odd integers. We determine infinitely many spaces for which the primes $3 \le \ell \le 37$ are not absolute values of coefficients of newforms with integer coefficients. For $\tau(n)$ we show that $\tau(n) \neq \pm\ell$ for any odd

prime $\ell < 100$. We also obtain sharp lower bounds for the number of prime factors of such newform coefficients. In the weight aspect, for powers of odd primes ℓ, we prove that $\pm\ell^m$ is not a coefficient of any such newform f with weight $2k > M^{\pm}(\ell, m) = O_\ell(m)$ and even level coprime to ℓ, where $M^{\pm}(\ell, m)$ is effectively computable. (Received August 10, 2020)

1160-11-194 **Lillian B Pierce***, Mathematics Department, Durham, NC 27514. *Burgess bounds in new settings.*

We will survey recent results that prove Burgess-type bounds in new settings. In particular the presentation will highlight the role played by two deep inputs: new bounds for complete character sums, and variations on the Vinogradov Mean Value Theorem. (Received August 10, 2020)

1160-11-197 **John Bergdall*** (jbergdall@brynmawr.edu), **Brandon Levin** and **Tong Liu**. *Reductions of local Galois representations for eigenforms with large $\mathcal{L}$-invariant.*

The $\mathcal{L}$-invariant is an important p-adic invariant associated with certain newforms of level Np, where N is co-prime to p. Interest in $\mathcal{L}$-invariants is driven by their role in the trivial zero phenomenon for p-adic L-functions, their connection to p-adic families, and their appearance in the parametrization of associated Galois representations. Little is known about the invariants' explicit influence, however, despite arising in so many ways. This talk focuses on $\mathcal{L}$-invarants and Galois representation modulo p. The main result is an exact description for the local mod p Galois representation whenever the $\mathcal{L}$-invariant has sufficiently large p-adic norm. Descriptions were previously known only for eigenforms of weight at most p; the new result is the first in higher weight. This is joint work with Brandon Levin and Tong Liu. (Received August 10, 2020)

1160-11-205 **Geoffrey Akers*** (gakers@gradcenter.cuny.edu). *On a universal deformation ring that is a discrete valuation ring.* Preliminary report.

I will talk about a universal deformation ring R of an n-dimensional, mod p Galois representation whose semisimplification is the direct sum of two non-isomorphic absolutely irreducible representations. Under some hypotheses, we obtain that R is a discrete valuation ring. The method examines the ideal of reducibility of R, which is used to construct extensions of representations in a Selmer group with specified dimension. (Received August 10, 2020)

1160-11-208 **Oguz Gezmis*** (gezmis@math.nthu.edu.tw), Department of Mathematics, National Tsing Hua, University, Hsinchu City, 30042, Taiwan, and **Federico Pellarin** (federico.pellarin@univ-st-etienne.fr), MR 5208, Site de Saint-Etienne, 23 rue du Dr., P. Michelon, 42023 Saint-Etienne, France. *Trivial multiple zeta values over Tate algebras.*

In 2016, Pellarin introduced the deformation of multiple zeta values over function fields defined by Thakur. In this talk, we study a special family of such values called trivial multiple zeta values. Moreover, we describe a module structure on the set of trivial multiple zeta values over a non-commutative polynomial ring and determine the generators under a certain condition on number of variables. Furthermore, we explain how one can detect linear relations among Thakur's multiple zeta values by using trivial multiple zeta values. This is a joint work with Federico Pellarin. (Received August 10, 2020)

1160-11-209 **Frank Garvan*** (fgarvan@ufl.edu). *A new approach to Dyson's rank conjectures.* Preliminary report.

In 1944 Dyson defined the rank of a partition as the largest part minus the number of parts, and conjectured that the residue of the rank mod 5 divides the partitions of $5n + 4$ into five equal classes. This gave a combinatorial explanation of Ramanujan's famous partition congruence mod 5. He made an analogous conjecture for the rank mod 7 and the partitions of $7n + 5$. In 1954 Atkin and Swinnerton-Dyer proved Dyson's rank conjectures by constructing several Lambert-series identities basically using the theory of elliptic functions. In 2016 the author gave another proof using the theory of weak harmonic Maass forms. In this talk we describe a new and more elementary approach using Hecke-Rogers series. (Received August 10, 2020)

1160-11-211 **Federico Pellarin*** (federico.pellarin@univ-st-etienne.fr), 21 b rue du Dr. Rafin, 69009 Lyon, France. *On Drinfeld modular forms with values in Tate algebras.* Preliminary report.

We discuss a generalisation of (scalar) Drinfeld modular forms. These have been introduced by David Goss in his Ph. D. thesis. In this work, we study infinity-adic families of them, where the weight is constant, but the level varies. Many of these families come from (vectorial) modular forms with values in Tate algebras for the full Drinfeld modular group. We describe the basic properties: analytic behaviour near the cusp infinity,

finite dimensionality of their spaces, Hecke operators. We end the talk with some conjectures on the generators. (Received August 10, 2020)

1160-11-232 **Alexander Berkovich*** (alexb@ufl.edu), University of Florida, Dept of Mathematics, Little Hall 496, Gainesville, FL 32653. *New q-series identities of the form Multisum = Product.*

In this talk I further explore new positivity preserving transformations for the q-binomial coefficients. I derive many new identities of the Rogers-Ramanujan type. In particular, I establish that for any integer $v > 1$ the following holds

$$\sum_{m,k,\mathbf{n}} \frac{q^{T(m)+T(m+k)+\sum_{j=1}^{v-1} N_j^2}}{(q)_m (q)_k} \begin{bmatrix} \lfloor \frac{k}{2} \rfloor - \sum_{t=1}^{v-2} N_t \\ n_{v-1} \end{bmatrix}_{q^2} \prod_{j=1}^{v-2} \begin{bmatrix} n_j + k - 2\sum_{t=1}^{j} N_t \\ n_j \end{bmatrix}_q = \frac{(q^{4v^2+2v}, q^{2v^2}, q^{2v^2+2v}; q^{4v^2+2v})_\infty}{(q)_\infty}.$$

Here $T(x) := \frac{x(x+1)}{2}$ and $N_j = \sum_{t=j}^{v-1} n_t$, $1 \le j < v$ with the rest of notations being standard. (Received August 10, 2020)

1160-11-233 **Floric Tavares Ribeiro*** (floric.tavares-ribeiro@unicaen.fr), Laboratoire Nicolas Oresme, Campus 2, Boulevard du Marechal Juin, 14000 Caen, France. *A class formula for Drinfeld modules.*

In 2012, Lenny Taelman proved an analogue of the class number formula for Drinfeld A-modules where A is a polynomial ring. It relates an L-series with quantities coming from a regulator of units and from a class module.

For an arbitrary coefficient ring A, several deep results generalizing this class formula have been obtained by various authors for families of Drinfeld A-modules, or even for some Anderson A-modules. We will present in this talk a new approach based on the notion of Stark units, and establish the class formula for all Drinfeld modules. This method in fact applies to a large family of Anderson modules containing in particular all the Drinfeld modules. This is a joint work with Bruno Anglès (Caen, France) and Tuan Ngo Dac (Lyon, France). (Received August 10, 2020)

1160-11-242 **Andrea Conti**, **Anna Medvedovsky** and **Jaclyn Lang*** (jaclyn.lang@maths.ox.ac.uk). *Images of two-dimensional pseudorepresentations.*

There is a general philosophy that the image of a Galois representation should be as large as possible, subject to its symmetries. This can be seen in Serre's open image theorem for non-CM elliptic curves, Ribet and Momose's work on Galois representations attached to modular forms, and recent work of the speaker and Conti-Iovita-Tilouine on Galois representations attached to p-adic families of modular forms. Recently, Bellaïche developed a way to measure the image of an arbitrary pseudorepresentations taking values in a local ring A. Under the assumptions that A is a domain and the residual representation is not too degenerate, we explain how the symmetries of such a pseudorepresentation are reflected in its image. This is joint work with Andrea Conti and Anna Medvedovsky. (Received August 10, 2020)

1160-11-248 **M Mornev*** (maxim.mornev@math.ethz.ch), Rämistrasse 101, 8092 Zürich, Switzerland. *Wild inertia of Drinfeld modules and t-motives.* Preliminary report.

In contrast to abelian varieties the image of wild inertia on ℓ-adic Tate modules of Drinfeld modules is almost always infinite. Nonetheless we shall prove that the action of wild inertia can not be arbitrarily complicated: The image of a sufficiently deep ramification subgroup is trivial. To this end we establish a z-adic analog of Grothendieck's ℓ-adic monodromy theorem. We shall also discuss preliminary results on the z-adic counterparts of Weil-Deligne representations and the ℓ-independence conjecture. (Received August 10, 2020)

1160-11-253 **Catherine M Hsu***, chsu2@swarthmore.edu. *Small Eisenstein congruences and explicit non-Gorenstein R=T.*

In his seminal work on modular curves and the Eisenstein ideal, Mazur studied the existence of congruences between certain Eisenstein series and newforms, proving that Eisenstein ideals associated to weight 2 cusp forms of prime level are locally principal. In this talk, we'll explore generalizations of Mazur's work to squarefree level, focusing on some work in progress, joint with Preston Wake and Carl Wang-Erickson, that seeks to establish an algebraic criterion for having R=T in a certain non-Gorenstein setting. (Received August 10, 2020)

1160-11-254 **Jennifer Berg*** (jsb047@bucknell.edu) and **Masahiro Nakahara**. *Conic bundles over positive rank elliptic curves.*

In this talk, we will explore the arithmetic of conic bundles $X \to E$ over elliptic curves of positive Mordell-Weil rank over a number field k. We will consider questions regarding the distribution of the rational points of X by examining the image of $X(k)$ inside of the rational points of the base elliptic curve E. In the process, we will mention a result on a local-to-global principle for torsion points on elliptic curves over the rational numbers. This is joint work with Masahiro Nakahara. (Received August 10, 2020)

1160-11-256 **Ahmad El-Guindy**, **Riad Masri** and **Matthew Papanikolas*** (papanikolas@tamu.edu), Department of Mathematics, 3368 TAMU, Texas A&M University, College Station, TX 77843, and **Guchao Zeng**. *Traces of singular moduli over the rational function field.*

In the 1990's and 2000's, Zagier proved remarkable results on traces of singular moduli that showed they satisfy various simple identities on average and that ultimately they constitute the Fourier coefficients of a particular half-integral weight modular form. In the present talk we will consider ways to phrase these same questions over the rational function field in one variable over a finite field. In this setting singular moduli are defined as values of the Drinfeld modular j-invariant on Heegner points in the Drinfeld upper half-space. Through work of Bae, Hsia, Wang, J. Yu, and J.-K. Yu, it is possible to define class polynomials over the rational function field whose roots are singular moduli but which also satisfy explicit connections with modular polynomials. Building on these constructions we devise average results for traces of singular moduli and recover formulas for Hurwitz class numbers that align with Zagier's results. (Received August 10, 2020)

1160-11-257 **Marie Jameson*** (mjameso2@utk.edu). *Cusp forms and p-adic limits.* Preliminary report.

In 2016, S. Ahlgren and D. Samart used the theory of modular forms to give strong results expressing certain cusp forms as p-adic limits of weakly holomorphic modular forms. This gives a strengthened version of a previous result of P. Guerzhoy, Z. Kent, and K. Ono in certain cases. Here, we search for other examples of this behavior, and explore some duality results that naturally appear along the way. (Received August 10, 2020)

1160-11-260 **Armand Brumer** (brumer@fordham.edu) and **Kenneth Kramer*** (kkramer@qc.cuny.edu). *Hyperelliptic S_7-Curves of Prime Conductor.* Preliminary report.

Let A be a g-dimensional abelian variety of prime conductor N. We say that A is favorable if its 2-division field $F = \mathbb{Q}(A[2])$ is the Galois closure of a degree $2g+1$ extension of $\mathbb{Q}$, totally ramified over 2, with discriminant $\pm 2^{2g}N$. If so, $\mathrm{Gal}(F/\mathbb{Q})$ is isomorphic to the symmetric group S_{2g+1}. Examples include the Jacobians of hyperelliptic curves $C\colon\ y^2 + y = f(x)$, with $f(x)$ monic in $\mathbb{Z}[x]$ and $\deg f = 2g+1$, provided that if θ is a root of $h(x) = 1 + 4f(x)$, then the discriminant of $\mathbb{Q}(\theta)$ is $\pm 2^{2g}N$. We treated favorable abelian surfaces in [Algebra and Number Theory (2018), 1027-1071].

Let A be a favorable abelian threefold and let B be an abelian variety of dimension $3d$ and conductor N^d such that $B[2]$ is filtered by copies of $A[2]$. We prove a sufficient class field theoretic criterion on $\mathbb{Q}(A[2])$ to guarantee that B is isogenous to A^d. The criterion can be tested numerically by bringing it down to an extension of degree 21 whose Galois closure is F. Numerical experiments suggest that this criterion implies uniqueness of the isogeny class of favorable abelian threefolds for infinitely many prime conductors. (Received August 10, 2020)

1160-11-272 **Pavel Guerzhoy**, **Michael Mertens** and **Larry Rolen*** (larry.rolen@vanderbilt.edu). *Periodicities for Taylor coefficients of half-integral weight modular forms.*

Congruences of Fourier coefficients of modular forms have long been an object of central study. By comparison, the arithmetic of other expansions of modular forms, in particular Taylor expansions around points in the upper-half plane, has been much less studied. Recently, Romik made a conjecture about the periodicity of coefficients around $au = i$ of the classical Jacobi theta function. Here we prove this conjecture and generalize the phenomenon observed by Romik to a general class of modular forms of half-integral weight. (Received August 10, 2020)

1160-11-274 **A.C. Cojocaru*** (cojocaru@uic.edu), 851 S Morgan St, SEO 415, Chicago, IL 60607, and **Mihran Papikian**. *The growth of the discriminant of the endomorphism ring of the reduction of a rank 2 generic Drinfeld module.*

For q an odd prime power, $A = \mathbb{F}_q[T]$, and $F = \mathbb{F}_q(T)$, let $\psi : A \to F\{\tau\}$ be a Drinfeld A-module over F of rank 2 and without complex multiplication, and let $\mathfrak{p} = pA$ be a prime of A of good reduction for ψ, with residue field $\mathbb{F}_\mathfrak{p}$. We discuss the growth of the absolute value $|\Delta_\mathfrak{p}|$ of the discriminant of the $\mathbb{F}_\mathfrak{p}$-endomorphism ring of the reduction of ψ modulo $\mathfrak{p}$. (Received August 11, 2020)

1160-11-277 **Jing-Jing Huang** (jingjingh@unr.edu) and **Huixi Li*** (huixil@unr.edu). *On the Connection between the Goldbach Conjecture and the Elliott-Halberstam Conjecture.*

In this presentation we show that the binary Goldbach conjecture for sufficiently large even integers would follow under the assumptions of both the Elliott-Halberstam conjecture and a variant of the Elliott-Halberstam conjecture twisted by the Möbius function, provided that the sum of their level of distributions exceeds 1. This continues the work of Pan. An analogous result for the twin prime conjecture is obtained by Ram Murty and Vatwani. This is joint work with Jing-Jing Huang. (Received August 11, 2020)

1160-11-280 **Dirk Basson*** (djbasson@sun.ac.za), Stellenbosch University, South Africa. *Fields of definition for Drinfeld modular forms of higher rank and their u-expansions.*

In joint work with Florian Breuer and Richard Pink, we proved a correspondence between Drinfeld modular forms of higher rank defined algebraically as sections of an ample invertible sheaf over a compactification of a Drinfeld modular variety and Drinfeld modular forms defined analytically as functions on a period domain satisfying a modular functional equation. For this correspondence we work over a large field of positive characteristic which is both complete and algebraically closed. In proving this correspondence, we defined a power series associated to an analytic Drinfeld modular form, called its u-expansion, and proved that its coefficients are themselves (weak) Drinfeld modular forms of lower rank. This opens the possibility of studying the field of definition of a Drinfeld modular form inductively as the composite of the fields of definition of its coefficients. In this talk I will report on some progress toward a u-expansion principle that relates this inductively defined field of definition with the more natural algebraic counterpart. (Received August 11, 2020)

1160-11-285 **William L Craig*** (wlc3vf@virginia.edu). *Distribution Properties for t-hooks in Partitions (Part 2).*

Partitions, the partition function $p(n)$, and the hook lengths of their Ferrers-Young diagrams are important objects in combinatorics, number theory and representation theory. For positive integers n and t, we study $p_t^e(n)$ (resp. $p_t^o(n)$), the number of partitions of n with an even (resp. odd) number of t-hooks. Using the Rademacher circle method, we find an exact formula for $p_t^e(n)$ and $p_t^o(n)$.

In this talk, we will discuss the distribution properties of the number of partitions having a certain quantity of t-hooks with respect to other moduli. (Part 1 of this talk is given by Anna Pun) (Received August 11, 2020)

1160-11-289 **Robert W Dougherty-Bliss*** (robert.w.bliss@gmail.com), 110 Frelinghuysen Rd, Piscataway, NJ 08854, and **Doron Zeilberger** (doronzeil@gmail.com), 110 Frelinghuysen Rd, Piscataway, NJ 08854. *Automatic Conjecturing and Proving of Exact Values of Some Infinite Families of Infinite Continued Fractions.*

"The Ramanujan Machine," an inverse symbolic calculator for continued fractions, recently garnered some attention with its automated *conjectures*. Inspired, we automatically (and rigorously) *proved* some of these conjectures which generalized to infinite families of continued fractions. This demonstrates a methodology by which new and accelerated continued fractions may be discovered and proved. (Received August 11, 2020)

1160-11-307 **Ashay Burungale***, 1200 E California Blvd, Caltech Department of Mathematics, Caltech Department of Mathematics, Pasadena, CA 91125, and **Ye Tian**. *An even parity instance of the Goldfeld conjecture.*

We show that the even parity case of the Goldfeld conjecture holds for the congruent number elliptic curve. We plan to outline the set-up (joint with Ye Tian). (Received August 11, 2020)

1160-11-318 **Gebhard Boeckle**, **Peter Graef** and **Rudolph Perkins*** (rudolph.perkins@gcccd.edu). *Toward a Maeda Conjecture for Drinfeld Modular Forms.*

Drinfeld modular forms are an analog for global function fields of classical modular forms. In a joint work with G. Boeckle and P. Graef, we lay the groundwork for the formulation of a conjecture like that of Maeda's but in the setting of Drinfeld modular forms of full level. Via an isomorphism due to Teitelbaum, we explain how Hecke stable subquotients of the spaces of modular forms of a given weight and type arise via representation theory. We find in examples that even when the representations under consideration are irreducible, the subquotients of modular forms may still fail to be irreducible as Hecke modules. In particular, we discover new families of rational Hecke eigenforms which as of yet have no theoretical explanation. We also discuss our investigation into the dimension of the subquotients arising from the irreducible representations mentioned above. (Received August 11, 2020)

1160-11-331 **Bruce C Berndt*** (berndt@illinois.edu), Professor, IL 61801. *Two- and Three-variable generalizations of the third order mock theta functions $\omega(q)$ and $\nu(q)$.*

Ramanujan's third order mock theta functions $\omega(q)$ and $\nu(q)$ are defined by

$$\omega(q) := \sum_{n=0}^{\infty} \frac{q^{2n^2+2n}}{(q;q^2)_{n+1}^2} \qquad \text{and} \qquad \nu(q) := \sum_{n=0}^{\infty} \frac{q^{n^2+n}}{(-q;q^2)_{n+1}},$$

where we use the familiar notation $(a;q)_n := \prod_{m=0}^{n-1}(1-aq^m)$. Identities for two and three variable extensions of $\omega(q)$ and $\nu(q)$ are established. The results are in the spirit of papers by Y.–S. Choi; G. E. Andrews and A. J. Yee; Andrews, Dixit, and Yee; and F. Z. K. Li and J. Y. X. Yang. This is joint work with Atul Dixit and Rajat Gupta. (Received August 11, 2020)

1160-11-342 **Daniel J Kriz*** (dkriz@mit.edu), MIT Department of Mathematics, 77 Massachusetts Avenue, MIT Department of Mathematics, 77 Massachuset, Cambridge, MA 02139. *Supersingular p-adic L-functions and converse theorems.*

We give constructions of new p-adic L-functions and describe their relation to Iwasawa theory for imaginary quadratic fields K in which p is not split. We also describe some applications to converses of the results of Gross-Zagier and Kolyvagin for elliptic curves with complex multiplication by K. (Received August 11, 2020)

1160-11-347 **Catherine Hsu** (chsu2@swarthmore.edu), **Preston Wake** (wakepres@msu.edu) and **Carl Wang-Erickson*** (carl.wang-erickson@pitt.edu). *Small Eisenstein congruences and explicit non-Gorenstein R=T, Part 2.*

In his seminal work on modular curves and the Eisenstein ideal, Mazur studied the existence of congruences between certain Eisenstein series and newforms, proving that Eisenstein ideals associated to weight 2 cusp forms of prime level are locally principal. In this talk, we'll explore generalizations of Mazur's work to squarefree level, focusing on some work in progress, joint with Catherine Hsu and Preston Wake, that seeks to establish an algebraic criterion for having R=T in a certain non-Gorenstein setting. This is a continuation of the talk by Hsu. (Received August 11, 2020)

13 ► Commutative rings and algebras

1160-13-13 **Vaibhav Pandey***, 947 East, 400 South, Apt #13, Salt Lake CIty, UT 84102. *Cohomological dimension of ideals defining Veronese subrings.* Preliminary report.

Given a standard graded polynomial ring over a commutative Noetherian ring R, we prove that the cohomological dimension and the height of the ideals defining any of its Veronese subrings are equal. This result is due to Ogus when R is a field of characteristic zero, and follows from a result of Peskine and Szpiro when R is a field of positive characteristic; our result applies, for example, when R is the ring of integers. (Received May 12, 2020)

1160-13-34 **Nicholas R Baeth*** (nicholas.baeth@fandm.edu) and **Daniel Smertnig** (daniel@smertnig.at). *Lattices over Bass Rings and Graph Agglomerations.*

A commutative ring R is a *Bass ring* if it is noetherian, reduced, has module-finite integral closure, and every ideal of R is 2-generated. To study direct-sum decompositions of finitely generated torsion-free modules over Bass Rings, we study the arithmetic of the semigroup $\mathcal{T}(R)$ of isomorphism classes of modules with operation induced by the direct sum.

To a Bass ring R we associate the graph $\mathcal{G}_R = (V, E, r)$ of *prime ideal intersections* with set of vertices the minimal prime ideals of R, and an edge between two minimal prime ideals $\mathfrak{p}$ and $\mathfrak{q}$ for every maximal ideal containing both $\mathfrak{p}$ and $\mathfrak{q}$. An *agglomeration* on $\mathcal{G}_R$ is an assignment $a\colon V \cup E \to \mathbb{N}_0$ such that $a(v) \geq a(e)$ whenever a vertex v is incident with an edge e. The *monoid of agglomerations on* $\mathcal{G}_R$, denoted $\mathbf{A}(\mathcal{G}_R)$, is the additive submonoid of $\mathbb{N}_0^{V\cup E}$ consisting of all agglomerations on $\mathcal{G}_R$.

We construct a transfer homomorphism from $T(R)$ to $\mathbf{A}(\mathcal{G}_R)$, study the arithmetic of this simpler object, and pull the information back to obtain results about the nonuniquess of direct-sum decompositions over R. (Received July 29, 2020)

1160-13-36 **Benjamin Briggs** and **Eloísa Grifo*** (eloisa.grifo@ucr.edu), eloisa.grifo@ucr.edu, and **Josh Pollitz**. *Constructing non-proxy small modules.* Preliminary report.

By Auslander-Buchsbaum-Serre, a local ring (R, m, k) is regular if and only if every finitely generated R-module has finite projective dimension. Moreover, the residue field k can be used as a test module: R is regular if and only if k has finite projective dimension. This characterization can be extended to the derived category $D^f(R)$, where every object is small if and only if R is regular.

Recent results of Pollitz extend this to complete intersections: R is a complete intersection if and only if every object in $D^f(R)$ is proxy small. In this talk, we will discuss a return to the world of R-modules, and search for finitely generated R-modules that are not proxy small whenever R is not a complete intersection. We give an algorithm to construct such modules in certain settings, including when R is equigenerated. (Received July 30, 2020)

1160-13-46 **Naoki Endo*** (nendo@purdue.edu), West Lafayette, IN 47907. *Weakly Arf rings.*

In this talk, we introduce weakly Arf rings, which is a generalization of Arf rings, defined by J. Lipman in 1971. We provide characterizations of weakly Arf rings and explore the relation between these rings, the Arf rings, and the strict closedness of rings. In addition, we give various examples of weakly Arf rings that come from idealizations, fiber products, determinantal rings, invariant subrings, and F-pure rings. This talk is based on the recent researches jointly with E. Celikbas, O. Celikbas, C. Ciupercă, S. Goto, R. Isobe, and N. Matsuoka. (Received July 31, 2020)

1160-13-58 **Shinya Kumashiro*** (shinyakumashiro@gmail.com), Chiba University, Yayoi-cho 1-33, Inage-ku, Chiba 263-8522, Japan. *The Auslander-Reiten conjecture for certain non-Gorenstein Cohen-Macaulay rings.* Preliminary report.

The Auslander-Reiten conjecture is one of the notorious conjectures about the vanishing. It is stated that, for a Noetherian ring R and a finitely generated R-module M, $\mathrm{Ext}^{>0}_R(M, M \oplus R) = 0$ implies that M is projective. In this talk we assume that R is commutative. Then the following fact is fundamental: "Let R be a commutative Noetherian local ring and Q an ideal of R generated by a regular sequence. Then the Auslander-Reiten conjecture holds for R if and only if it holds for R/Q"

The fact immediately follows that the conjecture holds for complete intersections. Furthermore it is known that the conjecture holds for several classes of rings, say Gorenstein normal domain, Cohen-Macaulay normal domain $\mathbb{Q}$-algebras, Golod rings, and so on.

Motivated by these results, we explore the Auslander-Reiten conjecture for R/Q^ℓ in connection with that for R, where $\ell > 0$. It is one of the natural questions that arise from the above fact. On the other hand, R/Q^ℓ is neither a Gorenstein ring nor a domain if $\ell > 1$. Therefore, the affirmative answer of the problem provides a new class of rings which satisfy the Auslander-Reiten conjecture. In this talk we give a partial affirmative answer to the question. (Received August 02, 2020)

1160-13-59 **Federico Galetto*** (f.galetto@csuohio.edu). *Finite group actions on free resolutions.* Preliminary report.

Many ideals inside polynomial rings come naturally equipped with group actions. Under mild assumptions, such actions extend to minimal free resolutions of the ideals, providing interesting interpretations of the associated Betti numbers. While describing these actions can in general be a difficult problem, I will present an algorithm that makes the computational approach a viable solution for finite groups. (Received August 02, 2020)

1160-13-64 **Federico Castillo**, **Yairon Cid-Ruiz**, **Binglin Li**, **Jonathan Montaño*** (jmon@nmsu.edu) and **Naizhen Zhang**. *When are multidegrees positive?*

Let k be an arbitrary field, P a multiprojective space over k, and X a closed subscheme of P. The multidegrees of X are the analogues of the notion of degree in the multiprojective setting and are fundamental invariants that describe algebraic and geometric properties of X. In this talk we present recent results that provide necessary and sufficient conditions for the positivity of the multidegrees of X. As a consequence of our methods, we show that when X is irreducible, the support of multidegrees forms a discrete algebraic polymatroid. This is joint work with Federico Castillo,Yairon Cid-Ruiz, Binglin Li, and Naizhen Zhang. (Received August 03, 2020)

1160-13-66 **Selvi K. Beyarslan**, **Jennifer Biermann*** (biermann@hws.edu), **Kuei-Nuan Lin** and **Augustine O'Keefe**. *Algebraic invariants of weighted oriented graphs.*

Let $\mathcal{D}$ be a weighted oriented graph and let $I(\mathcal{D})$ be its edge ideal in a polynomial ring R. We give a formula for the Castelnuovo-Mumford regularity of $R/I(\mathcal{D})$ when $\mathcal{D}$ is a weighted oriented path or cycle such that edges of $\mathcal{D}$ are oriented in one direction. Additionally, we compute the projective dimension for this class of graphs. (Received August 03, 2020)

1160-13-67 **Giulio Caviglia** (gcavigli@math.purdue.edu), Department of Mathematics, Purdue University, West Lafayette, IN 47907, and **Yihui Liang*** (liang226@purdue.edu), Department of Mathematics, Purdue University, West Lafayette, IN 47907. *Explicit Stillman bounds for all degrees.* Preliminary report.

In 2016 Ananyan and Hochster proved Stillman's conjecture by showing the existence of a uniform upper bound for the projective dimension of all homogeneous ideals generated by n forms of degree at most d. Explicit bounds for forms of degrees 5 and higher are not yet known. We give explicit such bounds for all degrees through a recurrence relation, which behave like power towers of height d^3. This is done by proving a bound $D(k, d)$, which controls the number of generators of a minimal prime over an ideal of a regular sequence of k forms with degree d, and supplementing it into Ananyan and Hochster's proof. (Received August 03, 2020)

1160-13-92 **Michael Gekhtman*** (mgekhtma@nd.edu). *Generalized cluster structures related to the Drinfeld double of GL(n).*

I will present a unifying approach to a construction of several generalized cluster structures of geometric type. Examples include the Drinfeld double of GL(n), spaces of periodic difference operators and generalized cluster structures in GL(n) compatible with a certain subclass of Belavin-Drinfeld Poisson-Lie brackets. Based on a joint work with M. Shapiro and A. Vainshtein. (Received August 06, 2020)

1160-13-95 **Farrah C Yhee*** (fyhee@umich.edu). *Ulrich modules do not always exist.* Preliminary report.

Ulrich modules were introduced by Bernd Ulrich in 1984 and has since been a very active area of research. The existence of Ulrich modules for complete local domains have powerful applications. For example, existence implies Lech's conjecture: given a flat local map of local rings from R to S, the Hilbert-Samuel multiplicity of S is at least the Hilbert-Samuel multiplicity of R. Until recently, it was unknown if there were any counterexamples to the existence of Ulrich modules for local domains. In this talk, I will introduce the notion of Ulrich modules and discuss implications of their existence. Then I will give a counterexample to the existence of Ulrich modules for complete local domains. (Received August 06, 2020)

1160-13-136 **Zhan Jiang*** (zoeng@umich.edu), 2074 East Hall, 530 Church St, Ann Arbor, MI 48109. *Closure operations in complete local rings of mixed characteristic.*

Extended plus (epf) closure and rank 1 (r1f) closure are two closure operations introduced by Raymond C. Heitmann for rings of mixed characteristic. Recently, he and Linquan Ma proved that epf closure satisfies the usual colon-capturing property under mild conditions. In this talk, we extend their result and prove that epf closure satisfies what we call the p-colon-capturing property. Based on that, we define a new closure notion called "weak epf closure," and prove that it satisfies the generalized colon-capturing property and some other colon-capturing properties. We also show that any module-finite extension of a complete local domain is epf-phantom, which generalizes a result of Mel Hochster and Craig Huneke about "phantom extensions." Finally, we present some related results in characteristic p. (Received August 08, 2020)

1160-13-139 **Monica Lewis*** (malewi@umich.edu). *The local cohomology of a parameter ideal with respect to an arbitrary ideal.*

Let R be a regular ring, let J be an ideal generated by a regular sequence of codimension at least 2, and let I be an ideal containing J. We give an example of a module $H^3_I(J)$ with infinitely many associated primes, answering a question of Hochster and Núñez-Betancourt in the negative. In fact, for $i \leq 4$, we show that under suitable hypotheses on R/J, $\operatorname{Ass} H^i_I(J)$ is finite if and only if $\operatorname{Ass} H^{i-1}_I(R/J)$ is finite. Our proof of this statement involves a novel generalization of an isomorphism of Hellus, which may be of some independent interest. The finiteness comparison between $\operatorname{Ass} H^i_I(J)$ and $\operatorname{Ass} H^{i-1}_I(R/J)$ tends to improve as our hypotheses on R/J become more restrictive. To illustrate the extreme end of this phenomenon, at least in the prime characteristic $p > 0$ setting, we show that if R/J is regular, then $\operatorname{Ass} H^i_I(J)$ is finite for all $i \geq 0$. (Received August 08, 2020)

1160-13-144 **Eric Canton** and **Monica Lewis*** (malewi@umich.edu). *The Fedder action and a simplicial complex of local cohomologies.*

Let R be a regular local ring of prime characteristic $p > 0$ and let J be an ideal generated by a regular sequence of codimension c. We describe a complex of $R\langle F\rangle$-modules, denoted $\mathbf{\Delta}^\bullet_J(R)$, whose terms include $\mathbf{\Delta}^0_J(R) = R/J$, equipped with its natural Frobenius action, and $\mathbf{\Delta}^c_J(R) = H^c_J(R)$, equipped with a Frobenius action we refer to as the Fedder action. We show that $H^i(\mathbf{\Delta}^\bullet_J(R)) = 0$ for all $i < c$, and that $H^c(\mathbf{\Delta}^\bullet_J(R))$ is a copy of $H^c_J(R)$ equipped with the usual Frobenius action. Using the $\mathbf{\Delta}^\bullet_J(R)$ complex, we show that if I is an ideal containing

J such that $H^i_I(R) = 0$ for all $\operatorname{ht}(I) < i < \operatorname{ht}(I) + c$ (which is automatic if R/I is Cohen-Macaulay or if $c = 1$), then the module $H^{\operatorname{ht}(I/J)+c}_I(R/J)$ has Zariski closed support. (Received August 08, 2020)

1160-13-146 **Beihui Yuan*** (by238@cornell.edu), 120 Mallot Hall, Ithaca, NY 14853. *A counter-example to the Schenck-Stiller "$2r + 1$" conjecture.*

For a planar simplicial $\Delta \subseteq \mathbb{R}^2$, Schumker proved in 1979 that a lower bound on the dimension of the space $C^r_k(\Delta)$ of planar splines of smoothness r and degree k on Δ is given by a polynomial $P_\Delta(r, k)$. Alfeld and Schumaker show that $P_\Delta(r, k) = \dim C^r_k(\Delta)$ when $k \geq 4r + 1$. It was conjectured by Schenck-Stiller that this equality holds for $k \geq 2r+1$. We have found a counter-example to this conjecture. (Received August 08, 2020)

1160-13-161 **Ela Celikbas**, **Emilie Dufresne**, **Louiza Fouli*** (lfouli@nmsu.edu), **Elisa Gorla**, **Kuei-Nuan Lin**, **Claudia Polini** and **Irena Swanson**. *The defining equations of the Rees algebra of determinantal ideals.* Preliminary report.

Let $X = (x_{ij})$ be an $m \times n$ sparse matrix of indeterminates, that is a matrix whose entries are either zero or indeterminates, and let I be the ideal of maximal minors of X in a polynomial ring $R = k[x_{ij}]$ over a field k. We present some recent progress in obtaining the defining equations of the Rees algebra of such an ideal I when X is any $2 \times n$ sparse matrix. Moreover, we will discuss various properties of the Rees algebra and the special fiber cone of such an ideal. This is joint work with Ela Celikbas, Emilie Dufresne, Elisa Gorla, Kuei-Nuan Lin, Claudia Polini, and Irena Swanson. (Received August 09, 2020)

1160-13-168 **Patricia Klein*** (klein847@umn.edu) and **Jenna Rajchgot**. *Geometric vertex decomposition and liaison.*

Geometric vertex decomposition and liaison are two frameworks that have been used to produce similar results about similar families of algebraic varieties. In this talk, we will describe an explicit connection between these approaches. In particular, we describe how each geometrically vertex decomposable ideal is linked by a sequence of elementary G-biliaisons of height 1 to an ideal of indeterminates and, conversely, how every G-biliaison of a certain type gives rise to a geometric vertex decomposition.

As a consequence, we can immediately conclude that several well-known families of ideals are glicci, including Schubert determinantal ideals, defining ideals of varieties of complexes, and defining ideals of graded lower bound cluster algebras. This connection also gives us a framework for implementing with relative ease Gorla, Migliore, and Nagel's strategy of using liaison to establish Gröbner bases. We describe briefly, as an application of this work, a proof of a recent conjecture of Hamaker, Pechenik, and Weigandt on diagonal Gröbner bases of Schubert determinantal ideals. (Received August 09, 2020)

1160-13-179 **Paolo Mantero** and **Matthew Mastroeni*** (mmastro@okstate.edu). *The structure of Koszul algebras defined by four quadrics.*

A question of Avramov, Conca, and Iyengar asks whether the Betti numbers of $R = S/I$ over S can be bounded above by binomial coefficients on the minimal number of generators of I if R is Koszul. Building on previous affirmative answers for Koszul algebras defined by three quadrics and Koszul almost complete intersections with any number of generators, we give an affirmative answer to the above question in the case of four quadrics. In the process, we prove structure theorems allowing for a complete description of the possible ideals of four quadrics defining Koszul algebras over an algebraically closed field. We will attempt to highlight the main ingredients that go into this classification. (Received August 10, 2020)

1160-13-180 **David Eisenbud** and **Irena Peeva*** (irena@math.cornell.edu). *MCM modules over a complete intersection.*

We provide a structural description of all MCM modules over a local complete intersection ring. We give an application in the codimension 2 case. This is a joint work with David Eisenbud. (Received August 10, 2020)

1160-13-183 **Alessandra Costantini*** (alessanc@ucr.edu). *Residual intersections and modules with Cohen-Macaulay Rees algebra.* Preliminary report.

Let R be a Noetherian ring, and E a finitely generated R-module with a rank. The Rees algebra $\mathcal{R}(E)$ of E is defined as the symmetric algebra $\mathcal{S}(E)$ modulo its R-torsion submodule. In 2003, Simis, Ulrich and Vasconcelos proved that the study of the Cohen-Macaulay property of Rees algebras of modules can be reduced to the case of Rees algebras of ideals: the Rees algebra $\mathcal{R}(E)$ of a module E is Cohen-Macaulay if and only if the Rees algebra $\mathcal{R}(I)$ of a *generic Bourbaki ideal* I of E is Cohen-Macaulay. However, the problem of determining under what conditions the Rees algebra of a module is Cohen-Macaulay is still object of current research.

In this talk, we will provide sufficient conditions for a module E to have Cohen-Macaulay Rees algebra. Our results are obtained by exploiting nice properties of the residual intersections of a generic Bourbaki ideal I of E.

We recover a well-known results of Johnson and Ulrich and of Goto, Nakamura and Nishida for Rees algebras of ideals, as well as previous work of Lin on a more restrictive class of modules. The talk is based on a preprint available on the arXiv (arXiv:1811.08402v1). (Received August 10, 2020)

1160-13-187 **Graham Keiper*** (keipergt@mcmaster.ca), 203 Dawlish Avenue, Toronto ON M4N 1H9, Toronto, ON M4N 1H9, Canada, and **Giuseppe Favacchio** and **Adam Van Tuyl**.
Regularity and h-polynomials of toric ideals of graphs.

I will discuss recent work of Favacchio, Van Tuyl and myself which demonstrates that we may construct toric ideals associated with graphs which have regularity r and h-polynomials of degree d for any $4 \leq r \leq d$. The ideas involved rely on Gröbner bases and combinatorics. I will also discuss how we can use this to recover a result of Hibi, Higashitani, Kimura, and O'Keefe. (Received August 10, 2020)

1160-13-189 **Lindsey Hill** and **Rachel Lynn***, rvlynn@purdue.edu. *Specialization of Coefficient Ideals.* Preliminary report.

Given an m-primary ideal I in a local ring (R, m), Kishor Shah introduced a sequence of ideals called coefficient ideals which generalize the integral closure and tell us about the growth of powers of I. The integral closure and Ratliff-Rush closure are the largest and smallest coefficient ideals, respectively. In this talk, I will discuss what is known about which coefficient ideals specialize and which do not, including a modification of a result of Hong and Ulrich which shows that an ideal of height at least two remains integrally closed after specialization by a general element of the ideal. (Received August 10, 2020)

1160-13-195 **Hugh R Geller*** (hgeller@clemson.edu). *DG-Structures for Fiber Products.* Preliminary report.

A construction of Tate shows that every algebra over a ring R possesses a DG-algebra resolution over R. These resolutions are not always minimal and Avramov even shows that certain algebras cannot have a minimal resolution with a DG-algebra structure. This talk gives an explicit construction of a DG-structure for certain fiber products and criteria for determining when the structure is a DG-module, DG-algebra, or minimal DG-algebra. (Received August 10, 2020)

1160-13-200 **Brian Harbourne*** (bharbourne1@unl.edu), Math Department, University of Nebraska-Lincoln, Lincoln, NE 68588, and **Jake Kettinger** and **Frank Zimmitti**.
Various versions of the resurgence and how they are related.

Recent research has highlighted the importance of integral closure to understanding asymptotic resurgences and has led to important advances and new questions. We discuss these advances and some of these questions. This talk is based on joint work with Jake Kettinger and Frank Zimmitti. (Received August 10, 2020)

1160-13-203 **Ayah Almousa**, **Kuei-Nuan Lin** and **Whitney Liske*** (whitney.liske@stvincent.edu). *Rees Algebras of Closed Determinantal Facet Ideals.* Preliminary report.

Let $X = (x_{ij})$ be an $m \times n$ matrix of indeterminates where $m \leq n$, and let $R = \mathbb{K}[X]$ be the polynomial ring over a field $\mathbb{K}$ in the indeterminates x_{ij}. Let Δ be a simplicial complex on the vertex set $V = [n]$. A determinantal facet ideal $J_\Delta \subseteq R$ is the subideal of $I_m(X)$ generated by determinants of the form $[\mathbf{a}]$ where $\mathbf{a}$ supports an $m-1$ face of Δ; that is, the columns of $[\mathbf{a}]$ correspond to the vertices of some facet $F \in \Delta$.

In this talk we will study the initial ideal of closed determinantal facet ideals and use SAGBI basis techniques to find a Gröbner basis for the defining ideal of the Rees algebra. In particular, we show that these ideals are of fiber type. (Received August 10, 2020)

1160-13-214 **Justin Lyle*** (justin.lyle@ku.edu), **Jonathan Montaño** and **Sean K. Sather-Wagstaff**. Tor*-persistence and related conjectures.*

A commutative Noetherian ring R is said to be Tor-persistent if, for any finitely generated R-module M, the vanishing of $\operatorname{Tor}_i^R(M, M)$ for all $i \gg 0$ implies that M has finite projective dimension. An open question of Avramov, Iyengar, Nasseh, and Sather-Wagstaff asks whether every commutative Noetherian ring is Tor-persistent. We show that any local ring $(R, \mathfrak{m})$ with $\mathfrak{m}^3 = 0$ is Tor-persistent, and we provide some partial results in the graded case. Our main approach is to exploit properties of exterior and symmetric squares of both modules and complexes. We discuss connections between these problems and a well-known open conjecture of Tachikawa. (Received August 10, 2020)

1160-13-215 **Paolo Mantero*** (pmantero@uark.edu). *Interpolation problems.* Preliminary report.
Homogeneous polynomial interpolation problems are (often) challenging problems in Algebraic Geometry. They ask for information regarding hypersurfaces of given degrees passing with given multiplicity through a given set X of points in $\mathbb{P}^n_{\mathbb{C}}$.

A theorem by Zariski and Nagata allows one to translate the problem into a question about the Hilbert function of symbolic powers of the defining ideal I_X, so the problem can be tackled with Commutative Algebra tools as well.

In the present talk we will review the history of some interpolation problems, state classical and recent results and discuss open problems. (Received August 10, 2020)

1160-13-223 **Andrew R. Kustin**, **Rebecca R.G.*** (rrebhuhn@gmu.edu) and **Adela Vraciu**. *The resolution of (x^N, y^N, z^N, w^N) over the hypersurface $k[x, y, z, w]/(x^n + y^n + z^n + w^n)$.*
We find the minimal graded resolution of $R/(x^N, y^N, z^N, w^N)$ over the hypersurface ring $R = k[x, y, z, w]/(x^n + y^n + z^n + w^n)$ as well as the related resolution of $P/(x^N, y^N, z^N, w^N) : (x^n + y^n + z^n + w^n)$ over $P = k[x, y, z, w]$, where k is a field of characteristic zero. To compute these resolutions, we make some useful reductions to the case of fewer variables and apply results on compressed Gorenstein rings. This is joint work with Andrew R. Kustin and Adela Vraciu. (Received August 10, 2020)

1160-13-230 **Nicholas R Packauskas*** (nicholas.packauskas@cortland.edu), **Luchezar L. Avramov** and **Mark E. Walker**. *Quasi-Polynomial Growth of Betti Sequences over Complete Intersection Rings.*
Let Q be a regular local ring and I an ideal generated by a regular sequence of c elements in the square of the maximal ideal. It is known that over the complete intersection $R = Q/I$ that any finitely generated module M has Betti numbers eventually given by quasi-polynomial of degree less than c. That is, there are integer-valued polynomial functions p^M_+ and p^M_- with the same leading term such that $\beta^R_{2i}(M) = p^M_+(2i)$ and $\beta^R_{2i+1}(M) = p^M_-(2i+1)$ for i sufficiently large. We will show that if q is the height of the ideal generated by the quadratic initial forms of I in the associated graded ring of Q, then the degree of $p^M_+ - p^M_-$ is less than $c - q - 1$. (Received August 10, 2020)

1160-13-236 **Michael DiPasquale*** (michael.dipasquale@colostate.edu), Mathematics Department, Colorado State University, 1874 Campus Delivery, Fort Collins, CO 80523, and **Nelly Villamizar**. *Regularity of uniform power ideals and the Waldschmidt constant.*
A power ideal is an ideal generated by powers of linear forms; a uniform power ideal is one which is generated by the same power of every linear form. Power ideals appear in a variety of contexts; our motivation for studying these ideals is drawn from applications to multivariate splines.

In this talk we present a bound on the regularity of a uniform power ideal which is linear in the common power of the linear forms; the slope of this linear bound is given in terms of the Waldschmidt constant of the dual set of points. The Waldschmidt constant is an asymptotic measure of the minimum degree a polynomial must have in order to vanish to high order at the set of points. Our bound is derived using apolarity (or Maucalay/Matlis duality) and is asymptotically sharp: as the common power on the linear forms increases, the bound converges to the regularity of the power ideal. While the Waldschmidt constant of a set of points is extremely difficult to compute, there are several lower bounds on the Waldschmidt constant which show that the regularity of a uniform power ideal is closely linked to the smallest degree of a polynomial vanishing at the dual set of points. This is part of a joint work with Nelly Villamizar. (Received August 10, 2020)

1160-13-244 **Janet Striuli*** (jstriuli@fairfield.edu), 18 Glen Ridge Rd, Fairfield, CT 06825-2502, and **Claudia Miller** and **Hamidreza Rahmati**. *The Koszul homology of Gorenstein Ideals.*
In this preliminary report we will present some results connected to the annihilator of the first Koszul homology for an ideal that is Gorenstein. (Received August 10, 2020)

1160-13-252 **Rachel N. Diethorn*** (rachel.diethorn@yale.edu). *A self-dual complete resolution.*
A classical result of Buchsbaum and Eisenbud states that the minimal free resolution of the quotient by a Gorenstein ideal is self-dual. However, when one moves from finite free resolutions to infinite complete resolutions, the idea of self-duality is not so well understood. In this talk, I will give a class of modules that have self-dual complete resolutions, and I will speak briefly about the construction. I will also highlight some (co)homological consequences of the existence of such resolutions for these modules. (Received August 10, 2020)

1160-13-269 **Oana Veliche*** (o.veliche@northeastern.edu), 360 Huntington Ave, Boston, MA 02115-5005, and **Lars W. Christensen** and **Jerzy Weyman**. *On Almost Complete Intersection of Grade 3 Perfect Ideals.* Preliminary report.

The combinatorial approach of Kunth (1996) to overlapping Pfaffians of a skew-symmetric matrix allowed us to reprove and clarify a formula of Brill (1904) on the minors of a skew-symmetric matrix. As a consequence, we give a new proof to the structure of grade 3 perfect almost complete intersection ideals. The original, independent, proofs were given by Avramov (1981) and Brown (1984) using linkage and the structure of grade 3 Gorenstein ideals given by Buchsbaum and Eisenbud (1977). (Received August 10, 2020)

1160-13-290 **John Myers*** (john.myers@oswego.edu). *Three algebras and three definitions of Koszulness.* Preliminary report.

Let R be a standard graded commutative algebra over a field k, let K be the Koszul complex on a minimal set of generators of the irrelevant ideal of R, and let H be the homology of K. Recall that R is said to be *Koszul* if k has a linear free resolution over R. We adapt this definition to apply to K (viewed as a DG algebra) and then to H (viewed as a bigraded algebra). We describe how these three Koszul properties transfer back and forth between the three algebras R, K, and H, and we give several examples of classes of algebras R for which H is Koszul. (Received August 11, 2020)

1160-13-293 **Gabriel E Sosa Castillo***, gsosacastillo@colgate.edu, and **Kuei-Nuan Lin** and **Selvi Kara**. *Koszulness of multi-Rees algebras of the direct sum strongly stable (0-Borel) Ideals.* Preliminary report.

In this talk we will survey results concerning the koszulness of Rees and multi-Rees algebras of strongly stable (0-Borel) ideals.

Particular attention will be given to (1) how the number, and degree, of the minimal Borel generators of the ideals involved affects the Koszulness, and (2) how combinatorial tools such as fiber graphs and Noetherian reduction relations allow us to detect Koszulness of these algebras. (Received August 11, 2020)

1160-13-310 **Haydee Lindo*** (hlindo@hmc.edu). *Trace modules and Rigidity.*

I will speak on some recent developments in the theory of trace modules over commutative Noetherian rings. This will include applications of trace modules in understanding endomorphism rings and a discussion of ongoing work examining the relationship between trace ideals and modules having no self-extensions. (Received August 11, 2020)

1160-13-315 **Rachelle R Bouchat*** (rbouchat@iup.edu), Indiana University of Pennsylvania, 210 S. Tenth Street, Stright Hall, Room 233, Indiana, PA 15701. *Splittings of Domino Ideals.*

Domino ideals are a class of monomial ideals arising from tilings of rectangular tableau by 2x1 dominoes. Results concerning the Betti numbers of these domino ideals will be determined using a natural splitting of the ideals. In particular, the Betti numbers of these domino ideals will be shown to be independent of the underlying field. Additionally, we will use this natural splitting to determine the projective dimension and the Castelnuovo-Mumford regularity for these domino ideals. (Received August 11, 2020)

1160-13-321 **Alexandre Tchernev*** (atchernev@albany.edu), atchernev@albany.edu. *Dynamical systems on chain complexes and canonical minimal resolutions.*

We will discuss how to introduce a notion of vector field and its (discrete time) flow on any chain complex. The resulting dynamical systems theory allows to replace the chain complex with a "smaller" one of the same homotopy type. In many cases of interest this can be accomplished in a functorial manner. As applications we will explain how this theory can be used to construct explicit canonical and symmetry-preserving minimal free resolutions of monomial ideals and of toric rings, thus providing a solution to two long-standing open problems in commutative algebra. (Received August 11, 2020)

1160-13-323 **Zach Greif*** (zsgreif@iastate.edu) and **Jason McCullough**. *Linear Syzygies of Toric Edge Ideals of Bipartite Graphs.*

Given a field k and a finite, simple graph G, one can construct the toric edge ideal I_G of G as the kernel of a surjective ring homomorphism $k[E]$ to $k[G]$, where $k[E]$ is the polynomial generated by variables associated to the edges of G and $k[G]$ is the subring of the polynomial ring with variables associated to the vertices of G and generators associated to the edges. In the case that $G = K_{m,n}$ is a complete bipartite graph, this corresponds to the Segre embedding of two projective spaces into a larger projective space. There is a combinatorial description of those bipartite graphs G such that I_G is generated by quadrics due to Ohsugi and Hibi. We give a combinatorial

classification for bipartite graphs G, of when I_G has a resolution which is linear for a fixed number of steps. (Received August 11, 2020)

1160-13-330 **Yu Xie*** (yxie@widener.edu), 21 Central Avenue, Berwyn, PA 19312. *Generalized Mixed Multiplicities.*

We define the concepts of mixed j-multiplicities of two arbitrary ideals in a Noetherian local ring using the 0-th local cohomology functor. Our definition generalizes classical mixed multiplicities and coincides with that of Achilles, Manaresi, and Pruschke (who used a different approach to define mixed multiplicities of two arbitrary ideals). We show that these invariants are preserved under general linear intersections and provide formulas to compute them. Under certain conditions, we prove that mixed j-multiplicities are the j-multiplicities of joint reductions of ideals. We also prove that the j-multiplicities of the Rees algebra and the extended Rees algebra can be expressed as sums of mixed j-multiplicities. One can also interpret the degrees of Stückrad-Vogel intersection cycles and Segre cycles using mixed j-multiplicities. (Received August 11, 2020)

1160-13-338 **Claudia Miller**, **Hamid Rahmati*** (hrahmati2@unl.edu), **Janet Striuli** and **Zheng Yang**. *Weak Complete intersection ideals.*

We introduce the notion of weak complete intersection ideals; these are the ideals with the property that every differential in their minimal free resolution can be represented by a matrix whose entries are in the ideal itself. We provide a family of these ideals and present some applications. We also discuss algebra structures on free reclusions of such ideals. This talk is based on joint work with Janet Striuli and Zheng Yang and with Claudia Miller and Zheng Yang. (Received August 11, 2020)

1160-13-343 **Jenny Kenkel*** (jennifer.kenkel@gmail.com). *A Construction of Local Cohomology Modules With an FI Structure.* Preliminary report.

Abstract: Let R_n be a polynomial ring in n variables, and I_n an ideal of R_n. There are many ways to embed R_n into R_{n+1}, and, depending on the situation, and many ways to embed I_n compatibly into I_{n+1}. Enforcing certain structure on these embeddings allows us to consider this whole class of rings as one object, known as an FI-algebra. In this talk, I will explain this structure, with lots of examples, and explain how to construct local cohomology modules with an FI-Structure. (Received August 11, 2020)

1160-13-359 **Tsutomu Nakamura**, **Sean Sather-Wagstaff*** (ssather@clemson.edu), **Peder Thompson** and **Jonathan Totushek**. *When Does Ext-Vanishing Imply Vanishing?* Preliminary report.

For a chain complex X over a commutative noetherian ring R of finite Krull dimension, we show that if $\operatorname{Ext}^i_R(R_P, X) = 0$ for all prime ideals P and all $i \gg 0$, then $X_i = 0$ for all $i \ll 0$. Applications for CI-dimensions will be given. (Received August 12, 2020)

14 ► Algebraic geometry

1160-14-10 **Ashley K. Wheeler*** (awheeler@mtholyoke.edu), Mount Holyoke College, **Jessica Sidman** (jsidman@mtholyoke.edu), Mount Holyoke College, and **Will Traves** (traves@usna.edu), US Naval Academy. *Geometric equations for matroid varieties*

Each point x in $\operatorname{Grass}(r, n)$ corresponds to an $r \times n$ matrix A_x which gives rise to a matroid $\mathcal{M}_x$ on its columns. Gel'fand, Goresky, MacPherson, and Serganova showed that the sets $\{y \in \operatorname{Grass}(r, n) \text{ — } \mathcal{M}_y = \mathcal{M}_x\}$ form a stratification of $\operatorname{Grass}(r, n)$ with many beautiful properties. However, results of Mnëv and Sturmfels show that these strata can be quite complicated, and in particular may have arbitrary singularities. We study the ideals I_x of matroid varieties, the Zariski closures of these strata. We construct several classes of examples based on theorems from projective geometry and describe how the Grassmann-Cayley algebra may be used to derive non-trivial elements of I_x geometrically when the combinatorics of the matroid is sufficiently rich. (Received February 21, 2020)

1160-14-19 **Nathan Chen*** (nathan.chen@stonybrook.edu), 100 Nicolls Road, Stony Brook, NY 11794. *Measures of irrationality for projective varieties.*

Given a smooth projective variety, there are two natural questions that can be asked: (1) How can we determine when it is rational? and (2) If it is not rational, can we measure how far it is from being rational? There are several invariants that have been developed and studied to try to answer the second question. For instance, the *degree of irrationality* of an algebraic variety is the minimal degree of a dominant rational map to a projective

space of the same dimension. We will survey some of the techniques that have been established to give lower bounds on these "measures of irrationality". This is joint work with David Stapleton. (Received July 11, 2020)

1160-14-40 **Dmitrii Pedchenko*** (dzp5326@psu.edu). *The Picard group of the moduli space of bundles on a quadric surface.*

In this talk, I will discuss the computation of the Picard group of the moduli space of bundles on a smooth quadric surface. I will focus especially on moduli spaces of bundles of small discriminant, where we observe new and interesting behavior. I will outline methods for constructing certain resolutions for semistable bundles on the quadric and applying techniques of geometric invariant theory to the resulting families of bundles to study the moduli space. (Received July 30, 2020)

1160-14-48 **Ruijie Yang***, ruijie.yang@stonybrook.edu. *The space of nodal elliptic curves on polarized K3 surfaces.*

In algebraic geometry, it is a common phenomenon that moduli space of geometric objects tend to become more and more complicated as invariants parametrizing them go to infinity. In this talk, I would like to discuss this phenomenon for the space of singular curves on K3 surfaces.

Let (X, L) be a very general polarized K3 surface of genus g so that $L^2 = 2g - 2$. A curve C in the linear system $|L|$ is a rational curve if and only if it has g nodes. The Yau-Zarslow formula implies that the number of rational curves goes to infinity as g goes to infinity.

We would like to explore the next case: curves in the linear system $|L|$ with $(g-1)$-nodes. They are singular elliptic curves and let M be the space of such curves. It can be shown that M is one dimensional, hence we can talk its geometric genus. With Nathan Chen and Francois Greer, we show that the geometric genus of M goes to infinity as g goes to infinity. (Received July 31, 2020)

1160-14-65 **Daniel Duarte** and **Luis Núñez-Betancourt*** (luisnub@cimat.mx), Guanajuato, Mexico. *Nash blowups in prime characteristic.*

The Nash blowup is a natural modification of algebraic varieties that replace singular points by limits of certain vector spaces associated to the variety at non-singular points. For several decades, It has been studied whether it is possible to resolve singularities of algebraic varieties by iterating Nash blowups. This problem has mostly been treated in characteristic zero due to an example given by Nobile. In this talk, we will discuss a new approach in prime characteristic using differential operators. (Received August 03, 2020)

1160-14-82 **Sayanta Mandal*** (sayanta.cmi@gmail.com), Chicago, IL 60607. *Betti numbers of the moduli space of stable sheaves on $\mathbb{P}^2$.*

We discuss stabilization of the Betti numbers and provide explicit lower bound on the second Chern class for stabilization of the Betti numbers of the moduli space of stable sheaves on $\mathbb{P}^2$. (Received August 05, 2020)

1160-14-97 **Takumi Murayama*** (takumim@math.princeton.edu), Princeton University, Department of Mathematics, Fine Hall, Washington Road, Princeton, NJ 08544-1000. *A uniform treatment of Grothendieck's localization problem.*

Let $f\colon Y \to X$ be a flat morphism of algebraic varieties. A theorem of Michel André implies that f is smooth if and only if the fibers of f over closed points are smooth. We prove that the same statement holds for other local properties of morphisms satisfying a list of axioms. This gives a uniform treatment of what is known as Grothendieck's localization problem, which has previously been shown for specific local properties of morphisms using a variety of methods. (Received August 06, 2020)

1160-14-106 **Mihai Fulger*** (mihai.fulger@uconn.edu), 341 Mansfield Rd, Unit 1009, Storrs, CT 06269, and **Takumi Murayama**. *Cones of divisors on self products of curves.*

Self-products of curves are very natural examples of surfaces and higher dimensional varieties. At least for curves with very general moduli, the space of divisors is easy to understand: it is generated by fibers and diagonals. However already in the surface case computing the nef cone of divisors is an open problem. We review what is known about the question, in particular its connection to the Nagata conjecture. We relate the conjectural boundary of the nef cone to the semistability of higher conormal sheaves, a generalization of Lazarsfeld-Mukai bundles. Using this approach, we construct new examples of nef classes in some cases. This is in joint work with Takumi Murayama. (Received August 07, 2020)

1160-14-140 **Honghao Gao**, **Linhui Shen** and **Daping Weng*** (wengdap1@msu.edu). *Augmentations, Fillings, and Clusters of Positive Braid Closures.*

By using an enhanced version of the CE dga, we construct a cluster K_2 structure on the augmentation variety of the rainbow closure Λ_β for any positive braid β, and prove that the algebraic tori arisen from exact Lagrangian fillings constructed by pinching crossings are cluster charts in this cluster structure. We also relate a cyclic rotation to the cluster Donaldson-Thomas transformation on these augmentation varieties. As an application of this new cluster structure, we give a sufficient condition on which Λ_β admits infinitely many non-Hamiltonian-isotopic exact Lagrangian fillings based on the order of DT, solving a conjecture on the existence of Legendrian links that admit infinitely many fillings. This is joint work in progress with Honghao Gao and Linhui Shen. (Received August 08, 2020)

1160-14-182 **Tim Ryan*** (rtimothy@umich.edu). *Irreducibility of Nested Hilbert Schemes of Points.*

Hilbert schemes of points on surfaces are some of the most well studied moduli spaces in algebraic geometry. They have been viewed from many perspectives and have proved useful in fields such as combinatorics, symplectic geometry, and representation theory. They are known to be smooth irreducible varieties. Nested Hilbert schemes of points are a less well studied generalization of Hilbert schemes which parametrize tuples of collections of points contained inside of each other. Cheah classified when these nested Hilbert schemes of points are smooth. However, irreducibility is known for very few cases, and there are no known examples of reducible nested Hilbert schemes. In this talk, I will discuss joint work with Gregory Taylor proving irreducibility in some cases. Time permitting, I will discuss related results. (Received August 10, 2020)

1160-14-188 **Giulia Saccà***, 2990 Broadway, New York, NY 10027, and **Laure Flapan**, **Emanuele Macrì** and **Kieran O'Grady**. *Fano manifolds associated to hyperkahler manifolds.* Preliminary report.

It is known that to many Fano manifolds whose cohomology looks like that of a K3 surface, one can associate, via geometric constructions, examples of hyperkahler manifolds. In this talk I will report on the first steps of a program whose aim is to reverse this construction: starting from a hyperkahler manifold how to recover geometrically a Fano manifold? This is joint work in progress with L. Flapan, E. Macrì, and K. O'Grady. (Received August 10, 2020)

1160-14-210 **Raymond Cheng*** (rcheng@math.columbia.edu). *q-bic Hypersurfaces.* Preliminary report.

In positive characteristic p, some hypersurfaces in projective space behave as if they are of lower degree than they really are. The prototypical example is the Fermat hypersurface of degree q 1, where q is a power of p: these are unirational, they have interesting Gauss maps, moduli of rational curves contained in them behave exceptionally, and so forth. In a sense, the global geometry of these hypersurfaces mimics that of cubic hypersurfaces over the complex numbers. This simplification of global geometry is balanced by the appearance of many interesting singularities in their fine geometry. In this talk, I will sketch this analogy between a class of hypersurfaces I call q-bic hypersurfaces and cubic hypersurfaces over the complex numbers, and suggest a perspective to push this analogy further. (Received August 10, 2020)

1160-14-216 **Gabriele Di Cerbo*** (gdi@math.princeton.edu). *Birational boundedness of elliptic Calabi-Yau manifolds.*

The minimal model program predicts that, up to a special class of birational equivalences, each projective variety decomposes into iterated fibrations with general fibers of 3 basic types: Fano varieties, Calabi-Yau varieties, and varieties of general type. Our understanding of the boundedness of Fano varieties and varieties of general type is quite solid but Calabi-Yau varieties are still elusive. In this talk, I will discuss recent results on the birational boundedness of elliptic Calabi-Yau varieties with a section. As a consequence, we obtain that there are finitely many possibilities for the Hodge diamond of such varieties. This is joint work with Caucher Birkar and Roberto Svaldi. (Received August 10, 2020)

1160-14-219 **Jingjun Han** and **Yuchen Liu*** (yuchen.liu@yale.edu), 10 Hillhouse Ave, New Haven, CT 06511, and **Lu Qi**. *On local volumes and boundedness of singularities.*

The local volume of a klt singularity, introduced by Chi Li, is an invariant that encodes interesting algebraic, geometric, and topological information. Conjecturally, if the coefficient set is finite (resp. DCC), the local volume in a fixed dimension is discrete away from zero (resp. ACC). We verify this conjecture in the following three settings: dimension 2, dimension 3 terminal with no boundary divisors, or analytically bounded germs. We also

investigate the relation between local volumes and the existence of δ-plt blow-ups. This talk is based on joint work in progress with Jingjun Han and Lu Qi. (Received August 10, 2020)

1160-14-231 **James B Collins*** (jbcolli2@gmail.com) and **Jonathan Hauenstein**. *Singular Value Homotopy for Finding Critical Parameters in Differential Equations.*

Many nonlinear differential equations are parameterized in some form. Often the value of this parameter can affect the number of solutions to the equation. In this talk I will describe a numerical algebraic geometry technique for finding the critical parameter value where the number of solutions of a differential equation changes. This technique involves forming a homotopy that drives the smallest singular value of the Jacobian of a system to zero. (Received August 10, 2020)

1160-14-235 **Lena Ji*** (lji@math.princeton.edu). *Geometrically non-reduced varieties.*

In positive characteristic, there exist fibrations between smooth varieties where every fiber is singular or even non-reduced. In the latter case, the generic fiber of the fibration is geometrically non-reduced. We study the failure of generic smoothness and obtain a structural result about geometrically non-reduced varieties, with applications to Fano varieties. This is joint work with Joe Waldron. (Received August 10, 2020)

1160-14-238 **Devlin Mallory*** (malloryd@umich.edu). *D-simplicity in characteristic 0 and characteristic p.*

Differential operators on singular rings have found many applications, but are generally quite hard to describe explicitly. A natural question is "how transitive" the action of the differential operators on the ring is, and specifically if there's a nontrivial submodule. If not, the ring is called D-simple. In many cases, D-simplicity corresponds to "milder" singularities: for example, smooth rings are D-simple, as are their invariant subrings under finite group actions. In this talk, we'll review results connecting D-simplicity and mildness of singularities in characteristic p, which lead to a natural expectation for a connection in characteristic 0. However, we'll show that this natural expectation is unfortunately not true. To prove this, we'll discuss the connection of D-simplicity of graded rings with mild singularities to the global geometry of Fano varieties, and point to some outstanding problems on positivity of their tangent bundles. (Received August 10, 2020)

1160-14-239 **Yajnaseni Dutta*** (ydutta@uni-bonn.de), Hausdorff Center for Mathematics, Endenicher Allee 62, Villa Maria, 53115 Bonn, Germany, and **Feng Hao** and **Yongqiang Liu**. *Holomorphic 1-forms and geometry.* Preliminary report.

In this talk I will discuss some old and new results dealing with the consequences of the existence of nowhere vanishing global holomorphic 1-forms on smooth projective varieties over the complex numbers. For instance, existence of such forms has deep implications on the geometry of algebraic maps to abelian varieties and hence in particular on the albanese morphisms. In a different direction, inspired by the work of Carell and Lieberman who show that the set of global holomorphic vector fields admitting zeros form a subspace of the space of global sections of the tangent bundle, I will present a result of similar flavour for global holomorphic 1-forms. This is a joint work with Feng Hao and Yongqiang Liu. (Received August 10, 2020)

1160-14-246 **Laure Flapan*** (flapanla@msu.edu), **Emanuele Macrì**, **Kieran O'Grady** and **Giulia Saccà**. *Antisymplectic involutions on hyperkähler manifolds.* Preliminary report.

We study the fixed locus of antisymplectic involutions on projective hyperkähler manifolds of K3 type that are induced by an ample class of square 2. We describe the number of components and show that when the divisibility of the hyperkähler is 2, one of the components is Fano, generalizing results of Lehn-Lehn-Sorger-van Straten about certain hyperkähler 8-folds. (Received August 10, 2020)

1160-14-255 **Charles Godfrey*** (cgodfrey@uw.edu) and **Takumi Murayama** (takumim@math.princeton.edu). *Du Bois singularities in families.* Preliminary report.

Du Bois singularities encompass several classes of singularities coming from the minimal model program, and have applications to moduli of varieties. We show that Du Bois singularities are preserved under descent via faithfully flat morphisms, and ascend in flat families. As an application, we show that the Du Bois locus is open in flat proper families over arbitrary base schemes. This extends a result of Kovács and Schwede, who proved openness of the Du Bois locus in the case where the base scheme is smooth. We will also discuss possible analogues of Du Bois singularities in positive characteristic. (Received August 11, 2020)

1160-14-284 **Kenneth Ascher*** (kascher@princeton.edu), **Kristin DeVleming** and **Amos Turchet**. *Hyperbolicity of varieties of log general type.*

This talk will survey some results regarding hyperbolicity of varieties of log general type. There are several classical results showing that positivity of the cotangent bundle implies various notions of hyperbolicity for projective varieties coming from algebraic, arithmetic, and differential geometry. The goal of this talk is to review these results and discuss generalizations of these results for log pairs / quasi-projective varieties. This is joint work with K. DeVleming and A. Turchet. (Received August 11, 2020)

1160-14-314 **Brooke Ullery*** (bullery@emory.edu). *Measures of irrationality for algebraic varieties.*

A smooth algebraic curve is said to be rational if it is isomorphic to $\mathbb{P}^1$, the projective line. More generally, the gonality of a smooth projective curve is the smallest degree of a map from the curve to the projective line. The intuition is that the higher the gonality, the further the curve is from being rational. A classical theorem of Noether says that if C is a smooth plane curve of degree d, then the gonality of C is $d-1$, and it is obtained by projecting away from a point on the curve. A natural question is: does Noether's theorem generalize in some way to curves in larger projective spaces? What about to higher dimensional varieties? We will explore these questions, focusing on the examples of hypersurfaces and, more generally, complete intersections in projective space. (Received August 11, 2020)

1160-14-317 **Jakub Witaszek*** (jakubw@umich.edu). *Relative four dimensional Minimal Model Program in positive characteristic.*

This talk will cover recent developments on the four dimensional Minimal Program Program in positive characteristic (joint with Christopher Hacon). (Received August 11, 2020)

1160-14-325 **John Sheridan*** (john.sheridan4579@gmail.com), Princeton, NJ 08544. *Divisor varieties of symmetric products of curves.*

For a smooth, projective variety X, one can parametrize effective divisors of a given numerical class in X by what we call a "divisor variety" which will be a projective scheme. Classically, when X is a curve, the divisor variety is smooth and irreducible, but these properties fail in higher dimension. We study the case when X is the symmetric product of a curve and give a rather complete description of the corresponding divisor varieties, indicating how the new phenomena in this setting relate to the concrete geometry of the underlying curve. (Received August 11, 2020)

15 ► Linear and multilinear algebra; matrix theory

1160-15-276 **Christina Pospisil*** (pospisil.christina@gmx.de), pospisil.christina@gmx.de. *Generalization Theory of Linear Algebra I+ II.*

An algorithm for multiplying and adding matrices regardless of dimensions via an embedding and inverses for non-injective mappings in one dimension are presented (this first part of the project Generalization Theory of Linear Algebra I was presented at JMM 2019). Then the talk continues and presents inverses for non-injective mappings in multiple dimensions, inverses for non-surjective mappings in one and multiple dimensions and introduces a general determinant theory (second part was presented at JMM 2020). In future work there will be further operations and applications to physics and other natural sciences be explored. (Received August 11, 2020)

16 ► Associative rings and algebras

1160-16-21 **Nicholas Ovenhouse*** (ovenh001@umn.edu), 206 Church Street SE, Minneapolis, MN 55455, and **Semeon Artamonov** and **Michael Shapiro**. *Noncommutative Integrable Systems on a Cylinder.* Preliminary report.

We consider generalizations of plabic graphs on a cylinder, whose edges are weighted by formal noncommuting variables. Generalizing work of Gekhtman, Shapiro, and Vainshtein, we define a noncommutative Poisson structure (using the double brackets of Van den Bergh), and show that this bracket can be expressed in terms of an R-matrix. Furthermore, we exhibit a collection of mutually Poisson-commuting functions in these noncommutative variables. (Received July 15, 2020)

1160-16-49 **Luigi Ferraro*** (lferraro@ttu.edu), **Desiree L. Martin** and **Frank W. Moore**. *The Taylor Resolution over a Skew Polynomial Ring.*

Let I be a monomial ideal in the polynomial ring $R = k[x_1, \ldots, x_n]$ over a field k. In her thesis, Taylor introduced a complex which provides a multi-graded free resolution for R/I as an R-module. Later, Gemeda provided a differential graded structure on Taylor's complex. We generalize each of these results to monomial ideals in a skew polynomial ring. (Received July 31, 2020)

1160-16-134 **Pierre-Guy Plamondon*** (pierre-guy.plamondon@universite-paris-saclay.fr) and **Toshiya Yurikusa** (toshiya.yurikusa.d8@tohoku.ac.jp). *On the density of g-vector fans.*

The g-vector fan of a cluster algebra is known to be complete precisely when then algebra is of finite cluster type. The next best thing would be that the g-vector fan is dense in the real vector space in which it lives. For skew-symmetric types, I will present a near-classification of cluster algebras whose g-vector fans are either dense or whose closure is a half-space, as well as a conjecture on what the full classification should be. Many cases were previously known using various methods; we provide a uniform approach using the representation theory of associative algebras. This is a report on a joint work with Toshiya Yurikusa (University of Tohoku). (Received August 08, 2020)

1160-16-173 **Bach Nguyen**, **Kurt Trey Trampel*** (ktrampel@nd.edu) and **Milen Yakimov**. *Quantum cluster algebras at roots of unity.*

Quantum cluster algebras at roots of unity are introduced, and standard cluster theory results such as the quantum Laurent phenomenon are obtained. Examples of these algebras include quantum cluster algebras specialized to a root of unity. For roots of unity with nice compatibility, we present a canonical central subalgebra. We also present a general theorem on the structure of discriminants of these quantum algebras. These discriminants encode representation theoretic information about these algebras. The discriminants of quantum Schubert cells for any symmetrizable Kac-Moody algebra are found as an example of the general theorem. (Received August 09, 2020)

1160-16-313 **Luigi Ferraro**, **W. Frank Moore*** (moorewf@wfu.edu) and **Josh Pollitz**. *Derived Hochschild cohomology over quotients of skew polynomial rings by normal elements.* Preliminary report.

Building on ideas present in work of Avramov, Buchweitz, Iyengar, and Pollitz, we use color differential graded homological algebra to compute derived Hochschild cohomology of a skew complete intersection ring R. Our calculation uses derivations, which seems to be new, even in the commutative case. In addition, we prove that for color modules M and N over R, $mathrmExt_R(M, N)$ is a finitely generated module over a (potentially different) skew polynomial ring. When the parameters defining the original skew polynomial ring are roots of unity, this allows us to define the support variety of a pair of color modules over such a ring, and we extend many commutative results to this new context. (Received August 11, 2020)

1160-16-364 **Steven Jin*** (sjin6816@umd.edu) and **Jooyoung Shin** (jshin5@kent.edu). *Iterated Differential Polynomial Rings over Locally Nilpotent Rings.*

We first survey the radical properties of differential polynomial rings over locally nilpotent rings. Next, we discuss recent work in which we show that a large of class of iterated differential polynomial rings over locally nilpotent rings are Behrens radical. This extends results of Chebotar and Chen, et al. (Received August 12, 2020)

18 ► Category theory; homological algebra

1160-18-212 **Emily M Rudman*** (walthere@indiana.edu). *The cyclic homology of* $k[x_1, x_2, \ldots, x_d]/(x_1, x_2, \ldots, x_d)^2$. Preliminary report.

The Hochschild homology of the ring $k[x_1, x_2, \ldots, x_d]/(x_1, x_2, \ldots, x_d)^2$ has been known and calculated several ways. Hochschild homology of rings is of interest as the target of the Dennis trace from algebraic K-theory, which involves a circle action on Hochschild homology. In an appropriate sense, cyclic homology is the homology of the quotient by this circle action. The calculation of the cyclic homology of $k[x_1, x_2, \ldots, x_d]/(x_1, x_2, \ldots, x_d)^2$ is relatively straightforward for $k = \mathbb{Q}$, but we see interesting torsion phenomena over $k = \mathbb{Z}$. (Received August 10, 2020)

20 ► Group theory and generalizations

1160-20-90 **Alexander J Hulpke*** (hulpke@colostate.edu), 1874 Campus Delivery, Department of Mathematics, Fort Collins, CO 80523-1874. *The perfect groups of order up to a million.* Preliminary report.

The perfect groups of most orders up to a million have been classified by Holt and Plesken in 1989. However for 11 orders (which amount to over 80% of these groups) the classification had not been completed since the calculation problems were beyond the capabilities of computers for close to 30 years. Using new computational tools for the calculation of 2-cohomology, permutation representations, and current isomorphism test functionality, we are completing this classfication. The construction process can be considered as a generalization of the Frattini extension method of Besche and Eick, used successfully for the construction of solvable groups of small orders. (Received August 06, 2020)

1160-20-91 **Daniel Berlyne*** (dberlyne@gradcenter.cuny.edu) and **Jacob Russell**. *Hierarchical hyperbolicity of graph products.*

Direct products and free products are two of the simplest ways of combining groups, yet they still give rise to interesting geometry. Direct products of the integers are commonly realised as lattice points in Euclidean space, while free products of the integers exhibit hyperbolic geometry. Combining free and direct products of the integers gives rise to right-angled Artin groups, which have geometry with both Euclidean and hyperbolic behaviour; they are "hierarchically hyperbolic", a tool developed by Behrstock, Hagen, and Sisto to describe the geometry of a space in terms of its projections to various associated hyperbolic spaces.

We show that Behrstock, Hagen, and Sisto's result for right-angled Artin groups can be generalised to any graph product of hierarchically hyperbolic groups. That is, if the copies of the integers in a right-angled Artin group are replaced with hierarchically hyperbolic groups, then the resulting group is still hierarchically hyperbolic. We construct this geometry explicitly, answering two questions of Behrstock, Hagen, and Sisto. (Received August 06, 2020)

1160-20-116 **Nguyen Ngoc Hung**, The University of Akron, Akron, OH 44325, **Thomas Michael Keller***, Department of Mathematics, 601 University Drive, San Marcos, TX 78666, and **Yong Yang**, Department of Mathematics, 601 University Drive, San Marcos, TX 78666. *Bounding $k(GV)$ from below.*

Let G be a finite group acting faithfully, irreducibly, and coprimely on the finite G-module V. Finding upper and lower bounds for $k(GV)$, the number of conjugacy classes of the corresponding semidirect product GV, has been an important (but difficult) topic of study in recent years. In this paper we study a new such conjecture: $k(GV)$ is bounded below by the dimension of V. We present a first result on it which has applications to counting odd-degree characters of finite groups. (Received August 07, 2020)

1160-20-127 **Nguyen Hung*** (hungnguyen@uakron.edu) and **Attila Maroti**. *p-Rational characters and p-regular classes.*

Let G be a finite group of order divisible by a prime p. We show that the number of p-regular and p'-regular conjugacy classes of G is at least $2\sqrt{p-1}$. Also, the number of p-rational and p'-rational irreducible characters of G is at least $2\sqrt{p-1}$. To achieve these bounds, we prove a uniform lower bound for the number of p-regular classes in a finite simple group of Lie type in terms of its rank and size of the underlying field. (Received August 07, 2020)

1160-20-149 **Aparna Upadhyay*** (aparnaup@buffalo.edu), 244 Mathematics Building, University at Buffalo, SUNY, Buffalo, NY 14260. *The Benson - Symonds invariant for ordinary and signed permutation modules.*

Signed permutation modules are a simultaneous generalization of the ordinary permutation modules and the twisted permutation modules of the symmetric group. In a recent paper, Dave Benson and Peter Symonds defined an invariant $\gamma_G(M)$ for a finite dimensional module M of a finite group G which quantifies the non-projective part of the tensor powers of M. This invariant is difficult to compute but captures interesting asymptotic properties of tensor products. In this talk, we will determine this invariant for all the signed permutation modules of the symmetric group using tools from representation theory and combinatorics. (Received August 09, 2020)

1160-20-156 **Robert Boltje*** (boltje@ucsc.edu), **Radha Kessar** and **Markus Linckelmann**. *Picard groups of block algebras of finite groups.*

The Picard group $\mathrm{Pic}(B)$ of a block algebra B of a finite group is the set of isomorphism classes of (B,B)-bimodules which induce a Morita self-equivalence of B. Very little is known about this group in general. We

consider three subgroups of $\mathrm{Pic}(B)$, arising from bimodules whose sources are endo-permutation modules, rank-one modules, and trivial modules, respectively. This is joint work with Radha Kessar and Markus Linckelmann. (Received August 09, 2020)

1160-20-166 **Gabriel Navarro**, **Noelia Rizo** and **Mandi A. Schaeffer Fry*** (aschaef6@msudenver.edu), Denver, CO 80217, and **Carolina Vallejo**. *Galois action on the principal block and generation of Sylow subgroups.*

We show that the character table (in fact, we only require the principal block) of a finite group G determines certain generation properties of a Sylow subgroup, in terms of the Galois action on characters. Specifically, in joint work with Noelia Rizo and Carolina Vallejo, we show that one can determine from the principal block whether a Sylow 2- or 3-subgroup of G is cyclic. With Gabriel Navarro, Noelia Rizo, and Carolina Vallejo, we additionally show that one can determine whether a Sylow 2-subgroup is generated by two elements. Our proofs of these results require the use of the Classification of Finite Simple Groups and provide evidence for the so-far elusive Alperin–McKay–Navarro conjecture. (Received August 09, 2020)

1160-20-192 **Aaron Calderon*** (aaron.calderon@yale.edu) and **Nick Salter**. *Mapping class groups, vector fields, and holomorphic differentials.*

The mapping class group of a surface is the set of its topological symmetries. When the surface is equipped with a vector field, one would like to understand which symmetries preserve (the isotopy class of) this vector field. Despite the fundamental nature of this question, these "framed mapping class groups" remain poorly understood. In this talk I will explain the discovery, joint with Nick Salter, of explicit, finite generating sets for certain framed mapping class groups. I will also discuss the implications of this result for the topology of moduli spaces of holomorphic 1-forms. (Received August 10, 2020)

1160-20-222 **Robert Boltje**, **Cisil Karaguzel** and **Deniz Yilmaz*** (deyilmaz@ucsc.edu). *Fusion systems of blocks of finite groups over arbitrary fields.*

Let k be a field of characteristic $p > 0$ and let G be a finite group. To any block idempotent b of the group algebra kG, Puig associated a fusion system: a category that underpins the local side of the local-global considerations in modular representation theory. Puig also showed that the fusion system is saturated if the k-algebra $kC_G(P)e$ is split, where (P, e) is a maximal b-Brauer pair. In this talk, we investigate the block fusion system in the non-split case by describing it in an explicit way as being generated by the fusion system of a related block idempotent over a larger field together with a single automorphism of the defect group. (Received August 10, 2020)

1160-20-228 **Sara Jensen*** (sjensen1@carthage.edu), WI, and **Mark Lewis** (lewis@math.kent.edu) and **Mary Phillips** (mphillips@carthage.edu). *Using the Sequence of k-th Powers of a Group to Determine the Isomorphism Type of a Frobenius Complement.*

For a finite group G and positive integer k, define $G^k = \{x^k \mid x \in G\}$. In a 2015 work, Cocke, Isaacs, and Skabelund showed that if k is such that $|G^k| < |G|$, we can bound $|G|$ in terms of $|G^k|$. More recent work has asked what stronger properties can be determined about G if we are given information about the sequence $(|G^k|)_{k\in\mathbb{N}}$. For example, Cocke and Jensen showed in a 2018 paper that this sequence determines whether or not G has a normal Hall π-subgroup for any set of primes π. It follows that whether or not a group is nilpotent is determinable from this sequence. Frobenius groups are an interesting collection of groups and the structure of Frobenius complements has been well studied. In this work, the authors show that whether or not a group is a Frobenius group is determinable from $(|G^k|)_{k\in\mathbb{N}}$. Furthermore, if a group G is a Frobenius group, we show that the isomorphism type of the Frobenius complement of G is determined by $(|G^k|)_{k\in\mathbb{N}}$. Finally, we show that this result cannot be strengthened to say that G is determined up to isomorphism by presenting examples of non-isomorphic Frobenius groups with identical sequences of k-th powers. (Received August 10, 2020)

1160-20-251 **Robert Kropholler** and **Stefano Vidussi** (genevieve.walsh@gmail.com), Somerville, MA 02144, and **Genevieve Walsh*** (genevieve.walsh@tufts.edu). *Incoherence of free-by-free and surface-by-free groups.*

One of the many beautiful features of the fundamental groups of compact three-manifolds is that these groups are coherent (due to Peter Scott). This means that every finitely generated subgroup is finitely presented. We show that, in contrast, free-by-free groups and surface-by-free groups are incoherent except in exceptional circumstances. The methods are very similar to techniques in 3-manifold theory. In contrast to hyperbolic 3-manifolds, we show that these groups do not always virtually algebraically fiber. (Received August 10, 2020)

1160-20-266 **Amanda A. Schaeffer Fry**, **Stephen J Trefethen** and **C. Ryan Vinroot*** (vinroot@math.wm.edu). *Jordan decomposition map and Galois action for finite reductive groups in the disconnected center case.* Preliminary report.

Let $\mathbf{G}$ be a connected reductive group over the algebraic closure of a finite field, F a Frobenius endomorphism of $\mathbf{G}$, and let $G = \mathbf{G}^F$ be the associated finite reductive group. In the case that the center of $\mathbf{G}$ is connected, Digne and Michel described a Jordan decomposition map for the complex characters of G which is unique with respect to a certain list of properties. We extend this result to the case that the center of $\mathbf{G}$ is not necessarily connected, but for the Lusztig series for semisimple classes (s), where s is in the dual group $\mathbf{G}^*$ such that the centralizer $C_{\mathbf{G}^*}(s)$ is connected. We show that this Jordan decomposition map is well-behaved with respect to the action of the absolute Galois group acting on the complex irreducible characters. (Received August 10, 2020)

1160-20-273 **Lucas Gagnon*** (lucas.gagnon@colorado.edu). *Supercharacter theories, normal subgroups, and Galois connections.*

It is believed that many groups have a large number of supercharacter theories, but there are relatively few known constructions which produce a supercharacter theory for any given group. We will describe a new method for constructing supercharacter theories of any finite group, which has non-trivial results when the given group has many normal subgroups. In the construction a supercharacter theory is specified by a lattice of normal subgroups and some (super)character theoretic data for certain sections (subquotients) in this lattice, and the output is either a set of supercharacters or superclasses. When the lattice sections arise from a Galois connection it is easy to describe both the supercharacters and superclasses. We also give a straightforward way to compare the result of our construction with other supercharacter theories. (Received August 10, 2020)

1160-20-298 **Carolyn R Abbott*** (abbott@math.columbia.edu), **Nick Miller** and **Priyam Patel**. *Actions of big mapping class groups on the arc graph.*

Given a finite-type surface (i.e. one with finitely generated fundamental group), there are two important objects naturally associated to it: a group, called the mapping class group, and an infinite-diameter hyperbolic graph, called the curve graph. The mapping class group acts by isometries on the curve graph, and this action has been extremely useful in understanding the algebraic and geometric properties of mapping class groups. One particularly important class of elements of the mapping class group are those which act as loxodromic isometries of the curve graph; these are called "pseudo Anosov" elements. Given an infinite-type surface with an isolated puncture, one can associate two analogous objects: the so-called big mapping class group, and the (relative) arc graph. In this talk, we will consider the action of big mapping class groups on the arc graph, and, in particular, we will construct an infinite family of "infinite-type" elements that act as loxodromic isometries of the arc graph, where (roughly) an infinite-type element is one which is not supported on any finite-type subsurface. This is joint work with Nick Miller and Priyam Patel. (Received August 11, 2020)

1160-20-301 **Rob Carman***, wrcarman@wm.edu. *Characterizing the Orthogonal Units of the Trivial Source Ring.* Preliminary report.

Trivial source modules (aka p-permutation modules) are an oft studied classes of modules in the modular representation theory of a finite group. The corresponding representation ring, the trivial source ring, can be seen as an intermediate between the Burnside ring and Brauer character ring of the group. The torsion units of this ring, referred to as orthogonal units due to their inverses necessarily coming from a dual module construction, are of interest due to their connection with equivalences of blocks of the group algebra. I will present some results on characterizing these orthogonal unit groups, including their relation to unit groups of other representation rings, notably the Burnside ring of the underlying p-fusion system. (Received August 11, 2020)

1160-20-324 **Farid Aliniaeifard** and **Nathaniel Thiem*** (thiemn@colorado.edu). *Categorifying combinatorics with towers of groups.* Preliminary report.

Supercharacter theories have given us new ways to tailor the representations of our favorite groups to specific applications. My current favorite such application is recovering combinatorial Hopf structures via the representation theory of a tower of groups. This talk describes joint work with Aliniaeifard on a program to use a supercharacter theory of his (constructed from a lattice of normal subgroups) to not only realize combinatorial Hopf algebras but to begin finding algebraic explanations for how they are related. (Received August 11, 2020)

1160-20-353 **Jeffrey M Riedl*** (riedl@uakron.edu), University of Akron, 302 Buchtel Common, Akron, OH 44325. *Few orbits in automorphic actions on subgroups of wreath 2-groups.* Preliminary report.

Let $W = C \wr E$ be the regular wreath product where C is cyclic of order 2^e and E is nonabelian of order 4. Thus W is the semidirect product $B \rtimes E$ where B is direct product of 4 copies of C which E regularly permutes. Let $\mathcal{N}$ be the set of all normal subgroups of W contained in B. (Note $|\mathcal{N}| = 280$ for $e = 3$ and $|\mathcal{N}|$ grows rapidly as e increases.) For each $N \in \mathcal{N}$ let $\mathcal{H}(N)$ be the set of all subgroups H of W such that $BH = W$ and $H \cap B = N$.

We wish to study these groups H. Computations suggest $|\mathcal{H}(N)|$ tends to get quite large for most N as e increases, which is discouraging. Nevertheless we have identified a fairly small subgroup A_E of the automorphism group $\mathrm{Aut}(W)$ that acts naturally on the set $\mathcal{H}(N)$ for arbitrary $N \in \mathcal{N}$ and has the following property.

For all those $N \in \mathcal{N}$ having rank at most 2 (without any restriction on e), if we let $f(N)$ denote the number of orbits in the action of A_E on the set $\mathcal{H}(N)$, our extensive computations suggest that always $f(N) = 2^m$ for $m \in \{1, 2, 3\}$. This would mean that there are at most 8 distinct groups in $\mathcal{H}(N)$ up to isomorphism, for each such normal subgroup N. (Received August 11, 2020)

1160-20-360 **Ellen Henke**, **Assaf Libman** and **Justin Lynd*** (lynd@louisiana.edu). *Punctured groups for exotic fusion systems.* Preliminary report.

The fusion system of a finite group G at a prime p is a category whose objects are the subgroups of a fixed Sylow p-subgroup S, and where the morphisms are the conjugation homomorphisms induced by the elements of G. The notion of a saturated fusion system is abstracted from this standard example, and provides a coarse representation of what is meant by the p-local structure of a finite group. Once the group G is abstracted away, there appear many exotic fusion systems, namely fusion systems which do not arise in the above fashion. Given an exotic fusion system, one might like to ask: just how "group-like" is it? I'll present one answer to this question, look at examples such as the Benson-Solomon exotic fusion systems at the prime 2, and mention an application to the topology of classifying spaces. (Received August 12, 2020)

33 ► Special functions

1160-33-50 **Jonathan Gabriel Bradley-Thrush*** (j.bradleythrush@ufl.edu), Department of Mathematics, University of Florida, 358 Little Hall, PO Box 118105, Gainesville, FL 32611-8105. *A derivation of some expansions of Hecke-Rogers type.*

The expression

$$\left(\sum_{n=-\infty}^{\infty} (-1)^n q^{\frac{n(n-1)}{2}} y^n\right)\left(\sum_{n=-\infty}^{\infty} \frac{(-1)^n p^{\frac{n(n+1)}{2}} x^n}{1 - yq^{kn}}\right),$$

in which $|p| < 1$, $|q| < 1$, $x \neq 0$ and k is any positive integer, can be expanded about the origin as a Laurent series in powers of the variable y. Moreover, this expansion is valid for all $y \neq 0$ since each simple pole of the second series is cancelled by a simple zero of the first. I will present several methods that can be used to determine the coefficients of the Laurent series and show how special cases of the series give rise to certain double series expansions of Hecke-Rogers type. Examples will be given which relate to the mock theta functions and to the generating function for the rank of a partition. (Received July 31, 2020)

34 ► Ordinary differential equations

1160-34-37 **Anthony Michael Bloch*** (abloch@umich.edu) and **Matthew Kvalheim**. *Geometry and Dynamics of Circulant Systems.*

We analyze the dynamics and geometry of so-called circulant systems of ODE's. Such equations are invariant under the action of a cyclic group and are inspired by equations studied in the synthetic biology literature. Examples include the repressilator and the the Sprott system. We analyze invariant sets, global bifurcation behavior, long time asymptotics and the existence of periodic orbits. We develop the theory and present simulations. (Received July 29, 2020)

1160-34-63 **C. Eugene Wayne*** (cew@math.bu.edu). *Metastable states in weakly damped systems of coupled oscillators.* Preliminary report.

Metastability in weakly dissipative systems refers to a common phenomenon in which certain prototypical structures (e.g. vortex dipoles, in 2D turbulence) appear in solutions of these systems and then dominate the

subsequent evolution of the system for very long periods of time. These structures typically emerge on a time scale whose origin is not evident from a naive inspection of the underlying equations of motion. Because of their importance in physical circumstances as disparate as turbulence and heat transport, understanding their origin and properties is important. In this talk I will discuss the appearance of metastable states in weakly damped chains of Hamiltonian oscillators. In certain regions of phase space, the system undergoes a very regular evolution, remaining near a family of breather solutions for the undamped, Hamiltonian system. Although these breathers no longer exist in the damped problem, there is a family of exact breathers in a related damped and driven problem which seem to provide an very accurate approximation to these metastable states. This is joint work with Jean-Pierre Eckmann (Univ. of Geneva) and Daniel Caballero (Boston Univ.) (Received August 03, 2020)

1160-34-193 **joel c langer*** (joel.langer@case.edu), Department of Mathematics, Applied Mathematics and Statistics, CWRU, Cleveland, OH 44106. *A quick tour of planar foliations by pencils of cubics.* Preliminary report.

A pencil of real algebraic plane curves P=aG+bH determines a locally orientable foliation F of the real projective plane with finitely many singularities (when G, H have no common components). The index of F at a point p depends only on the numbers of distinct real tangent lines of G and H at p; by the Poincare-Hopf Theorem, these indices sum to one.

Such pencils provide a versatile family of planar phase portraits for associated polynomial vector fields. To what extent one may "prescribe" singularities of such fields of a given degree d is a matter of enumerative geometry, but low degree cases provide opportunities for more elementary investigations.

This talk will present results of a student capstone project (Jeremy Wall, CWRU 20) exploring the case d=3. E.g., letting cubics G, H be "triangles" (and degenerations), examples with N singularities were constructed, for all numbers from N=1 to N=21—with the one exception N=20! (Received August 10, 2020)

1160-34-345 **Zhaopeng Hao*** (zhaopenghao2015@gmail.com), 150 N. University Street, West Lafayette, IN 47906, and **Zhongqiang Zhang**. *Numerical approximation of optimal convergence for fractional elliptic equations with additive noise.* Preliminary report.

We consider a spectral approximation for fractional elliptic equations with additive Gaussian noise including fractional Brownian noise and one over f noise. For a wide range of parameters for the order of fractional operator and the power-law noise. We present optimal convergence of our method. The proof of the convergence relies on the well-posedness of Petrov-Galerkin formulation and the regularity of solutions in Sobolev spaces. We also present several numerical examples to illustrate optimal convergence orders in the mean-square sense. (Received August 11, 2020)

35 ► Partial differential equations

1160-35-42 **Katelyn Plaiser Leisman** and **Jared C Bronski*** (bronski@illinois.edu), University of Illinois, Urbana, IL 61801, and **Mat Johnson** (matjohn@ku.edu) and **Robert Marangell**. *Stability and Modulation theory for Hamiltonian PDEs.*

We present a rigorous version of Whitham modulation theory at the level of the linearized operator. Generically for a Hamiltonian PDE on the torus with d additional conserved quantities the linearized operator has a $2d$ dimensional generalized kernel with a non-trivial Jordan structure. As one varies the Floquet parameter this Jordan block breaks up into $2d$ small eigenvalues. We show that the normal form for these small eigenvalues is given by the formal Whitham modulation equations for the traveling wave. (Received July 30, 2020)

1160-35-44 **Theodore D Drivas***, Fine Hall – Washington Road. *Flexibility, rigidity and stability of steady fluid motion.*

We discuss the possibility and the impossibility of smooth deformations of equilibrium fluid states in the presence of symmetries. We will also make some remarks about the stability/instability of stationary states. (Received July 30, 2020)

1160-35-53 **Antonio De Rosa*** (derosa@cims.nyu.edu), Department of Mathematics, William E. Kirwan, 4176 Campus Dr, College Park, MD 20742. *Solutions to two conjectures in branched transport: stability and regularity of optimal paths.*

Models involving branched structures are employed to describe several supply-demand systems. The transportation cost in these models is proportional to a concave power $\alpha \in (0,1)$ of the intensity of the flow. We focus on the stability of the optimal transports in $\mathbb{R}^n$, with respect to variations of the source and target measures. The

stability was known when α is bigger than the critical threshold $1 - \frac{1}{n}$. We prove it holds for every exponent $\alpha \in (0, 1)$ (and for general lower semicontinuous cost functionals) and we provide a counterexample for $\alpha = 0$. Thus we completely solve a conjecture of Bernot, Caselles and Morel. Furthermore, we prove the stability for the mailing problem, which was completely open in the literature and we apply it to obtain the regularity of the optimal networks. (Received August 01, 2020)

1160-35-56 **Geng Chen**, **Tao Huang*** (taohuang@wayne.edu) and **Weishi Liu**. *Poiseuille Flow of Nematic Liquid Crystals via the Full Ericksen-Leslie Model.*

We study the Cauchy problem of the Poiseuille flow of full Ericksen-Leslie model for nematic liquid crystals. The model is a coupled system of a parabolic equation for the velocity and a quasilinear wave equation for the director. For a particular choice of several physical parameter values, we construct solutions with smooth initial data and finite energy that produce, in finite time, cusp singularities-blowups of gradients. The formation of cusp singularity is due to local interactions of wave-like characteristics of solutions, which is different from the mechanism of finite time singularity formations for the parabolic Ericksen-Leslie system. We are also able to establish the global existence of weak solutions that are Hölder continuous and have bounded energy. (Received August 01, 2020)

1160-35-70 **Siddhant Agrawal*** (agrawal@math.umass.edu) and **Andrea Nahmod** (nahmod@math.umass.edu). *Uniqueness of the 2D incompressible Euler equation on corner domains.*

We consider the 2D incompressible Euler equation on a corner domain with an angle between $\pi/2$ and π. In this setup, the uniqueness of solutions in the Yudovich class is not known in general due to the fact that the velocity is very far from being Lipschitz. The only case known of uniqueness is the result of Lacave and Zlatos (2018) where they prove uniqueness if the initial vorticity is constant around the boundary.

We prove that if the initial vorticity is non-negative and supported on one side of the angle bisector of the domain, then solutions in the Yudovich class are unique. This is the first uniqueness result which allows nontrivial vorticity around the corner. This is joint work with Andrea Nahmod. (Received August 05, 2020)

1160-35-96 **Jiahong Wu*** (jiahong.wu@okstate.edu), 401 Mathematical Sciences, Department of Mathematics, Stillwater, OK 74078. *Stability and large-time behavior for the 2D anisotropic Boussinesq equations.*

The stability and large-time behavior problem on many anisotropic PDEs modeling fluids is not well-understood. One such example is the 2D incompressible Navier-Stokes equation with only horizontal (or vertical) dissipation. When the spatial domain is the whole space $\mathbb{R}^2$, the global regularity relies on the Yudovich approach while the stability of perturbations near the trivial solution is an open problem. Another important example is the 2D anisotropic Boussinesq system. When the velocity dissipation and the thermal diffusion are only in the horizontal (or vertical) direction, the stability problem on perturbations near the hydrostatic equilibrium is not trivial. This talk presents recent work dealing with the aforementioned stability and large-time behavior problems on the 2D Navier-Stokes and the Boussinesq equations. In particular, our results on the anisotropic Boussinesq system reveal that the temperature (via the coupling and interaction in the system) actually tames and stabilizes the fluids. (Received August 06, 2020)

1160-35-123 **Changyu Ren**, **Zhizhang Wang** and **Ling Xiao*** (ling.2.xiao@uconn.edu). *Entire spacelike constant σ_k curvature hypersurfaces in Minkowski space.*

We prove that, in the Minkowski space, if a spacelike, (n-1) -convex hypersurface M with constant σ_{n-1} curvature has bounded principal curvatures, then M is convex. Moreover, if M is not strictly convex, after an rigid motion, M splits a line. We also construct nontrivial examples of strictly convex, spacelike hypersurface M with constant σ_k curvature and bounded principal curvatures. This is a joint work with Changyu Ren and Zhizhang Wang. (Received August 07, 2020)

1160-35-132 **John K. Hunter**, **Ryan Christopher Moreno-Vasquez**, **Jingyang Shu*** (jyshu@temple.edu) and **Qingtian Zhang**. *Surface waves on vorticity fronts.*

Surface waves are waves that propagate along an interface, with energy that is localized near the surface. A planar discontinuity in vorticity in two-dimensional inviscid, incompressible fluid flows supports surface waves. However, the formal contour dynamics equations for the infinite vorticity fronts in the two-dimensional incompressible Euler equations do not converge. In this talk, we will talk about how to overcome this difficulty to obtain meaningful well-formulated contour dynamics equations when the fronts are described as a graph. We will also discuss some properties of this equation, including an approximation result by the Burgers-Hilbert equation on cubically nonlinear timescales. (Received August 07, 2020)

1160-35-150 **Javier Gómez-Serrano** (jg27@math.princeton.edu), **Jaemin Park** (jpark776@gatech.edu), **Jia Shi** (jiashi@princeton.edu) and **Yao Yao*** (yaoyao@math.gatech.edu). *Radial symmetry of stationary and uniformly-rotating solutions in 2D incompressible fluid equations.*

In this talk, I will discuss some results on radial symmetry property for stationary or uniformly-rotating solutions for 2D Euler and SQG equation, where we aim to answer the question whether every stationary/uniformly-rotating solution must be radially symmetric, if the vorticity is compactly supported. This is a joint work with Javier Gómez-Serrano, Jaemin Park and Jia Shi. (Received August 09, 2020)

1160-35-191 **Huy Q Nguyen***, 151 Thayer Street, PROVIDENCE, RI 02906, and **Walter A Strauss** (wstrauss@math.brown.edu), 151 Thayer Street, Providence, RI 02906. *Proof of modulational instability of Stokes waves in deep water.*

It is proven that small-amplitude steady periodic water waves with infinite depth are unstable with respect to long-wave perturbations. This modulational instability was first observed more than half a century ago by Benjamin and Feir. It has never been proven rigorously except in the case of finite depth. We provide a completely different and self-contained approach to prove the spectral modulational instability for water waves in both the finite and infinite depth cases. Our linearization retains the physical variables and is compatible with energy estimates for the nonlinear problem. (Received August 10, 2020)

1160-35-198 **Marcelo M. Disconzi*** (marcelo.disconzi@vanderbilt.edu), **Mihaela Ifrim** and **Daniel Tataru**. *The relativistic Euler equations with a physical vacuum boundary.*

We consider the relativistic Euler equations with a physical vacuum boundary and an equation of state $p(\varrho) = \varrho^\gamma$, $\gamma > 1$. We establish the following results. (i) local well-posedness in the Hadamard sense, i.e., local existence, uniqueness, and continuous dependence on the data; (ii) low regularity solutions: our uniqueness result holds at the level of Lipschitz velocity and density, while our rough solutions, obtained as unique limits of smooth solutions, have regularity only a half derivative above scaling; (iii) stability: our uniqueness in fact follows from a more general result, namely, we show that a certain nonlinear functional that tracks the distance between two solutions (in part by measuring the distance between their respective boundaries) is propagated by the flow; (iv) we establish sharp, essentially scale invariant energy estimates for solutions; (v) a sharp continuation criterion, at the level of scaling, showing that solutions can be continued as long as the the velocity is in $L^1_t Lip$ and a suitable weighted version of the density is at the same regularity level. (Received August 10, 2020)

1160-35-206 **ALVIS ZHAODH*** (alvis@live.unc.edu), Chapel Hill, NC 27514, and **Mark Williams**. *Hyperbolic boundary problems with large oscillatory coefficients on small frequency region.*

We study weakly stable hyperbolic boundary problems with highly oscillatory coefficients that are large, $O(1)$, compared to the small wavelength ϵ of oscillations. Such problems arise, for example, in the study of classical questions concerning the stability of Mach stems and compressible vortex sheets. For such applications one seeks to prove energy estimates that are in an appropriate sense "uniform" with respect to the small wavelength ϵ, but the large oscillatory coefficients are a formidable obstacle to obtaining such estimates. In this paper we analyze a simplified form of the linearized problems that are relevant to the above stability questions, and obtain positive results in the small frequency region with the approach presented in [Wi19] under "unfavorable" conditions. We also construct geometric optics solutions for those systems as applications to the main estimation theorem. (Received August 10, 2020)

1160-35-207 **Andrew Lawrie*** (alawrie@mit.edu) and **Jacek Jendrej**. *Asymptotically stationary nonlinear waves.*

A generic global solution to a nonlinear wave equation exhibits oscillatory behavior, which is reflected in the fact that its kinetic energy does not tend to zero in infinite time. The same is true of solutions to Newton's equation, which is the finite dimensional analogue. However, there are special solutions whose kinetic energy converges to zero (in both cases) and we call such solutions asymptotically stationary. These play an important role in the description of the phase portrait, whether they are stable or not. Familiar examples include stationary solutions (corresponding to critical points of the potential energy) or their stable manifolds. In this talk, we discuss asymptotically stationary solutions for two canonical nonlinear wave equations admitting topological solitons: scalar field models in 1d, and wave maps in 2d. We classify and describe all asymptotically stationary solutions exhibiting two-soliton structure for these equations. This is joint work with Jacek Jendrej (CNRS and U. Paris 13) and partly with Michal Kowalczyk (U. Chile). (Received August 10, 2020)

1160-35-213 **Xiaoqian Xu*** (xiaoqian.xu@dukekunshan.edu.cn), No. 8 Duke Avenue, Kunshan, Duke Kunshan University, IB 3052, Kunshan, Jiangsu 215316, Peoples Rep of China. *Suppression of chemotactic explosion by mixing.*

Keller-Segel equation is one of the most studied PDE models of processes involving chemical attraction. However, solution of Keller-Segel equation can exhibit dramatic collapsing behavior, where the population density of bacteria concentrates positive mass in a measure zero region. In other words, there exist initial data leading to finite time blow up. In this talk, we will discuss the possible effects resulting from interaction of chemotactic and fluid transport processes, namely we will consider the Keller-Segel equation with additional advection term modeling ambient fluid flow. We will prove that the presence of fluid can prevent the singularity formation. We will discuss two classes of flows that have the explosion arresting property. (Received August 10, 2020)

1160-35-225 **Jared Speck*** (jared.speck@vanderbilt.edu), Vanderbilt University, Department of Mathematics, 1326 Stevenson Center Ln, Nashville, TN 37212. *Advances in the mathematical theory of shock waves.*

Roughly, a shock singularity in a quasilinear hyperbolic PDE solution is such that the solution remains bounded but one of its derivatives blows up due to the intersection of the characteristics. While the theory of shocks is relatively well-developed in one spatial dimension, it is only in recent years that a rigorous theory in multiple spatial dimensions has finally started to emerge. The recent advances are due to new analytic techniques that allow for a detailed understanding of singularities, as well as new geometric formulations of the equations, inspired by results in mathematical general relativity, which reveal crucial hidden structures. In this talk, I will describe some of the recent progress, with a special focus on the compressible Euler equations. I will also paint a picture of the direction of the field and its prospects for achieving the main long-term goal: obtaining a rigorous mathematical theory for the long-time behavior of solutions that is able to accommodate the formation of shocks as well as their subsequent interactions. Some of the works I will discuss are joint with L. Abbrescia, G. Holzegel, J. Luk, and W. Wong. (Received August 10, 2020)

1160-35-227 **Daniel Tataru** (tataru@math.berkeley.edu) and **Mihaela Ifrim*** (ifrim@wisc.edu). *The compressible Euler equations in a physical vacuum: a comprehensive Eulerian approach.*

This article is concerned with the local well-posedness problem for the compressible Euler equations in gas dynamics. For this system we consider the free boundary problem which corresponds to a physical vacuum. Despite the clear physical interest in this system, the prior work on this problem is limited to Lagrangian coordinates, in high regularity spaces. Instead, the objective of the present work is to provide a new, fully Eulerian approach to this problem, which provides a complete, Hadamard style well-posedness theory for this problem in low regularity Sobolev spaces. In particular we give new proofs for both existence, uniqueness, and continuous dependence on the data with sharp, scale invariant energy estimates, and continuation criterion. (Received August 10, 2020)

1160-35-237 **Kris K Jenssen*** (hkjenssen@gmail.com) and **Charis Tsikkou**. *Unbounded multi-d compressible Euler flows.*

In 1942 Guderley showed that the Euler system describing the flow of an ideal polytropic gas admits solutions in which a symmetric shock wave collapses at the origin and reflects a new outgoing spherical shock. The solutions in question blow up in amplitude and describe purely radial (quasi-1-d), self-similar flows.

In this talk we first report on recent work showing that Guderley's solutions provide bona fide weak solutions of the original, multi-d Euler system.

On the other hand, these same solutions include regions of zero pressure due to vanishing temperature; these solutions are therefore of border-line physicality.

We demonstrate that the Euler model admits other types of radial, self-similar solutions which blow up in amplitude, and with an everywhere strictly positive pressure field. This demonstrates that the mechanism of wave focusing is sufficiently strong, on its own, to generate unbounded amplitudes.

Joint work with Charis Tsikkou (University of West Virginia). (Received August 10, 2020)

1160-35-250 **Siming He*** (simhe@math.duke.edu), Room 218 Physics Building, 120 Science Drive, Room 218 Physics Building, 120 Science Drive, Durham, NC 27708. *Two applications of the shear flow-induced enhanced dissipation.* Preliminary report.

In this talk, I will discuss two applications of the enhanced dissipation effect induced by shear-flow mixing. The first application involves the suppression of nonlinear blow-up in the Patlak-Keller-Segel type equations. It is well-known that classical Patlak-Keller-Segel equations form Dirac type singularities when the total mass is above a critical threshold. By introducing external shear flow, it is possible to suppress this potential blow-up

scenario. The second application is on reaction enhancement through the shear flow. (Received August 10, 2020)

1160-35-258 **Tien Khai Nguyen*** (khai@math.ncsu.edu), Department of Mathematics, North Carolina State University, Raleigh, NC 27695. *Metric entropy for Hamilton-Jacobi equation with uniformly directional convex Hamiltonian.*

Inspired by a question posed by Lax in 2002, in recent years it has received an increasing attention the study on the metric entropy (epsilon entropy) for nonlinear PDEs. In this talk, I will present the BV-type regularity for viscosity solutions of the Hamilton-Jacobi equation

$$u_t(t,x) + H\big(D_x u(t,x)\big) \;=\; 0 \qquad \forall (t,x) \in]0,\infty[\times\mathbb{R}^d$$

with coercive and uniformly directional convex Hamiltonian $H \in \mathcal{C}^1(\mathbb{R}^d)$. More precisely, we establish a BV bound on the slope of backward characteristics $DH(u(t,\cdot))$ starting at a positive time $t > 0$. Relying on the BV bound, we quantify the metric entropy (the degree of compactness) in $\mathbf{W}^{1,1}_{\rm loc}(\mathbb{R}^d)$ for the map S_t that associates to every given initial data $u_0 \in \mathbf{Lip}\big(\mathbb{R}^d\big)$ the corresponding solution $S_t u_0$. (Received August 10, 2020)

1160-35-259 **Alberto Bressan** and **Maria Teresa Chiri*** (mxc6028@psu.edu), STATE COLLEGE, and **Wen Shen**. *A posteriori Error Estimates for Numerical Solutions to Hyperbolic Conservation Laws.*

Consider the Cauchy problem for a strictly hyperbolic system of conservation laws

$$u_t + f(u)_x \;=\; 0, \qquad u(0,x) \;=\; \bar u(x).$$

Convergence results for approximate solutions of this problem (proved so far) rely on uniform bounds on the total variation. Without an a priori bound on the total variation, one cannot compare an approximate solution with trajectories of the semigroup of exact solutions. Therefore we focus on *a posteriori* error estimates. The result we show is the following.

Let u^{approx} be an approximate solution produced by a conservative scheme which dissipates entropy, and assume that

(i) the total variation of $u^{approx}(t,\cdot)$ is uniformly bounded,

(ii) outside a finite number of narrow strips in the domain $[0,T]\times\mathbb{R}$, the local oscillation of u^{approx} remains small.

Then the $\mathbf{L}^1$ distance

$$\|u^{approx}(T,\cdot) - u^{exact}(T,\cdot)\|_{\mathbf{L}^1(\mathbb{R})}\,.$$

is small. Our estimates do not require any regularity of the exact solution. We provide an error bounds which can be applied to a wide class of approximation schemes. (Received August 10, 2020)

1160-35-281 **Michele Coti Zelati*** (m.coti-zelati@imperial.ac.uk), **Tarek M Elgindi** and **Klaus Widmayer**. *Stationary Euler flows near Kolmogorov and Poiseuille.*

We exhibit a large family of new, non-trivial stationary states of analytic regularity, that are arbitrarily close to the Kolmogorov flow on the square torus. Our construction of these stationary states builds on a degeneracy in the global structure of the Kolmogorov flow. This is in contrast with both the Kolmogorov flow on a rectangular torus and the Poiseuille flow in a channel, for which we can show that the only stationary states near them must be shears. This has surprising consequences in the context of inviscid damping in 2D Euler and enhanced dissipation in Navier-Stokes. (Received August 11, 2020)

1160-35-286 **Gianluca Crippa*** (gianluca.crippa@unibas.ch). *On the local limit for nonlocal continuity equations.*

I will discuss a question posed by P. Amorim, R. Colombo and A. Teixeira, which can be loosely speaking formulated as follows. Consider a family of continuity equations where the velocity field is given by the convolution of the solution with a regular kernel. In the singular limit where the convolution kernel is replaced by a Dirac delta, one formally recovers a conservation law: can we rigorously justify this formal limit? We exhibit counterexamples showing that, despite numerical evidence suggesting a positive answer, one in general does not have convergence of the solutions. We also show that the answer to the above question is positive if we consider viscous perturbations of the nonlocal equations. In this case, in the singular local limit the solutions converge to the solution of the viscous conservation law. We will also discuss the possible role of numerical viscosity in numerical simulations, as well as some more recent results dealing with the case of an anisotropic convolution kernel, a more realistic case for applications to traffic modelling. The talk will be based on some joint works with Maria Colombo, Marie Graff, Elio Marconi, and Laura Spinolo. (Received August 11, 2020)

1160-35-303 **Robin Young*** (young@math.umass.edu) and **J. Blake Temple**. *Inversion of Non-uniform Difference Operators.*

We introduce and solve a novel problem, that of inverting a non-uniform difference operator, or equivalently of solving the (linear) difference equation D[v](t) = v(f(t)) - v(t) = w(t). This is the simplest form of an equation that forms an essential step in our long-term project of proving the existence of time- and space-periodic solutions of the compressible Euler equations of fluid dynamics. We demonstrate the need to use direct rather than Fourier methods, find a solvability condition, and provide bounds for the solution. We also discuss stability under small perturbations of appropriate parameters.

Joint work with Blake Temple (UC Davis) (Received August 11, 2020)

1160-35-304 **Javier Gomez-Serrano***, jg27@math.princeton.edu. *Highest waves of some dispersive equations.*

In this talk I will explain how to construct travelling waves of greatest height for equations which are dispersive perturbations of Burgers' equation. In contrast to bifurcation methods, this method also yields quantitative information of the solution. The proof combines careful analysis with computer-assisted estimates. (Received August 11, 2020)

1160-35-316 **Thomas A. Ivey*** (iveyt@cofc.edu) and **Annalisa Calini** (calinia@cofc.edu). *New Integrable Curve Flows in the Pseudoconformal 3-Sphere.* Preliminary report.

The pseudoconformal 3-sphere is the projectivization of the null cone in $\mathbb{C}^3$ with the standard pseudo-Hermitian inner product. The group $SU(2,1)$ fixing this metric naturally acts on S^3, preserving a contact structure, and can be identified with the pseudoconformal frame bundle of S^3. By normalizing lifts to the frame bundle, we define scalar geometric invariants for Legendrian curves (L-curves) in S^3, and for curves transverse to the contact planes (T-curves).

We seek invariant geometric flows for these parametrized curves that induce integrable evolution systems for the invariants. While there is an infinite sequence of geometric flows for L-curves inducing the Boussinesq hierarchy, for T-curves there is another infinite sequence of flows that induces a sequence of 3-component evolution systems for the invariants, evidently a novel integrable bi-Hamiltonian hierarchy. This closely resembles the NLS hierarchy, itself realized by a sequence of curve flows in Euclidean 3-space, including the vortex filament equation. We discuss some common features of these hierarchies, describe the geometry and dynamics of travelling wave solutions (also arising as critical curves for Lagrangians derived from the conserved densities) and conclude with some open questions. (Received August 11, 2020)

1160-35-319 **Ming Chen*** (mingchen@pitt.edu), 301 Thackeray, Department of Mathematics, University of Pittsburgh, Pittsburgh, PA 15260, and **Dmitry E Pelinovsky** (dmpeli@math.mcmaster.ca), Department of Mathematics and Statistics, McMaster University, Hamilton, ON L8S 4K1, Canada. *Strong instability of Novikov peakons.*

In this talk we consider a quasilinear dispersive equation with cubic nonlinearities which arises from integrable systems and shallow water modeling. A characteristic feature of this equation is its ability to support solitary waves with corner singularities, called peakons. These peakons have been shown to be orbitally and asymptotically stable in H^1. However it is also known that the equation loses continuous dependence on data (and hence is not well-posed in the Hadamard sense) in H^1. We are able to find a function space more suitable for the well-posedness theory for the peakons, and prove that a single peakon fails to be stable in this finer topology. Indeed we show that perturbation in this class leads to finite-time blow-up of the corresponding solutions. This is a joint work with Dmitry Pelinovsky. (Received August 11, 2020)

1160-35-329 **Huijing Du*** (hdu5@unl.edu). *Well-Balanced Discontinuous Galerkin Method for Shallow Water Equations with Constant Subtraction Techniques on Unstructured Meshes.*

The classical Saint–Venant shallow water equations on complex geometries have wide applications in many areas including coastal engineering and atmospheric modeling. The main numerical challenge in simulating Saint–Venant equations is to maintain the high order of accuracy and well-balanced property simultaneously. In this paper, we propose a high-order accurate and well-balanced discontinuous Galerkin (DG) method on two dimensional (2D) unstructured meshes for the Saint–Venant shallow water equations. The technique used to maintain well-balanced property is called constant subtraction and proposed in Yang et al. (J Sci Comput 63:678–698, 2015). Hierarchical reconstruction limiter with a remainder correction technique is introduced to control numerical oscillations. Numerical examples with smooth and discontinuous solutions are provided to demonstrate the performance of our proposed DG methods. (Received August 11, 2020)

1160-35-349 **Samuel Punshon-Smith*** (punshs@brown.edu), **Samuel Punshon-Smith** and **Alex Blumenthal**. *Chaotic mixing of passive scalars in stochastic fluid mechanics.*

The long-time behavior of a passive scalar in an incompressible fluid has long been of interest in physics. In this talk I will discuss several recent rigorous results in this area for a passive scalar that is advected by a number of stochastic fluid models, including the stochastic Navier-Stokes equations. We will see how tools from theory of random dynamics and the ergodic and hypoelliptic theory for stochastic PDE can be used to show that the associated Lagrangian flow has a positive Lyapunov exponent. I will then discuss how to use this chaos to show that the passive scalar almost surely mixes exponentially fast, uniformly in the diffusivity, a phenomenon known in physics as chaotic mixing. Joint work with Jacob Bedrossian and Alex Blumenthal. (Received August 11, 2020)

1160-35-363 **Yuxi Zheng*** (yuz2@psu.edu), Department of Mathematics, The Pennsylvania State University, University Park, PA 16802. *Large time existence of weak solutions to the initial value problems for Poiseuille nematic liquid crystal flows in one space dimension.* Preliminary report.

This will generalize a recent result on the subject by Geng Chen, Tao Huang and Weishi Liu, so that the large time existence of weak solutions can be extended for all physically relevant parameter ranges. The aforementioned paper is "Poiseuille Flow of Nematic Liquid Crystals via the Full Ericksen-Leslie Model" available on arxiv. (Received August 12, 2020)

37 ► Dynamical systems and ergodic theory

1160-37-18 **Konstantin Khanin*** (kkhanin@gmail.com), University of Toronto, Toronto, Canada. *The symplectic structure for renormalization of circle diffeomorphisms with breaks.*

The talk is based on a recent paper joint with Selim Ghazouani. In the first part of the talk I'll present the main results of the renormalization theory for circle homeomophisms with one break. Then we shall discuss the symplectic structure related to renormalisation of circle maps with many breaks. The invariant symplectic form which we construct is related to the symplectic form introduced by Goldman back in 1984. (Received July 10, 2020)

1160-37-160 **Christophe Golé*** (cgole@smith.edu), Smith College, Northampton, MA 01063-6303, and **Stephane Douady**. *Pre-periodic tilings in disk stacking on the cylinder.*

The process of stacking disks of constant radius one by one on the surface of a cylinder has surprising asymptotic properties: it yields in finite or infinite time lattice-like configurations of disks, that we call rhombic tilings. Understanding the topology of the set of cylindrical rhombic tilings, and the mechanism of convergence to it, helps explain the prevalence of consecutive Fibonacci numbers of helices in plants. It also illuminates the mechanism of formation of the more recently discovered quasi-symmetric patterns in plants, where the numbers of helices are equal, or almost equal. (Received August 09, 2020)

1160-37-264 **Ariel Lerman** and **Vadim Zharnitsky*** (vadimnz@gmail.com), Champaign, IL 61822-7567. *Whispering gallery orbits in Sinai oscillator trap.*

Experimental realizations of trapping Bose-Einstein condensate lead to a Hamiltonian system of a classical particle bouncing off a convex scatterer in the field of an attracting potential. It is shown by application of KAM theory that under some natural conditions there exists positive measure of quasiperiodic solutions near the scatterer's boundary. (Received August 10, 2020)

1160-37-335 **Annalisa Calini*** (calinia@cofc.edu) and **Gloria Mari-Beffa**. *Integrable evolutions of twisted polygons in centro-affine R^n.*

This talk focuses on a natural geometric flow for polygons in centro-affine geometry derived from discretizations of the Adler-Gel'fand-Dikii flows for curves in projective space. We prove the compatibility of its pair of Hamiltonian structures in arbitrary dimension by lifting them to a pair of pre-symplectic forms on the moduli space of centro-affine arc length parametrized polygons. We also describe their kernels and discuss the integrability of the polygonal flows. (Received August 11, 2020)

1160-37-337 **Tadashi Tokieda***, Department of Mathematics, Stanford University, Stanford, CA 94305-2125. *The simplest nontrivial example of renormalization.*

I present what I believe is the simplest nontrivial example of renormalization. Unlike public lectures of mine that you may have seen, this is technique-heavy, but I shall try to let as many people as possible cross the desert. (Received August 11, 2020)

1160-37-350 **Adrian P Bustamante***, apb7@math.gatech.edu, and **Rafael de la Llave**. *Gevrey estimates for asymptotic expansions of tori in weakly dissipative systems.*

We consider a singular perturbation for a family of analytic symplectic maps possessing a KAM torus. The perturbation introduces dissipation and contains an adjustable parameter, making the family conformally symplectic (the pull-back of the symplectic form is equal to a constant times itself – $f^*\Omega = \lambda\Omega$). By choosing the adjustable parameter one can ensure that, for a fixed frequency, the torus persist under perturbation. Such models are common in celestial mechanics. We prove that the asymptotic expansions of the quasi periodic orbits satisfy Gevrey estimates, that is, the n-th term of the expansion is bounded by a power of $n!$. (Received August 11, 2020)

46 ► Functional analysis

1160-46-84 **Geraldo Botelho**, **Qingying Bu**, **Donghai Ji** and **Khazhak Varazdat Navoyan***, Khazhakanush@gmail.com. *The positive Schur property on spaces of regular multilinear operators.* Preliminary report.

In this paper we give necessary and sufficient conditions for the space of regular multilinear operators from the product of Banach lattices to a Dedekind complete Banach lattice to have the positive Schur property. We also characterize the positive Schur property on the positive projective m-fold tensor product of Banach lattices, $m \in \mathbb{N}$, and on its dual. (Received August 06, 2020)

49 ► Calculus of variations and optimal control; optimization

1160-49-241 **Thiago Serra*** (thiago.serra@bucknell.edu), 1 Dent Drive, Lewisburg, PA 17837, and **Abhinav Kumar** and **Srikumar Ramalingam**. *Lossless Compression of Deep Neural Networks.*

Deep neural networks have been successful in many predictive modeling tasks, such as image and language recognition, where large neural networks are often used to obtain good accuracy. Consequently, it is challenging to deploy these networks under limited computational resources, such as in mobile devices. In this work, we introduce an algorithm that removes units and layers of a neural network while not changing the output that is produced, which thus implies a lossless compression. This algorithm, which we denote as LEO (Lossless Expressiveness Optimization), relies on Mixed-Integer Linear Programming (MILP) to identify Rectified Linear Units (ReLUs) with linear behavior over the input domain. By using L1 regularization to induce such behavior, we can benefit from training over a larger architecture than we would later use in the environment where the trained neural network is deployed. (Received August 10, 2020)

51 ► Geometry

1160-51-176 **Anton Izosimov*** (izosimov@math.arizona.edu), AZ. *Intersecting the sides of a polygon.*

Consider the map T which sends a planar polygon P to a new polygon P' whose vertices are the intersection points of second nearest sides of P. This map is the inverse of the famous pentagram map. In this talk we will investigate the dynamics of the map T. Namely, we will address the question of whether a convex polygon stays convex under iterations of T. It turns out that this almost never happens. More precisely, we will see that the set of polygons which remain convex under iterations of T is an algebraic subvariety of codimension two. We will also discuss the equations cutting out this subvariety, as well their geometric meaning in the case of pentagons. (Received August 09, 2020)

1160-51-291 **Tarik Aougab***, Mathematics Department, Haverford College, Haverford, PA 19041, and **Jonah Gaster**. *Curves on the torus intersecting at most k times.*

We consider the following well known question: given a closed orientable surface of genus g and a natural number k, what is the largest size $N(k, g)$ of a collection of pairwise non-homotopic simple closed curves on S which pairwise intersect at most k times? Surprisingly, this question remains open even on the torus. A lemma due to Agol states that $N(k, 1) \leq 1 + k_p$, where k_p denotes the smallest prime greater than k. The prime number theorem then implies that $N(k, 1)$ is asymptotic to k. The Riemann Hypothesis would imply a bound on the error of $O(\sqrt{k}\log(k))$. Using combinatorial and elementary geometric techniques, we obtain this same bound. This is joint work with Jonah Gaster. (Received August 11, 2020)

1160-51-357 **RONALD K pERLINE*** (perlinrk@drexel.edu), PHILADELPHIA, PA 19144, and **Jonah Smith**. *Backlund transformations for Bertrand curves and related geometric evolution equations.* Preliminary report.

Backlund transformations of curves ,inheritable from Backlund transformations for surfaces, can have interesting geometric properties worth studying in their own right. Two examples are transformations of curves if constant torsion (coming from Backlund transformations of pseudospherical surfaces) and the bicycle transformation (coming from Backlund transformations of Hasimoto surfaces). In these two cases the transformations on curves satisfy a constant distance point to point correspondence, and preserves constant speed parameterization.

Razzaboni surfaces are defined as surfaces for which a family of geodesics are Bertrand curves (satisfying $a\tau + b\kappa = c$, a, b, c are constants and τ, κ are torsion and curvature respectively. These surfaces also enjoy the existence of Backlund transformations (introduced by Razzaboni, and clarified by Schief). The corresponding point to point transformations preserve the Bertrand condition, are constant distance, but do not preserve arclength parameterization. We investigate this transformation, and illustrate its connection to a novel evolution equation on constant torsion curves. (Received August 12, 2020)

1160-51-361 **Goncalo Oliveira*** (galato97@gmail.com), Niterói, Brazil , Brazil. *The geometry of the Landau–Ginzburg functional.* Preliminary report.

I will report on new results obtained in collaboration with Ákos Nagy and Casey Kelleher on the existence of solutions to the Landau–Ginzburg equations. (Received August 12, 2020)

52 ► Convex and discrete geometry

1160-52-278 **Benjamin Schröter***, KTH - Royal Institute of Technology, Stockholm, Sweden. *The Geometry of Parameterized Shortest Paths.* Preliminary report.

One of the most basic classes of algorithmic problems in combinatorial optimization is the computation of shortest paths. Tropical geometry is a natural language to analyze parameterized versions where some of the arc weights are unknown. I will introduce this point of view, with a link to polyhedral geometry. Moreover, I will present an algorithm that computes these parameterized shortest paths and apply it to real world data from traffic networks.

This is joint work with Michael Joswig. (Received August 11, 2020)

53 ► Differential geometry

1160-53-15 **Akashdeep Dey*** (adey@math.princeton.edu). *A comparison of the Almgren-Pitts and the Allen-Cahn min-max theory.*

Min-max theory for the Allen-Cahn equation was developed by Guaraco and Gaspar-Guaraco. Allen-Cahn min-max theory provides a PDE based alternative to the Almgren-Pitts min-max theory for the min-max construction of minimal hypersurfaces. In my talk, I will briefly describe the Almgren-Pitts and the Allen-Cahn min-max theory and discuss the question to what extent these two theories agree. Part of the talk will be based on the above mentioned works of Guaraco and Gaspar-Guaraco. (Received July 02, 2020)

1160-53-16 **David N Pham*** (dnpham@qcc.cuny.edu) and **Fei Ye**. *Left Invariant Complex Structures on Double Lie Groups.* Preliminary report.

Let $(\mathfrak{g}, \gamma)$ be a Lie bialgebra and let $D(\mathfrak{g})$ be its associated double. A double Lie group is then defined to be a Lie group whose Lie algebra is $D(\mathfrak{g})$. In this talk, I discuss recent work on the construction of left invariant

complex structures on double Lie groups which are 'totally real' in the sense of Cleyton, Lauret, and Poon. This project is joint work with Fei Ye (QCC CUNY). (Received July 06, 2020)

1160-53-60 **Yueh-Ju Lin*** (lin@math.wichita.edu), Department of Mathematics, Statistics, and Physics, Wichita State University, Wichita, KS 67260, and **Wei Yuan** (gnr-x@163.com). *Volume Comparison of Q-curvature.* Preliminary report.

Classical volume comparison for Ricci curvature is a fundamental result in Riemannian geometry. In general, scalar curvature as the trace of Ricci curvature, is too weak to control the volume. However, with the additional stability assumption on the closed Einstein manifold, one can obtain a volume comparison for scalar curvature. In this talk, we investigate a similar phenomenon for Q-curvature, a fourth-order analogue of scalar curvature. In particular, we prove a volume comparison result of Q-curvature for metrics near stable Einstein metrics by variational techniques and a Morse lemma for infinite dimensional manifolds. This is a joint work with Wei Yuan. (Received August 02, 2020)

1160-53-74 **Xuemiao Chen*** (xuemiaochen1991@gmail.com) and **Song Sun**. *Reflexive sheaves, Hermitian-Yang-Mills connections, and tangent cones.*

The Donaldson-Uhlenbeck-Yau theorem proves the existence of a Hermitian-Yang-Mills connection on a given slope stable holomorphic vector over a compact Kahler manifold. Later, Bando and Siu generalized this to stable reflexive sheaves by using natural singular Hermitian-Yang-Mills connections. In this talk, I will talk about how to characterize the analytic tangent cones of a singular Hermitian-Yang-Mills connections by using certain new local complex geometric data extracted from the corresponding reflexive sheaves. (Received August 05, 2020)

1160-53-138 **Yangyang Li*** (yl15@math.princeton.edu). *Min-max minimal hypersurfaces in higher dimensions.*

Given a closed Riemannian manifold, the celebrated Almgren-Pitts Min-max theory provides a framework for generating embedded minimal hypersurfaces. However, in higher dimensions ($n + 1 \geq 8$), the existence of singularities makes it difficult to obtain more information of these min-max minimal hypersurfaces. In this talk, I'll discuss how to overcome this difficulty and to obtain results as in the low-dimensional case, e.g., the generic Yau's conjecture, Morse index estimates, multiplicity estimates, etc. (Received August 08, 2020)

1160-53-204 **Demetre Kazaras*** (dkazaras@gmail.com), 530 Foster St., Durham, NC 27701, and **Sven Hirsch** and **Marcus Khuri**. *Spacetime harmonic maps on asymptotically flat initial data sets.*

We give a new lower bound for the ADM mass of 3-dimensional asymptotically flat initial data sets. The bound is given in terms of linear growth 'spacetime harmonic functions' in addition to the energy-momentum density of matter fields, and is valid regardless of whether the dominant energy condition holds or whether the data possess a boundary. A corollary of this result is a new proof of the spacetime positive mass theorem and includes the rigidity statement which asserts that the mass vanishes if and only if the data arise from Minkowski space. Some further applications are discussed. (Received August 10, 2020)

1160-53-240 **Franco E Vargas Pallete*** (franco.vargaspallete@yale.edu) and **James Farre**. *Some curvature bounds for least area fibers in fibered hyperbolic 3-manifolds.*

Due to work of Uhlenbeck, we know that for a least area fiber on a fibered hyperbolic 3-manifold the maximum of the norm of its shape operator $|A|_\infty$ is strictly bigger than 2. On this talk we will see that for certain families of fibered hyperbolic 3-manifolds there is a uniform lower bound for $|A|_\infty$ which is greater than 2. We will see how this relates to pants distance between drilling geodesics along a sequence in our family. This is joint work with James Farre. (Received August 10, 2020)

1160-53-311 **Gil Bor***, CIMAT, A.P. 402, 36000 Guanajuato, Guanajuato, Mexico, and **Connor Jackman**. *Orbital symmetries and projective geometry of the Kepler problem.*

The *Kepler orbits* form a 3-parameter family of (unparametrized) plane curves, consisting of all conics sharing a focus. We study the geometry of this family, as well as certain natural 2-parameter subfamilies. Here is a sample of our results:

- The Kepler orbits form a *flat* family of curves, admitting a local action of the 7-dimensional conformal group of Minkowski's 2+1 spacetime, the maximum dimension possible for a 3-parameter family of plane curves.
- Kepler orbits with fixed non-zero energy (Kepler ellipses or hyperbolas with fixed major axis) form a non-flat 2-parameter locally homogeneous family, admitting a local action of the Möbius group $PGL(2, R)$ (for negative energy the action is global). All these families are locally equivalent.

- Kepler orbits with zero energy (Kepler parabolas), or fixed non-zero angular momentum, form a flat 2-parameter family, admitting a local action of the 8-dimensional projective group $PGL(3, R)$.
- All these symmetries and equivalences can be realized *globally*, by first passing to a double cover of the Kepler plane by the cone $x^2 + y^2 = z^2$ in euclidean 3-space, then compactifying this cone in projective 3-space.

(Received August 11, 2020)

1160-53-327 **Zhihan Wang*** (zhihanw@math.princeton.edu), Taylor Court 151, Unit 307, Princeton, NJ 08540. *Deformations of Singular Minimal Hypersurfaces.*

Minimal hypersurfaces has been frequently studied in recent years in dimension ≤ 7; In higher dimensions, singularities are known to exist in general, locally modeled on stable and area-minimizing cones in the Euclidean spaces. In this talk, we present different aspects of how these singularities may affect the local behavior of minimal hypersurfaces. In particular, given a non-degenerate minimal hypersurface with strictly stable and strictly minimizing regular tangent cone at each singular point, under any small perturbation of the metric, we show the existence of a nearby minimal hypersurface under new metric. For a generic choice of perturbation direction, we show the entirely smoothness of the resulting minimal hypersurface. (Received August 11, 2020)

1160-53-362 **Yanyan Li*** (yyli@math.rutgers.edu), Rutgers University Department of Mathematics. *On the σ_k-Nirenberg problem and related topics.* Preliminary report.

The following problem was raised by Nirenberg: Which function on the standard 2-sphere is the Gauss curvature of a metric conformally equivalent to the standard metric. Naturally one may ask a similar question in the higher dimensional case, namely, which function on the standard $n-$sphere is the scalar curvature of a metric conformally equivalent to the standard metric.

An analogous question can be asked for the σ_k-curvature instead of the scalar curvature, and we call it the σ_k-Nirenberg problem. We will present some results on the existence and compactness of solutions of the σ_k-Nirenberg problem for $n \geq 3$ and $k \geq n/2$. The results for $n = 4$ and $k = 2$ were established by Alice Chang, Zheng-Chao Han and Paul Yang in 2011. We will also present some recent results on the σ_k-Loewner–Nirenberg problem, and on nonlinear equations invariant under Mobius transformations in two dimension. These results are from a couple of joint works with Maria del Mar Gonzalez, Han Lu, Siyuan Lu, Luc Nguyen, Bo Wang, and Jingang Xiong. (Received August 12, 2020)

1160-53-365 **Eric Chen** and **Yi Wang*** (ywangjhu@gmail.com), 3400 N. Charles St., Baltimore, MD 21218. *Yamabe flow of asymptotically flat metrics.* Preliminary report.

In this talks, we will discuss the behavior of the Yamabe flow on an asymptotically flat (AF) manifold. We will first show that any AF manifold $(M^n; g_0)$ with the Yamabe flow existing on a time interval $[0; T]$ must satisfy a uniform L^2 Euclidean-type Sobolev inequality. This would allow us to control the $L^{n/2}$-norm of the scalar curvature along the Yamabe flow. After that, we will obtain the estimates on higher order integral norms of the scalar curvature, which will eventually lead to the long-time existence of the Yamabe flow starting from an AF manifold. If time permits, we will also discuss the convergence along the Yamabe flow. This is joint work with Eric Chen. (Received August 12, 2020)

54 ► General topology

1160-54-226 **Joseph Boninger*** (jboninger@gmail.com). *New Quantum Invariants for Links in the Thickened Torus with Volume Conjecture Behavior—Preliminary Report.* Preliminary report.

We introduce new quantum invariants for links in $T^2 \times I$, analogous to the colored Jones polynomial of links in S^3. Our interest in these invariants stems from the volume conjecture of Kashaev, Murakami, and Murakami—for at least one link in the thickened torus, the asymptotic growth of our invariants can be shown to give the hyperbolic volume of the complement in a way much like the conjecture predicts for the colored Jones polynomial. (Received August 10, 2020)

57 ► Manifolds and cell complexes

1160-57-29 **David Futer*** (dfuter@temple.edu), Mathematics Department, Suite 638, Wachman Hall, 1805 North Broad St., Philadelphia, PA 19122, and **Neil R. Hoffman**. *Infinitely many virtual geometric triangulations.*

Every cusped hyperbolic 3-manifold should admit a decomposition into a union of positively oriented ideal tetrahedra. Somewhat shockingly, the question of whether such a geometric triangulation exists is still open. Luo, Schleimer, and Tillmann proved that geometric ideal triangulations of this sort exist in some cover of every cusped 3-manifold. We extend their result by showing that every cusped hyperbolic 3-manifold has a single cover admitting an infinite sequence of geometric ideal triangulations. The proof involves double coset separability of peripheral subgroups. This is joint work with Neil Hoffman. (Received July 22, 2020)

1160-57-87 **Michael Landry**, **Yair Minsky** and **Samuel J. Taylor*** (samuel.taylor@temple.edu). *A polynomial invariant for veering triangulations.*

Veering triangulations form a rich class of ideal triangulations of cusped hyperbolic manifolds that were introduced by Agol and have connections to hyperbolic geometry, Teichmuller theory, and the curve complex. In this talk, we introduce a polynomial invariant of a veering triangulation that, when the triangulation comes from a fibration, recovers the Teichmuller polynomial introduced by McMullen. We show that in general, the polynomial determines a (typically non-fibered) face of the Thurston norm ball and that the classes contained in the cone over this face have representatives that are carried by the veering triangulation itself. (Received August 06, 2020)

1160-57-103 **Tommaso Cremaschi*** (cremach@usc.edu), 3620 S Vermont Ave, Kaprelian hall, Los Angeles, CA 90089. *Volumes and filling collections of multicurves.*

Given a link L in a Seifert-fibered space N over a surface S of negative Euler characteristic such that the fiberwise projection of L to S is a collection C of closed curves in minimal position, then N L is hyperbolic if and only if C is filling and N L is acylindrical. In this talk we will study the case where C is a pair of filling simple closed loops and N = PT(S) is the projectivized tangent bundle and L is the canonical lift of C. We will show that vol(N L) is quasi-isometric to the Weil-Petersson distance between the corresponding strata in Teichmuller space. In the more general setting, we show that vol(N L) is quasi-isometric to expressions involving distances in the pants graph whenever L is a stratified link. This is joint work with J. A. Rodriguez-Migueles and A. Yarmola. (Received August 07, 2020)

1160-57-111 **Michael Landry*** (mlandry@wustl.edu). *Veering triangulations and Thurston's norm.*

Thurston's norm on the second real homology of a hyperbolic three-manifold M enjoys broad interest in low-dimensional topology. The norm measures a type of complexity for homology classes, and its unit ball is a rational polyhedron. Work of Thurston, Fried, Mosher, McMullen, and others has shown that the combinatorics of this polyhedron are related in beautiful ways to the topology of M and to certain dynamical systems living in M. However, the understanding of this picture is mostly concentrated around fibrations of M over the circle and there is still much to discover. Veering triangulations, defined in 2010 by Agol, form a class of ideal triangulations of cusped hyperbolic three-manifolds. We show that we can use a veering triangulation to capture the data of a particular face of the norm ball after Dehn filling. We also show that the triangulation detects more refined information: the result can be summarized as "a veering triangulation sees a face at the level of isotopy, not just homology." (Received August 07, 2020)

1160-57-118 **Justin Lanier** and **Nicholas G Vlamis*** (nicholas.vlamis@qc.cuny.edu). *Mapping class groups with the Rokhlin property.*

A topological group with a dense conjugacy class is said to have the Rokhlin property. We will classify every orientable surface whose mapping class group has the Rokhlin property. We will explain how the Rokhlin property—a topological property—can be used to obstruct the existence of abstract homomorphisms with certain codomains. This work is joint with Justin Lanier. (Received August 07, 2020)

1160-57-159 **Colin Adams***, Bascom Hall, Williams College, 33 Stetson Ct., Williamstown, MA 01267, and **Michele Capovilla-Searle**, **Darin Li**, **Lily Li**, **Jacob McErlean**, **Alex Simons**, **Natalie Stewart** and **Xiwen Wang**. *Lower Bounds on Volumes of Hyperbolic 3-Manifolds.* Preliminary report.

Given a link in a 3-manifold that can be appropriately decomposed into tangles, we define certain types of hyperbolicity for the tangles and an associated volume. Using results of Agol-Storm-Thurston, we show that links obtained from the tangles are themselves hyperbolic. Further, we obtain surprisingly accurate lower bounds

on the volumes of the links from the sum of the volumes of the constituent tangles in various 3-manifolds, including S^3, $S^2 \times S^1$, $D^2 \times S^1$ and $S^1 \times S^1 \times (0,1)$. (Received August 09, 2020)

1160-57-164 **Donghao Wang*** (donghaow@mit.edu), MA. *Monopoles and Landau-Ginzburg Models.*
The monopole Floer homology of closed 3-manifolds was introduced by Kronheimer-Mrowka around 2007 and has greatly influenced the study of 3-manifold topology since its inception.

In this talk, we will generalize their construction and define the monopole Floer homology for any 3-manifolds with torus boundary. Its Euler charateristic recovers the Milnor-Turaev torsion invariant by a classical theorem of Meng-Taubes. We will also explain its relation with gauged Landau-Ginzburg models and the gauged Witten equations. (Received August 09, 2020)

1160-57-171 **Yi Xie** and **Boyu Zhang*** (bz@math.princeton.edu). *Several detection results of Khovanov homology on links.*
In this talk, I will present an excision formula for singular instanton Floer homology where the excision surfaces intersect the singular locus. This formula allows us to define an instanton Floer homology theory for sutured manifolds with tangles. Similar to the non-singular case, the instant Floer homology for tangles is non-vanishing on taut manifolds with tangles and detects trivial products. As applications, we prove several detection results for Khovanov homology. This is joint work with Yi Xie. (Received August 09, 2020)

1160-57-177 **Andrew Yarmola*** (yarmola@math.princeton.edu). *Searching for marked Kleinian groups.* Preliminary report.
We will give an overview of rigorous computational techniques and applications of parameter space searches for marked Kleinian groups. These tools open the door to several difficult results in the study of hyperbolic 3-manifolds, including questions about volumes, tubes, Dehn surgery, and Margulis constants. An important aspect of this approach is that it can answer definitive questions about infinite families of Kleinian groups. The talk will focus on the necessary ideas and constructs for implementing such searches, as well as a review of applications and a discussion of future projects. (Received August 09, 2020)

1160-57-196 **Thomas Koberda** and **Feng Luo*** (fluo@math.rutgers.edu), Dept of Math, Rutgers University, Piscataway, NJ 08854, and **Hongbin Sun**. *Some remarks on linear representations of the mapping class groups.*
We will discuss some of the observations we made on the linear representations of the mapping class groups of a surface of genus at least 3. This is a joint work with Thomas Koberda and Hongbin Sun. (Received August 10, 2020)

1160-57-220 **Pablo Portilla Cuadrado** and **Nick Salter***, Department of Mathematics, 2990 Broadway, New York, NY 10027. *Plane curve singularities and mapping class groups.*
The theory of plane curve singularities is a meeting ground for many topics in mathematics, including algebraic geometry, analysis, topology, and geometric group theory. I will discuss some recent work, joint with Pablo Portilla Cuadrado, that answers some basic topological questions about plane curve singularities. Our results make essential use of some developments in the theory of "framed mapping class groups", representing joint work between myself and Aaron Calderon. (Received August 10, 2020)

1160-57-288 **Diana Hubbard*** (diana.hubbard@brooklyn.cuny.edu), **K. Kawamuro**, **F. C. Kose**, **G. Martin**, **O. Plamenevskaya**, **K. Raoux**, **L. Truong** and **H. Turner**. *Braids and Slice Genus Bounds.*
Knots in the three-sphere can be represented by braids, which can in turn be thought of as diffeomorphisms of punctured disks. One can ask to what extent properties of such a diffeomorphism dictate topological properties of the corresponding knot. Intuitively, a number called the fractional Dehn twist coefficient measures how much a braid twists about the boundary of the punctured disk. In this talk I will discuss some observations relating the fractional Dehn twist coefficient of a braid to the slice genus of the corresponding knot. This work arose from a research group at the Women in Symplectic and Contact Geometry Workshop held at ICERM in Summer 2019. (Received August 11, 2020)

1160-57-297 **Edgar A. Bering IV*** (edgar.bering@temple.edu) and **David Futer**. *Effective special covers of alternating links.*
The "virtual conjectures" in low-dimensional topology, stated by Thurston in 1982, postulated that every hyperbolic 3-manifold has finite covers that are Haken and fibered, with large Betti numbers. These conjectures

were resolved in 2012 by Agol and Wise, using the machine of special cube complexes. Since that time, many mathematicians have asked how big a cover one needs to take to ensure one of these desired properties.

We begin to give a quantitative answer to this question, in the setting of alternating links in S^3. If a prime, alternating link L has a diagram with n crossings, we prove that the complement of L has a special cover of degree less than $12(n-1)!$. As a corollary, we bound the degree of the cover required to get Betti number at least k. We also quantify residual finiteness, bounding the degree of a cover where a closed curve fails to lift. (Received August 11, 2020)

1160-57-320 **Joel Hass**, **Abigail Thompson** and **Anastasiia Tsvietkova*** (n.tsvet@gmail.com), 34 Conforti Ave, West Orange, NJ 07052. *Tangle decompositions of alternating link complements.*

Decomposing knots and links into tangles is a useful technique for understanding their properties. The notion of prime tangles was introduced by Kirby and Lickorish; Lickorish proved that by summing prime tangles one obtains a prime link. In a similar spirit, summing two prime alternating tangles will produce a prime alternating link, if summed correctly with respect to the alternating property. Given a prime alternating link, we seek to understand whether it can be decomposed into two prime tangles each of which is alternating. We refine results of Menasco and Thistlethwaite to show that if such a decomposition exists either it is visible in an alternating link diagram or the link is of a particular form, which we call a pseudo-Montesinos link. This is joint work with Joel Hass and Abigail Thompson. (Received August 11, 2020)

1160-57-340 **William Menasco** and **Margaret Nichols*** (nicholsm@buffalo.edu). *Characterizing $S^2 \hookrightarrow \mathbb{R}^3$ via the crease set.*

The crease set of a surface embedded in $\mathbb{R}^3$ captures where the surface folds under a choice of projection $\mathbb{R}^3 \to \mathbb{R}^2$. In forthcoming work, we develop tools to characterize the crease set of an embedded sphere and its behavior under isotopy of the embedding. In this talk, we introduce some of these tools, explore some examples, and discuss potential applications. (Received August 11, 2020)

58 ► Global analysis, analysis on manifolds

1160-58-224 **Cheikh Birahim Ndiaye*** (cheikh.ndiaye@howard.edu), Washington, DC 20059. *Variational Theory for Yamabe type problems.*

In this talk, we will discuss a Variational Theory for Yamabe type problems. We will show how the combination of the Aubin-Schoen's minimizing technique and the Bahri-Coron 's algebraic topological argument leads to a natural solution for such problems without any Positive Mass type assumption. To emphasize the main ideas of the Theory, we will focus on the Fractional Yamabe problem with parameter $\gamma \in (1,2)$ discussing a joint work with Y. Sire and L. Sun. (Received August 10, 2020)

1160-58-229 **Daniel Offin*** (offind@queensu.ca), 196 White Lake Road, Godfrey,, Ontario K0H 1T0, and **Henry Kavle**. *Searching for stable orbits, via index theory.*

Starting with the geodesic flow on a tri-axial ellipsoid, we discuss variational methods and index theory for stable periodic orbits on an energy surface. (Received August 10, 2020)

1160-58-299 **Chris Kottke*** (ckottke@ncf.edu). *Large scale metric geometry of monopoles and L^2 cohomology.* Preliminary report.

The moduli spaces M_k of $SU(2)$ monopoles on R^3 of charge k are among the oldest studied objects in gauge theory, yet open questions still remain, such as Sen's conjecture for their L^2 cohomology. I will report on two interrelated projects which make progress in this direction. The first, with K. Fritzsch and M. Singer, concerns compactifications of these moduli spaces, exibiting their hyperKähler metrics as 'quasi fibered boundary' (QFB) metrics, a metric structure which generalizes the 'quasi asymptotically locally euclidean' (QALE) and 'quasi asymptotically conic' (QAC) structures introduced by Joyce and others. The second, with F. Rochon, concerns the analysis of L^2 cohomology on QFB manifolds, which under certain conditions may be identified with certain intersection cohomology groups of associated stratified spaces. (Received August 11, 2020)

60 ► Probability theory and stochastic processes

1160-60-38 **Andrew V Sills*** (asills@georgiasouthern.edu). *A q-Binomial Probability Distribution.* q-analogs of special functions play a central role in mathematics and physics. Starting with Box and Cox in 1964, a variety of q-analogs of (continuous) probability distributions have been introduced, where the role of q is invariably as a parameter that can assume a range of real values, additional to the parameters of the original distribution. Here, I propose a q-analog of a discrete probability distribution, namely the binomial distribution. Here the "q" is a formal variable that encodes and preserves information from the binomial experiment associated with the binomial random variable, in its exponent and coefficient. (Received July 29, 2020)

1160-60-78 **Jian Ding** and **Ewain Gwynne*** (ewain@uchicago.edu). *Tightness of supercritical Liouville first passage percolation.*

Liouville first passage percolation (LFPP) with parameter $\xi > 0$ is the family of random distance functions on the plane obtained by integrating $e^{\xi h_\epsilon}$ along paths, where h_ϵ for $\epsilon > 0$ is a mollification of the planar Gaussian free field. Previous work by Ding-Dubédat-Dunlap-Falconet and Gwynne-Miller showed that there is a $\xi_{\text{crit}} > 0$ such that for $\xi < \xi_{\text{crit}}$, LFPP converges under re-scaling to a random metric on the plane which induces the same topology as the Euclidean metric: the so-called γ-Liouville quantum gravity (LQG) metric for $\gamma \in (0, 2)$.

Recently, Jian Ding and I showed that LFPP admits non-trivial subsequential scaling limits for all $\xi > 0$. For $\xi > \xi_{\text{crit}}$, the subsequential limiting metrics do not induce the Euclidean topology. Rather, there is an uncountable, dense, Lebesgue measure-zero set of points $z \in \mathbb{C}$ such that $D_h(z, w) = \infty$ for every $w \in \mathbb{C} \setminus \{z\}$. We expect that the subsequential limiting metrics are related to LQG with matter central charge in $(1, 25)$.

I will discuss the subsequential limiting metrics, their connection to LQG, and several open problems. (Received August 05, 2020)

1160-60-170 **Dor Elboim**, **Danny Nam** and **Allan Sly***, Fine Hall, Department of Mathematics, Princeton University, Princeton, NJ. *Critical One-dimensional Multi-particle DLA.*

In multi-particle Diffusion Limited Aggregation (DLA) a sea of particles perform independent random walks until they run into the aggregate and are absorbed. In dimension 1, the rate of growth of the aggregate depends on lambda, the density of the particles. Kesten and Sidoravicius proved that when $\lambda < 1$ the aggregate grows like $t^{1/2}$. They furthermore predicted linear growth when $\lambda > 1$ (subsequently confirmed) and $t^{2/3}$ growth at the critical density $\lambda = 1$.

We address the critical case, confirming the $t^{2/3}$ rate of growth and show that aggregate has a scaling limit whose derivative is a self-similar diffusion process. Surprisingly this contradicts conjectures on the speed in the mildly supercritical regime when $\lambda = 1 + \epsilon$. (Received August 09, 2020)

1160-60-174 **Kavita Ramanan*** (kavita_ramanan@brown.edu), RI , and **Ankan Ganguly**. *Approximations for Interacting Particle Systems on Tree-like Graphs.* Preliminary report.

We consider interacting particle systems with particles indexed by the nodes of a locally finite graph, and whose evolution is described by a (possibly non-Markovian) jump process on a countable state space, where the transitions of each particle depends only on the states of particles in its neighborhood. Models that fall under this framework include the contact process, voter model and neuronal Hawkes process. We establish limit theorems for the dynamics and empirical measures when the graphs converge (in the local weak sense) to a limit infinite graph. Moreover, when the limit graph is tree-like, we provide an autonomous characterization of the dynamics of a typical particle and its neighbors in terms of certain local equations. These local equations can be viewed as the sparse analog of mean field equations, which are used to approximate interacting particle systems on dense graphs. Along the way we establish a Markov random field property of the dynamics and an extension of the Clifford-Hammersley theorem that may be of independent interest. (Received August 09, 2020)

1160-60-263 **Morris Ang**, **Nina Holden** and **Xin Sun*** (xinsun@sas.upenn.edu), 209 S 33rd Street, 4N24, Philadelphia, PA 19104-6395. *Conformal welding of quantum surfaces from Liouville CFT.*

Liouville conformal field theory (LCFT) defines a family of quantum surfaces with boundaries. We show that this family of surfaces is closed under the operation of conformal welding, where the interfaces are variants of SLE whose law can be described explicitly. This leads to exact results for both SLE and LCFT. Joint work with Morris Ang and Nina Holden. (Received August 10, 2020)

1160-60-270 **Ewain Gwynne** and **Nina Holden***, nina.holden@eth-its.ethz.ch, and **Joshua Pfeffer** and **Guillaume Remy**. *Liouville quantum gravity with matter central charge in* $(1, 25)$*: a probabilistic approach.*

Liouville quantum gravity (LQG) is a theory of random fractal surfaces with origin in the physics literature in the 1980s. Most literature is about LQG with matter central charge $\mathbf{c} \in (-\infty, 1]$. We introduce and study a discretization of LQG which makes sense for $\mathbf{c} \in (-\infty, 25)$. In particular, we show that the cardinality of a graph distance ball of radius r grows faster than any power of r for $\mathbf{c} \in (1, 25)$ (which suggests that the Hausdorff dimension of LQG surfaces for such $\mathbf{c}$ is infinite) and that the so-called KPZ formula extends to $\mathbf{c} \in (1, 25)$. Based on a joint work with Gwynne, Pfeffer, and Remy. (Received August 10, 2020)

1160-60-279 **Hugo Duminil-Copin*** (hugo.duminil@unige.ch), 2-4 rue du lievre, 1211 Geneve, Switzerland. *Marginal triviality of the scaling limits of critical 4D Ising and phi44 models.*

In this talk, we will discuss the scaling limits of spin fluctuations in four-dimensional Ising-type models with nearest-neighbor ferromagnetic interaction at or near the critical point are Gaussian and its implications from the point of view of Euclidean Field Theory. Similar statements will be proven for the $\lambda\phi 4$ fields over R^4 with a lattice ultraviolet cutoff, in the limit of infinite volume and vanishing lattice spacing. The proofs are enabled by the models' random current representation, in which the correlation functions' deviation from Wick's law is expressed in terms of intersection probabilities of random currents with sources at distances which are large on the model's lattice scale. Guided by the analogy with random walk intersection amplitudes, the analysis focuses on the improvement of the so-called tree diagram bound by a logarithmic correction term, which is derived here through multi-scale analysis. Joint work with Michael Aizenman. (Received August 11, 2020)

1160-60-283 **Reza Gheissari** (gheissari@berkeley.edu) and **Eyal Lubetzky*** (eyal@courant.nyu.edu). *Rigidity and large deviations of 3D Ising interfaces.*

We will present an overview of recent developments on the rigidity and shape of the interface of the classical 3D Ising model with Dobrushin's boundary conditions at low temperature, given parts of the surface that atypically propel its height. These generalize results of the authors on the tightness of the maximum of the 3D Ising interface, and have applications both static and dynamic. (Received August 11, 2020)

1160-60-309 **Julien Dubedat***, 2990 Broadway, Department of Mathematics, Columbia University, New York, NY 10027, and **Guillaume Remy**, 2990 Broadway, Columbia University, Department of Mathematics, New York, NY 10027. *A deformation perspective on Liouville Conformal Field Theory.*

The object of Conformal Field Theory (CFT) are models of statistical mechanics - in the continuum - defined on an underlying 2d domain or surface, and satisfying some conformal invariance, so that the laws of the fields encoding the random information of a given model depend only on the complex structure of the underlying surface. The study and description of variations (deformations) of complex structures is classical in complex geometry. This leads to a generic approach to representation-theoretic aspects of CFT, in particular degenerate highest-weight representations, leading to Belavin-Polyakov-Zamolodchikov (BPZ) differential equations. In recent years, Liouville CFTs have seen intense activity on the mathematical side, following in particular work by David, Kupiainen, Rhodes and Vargas. This gives a rich, but still tractable, family of CFTs. In the work presented here, we implement the deformation approach in the case of LCFT. (Received August 11, 2020)

1160-60-322 **Lionel Levine*** (levine@math.cornell.edu) and **Vittoria Silvestri** (silvestri@mat.uniroma1.it). *How far do Activated Random Walkers spread from a single source?* Preliminary report.

Activated Random Walk is the most promising candidate for a "universal model of self-organized criticality". I'll state several conjectures that aim to convert these buzzwords ("universal model of SOC") into mathematics. Among the conjectured properties are **hyperuniformity** (the variance of the number of particles in a ball scales sublinearly with volume), and **fast mixing** (cutoff in total variation when the system reaches its critical density). (Received August 11, 2020)

1160-60-339 **Philippe Sosoe*** (psosoe@math.cornell.edu), Malott Hall, 301 Tower Road, Ithaca, NY 14853, and **Christian Noack** (noack@cornell.edu), Malott Hall, 301 Tower Road, Ithaca, NY 14853. *Fluctuation bounds for integrable directed polymers and diffusions.*

We present a method to obtain bounds of near optimal order for the central moments of integrable polymers, including the log-Gamma polymer. We will also mention recent work on systems of diffusions with KPZ-like behavior to which some of the techniques may also be extended.

Joint work with Christian Noack. (Received August 11, 2020)

65 ► Numerical analysis

1160-65-31 **Yanxiang Zhao*** (yxzhao@email.gwu.edu). *Maximum Principle Preserving Schemes for Binary Systems with Long-range Interactions.*

We study some maximum principle preserving and energy stable schemes for the Allen-Cahn-Ohta-Kawasaki model with fixed volume constraint. With the inclusion of a nonlinear term in the Ohta-Kawasaki free energy functional, we show that the Allen-Cahn-Ohta-Kawasaki dynamics is maximum principle preserving. We further design some first order energy stable numerical schemes which inherit the maximum principle preservation in both semi-discrete and fully-discrete levels. Furthermore, we apply the maximum principle preserving schemes to a general framework for binary systems with long-range interactions. We also present some numerical results to support our theoretical findings. (Received July 26, 2020)

1160-65-61 **Duan Chen*** (dchen10@uncc.edu), 4026 Riseley Ln, Charlotte, NC 28270-1558. *Randomized matrix compression algorithms for fast scientific computations.*

In modern scientific computations, such as large scale differential-integral equations, optimizations, or machine learning algorithms, we face challenges from both "massive" and "complicated" data. When the number of variables N is large, iterative solvers or solution updating processes and data storage become extremely expensive. Further, generating matrix entries itself could be time consuming or subject to information lost and experimental uncertainty. Our research is motivated by two types of problems, the hypersingular volume integral equation in layered materials, and manifold-based optimization for predicting 3D structures of chromosomes from experimental data. We introduce randomized numerical linear algebra (RNLA) based algorithms to perform low-rank approximation of given data, such that the sequential matrix operations are highly efficient. Our algorithms (1) only use partial information of the original matrix entries and (2) the compression process itself is cheaper than the direct matrix-vector multiplication. Accuracy and statistical robustness are analyzed with respect to sampling strategies. Numerical results will also be displayed for the two applications. (Received August 02, 2020)

1160-65-80 **Margaret Beck**, **Eric Cooper**, **Gabriel Lord** and **Konstantinos Spiliopoulos*** (kspiliop@math.bu.edu), Department of Mathematics and Statistics, Boston University, Boston, MA 02215. *Selection of quasi-stationary states in the stochastically forced Navier-Stokes equation on the torus.*

The stochastically forced vorticity equation associated with the two dimensional incompressible Navier-Stokes equation on $D\delta := [0, 2\pi\delta]\times[0, 2\pi]$ is considered for $\delta \approx 1$, periodic boundary conditions, and viscocity $0 < \nu \ll 1$. An explicit family of quasi-stationary states of the deterministic vorticity equation is known to play an important role in the long-time evolution of solutions both in the presence of and without noise. Recent results show the parameter δ plays a central role in selecting which of the quasi-stationary states is most important. In this work, we aim to develop a finite dimensional model that captures this selection mechanism for the stochastic vorticity equation. This is done by projecting the vorticity equation in Fourier space onto a center manifold corresponding to the lowest eight Fourier modes. Through Monte Carlo simulation, the vorticity equation and the model are shown to be in agreement regarding key aspects of the long-time dynamics. Following this comparison, perturbation analysis is performed on the model via averaging and homogenization techniques to determine the leading order dynamics for statistics of interest for $\delta \approx 1$. (Received August 05, 2020)

1160-65-83 **Yukun Li***, 200 Isle of Sky Cir, Orlando, FL 32828, and **Yi Zhang**. *Analysis of adaptive two-grid finite element algorithms for linear and nonlinear problems.*

This talk discusses some efficient and accurate adaptive two-grid (ATG) finite element algorithms for linear and nonlinear partial differential equations (PDEs). The main idea of these algorithms is to use the information of the solutions on the k-th level adaptive meshes to find the solutions on the (k+1)-th level adaptive meshes which are constructed by performing adaptive element bisections to the k-th level adaptive meshes. These algorithms transform the non-symmetric positive definite (non-SPD) PDEs into symmetric positive definite (SPD) PDEs or transform the nonlinear PDEs into the linear PDEs. The proposed algorithms are both accurate and efficient due to the following advantages: they do not need to solve the non-symmetric or nonlinear systems; the degrees of freedom (d.o.f.) are very small; they are easily implemented; the interpolation errors are very small. Next, it constructs residue-type a posteriori error estimators, which are shown to be reliable and efficient. The key ingredient in proving the efficiency is to establish an upper bound of the oscillation terms, which may not be higher-order terms (h.o.t.) due to the low regularity of the numerical solution. Furthermore, the convergence of the algorithms is proved when bisection is used for the mesh refinement. (Received August 05, 2020)

1160-65-85 **Xiaoming He***, hex@mst.edu. *A diffuse interface model and semi-implicit energy stable finite element method for two-phase MHD flows.*

In this talk, we present a diffuse interface model and finite element approximation for two-phase magnetohydrodynamic (MHD) flows with different viscosities and electric conductivities. An energy stable scheme, which is based on the finite element method for the spatial discretization and first order semi-implicit scheme combined with convex splitting method for the temporal discretization, is proposed to solve this new model. The numerical scheme is proved to be mass-conservative and energy law preserving. By Leray-Schauder fixed point theorem, the existence of solutions to the numerical scheme is shown. The uniqueness of the numerical solutions is obtained. Utilizing the stability of the numerical scheme and the compactness method, the existence of the weak solutions to the two-phase MHD model is established as well. Furthermore, given more regularity on the weak solution, the convergence of the numerical scheme is derived. Finally, numerical experiments are provided to verify the theoretical results and validate the proposed model. (Received August 06, 2020)

1160-65-119 **Zhonggang Zeng*** (zzeng@neiu.edu), Department of Mathematics, Northeastern Illinois University, Chicago, IL 60625. *A Newton's Iteration on Zeros of Higher Dimensions.*

The textbook Newton's iteration is practically inapplicable on positive dimensional solutions of unregularized nonlinear systems. With a simple modification, a version of Newton's iteration regains its local quadratic convergence to nonisolated zeros of smooth mappings assuming the solutions are semiregular as properly defined regardless of whether the system is square, underdetermined or overdetermined. More importantly, the iteration serves as a de facto regularization mechanism for computing singular zeros from empirical data. Even if the given system is perturbed so that the nonisolated solution disappears, the iteration still locally converges to a stationary point that approximates a solution of the underlying system with an error bound in the same order of the data accuracy. Geometrically, the iteration approximately converges to the nearest point on the solution manifold. This extension simplifies a broad range of models by eliminating the zero isolation process and enables a wide scope of applications in algebraic computation. (Received August 07, 2020)

1160-65-128 **Constantin Bacuta*** (bacuta@udel.edu), 64 Avignon Drive, Newark, DE 19702, and **Jacob Jacavage** and **Daniel Hayes**. *Least Squares Discretization and Preconditioning for Mixed Variational Formulations.*

We introduce a least squares method for discretizing boundary value problems written as mixed variational formulations. The method could be extend to variational formulations that correspond to non-linear PDEs. At the continuous level we assume a stability LBB condition and data compatibility. The proposed discretization method is associated with standard discretization of an equivalent saddle point reformulation of the original mixed problem. We use the concepts of optimal and almost optimal test norm and provide estimates for the continuity and stability constants. A preconditioning strategy for solving the discrete mixed formulations is used in combination with the special test norms. We further provide a choice for a discrete trial space, that depends on the choice of a standard test space and leads to discrete stability, when using the appropriate test norm. Applications of the method include discretizations of first order systems of parametric PDEs, such as the time-hamonic Maxwell equations and second order PDEs with variable or discontinuous coefficients. (Received August 07, 2020)

1160-65-133 **Li Wang*** (wang8818@umn.edu), 206 Church St SE, Minneapolis, MN 55455. *A proximal gradient algorithm for crystal surface evolution.*

As a counterpoint to recent numerical methods for crystal surface evolution, which agree well with microscopic dynamics but suffer from significant stiffness that prevents simulation on fine spatial grids, we develop a new numerical method based on the macroscopic partial differential equation, leveraging its formal structure as the gradient flow of the total variation energy, with respect to a weighted H-1 norm. Inspired by the classical minimizing movements scheme, we develop a novel semi-implicit time discretization of the gradient flow. We then use a primal dual hybrid gradient (PDHG) method to compute each element of the semi-implicit scheme. In one dimension, we prove convergence of the PDHG method to the semi-implicit scheme, under general integrability assumptions on the mobility and its reciprocal. Finally, by taking finite difference approximations of our PDHG method, we arrive at a fully discrete numerical algorithm, with iterations that converge at a rate independent of the spatial discretization: in particular, the convergence properties do not deteriorate as we refine our spatial grid. (Received August 08, 2020)

1160-65-202 **Jinchao Xu*** (xu@math.psu.edu), 314 McAllister Building, Pennsylvania State University, State College, PA 16802. *Error Estimates for Deep Neural Networks with Applications to Numerical PDEs.*

In this talk, I will present some asymptotic approximation results for deep neural networks and and their applications to numerical solutions for partial differential equations. (Received August 10, 2020)

1160-65-265 **Andreas C. Aristotelous*** (a.aristotelous@gmail.com). *Energy Stable Discontinuous Galerkin Finite Element Solutions of Two-Phase Flows.*

Discontinuous Galerkin Finite Element methods for the solution of Cahn-Hilliard type equations will be presented. Numerical analysis of the schemes will be performed. Numerical tests verifying the theoretical results will be presented. (Received August 10, 2020)

1160-65-300 **Wenrui Hao*** (wxh64@psu.edu), 208 McAllister bldg, University Park, PA 16801. *Nonlinear computations in neural networks.*

It is well known that the nonlinearity plays important roles in deep neural networks but still remain unclear in the mathematical context. In this talk, I will study neural networks from a nonlinear computation point of view and propose a novel training and initialization strategy that is based on the nonlinear structure of neural networks. The proposed strategy is derived from the polynomial approximation of activation functions by using theories of numerical algebraic geometry to guarantee to find all the local minima. (Received August 11, 2020)

1160-65-302 **Dhagash Mehta*** (dhagashbmehta@gmail.com), **Edgar A Bernal**, **Jonathan D Hauenstein**, **Margaret Regan** and **Tingting Tang**. *Machine learning discriminant locus.*

Parameterized systems of polynomial equations arise in many applications in science and engineering with the real solutions describing, for example, equilibria of a dynamical system, linkages satisfying design constraints, and scene reconstruction in computer vision. Since different parameter values can have a different number of real solutions, the parameter space is decomposed into regions whose boundary forms the real discriminant locus. We view the problem of locating the real discriminant locus as a supervised classification problem in machine learning where the goal is to determine classification boundaries over the parameter space, with the classes being the number of real solutions. At each sample point on the parameter space, homotopy continuation is used to obtain the number of real solutions to the corresponding polynomial system. Machine learning techniques including nearest neighbor and deep learning are used to efficiently approximate the real discriminant locus. One application of having learned the real discriminant locus is to develop a real homotopy method that only tracks the real solution paths unlike traditional methods which track all complex solution paths. We will also discuss a few examples such as those arising from the equilibria of the Kuramoto model. (Received August 11, 2020)

1160-65-328 **Yuan Liu*** (yuan.liu@wichita.edu), 1845 Fairmount St., Wichita, KS 67260, and **Yingda Cheng**, **Shanqin Chen** and **Yong-Tao Zhang**. *Krylov implicit integration factor discontinuous Galerkin methods on sparse grids for high dimensional reaction-diffusion equations.*

Reaction diffusion system are most commonly used mathematical models for pattern formation in computational biology. In this talk, we will discuss a class of Krylov implicit integration factor (IIF) discontinuous Galerkin (DG) methods on sparse grids to solve reaction-diffusion equations on high spatial dimensions. The key ingredient of spatial DG discretization is the multiwavelet bases on nested sparse grids, which can significantly reduce the numbers of degrees of freedom. To deal with the stiffness of the DG spatial operator in discretizing reaction-diffusion equations, we apply the efficient IIF time discretization methods, which are a class of exponential integrators. Krylov subspace approximations are used to evaluate the large size matrix exponentials resulting from IIF schemes for solving PDEs on high spatial dimensions. (Received August 11, 2020)

1160-65-354 **Elliot Shi***, RISE Group. *Bio-Imaging Analysis Using Computational and Mathematical Analysis.*

A significant amount of work is required for the morphological reconstruction of certain bio-images in comparison to a mere quantitative bioimaging analysis. In the past, bio-images were enhanced by means of morphological image processing, preprocessing, and watershed segmentation; however, such methods did not equate to significant changes in image quality. This paper showed the use of spectrum modification algorithms applied to resolution enhancement and segmentation of stem cells and digital images. With the ultimate goal of producing a better bioimage, various enhancement methods were applied to the raw data and images of stem cells. Presented research was conducted as an analysis of the combination of the histogram analysis and proposed transformation. In this paper, a strategy to enhance bio-images is put forward as a result of linear and quadratic

transformations. With the assistance of histogram equalization of stem cells, we were able to obtain quantitative information about the cells: their shape, size, coloration, and overall condition. As a result, we observed the effects of an improved algorithm with the ability to enhance stem cell images and other digital images. (Received August 11, 2020)

68 ► Computer science

1160-68-32 **Wei Zhu*** (zhu@math.umass.edu), Department of Mathematics and Statistics, University of Massachusetts Amherst, 710 N. Pleasant Street, 1623D, Amherst, MA 01003, and **Qiang Qiu**, **Robert Calderbank**, **Guillermo Sapiro** and **Xiuyuan Cheng**. *Scale-Equivariant Neural Networks with Decomposed Convolutional Filters.*

Encoding the input scale information explicitly into the representation learned by a convolutional neural network (CNN) is beneficial for many vision tasks especially when dealing with multiscale input signals. I will explain, in this talk, a scale-equivariant CNN architecture with joint convolutions across the space and the scaling group, which is shown to be both sufficient and necessary to achieve scale-equivariant representations. To reduce the model complexity and computational burden, we decompose the convolutional filters under two pre-fixed separable bases and truncate the expansion to low-frequency components. A further benefit of the truncated filter expansion is the improved deformation robustness of the equivariant representation. Numerical experiments demonstrate that the proposed scale-equivariant neural network with decomposed convolutional filters (ScDCFNet) achieves significantly improved performance in multiscale image classification and better interpretability than regular CNNs at a reduced model size. (Received July 26, 2020)

1160-68-114 **Jonathan Gryak*** (gryakj@med.umich.edu), **Robert M Haralick** and **Delaram Kahrobaei**. *Solving Algorithmic Problems in Algebraic Structures via Machine Learning.*

Computer algebra systems such GAP and SageMath have been very successful in providing tools to explore groups, rings, and other algebraic structures. These systems take advantage of a number of rich mathematical theories at their disposal, including combinatorial/geometric group theory, representation theory, and algebraic geometry.

However, there are many classes of algebraic structures for which these techniques are inapplicable. For others, decision, search, and other algorithmic questions are known to be solvable, yet no efficient algorithms are known. In such cases machine learning may be able to elucidate these problems. There are a number of ways that machine learning can be used: to train models that solve these problems in most instances, to create heuristics to reduce computational time in search algorithms, to optimize algorithm selection for a given problem, and to provide another method of exploring the mathematical nature of these structures.

As a prototypical example, we show how the conjugacy decision problem can be solved in a variety of infinite groups using random forest and n-tuple neural networks. These models have the advantage of being discrete and easily interpretable, and can be utilized to explore a variety of algebraic structures beyond groups. (Received August 07, 2020)

1160-68-221 **Michael Gene Dobbins***, 4400 Vestal Pkwy E, Binghamton, NY 13902, and **Linda Kleist**, **Tillmann Miltzow** and **Paweł Rzążewski**. *Area Universality and the real polynomial-time hierarchy.*

A plane graph is area universal when, for every assignment of positive area to each inner face, there is a planar drawing of the graph such that each face has its assigned area. The Area Universality decision problem asks whether a given planar graph is area universal. How hard is this decision problem? Area Universality is conjectured to be complete for the universal-existential-real complexity class, which consists of problems that are polynomial-time Turing reducible to deciding the validity of a given formula in prenex normal form with a string of universally quantified real variables followed by a string of existentially quantified real variables. I will present some partial results related to this conjecture and say more about this class. This is joint work with Linda Kleist, Tillmann Miltzow, and Paweł Rzążewski. (Received August 10, 2020)

1160-68-326 **Alexander E. Holroyd** (holroyd@uw.edu), **Lionel Levine*** (levine@math.cornell.edu) and **Peter Winkler** (peter.winkler@dartmouth.edu). *Abelian Logic Gates.*

An **abelian processor** is an automaton whose output is independent of the order of its inputs. We prove that any finite abelian processor can be emulated by a network of certain simple gates. The most fundamental gate is a **toppler**, which absorbs input particles until their number exceeds some given threshold, at which point it emits one particle and returns to its initial state.

Our results can be reformulated in terms of the function computed by an abelian processor. One consequence is that any increasing function $f : \mathbb{N}^k \to \mathbb{N}^\ell$ that is the sum of a linear function and a periodic function can be expressed in terms of (possibly nested) sums of floors of quotients by integers, for example

$$f(m,n) = \left\lfloor \frac{m+1}{2} \right\rfloor + \left\lfloor \frac{\lfloor \frac{3n}{4} \rfloor + \lfloor \frac{5m+6n}{7} \rfloor + 8}{9} \right\rfloor .$$

(Received August 11, 2020)

1160-68-346 **Eva Tardos*** (eva.tardos@cornell.edu), Department of Computer Science, Cornell University, Ithaca, NY 14853. *Adversarial Perturbations of Opinion Dynamics in Networks.* People's opinions are fundamentally influenced by their interactions with other individuals. In the computer science and social science literature, the interplay between network structure and opinion formation has been extensively studied to understand phenomena of consensus and polarization, as well as the role of interventions to unidirectionally shift the overall sentiment on a particular issue. In recent years, however, concomitant with the prevalence of online social media platforms, external actors have exhibited somewhat more complicated behaviors with a more abstract goal to induce discord in a social network. In this joint work with Jason Gaitonde and Jon Kleinberg, we study this recent phenomenon using a classical model of opinion dynamics. The main questions we address in this work are (i) how to model such adversaries, (ii) what is the connection between network structure and the adversary's ability to induce discord in this model, and (iii) what can a network defender do to mitigate these actions? (Received August 11, 2020)

70 ► Mechanics of particles and systems

1160-70-17 **Richard W Montgomery*** (rmont@ucsc.edu), 223 Dickens Way, Santa Cruz, CA 95064-1058. *N body scattering and billiards.* Preliminary report.
For a large open set of solutions to the positive energy Newtonian N-body problem the $\binom{N}{2}$ mutual distances diverge linearly in both time directions. For these "hyperbolic solutions" we can set up a scattering map based on a compactification of phase space in the spirit of McGehee, or, alternatively, of Melrose, which gives the hyperbolic solutions a place to go. Those hyperbolic scattering solutions which never stray far from infinity will limit onto a non-deterministic "flow" at infinity described via a "point billiard system". The talk's goal is to state and sketch a proof of a shadowing lemma, in the spirit of Bolotin-MacKay, asserting that nearby to any such point billiard trajectory is a solution to the N-body equations. The talk is a synthesis of discussions and past works with involved Nathan Duignan, Jacques Fejoz, Andreas Knauf, Rafe Mazzeo, Rick Moeckel, and Guowei Yu. (Received July 09, 2020)

1160-70-356 **Andy Ruina*** (ruina@cornell.edu), 566 Upson Hall, Mechanical Engineering, Cornell U, Ithaca, NY 14853, Ithaca, NY 14853. *What is a reasonable definition of rotation for a non-rigid object?*
Even in 2D, there is no obvious simple definition for the rotation θ of a non-rigid object. Translation is sensibly defined by the motion of the system's center of mass. But there is no single accepted coordinate system (or frame), that is defined by the system's distribution of mass, whose rotation is generally accepted as the system rotation.

We usually think of rotation as something whose rate is related to angular momentum. But angular momentum is not the derivative of anything. For example, a cat can turn over (have net rotation) while always having zero angular momentum.

A sensible definition of rotation surely: 1)Agrees with the net rigid rotation if the system ever returns to a configuration with no strain; and 2) the rotation rate is finite if the mass is not all collapsed on to a line. These preclude defining rotation in terms of moments of inertia.

Any triad based on 3 or more non-colinear mass points can serve to define a system's rotational state and meets the minimal criteria above. Is there a reasonable way to pick a best from amongst these?

Who cares? For robotics we want minimal representation of the average orientation of a robot. (Received August 12, 2020)

74 ► Mechanics of deformable solids

1160-74-336 **L Mahadevan***, lmahadev@g.harvard.edu. *Geometry and dynamics in chemical precipitation and mechanical frustration.*

Form both enables and constrains function. Thus the understanding and control of geometric patterns in space-time is critical for all science and technology. I will provide two vignettes at the nexus of geometry and pattern formation. In the first example, I will consider the formation of hard precipitate patterns in a biomineralization experiment (involving Silica-Barium-Carbonate) that leads to vases, flowers, helicoids etc. The governing reaction-diffusion equations lead to a set of coupled geometric PDE with an approximate closure based on symmetry and scale separation arguments. I will discuss some simple aspects of the solutions of the PDE that explain our experimental observations. In the second example, I will consider the self-organization of soft sticky slender fibrils into a bundle (seen in a variety of amyloid diseases and synthetic systems) whose shape is determined by the geometric frustration induced by adhesion and elasticity. The governing equations are a generalization of the Euler rigid body equations (with time -> space) with both integrable and chaotic solutions that correspond to past observations. Along the way, in both problems, I will try and raise some interesting mathematical questions. (Received August 11, 2020)

76 ► Fluid mechanics

1160-76-24 **Eleftherios Gkioulekas*** (eleftherios.gkioulekas@utrgv.edu), University of Texas Rio Grande Valley, School of Mathematical and Statistical Scienc, 1201 West University Drive, Edinburg, TX 78539-2909. *The effect of the asymmetric Ekman term on the phenomenology of the two-layer quasigeostrophic model.*

In the two-layer quasi-geostrophic model, the friction between the flow at the lower layer and the surface boundary layer, placed beneath the lower layer, is modeled by the Ekman term, which is a linear dissipation term with respect to the horizontal velocity at the lower layer. The Ekman term appears in the governing equations asymmetrically; it is placed at the lower layer, but does not appear at the upper layer. A variation, proposed by Phillips and Salmon, uses extrapolation to place the Ekman term between the lower layer and the surface boundary layer, or at the surface boundary layer. We present theoretical results that show that in either the standard or the extrapolated configurations, the Ekman term dissipates energy at large scales, but does not dissipate potential enstrophy. It also creates an approximately symmetric stable distribution of potential enstrophy between the two layers. The behavior of the Ekman term changes fundamentally at small scales. (Received July 18, 2020)

1160-76-79 **Boris Khesin*** (khesin@math.toronto.edu), Dept of Mathematics, University of Toronto, 40 St. George Street, Toronto, Ontario M5S 2E4, Canada. *3D fluids and KAM.*

We show that the dynamical system defined by the hydrodynamical Euler equation on any closed Riemannian 3-manifold M is not mixing in the C^k topology for sufficiently high k. Furthermore, it turns out that this non-mixing property has a local nature: in any neighbourhood of a "typical" steady solution there is a generic set of initial conditions, such that the corresponding Euler flows will never enter a vicinity of the original steady one. The proof is based on a KAM-type argument which allows to generate knotted invariant tori from elliptic orbits in vorticity fields. This is a joint work with Sergei Kuksin and Daniel Peralta-Salas. (Received August 05, 2020)

1160-76-261 **Huan Lei***, 1501 Engineering Bldg., 428 S. Shaw Lane, East Lansing, MI 48823, and **Lei Wu** and **Weinan E**. *Machine learning based non-Newtonian fluid model with molecular fidelity.* Preliminary report.

We introduce a machine-learning-based framework for constructing a continuum non-Newtonian fluid dynamics model directly from a micro-scale description. To faithfully retain molecular fidelity, we establish a micro-macro correspondence via a set of encoders for the micro-scale polymer configurations and their macro-scale counterparts, a set of nonlinear conformation tensors. The dynamics of these conformation tensors can be derived from the micro-scale model and the relevant terms can be parametrized using machine learning. The final model, named the deep non-Newtonian model (DeePN^2), takes the form of conventional non-Newtonian fluid dynamics models, with a new form of the objective tensor derivative. Numerical results demonstrate the accuracy of DeePN^2, where empirical constitutive models show limitations. (Received August 10, 2020)

1160-76-262 **Sam G. Krupa*** (skrupa@math.utexas.edu). *New Developments in the Theory of A-Contraction and Shifts.*

This talk concerns the well-posedness of hyperbolic systems of conservation laws in one space dimension. We have in mind applications to physical systems, such as the compressible Euler system. In this talk, I review the current progress on using the theory of shifts and a-contraction to push forward the theory of well-posedness for systems of conservation laws in one space dimension. (Received August 10, 2020)

1160-76-292 **Tristan Buckmaster**, **Steve Shkoller** and **Vlad Vicol*** (vicol@cims.nyu.edu), Warren Weaver Hall, 251 Mercer St, New York, NY 10012. *Shock formation and vorticity creation for compressible Euler.*

We analyze the shock formation process for the 3D non-isentropic Euler equations with the ideal gas law, in which sound waves interact with entropy waves to produce vorticity. Building on our theory for isentropic flows, we give a constructive proof of shock formation from smooth initial data. (Received August 11, 2020)

1160-76-352 **Jacob Bedrossian*** (jacob@math.umd.edu), **Alex Blumenthal** and **Sam Punshon-Smith**. *The Batchelor spectrum of passive scalar turbulence in stochastic fluid mechanics.*

In 1959, Batchelor predicted that passive scalars advected in fluids at finite Reynolds number with small diffusivity kappa should display a -1 power spectrum over a small-scale inertial range in a statistically stationary experiment. This prediction has been experimentally and numerically tested extensively in the physics and engineering literature and is a core prediction of passive scalar turbulence. Together with Alex Blumenthal and Sam Punshon-Smith, we provide the first mathematically rigorous proof of Batchelor's prediction on the cumulative power spectrum in the kappa -> 0 limit. We consider fluids governed by the 2D Navier-Stokes equations and 3D hyperviscous Navier-Stokes equations in Td forced by sufficiently regular, nondegenerate stochastic forcing at fixed (arbitrary) Reynolds number. The scalar is subjected to a smooth-in-space, white-in-time stochastic source, and evolves by advection-diffusion with diffusivity kappa>0. Our results rely on the quantitative understanding of Lagrangian chaos and passive scalar mixing established in our recent works. (Received August 11, 2020)

1160-76-355 **Hyukjoo Chung***, RISE Group, Cresskill, NJ. *Study of Airfoils Using Computational Fluid Dynamics (CFD).*

In aerodynamic engineering, computational simulation is an effective approach to easily obtaining numerical data for an airfoil design. Calculating the forces acting on an airfoil requires a deep understanding of the aerodynamic feasibility of the airfoil's streamlined body as well as various forces (drag, lift, and primary forces) that act on the airfoil. Throughout this research, mechanical and computational simulations were conducted to examine the properties of different types of airfoils. Fluid, mechanical, and computational analyses were performed to determine the stability, as well as other important properties, of airfoils. The objective of this research utilizes the airfoil's geometrical factors and various Reynolds numbers to find the best-fit airfoil design as well as the highest dynamics efficiency. Data from the narrow airfoil shows that there is no significant change in the correlation between independent and dependent variables as the Reynold number is altered. Correlations including Cd (drag coefficient) vs Alpha (angle of attack), Cl (lift coefficient) vs Alpha, Cl/Cd vs Alpha, and Cl vs Cd are examined; differences and trends are also recorded. (Received August 11, 2020)

81 ► *Quantum theory*

1160-81-154 **Andrew B. Royston*** (abr84@psu.edu), 2201 University Drive, Lemont Furnace, PA 15456. *Accelerating Solitons.*

We present the semiclassical approximation for the effective Hamiltonian of the quantum kink to all orders in the time-derivative expansion. The approximation scheme requires, as input, solutions to a wave-like integro-differential equation that represents a soliton undergoing acceleration. We work in the context of models with kink solitons in two spacetime dimensions, but our methods are purely semiclassical and generalizable. This talk is based on work carried out in collaboration with Ilarion Melnikov and Costis Papageorgakis. (Received August 09, 2020)

82 ► Statistical mechanics, structure of matter

1160-82-75 **Simone Warzel***, Zentrum Mathematik, Boltzmannstr 3, Garching, Germany, and **Chokri Manai**, Zentrum Mathematik, Boltzmannstr 3, Garching, Germany. *Quantum mean-field spin glasses: some recent results.*

The theory of classical mean-field spin glasses is a well-established and celebrated field within probability theory. The addition of a constant perpendicular magnetic field introduces a non-commuting term into the energy of such spin glasses and hence causes quantum effects. In this talk, I will give a brief introduction and describe some recents results concerning the thermodynamic phase diagrams of quantum spin glasses. In particular, I will present a novel quantum version of a Parisi-type formula for the generalized random energy model and its relatives. (Received August 05, 2020)

1160-82-76 **Marcus Michelen** and **Will Perkins*** (math@willperkins.org). *Analyticity for classical gasses via recursion.*

We give a new criterion for a classical gas with a repulsive pair potential to exhibit uniqueness of the infinite volume Gibbs measure and analyticity of the pressure. Our improvement on the bound for analyticity is by a factor e^2 over the classical cluster expansion approach and a factor e over the known limit of cluster expansion convergence. The key ingredients in our proofs include an integral identity for the density of a Gibbs point process and an adaptation of the algorithmic correlation decay method from theoretical computer science. (Received August 05, 2020)

1160-82-98 **Ian Jauslin*** (ijauslin@princeton.edu) and **Elliott H. Lieb**. *Liquid crystals and the Heilmann-Lieb model.*

A liquid crystal is a phase of matter in which order and disorder coexist: for some degrees of freedom, there is order, whereas for others, disorder. Such materials were discovered in the late XIXth century, but it took over a century to understand, from microscopic models, how such phases form. In 1979, O. Heilmann and E.H. Lieb introduced an interacting dimer model with the goal of proving the emergence of such a liquid crystal phase. In this setting, this amounts to showing that dimers spontaneously align, but do not fully crystallize: there is no translational order. Heilmann and Lieb proved that dimers do, indeed, align, and conjectured that there is no translational order. In a joint work with E.H. Lieb, we have proved this conjecture, thus showing the spontaneous emergence of a liquid crystal phase in the Heilmann-Lieb model. (Received August 06, 2020)

1160-82-247 **Paul Dario**, **Matan Harel** and **Ron Peled*** (peledron@tauex.tau.ac.il). *Quantitative estimates for the effect of disorder on low-dimensional lattice systems.*

The addition of an arbitrarily weak random field to low-dimensional classical statistical physics models leads to the "rounding" of first-order phase transitions at all temperatures, as predicted in 1975 by Imry and Ma and proved rigorously in 1989 by Aizenman and Wehr. This phenomenon was recently quantified for the two-dimensional random-field Ising model (RFIM), proving that this model exhibits exponential decay of correlations at all temperatures. The RFIM analysis makes essential use of the model's monotonicity (FKG) properties. We investigate quantitative aspects of the phenomenon for general systems with discrete and continuous symmetries, including Potts, spin O(n), spin glass and height function models. Joint work with Paul Dario and Matan Harel. (Received August 10, 2020)

90 ► Operations research, mathematical programming

1160-90-218 **Yi Sun*** (yisun@math.sc.edu), University of South Carolina, 1523 Greene St., LeConte 411, Columbia, SC 29208, and **Changhui Tan** (tan@math.sc.edu), University of South Carolina, 1523 Greene St., Columbia, SC 29208. *Kinetic Monte Carlo simulations of traffic/pedestrian flows.*

We employ an efficient list-based kinetic Monte Carlo (KMC) method to study traffic flow models on 1D and 2D lattices based on the exclusion principle and Arrhenius microscopic dynamics. This model implements stochastic rules for cars' movement based on the configuration of the traffic ahead of each car. In particular, we compare two different look-ahead rules: one is based on the distance from the car under consideration to the car in front of it, the other one depends on the car density ahead. Both rules feature a novel idea of multiple moves, which plays a key role in recovering the nonconcave flux in the macroscopic dynamics. Our results show that the fluxes of the KMC simulations agree with the coarse-grained macroscopic averaged fluxes for the different look-ahead

rules under various parameter settings. If time permits, we will also show some KMC results of 2D pedestrian flow models. (Received August 10, 2020)

92 ► Biology and other natural sciences

1160-92-108 **Stanca M Ciupe*** (stanca@vt.edu), 416 McBryde Hall, Blacksburg, VA 24060. *Inoculum size, immune responses, and the nonlinear host-virus interactions.*

Experimental studies have shown that immune protection and disease severity do not correlate linearly with the size and infectious-stage of viral inoculum. We use mathematical models in connection with data to better represent the relationship between the inoculum dose and disease outcome. In this talk, I will present two case studies: simian immunodeficiency virus infection in rhesus macaques and hepatitis B virus infection in non-human primates to provide hypotheses on when different inoculum doses trigger immune responses that provide protection, induce immune tolerance and chronic disease, and/or lead to pathogenesis. Such results can guide our understanding of the virus-host dynamics that control virus infections or permit a transition to chronic disease. (Received August 07, 2020)

1160-92-186 **Marc Sturrock***, Royal College of Surgeons in Ireland, Dublin, Ireland. *Mathematical modelling of emergent gene expression.*

A number of recent studies have shown that natural selection occurs not only on genes (on genetic variation), but also at the level of gene expression (on heritable phenotypic variation). Emergent gene expression can be defined as the selective upregulation of a fitness conferring gene when a cell population is placed in a stressful environment. One of the clearest example of this phenomena is concerning the expression of the HisC gene in E. Coli after cells are placed in a histidine deprived environment. The HisC enzyme plays a crucial role in the synthesis of Histidine, an essential amino acid required for cell growth. In this talk, I will present a minimal mathematical model of HisC expression that links stochastic gene expression to cell growth and partitioning/dilution. I will then go on to present the minimal ingredients required for emergent gene expression to take place. I will end by changing contexts and present a similar model of MGMT expression in glioblastoma cells which may explain a possible mechanism by which chemotherapy resistance can develop in the absence of genetic mutations. (Received August 10, 2020)

1160-92-201 **Jane Heffernan*** (jmheffer@yorku.ca), 4700 Keele St, Mathematics & Statistics, York University, Toronto, ON M3J 1P3, Canada. *Models of waning immunity.*

Immunity is gained from infection and/or vaccination. An effect of immunity is in the ability of a host's body to resist infection. This effect, however, may be less likely to occur as time since vaccination increases – immunity can wane over time, allowing asymptomatic or mild infections, or severe infections if immunity fully wanes. In this talk I will introduce models of waning immunity applied to different childhood and respiratory diseases. I will then discuss the effect of immunity waning rates on herd immunity, and on infection observations in a population. (Received August 10, 2020)

1160-92-245 **Megan A. Greischar***, megan.greischar@cornell.edu, and **Nicholas J. Savill**, **Sarah E. Reece** and **Nicole Mideo**. *New approach for quantifying synchrony in malaria infections.* Preliminary report.

Malaria infection is often characterized by periodic fevers, caused when the parasite population within a host proliferates in a synchronized burst. Not all infections exhibit such synchrony, however, and whether infections are synchronized (or not) could influence host symptoms, parasite transmission and the efficacy of antimalarial drug treatment. Yet the causes of synchrony remain unclear, including to what extent hosts versus parasites influence the timing of infection dynamics. A key challenge is quantifying the degree of synchrony within an infection, and I use a heuristic model (a Leslie matrix) to show that existing methods to quantify synchrony are inadequate. I then use a system of delayed differential equations to simulate time series and use these simulated data to validate an improved approach. This new approach can quantify wholesale differences in synchrony as well as changes in synchrony over the course of an infection. Simulating 'experiments' reveals best practices for designing experiments to minimize the impact of sampling noise and discern true patterns of synchrony. (Received August 10, 2020)

1160-92-275 **Duc D Nguyen*** (ducnguyen@uky.edu). *Opportunities and challenges for AI and Math in drug discovery.*

Drug discovery is one of the most challenging tasks in the biological sciences since it requires over 10 years and costs more than $2.6 billion to put an average novel medicine on the marketplace. The abundant availability of biological data along with the flourishing advanced AI algorithms opens a future with great hope for discovering new drugs faster and cheaper. Unfortunately, AI faces an enormous obstacle in drug discovery due to the intricate complexity of biomolecular structures and the high dimensionality of biological datasets. In our lab, these challenges have been tackled mathematically. We have introduced multiscale modeling, differential geometry, algebraic topology, and graph theory-based models to systematically represent the diverse biological datasets in the low-dimensional spaces. Combining these mathematical representations with cutting edge deep neural networks, we arrived at novel models not only perform well on virtual-screening targeting important drug properties but also have the ability to design new drugs at an unprecedented speed. Our team has emerged as a top winner in D3R Grand Challenges, a worldwide annual competition series in computer-aided drug design, in the past few years. (Received August 11, 2020)

1160-92-294 **Naveen K Vaidya*** (nvaidya@sdsu.edu). *Persistence of HIV under antiretroviral therapy: The Role of the Brain as a Reservoir.*

It is not fully understood whether the brain plays a role of viral reservoir causing HIV persistence under antiretroviral therapy. In this talk, I will present a novel mathematical model describing virus dynamics under antiretroviral therapy to study the role of the brain in virus persistence. Using experimental data from SIV infected macaques, we identify key parameters related to the brain infection, including virus-transfer across blood-brain barrier. Our model predicts that the brain can be an important reservoir causing long-term virus persistence in the brain, despite successful control of viral load in the plasma by antiretroviral drugs. (Received August 11, 2020)

1160-92-295 **Libin Rong*** (libinrong@ufl.edu). *Modeling the role of macrophages in HIV persistence.*

HIV preferentially infects activated CD4+ T cells. Current antiretroviral therapy cannot eradicate the virus. Viral infection of other cells such as macrophages may contribute to viral persistence during antiretroviral therapy. In addition to cell-free virus infection, macrophages can also get infected when engulfing infected CD4+ T cells as innate immune sentinels. How macrophages affect the dynamics of HIV infection remains unclear. In this talk, I will present a new HIV model that includes the infection of CD4+ T cells and macrophages via cell-free virus infection and cell-to-cell viral transmission. I will show the analysis of the model and discuss the roles of macrophages in HIV persistence during treatment. (Received August 11, 2020)

1160-92-306 **James Watmough*** (watmough@unb.ca), Department of Mathematics and Statistics, University of New Brunswick, University of New Brunswick, Fredericton, NB E3B5A3, Canada. *Infections, invasions, sustained immunity, and alternate stable states.*

In many models, treatment works by lowering the reproduction number of the pathogen below a threshold, and the pathogen reestablishes itself if the treatment is stopped. In this talk, I present a simple model of a case where treatment can be stopped without reemergence of the pathogen. A situation referred to as sustained-immunity. This is possible because a stable infection-equilibrium coexists with a stable infection-free or low-infection state. Treatment, if applied for a sufficient time, can move the system into the basin of attraction for the infection-free equilibrium, and immunity is sustained after treatment is stopped. Although the main focus of the talk will be what is primarily an immune system model, references will be made to similar dynamics in ecological and epidemiological systems. (Received August 11, 2020)

1160-92-308 **Christiaan H van Dorp*** (chvandorp@lanl.gov), Los Alamos National Laboratory, Mailstop K710, Los Alamos, NM 87545, and **Jessica M Conway**, **Dan H Barouch**, **James B Whitney** and **Alan S Perelson**. *Modelling HIV rebound after antiretroviral treatment interruption using SDEs with jumps.*

HIV can be controlled by antiretroviral therapy (ART), suppressing the viral load (VL) to undetectable levels. But infection persists despite ART, due to a reservoir of latently infected cells. When ART is stopped, the VL rebounds (becomes detectable) typically within weeks. Therefore, a HIV infection requires life-long ART. Encouraged by the finding that some patients do not rebound for months or even years, novel treatments are being tested for their ability to reduce the reservoir size, aiming to prolong remission after ART interruption. These treatments are often first tested in macaques, and their efficacy is assessed by stopping ART and measuring the time to rebound. In support of this approach, mathematical models connect the size of the reservoir and the rebound time. The simplest model assumes that a single latent cell reactivation leads to rebound after a period

of exponential VL growth. In reality, multiple latently infected cells reactivate, leading to biased estimates. Therefore, we study a rebound model that takes multiple reactivations into account. We represent this model as a SDE with jumps and use the Kramers-Moyal expansion to derive a PDF for the rebound time. This parametric PDF can be used for inference, which we show with data from SIV infected macaques. (Received August 11, 2020)

1160-92-341 **Elsa Hansen*** (ekh13@psu.edu), PA. *Optimal removal of drug-sensitive bacteria to manage infection.*

Mathematical, animal and clinical studies suggest that we can improve patient outcomes by leveraging competition between drug-resistant and drug-sensitive bacteria. The best way to use this competition depends on the details of the bacterial populations. I will present a mathematical framework and corresponding optimization problem for determining the optimal removal of drug-sensitive bacteria. Although the optimization problem can be solved using impulsive control theory and Pontryagin's maximum principle, I will show that the main results can also be derived using simple graphical arguments. (Received August 11, 2020)

1160-92-348 **Jessica M Conway***, jmconway@psu.edu. *SIV infection dynamics in vaginal tissues.* Preliminary report.

Understanding the events that occur following HIV exposure is critical to the development of interventions, but these events are difficult to study experimentally. Nonhuman primate (NHP) experiments with SIV from the Keele Lab shed light on the dynamics of early viral replication and development of systemic dissemination following vaginal exposure. Importantly, the infecting inoculum was a synthetic viral swarm permitting distinction between genotypically separate, but phenotypically identical variants. The data reveals that SIV form foci of local infection in the female genital tract (FGT), with viral variants from multiple viral lineages were found within each focus. We use data on viral variants in foci to validate our FGT viral dynamics model, relying on multitype continuous time branching processes. From this formulation we will derive time-dependent probabilities of extinction and viral variant proportion. Using resulting model predictions we will explore in particular the role of the infected cell burst size to examine the survival of viral lineages in the vaginal tissues preceding systemic dissemination. Early results suggest that the variability in the burst size can explain the variability in proportions of viral variants observed within foci of infection. (Received August 11, 2020)

2010 MATHEMATICS SUBJECT CLASSIFICATION

Compiled in the Editorial Offices of MATHEMATICAL REVIEWS and ZENTRALBLATT MATH

00 General
01 History and biography
03 Mathematical logic and foundations
05 Combinatorics
06 Order, lattices, ordered algebraic structures
08 General algebraic systems
11 Number theory
12 Field theory and polynomials
13 Commutative rings and algebras
14 Algebraic geometry
15 Linear and multilinear algebra; matrix theory
16 Associative rings and algebras
17 Nonassociative rings and algebras
18 Category theory; homological algebra
19 *K*-theory
20 Group theory and generalizations
22 Topological groups, Lie groups
26 Real functions
28 Measure and integration
30 Functions of a complex variable
31 Potential theory
32 Several complex variables and analytic spaces
33 Special functions
34 Ordinary differential equations
35 Partial differential equations
37 Dynamical systems and ergodic theory
39 Difference and functional equations
40 Sequences, series, summability
41 Approximations and expansions
42 Fourier analysis
43 Abstract harmonic analysis
44 Integral transforms, operational calculus
45 Integral equations
46 Functional analysis
47 Operator theory
49 Calculus of variations and optimal control; optimization
51 Geometry
52 Convex and discrete geometry
53 Differential geometry
54 General topology
55 Algebraic topology
57 Manifolds and cell complexes
58 Global analysis, analysis on manifolds
60 Probability theory and stochastic processes
62 Statistics
65 Numerical analysis
68 Computer science
70 Mechanics of particles and systems
74 Mechanics of deformable solids
76 Fluid mechanics
78 Optics, electromagnetic theory
80 Classical thermodynamics, heat transfer
81 Quantum theory
82 Statistical mechanics, structure of matter
83 Relativity and gravitational theory
85 Astronomy and astrophysics
86 Geophysics
90 Operations research, mathematical programming
91 Game theory, economics, social and behavioral sciences
92 Biology and other natural sciences
93 Systems theory; control
94 Information and communication, circuits
97 Mathematics education

2689-4831(2020)41:3;1-J

AMS
www.ams.org